

# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

**Department of Mineral Resources**

Lynn D. Helms - Director

**North Dakota Industrial Commission**

[www.dmr.nd.gov/oilgas/](http://www.dmr.nd.gov/oilgas/)

April 6, 2018

Ms. Donna Patocka  
Continental Resources, Inc.  
P.O. Box 269000  
Oklahoma City, OK 73126

**RE: Tallahassee #3-16H  
SENE Sec. 16, T.153N., R.101W.  
McKenzie County, North Dakota  
Baker Field  
Well File No. 25157  
STRIPPER WELL DETERMINATION**

Dear Ms. Patocka:

Continental Resources, Inc. (Continental) filed with the North Dakota Industrial Commission – Oil and Gas Division (Commission) on March 9, 2018 an application for a Stripper Well Determination for the above captioned well.

Information contained in the application indicates that the above mentioned well is a stripper well pursuant to statute and rule, and Continental has elected to designate said well as a stripper well. The well produced from a well depth greater than 10000 feet and was completed after June 30, 2013. During the qualifying period, October 1, 2016 through September 30, 2017, the well produced at a maximum efficient rate or was not capable of exceeding the production threshold. The average daily production from the well was 32.0 barrels of oil per day during this period.

It is therefore determined that the above captioned well qualifies as a “Stripper Well” pursuant to Section 57-51.1-01 of the North Dakota Century Code. This determination is applicable only to the Bakken Pool in and under said well.

The Commission shall have continuing jurisdiction, and shall have the authority to review the matter, and to amend or rescind the determination if such action is supported by additional or newly discovered information. If you have any questions, do not hesitate to contact me.

Sincerely,

David J. McCusker  
Petroleum Engineer

Cc: ND Tax Department



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**  
INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)

Well File No. 25157

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <input type="checkbox"/> Notice of Intent                                                                                                       | Approximate Start Date                          |
| <input checked="" type="checkbox"/> Report of Work Done                                                                                         | Date Work Completed<br><b>December 29, 2017</b> |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                          |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>NDAC 43-02-03-49 Compliance</b>                |

|                                             |   |         |         |          |       |
|---------------------------------------------|---|---------|---------|----------|-------|
| Well Name and Number<br><b>See Attached</b> |   |         |         |          |       |
| Footages                                    |   | Qtr-Qtr | Section | Township | Range |
| F                                           | L |         |         |          |       |
| Field                                       |   | Pool    | County  |          |       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Continental Resources, Inc. has inspected all the sites listed on the attached list and all are in compliance with NDAC 43-02-03-49. All sites are in compliance based on the fact that our oil storage tanks, flow-through process vessels, recycle pumps, and load lines are all located within secondary containment.

*confirmed per RSD*

|                                                  |                                      |                                           |                          |
|--------------------------------------------------|--------------------------------------|-------------------------------------------|--------------------------|
| Company<br><b>Continental Resources, Inc.</b>    |                                      | Telephone Number<br><b>(405) 234-9020</b> |                          |
| Address<br><b>PO Box 268870</b>                  |                                      |                                           |                          |
| City<br><b>Oklahoma City</b>                     |                                      | State<br><b>OK</b>                        | Zip Code<br><b>73126</b> |
| Signature<br><i>Robert Sandbo</i>                | Printed Name<br><b>Robert Sandbo</b> |                                           |                          |
| Title<br><b>Regulatory Compliance Supervisor</b> | Date<br><b>February 8, 2018</b>      |                                           |                          |
| Email Address<br><b>robert.sandbo@clr.com</b>    |                                      |                                           |                          |

| FOR STATE USE ONLY                |                                              |
|-----------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received | <input checked="" type="checkbox"/> Approved |
| Date                              | <b>3/22/18</b>                               |
| By                                | <i>Robert E. Anderson</i>                    |
| Title                             | <i>Regulatory Compliance Supervisor</i>      |





# Oil and Gas Division

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North Dakota Industrial Commission

[www.dmr.nd.gov/oilgas](http://www.dmr.nd.gov/oilgas)

August 29, 2017

CONTINENTAL RESOURCES  
ATTENTION: BOB SANDBO  
P.O. BOX 268870  
OKLAHOMA CITY, OK 73126

RE: North Dakota Administrative Code (NDAC) Section 43-02-03-49  
Perimeter Berm Requirement

NDIC # Please see attached list of 103 Facilities

Dear Bob Sandbo:

Please be advised that the attached list of facilities require a perimeter berm to be constructed within 180 days of this notice because they have:

1. Storage tanks;
2. Daily throughput of more than one hundred barrels of fluid per day; and
3. Includes production equipment or load lines that are not contained within secondary containment dikes

The berm must be at least six inches in height, constructed of sufficiently impermeable material to provide emergency containment, and must be maintained until the facility is either granted a waiver or the site is reclaimed.

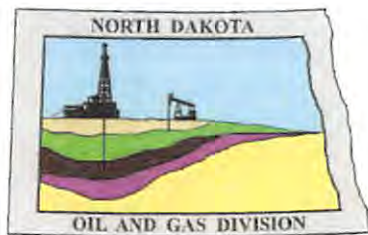
Pursuant to NDAC Section 43-02-03-49 - Within one hundred eighty days from the date the operator is notified by the Commission, a perimeter berm, at least six inches in height, must be constructed of sufficiently impermeable material to provide emergency containment and to divert surface drainage away from the site around all storage facilities and production sites that include storage tanks, have a daily throughput of more than one hundred barrels of fluid per day, and include production equipment or load lines that are not contained within secondary containment dikes. The Director may consider an extension of time to implement these requirements if conditions prevent timely construction, or modification of these requirements if other factors are present that provide sufficient protection from environmental impacts.

This perimeter berm requirement may be modified or waived if the operator can demonstrate that other factors are present that provide sufficient protection from environmental impacts. A Sundry Notice (Form 4) outlining any engineering controls or other factors must be submitted to the Commission for approval of this modification or waiver. Should you have any questions regarding this matter, feel free to contact me at 701-770-3554.

Sincerely,

  
Richard Dunn  
Field Inspector





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[www.dmr.nd.gov/oilgas](http://www.dmr.nd.gov/oilgas)

| NDIC # | Facility Name    | TB/CTB #  | NDIC # | Facility Name            | TB/CTB # |
|--------|------------------|-----------|--------|--------------------------|----------|
| 19126  | LANSING 1-25H    |           | 24834  | DURHAM 3X-2H             |          |
| 19578  | MISSOULA 1-21H   |           | 24837  | WAHPETON 2-16H2          |          |
| 19740  | SYRACUSE 1-23H   |           | 24840  | WAHPETON 4-16H1          |          |
| 19858  | JAMESTOWN 1-17H  |           | 24842  | WAHPETON 5-16H2          |          |
| 19915  | STEELE 1-24H     |           | 24843  | WAHPETON 6-16H           |          |
| 19918  | CHARLOTTE 1-22H  |           | 24844  | WAHPETON 7-16H3          |          |
| 20566  | MONTPELIER 1-14H |           | 24908  | CHARLOTTE 6-22H2         |          |
| 20629  | PATTERSON 1-13H  |           | 25116  | BJARNE 2-29H             |          |
| 20638  | KUHN 1-12H       |           | 25117  | BJARNE 3-29H             |          |
| 20676  | NORFOLK 1-1H     |           | 25156  | COLUMBUS FEDERAL 1-16H   |          |
| 21128  | CHARLOTTE 2-22H  |           | 25157  | TALLAHASSEE 3-16H        |          |
| 21511  | BOULDER 1-4H     |           | 25158  | TALLAHASSEE 2-16H        |          |
| 21600  | ROCHESTER 1-24H  |           | 25159  | COLUMBUS FEDERAL 2-16H   |          |
| 22155  | LANSING 2-25H    |           | 25160  | COLUMBUS FEDERAL 3-16H   |          |
| 22158  | KUHN 2-12H       |           | 25189  | PATTERSON FEDERAL 2-13H  |          |
| 22273  | STEELE 2-24H     |           | 25190  | PATTERSON FEDERAL 3-13H  |          |
| 22375  | CHICAGO 2-26H    |           | 25826  | AKRON 6-34H1             |          |
| 22891  | MONROE 1-2H      |           | 25827  | AKRON 5-34H1             |          |
| 23048  | CHICAGO 3-26H    |           | 26190  | MONTPELIER 4-14H         |          |
| 23049  | CHICAGO 4-26H    |           | 26191  | MONTPELIER 3-14H1        |          |
| 23050  | SYRACUSE 3-23H   |           | 26420  | MONTPELIER 2-14H         |          |
| 23051  | SYRACUSE 4-23H   |           | 26476  | ROCHESTER FEDERAL 6-24H  |          |
| 23086  | NORFOLK 2-1H     |           | 26477  | ROCHESTER FEDERAL 7-24H1 |          |
| 23087  | NORFOLK 3-1H     |           | 26525  | JERRY 2-8H               |          |
| 23351  | MISSOULA 2-21H   |           | 26526  | JERRY 3-8H               |          |
| 23352  | MISSOULA 3-21H   |           | 26530  | JERRY 5-8H               |          |
| 23427  | MISSOULA 7-21H   |           | 26531  | JERRY 4-8H               |          |
| 23428  | MISSOULA 6-21H   |           | 26535  | JERRY 7-8H               |          |
| 23429  | MISSOULA 5-21H   |           | 26536  | JERRY 6-8H               |          |
| 23430  | MISSOULA 4-21H   |           | 27418  | GARFIELD FEDERAL 7-5H1   |          |
| 23477  | DURHAM 2-2H      |           | 27419  | GARFIELD FEDERAL 6-5H    |          |
| 23608  | CHARLOTTE 5-22H  |           | 27420  | GARFIELD FEDERAL 5-5H1   |          |
| 23609  | AKRON 3-27AH     |           | 27421  | GARFIELD 4-5H            |          |
| 23610  | AKRON 2-27AH     |           | 27694  | BERLAIN 3-30H            |          |
| 23611  | AKRON 4-34H      |           | 27695  | BERLAIN 2-30H            |          |
| 23612  | CHARLOTTE 4-22H  |           | 28202  | JAMESTOWN FEDERAL 2-17H  |          |
| 23664  | CHARLOTTE 3-22H  |           | 28203  | JAMESTOWN FEDERAL 3-17H1 |          |
| 23747  | ROCHESTER 3-24H  |           | 28405  | HARRISBURG 1-34H         |          |
| 23748  | ROCHESTER 2-24H  |           | 28604  | JAMESTOWN FEDERAL 6-17H  |          |
| 23749  | ROCHESTER 5-24H1 |           | 28605  | JAMESTOWN FEDERAL 7-17H  |          |
| 23750  | ROCHESTER 4-24H  |           | 28735  | DURHAM 7-2H              |          |
| 24490  | DURHAM 4-2H      |           | 28736  | DURHAM 6-2H1             |          |
| 24491  | DURHAM 5-2H      |           | 28737  | UHLMAN 1-7H              |          |
| 24507  | NORFOLK 5-1H     |           | 28999  | NORFOLK 6-1H1            |          |
| 24508  | NORFOLK 4-1H     |           | 29000  | NORFOLK 7-1H             |          |
| 24804  | WAHPETON 14-16H2 | 224837-01 | 31508  | AKRON FEDERAL 7-27H      |          |
| 24805  | WAHPETON 13-16H  |           | 31838  | CHARLOTTE 7X-22H         |          |
| 24806  | WAHPETON 12-16H3 |           | 32033  | NORFOLK 11-1H            |          |
| 24807  | WAHPETON 11-16H1 |           | 32034  | NORFOLK 10-1H1           |          |
| 24808  | WAHPETON 10-16H2 |           | 32035  | NORFOLK 9-1H             |          |
| 24809  | WAHPETON 9-16H   |           | 32036  | NORFOLK 8-1H1            |          |
| 24810  | WAHPETON 8-16H1  |           |        |                          |          |



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)

**Received****OCT 05 2016**

Well File No.

**25157****ND Oil & Gas Division**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                              |                                               |                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/> Notice of Intent                                                                                                       | Approximate Start Date                       | <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report                        |
| <input checked="" type="checkbox"/> Report of Work Done                                                                                         | Date Work Completed<br><b>August 2, 2016</b> | <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                            |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                       | <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                           |
|                                                                                                                                                 |                                              | <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment                  |
|                                                                                                                                                 |                                              | <input type="checkbox"/> Supplemental History | <input checked="" type="checkbox"/> Change Production Method |
|                                                                                                                                                 |                                              | <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation                         |
|                                                                                                                                                 |                                              | <input type="checkbox"/> Other                |                                                              |

|                                                  |               |                 |              |              |  |
|--------------------------------------------------|---------------|-----------------|--------------|--------------|--|
| Well Name and Number<br><b>Tallahassee 3-16H</b> |               |                 |              |              |  |
| Footages                                         | Qtr-Qtr       | Section         | Township     | Range        |  |
| <b>2585 F N L 268 F E L</b>                      | <b>SENE</b>   | <b>16</b>       | <b>153 N</b> | <b>101 W</b> |  |
| Field                                            | Pool          | County          |              |              |  |
| <b>Baker</b>                                     | <b>Bakken</b> | <b>McKenzie</b> |              |              |  |

| 24-HOUR PRODUCTION RATE |         |       |          |
|-------------------------|---------|-------|----------|
| Before                  |         | After |          |
| Oil                     | 41 Bbls | Oil   | 44 Bbls  |
| Water                   | 75 Bbls | Water | 108 Bbls |
| Gas                     | 65 MCF  | Gas   | 68 MCF   |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Continental Resources, Inc. requests a change in production method on the above mentioned well. The well went from flowing to Rod Pump on 8/2/2016. New Tubing: 1 3/4 Depth: 9928

|                                            |                                   |                                           |                          |
|--------------------------------------------|-----------------------------------|-------------------------------------------|--------------------------|
| Company<br><b>Continental Resources</b>    |                                   | Telephone Number<br><b>(405) 234-9688</b> |                          |
| Address<br><b>P.O. Box 268870</b>          |                                   |                                           |                          |
| City<br><b>Oklahoma City</b>               |                                   | State<br><b>OK</b>                        | Zip Code<br><b>73126</b> |
| Signature<br>                              | Printed Name<br><b>Zach Green</b> |                                           |                          |
| Title<br><b>Regulatory Specialist</b>      | Date<br><b>September 30, 2016</b> |                                           |                          |
| Email Address<br><b>Zach.Green@clr.com</b> |                                   |                                           |                          |

**FOR STATE USE ONLY**

|                                                              |                                   |
|--------------------------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received                 | <input type="checkbox"/> Approved |
| Date<br><b>11-2-2016</b>                                     |                                   |
| By<br>                                                       |                                   |
| Title<br><b>TAYLOR ROTH</b><br><b>Engineering Technician</b> |                                   |

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - FORM 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5698 (03-2000)

|                               |
|-------------------------------|
| Well File No.<br><b>25157</b> |
| NDIC CTB No.<br><b>125157</b> |

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

|                                                  |                                         |                        |                           |                       |                           |
|--------------------------------------------------|-----------------------------------------|------------------------|---------------------------|-----------------------|---------------------------|
| Well Name and Number<br><b>Tallahassee 3-16H</b> | Qtr-Of.<br><b>SENE</b>                  | Section<br><b>2716</b> | Township<br><b>6132 N</b> | Range<br><b>101 W</b> | County<br><b>McKenzie</b> |
| Operator<br><b>Continental Resources Inc.</b>    | Telephone Number<br><b>405-234-9000</b> | Field<br><b>Baker</b>  |                           |                       |                           |
| Address<br><b>P.O. Box 268870</b>                | City<br><b>Oklahoma City</b>            | State<br><b>OK</b>     | Zip Code<br><b>73126</b>  |                       |                           |

|                                                               |                                         |                           |                                         |
|---------------------------------------------------------------|-----------------------------------------|---------------------------|-----------------------------------------|
| Name of First Purchaser<br><b>Continental Resources, Inc.</b> | Telephone Number<br><b>405-234-9000</b> | % Purchased<br><b>100</b> | Date Effective<br><b>April 10, 2014</b> |
| Principal Place of Business<br><b>P.O. Box 268870</b>         | City<br><b>Oklahoma City</b>            | State<br><b>OK</b>        | Zip Code<br><b>73126</b>                |
| Field Address                                                 | City                                    | State                     | Zip Code                                |
| Name of Transporter<br><b>Hiland Crude, LLC</b>               | Telephone Number<br><b>580-616-2053</b> | % Transported             | Date Effective<br><b>April 10, 2014</b> |
| Address<br><b>P.O. Box 3886</b>                               | City<br><b>Enid</b>                     | State<br><b>Ok</b>        | Zip Code<br><b>73702</b>                |

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

|                                                   |               |                |
|---------------------------------------------------|---------------|----------------|
| Other First Purchasers Purchasing From This Lease | % Purchased   | Date Effective |
| Other First Purchasers Purchasing From This Lease | % Purchased   | Date Effective |
| Other Transporters Transporting From This Lease   | % Transported | Date Effective |
| Other Transporters Transporting From This Lease   | % Transported | Date Effective |

|          |
|----------|
| Comments |
|----------|

|                                                                                                                                |                                            |                                                          |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. |                                            | Date<br><b>April 11, 2014</b>                            |
| Signature<br><i>Becky Barnes</i>                                                                                               | Printed Name<br><b>Becky Barnes</b>        | Title<br><b>Regulatory Compliance Specialist</b>         |
| Above Signature Witnessed By                                                                                                   |                                            |                                                          |
| Witness Signature<br><i>Terry Olson</i>                                                                                        | Witness Printed Name<br><b>Terry Olson</b> | Witness Title<br><b>Regulatory Compliance Specialist</b> |

FOR STATE USE ONLY

|                                                  |
|--------------------------------------------------|
| Date Approved<br><b>MAY 16 2014</b>              |
| By<br><i>Annette Tredger</i>                     |
| Title<br><b>Oil &amp; Gas Production Analyst</b> |

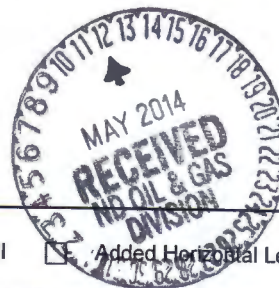




# WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 2468 (04-2010)

Well File No.  
**25157**



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                  |                                   |                                            |                                                                                                                                    |                                               |                                                  |
|--------------------------------------------------|-----------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| Designate Type of Completion                     |                                   |                                            |                                                                                                                                    |                                               |                                                  |
| <input checked="" type="checkbox"/> Oil Well     | <input type="checkbox"/> EOR Well | <input type="checkbox"/> Recompletion      | <input type="checkbox"/> Deepened Well                                                                                             | <input type="checkbox"/> Added Horizontal Leg | <input type="checkbox"/> Extended Horizontal Leg |
| <input type="checkbox"/> Gas Well                | <input type="checkbox"/> SWD Well | <input type="checkbox"/> Water Supply Well | <input type="checkbox"/> Other:                                                                                                    |                                               |                                                  |
| Well Name and Number<br><b>Tallahassee 3-16H</b> |                                   |                                            | Spacing Unit Description<br><b>Sec 4, 9, 16, &amp; 21-153-101</b>                                                                  |                                               |                                                  |
| Operator<br><b>Continental Resources, Inc.</b>   |                                   | Telephone Number<br><b>405-234-9000</b>    |                                                                                                                                    | Field<br><b>Baker</b>                         |                                                  |
| Address<br><b>P.O. Box 268870</b>                |                                   |                                            | Pool<br><b>Bakken</b>                                                                                                              |                                               |                                                  |
| City<br><b>Oklahoma City</b>                     | State<br><b>Ok</b>                | Zip Code<br><b>73126</b>                   | Permit Type<br><input type="checkbox"/> Wildcat <input checked="" type="checkbox"/> Development <input type="checkbox"/> Extension |                                               |                                                  |

## LOCATION OF WELL

|                                                                                                      |                                      |                                                        |  |                        |                      |                                  |                                      |                           |
|------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------|--|------------------------|----------------------|----------------------------------|--------------------------------------|---------------------------|
| At Surface<br><b>2585 F N L</b>                                                                      |                                      | <b>268 F E L</b>                                       |  | Qtr-Qtr<br><b>SENE</b> | Section<br><b>16</b> | Township<br><b>153 N</b>         | Range<br><b>101 W</b>                | County<br><b>McKenzie</b> |
| Spud Date<br><b>8/16/2013</b>                                                                        | Date TD Reached<br><b>11/21/2013</b> | Drilling Contractor and Rig Number<br><b>Cyclone 4</b> |  |                        |                      | KB Elevation (Ft)<br><b>1940</b> | Graded Elevation (Ft)<br><b>1920</b> |                           |
| Type of Electric and Other Logs Run (See Instructions)<br><b>CBL/GR, Mud Logs, Geological Report</b> |                                      |                                                        |  |                        |                      |                                  |                                      |                           |

## CASING & TUBULARS RECORD (Report all strings set in well)

| Well Bore | String Type  | Size (Inch) | Top Set (MD Ft) | Depth Set (MD Ft) | Hole Size (Inch) | Weight (Lbs/Ft) | Anchor Set (MD Ft) | Packer Set (MD Ft) | Sacks Cement | Top of Cement |
|-----------|--------------|-------------|-----------------|-------------------|------------------|-----------------|--------------------|--------------------|--------------|---------------|
| Lateral1  | Conductor    | 16          |                 | 101               | 20               |                 |                    |                    | 7 yds        |               |
|           | Surface      | 9 5/8       |                 | 1985              | 13 1/2           | 36              |                    |                    | 1008         |               |
|           | Intermediate | 7           |                 | 10830             | 8 3/4            | 26-29           |                    |                    | 1028         | 1800          |
|           | Liner        | 4 1/2       | 10066           | 18270             | 6                | 11.6            |                    |                    |              |               |
|           | Tubing       | 2 7/8       |                 |                   | 4                | 6.5             |                    |                    |              |               |
|           |              |             |                 |                   |                  |                 |                    |                    |              |               |
|           |              |             |                 |                   |                  |                 |                    |                    |              |               |
|           |              |             |                 |                   |                  |                 |                    |                    |              |               |

## PERFORATION & OPEN HOLE INTERVALS

| Well Bore | Well Bore TD Drillers Depth (MD Ft) | Completion Type | Open Hole/Perforated Interval (MD,Ft) |       | Kick-off Point (MD Ft) | Top of Casing Window (MD Ft) | Date Perfd or Drilled | Date Isolated | Isolation Method | Sacks Cement |
|-----------|-------------------------------------|-----------------|---------------------------------------|-------|------------------------|------------------------------|-----------------------|---------------|------------------|--------------|
| Lateral1  | 18434                               | Perforations    | 10830                                 | 18434 | 10137                  | 1800                         | 11/23/2013            |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |
|           |                                     |                 |                                       |       |                        |                              |                       |               |                  |              |

## PRODUCTION

|                                                                                                                                  |                           |                                    |                     |                             |                          |                            |                                                                  |                                   |  |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------|---------------------|-----------------------------|--------------------------|----------------------------|------------------------------------------------------------------|-----------------------------------|--|
| Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)<br><b>Bakken 10,830'/18,434'</b> |                           |                                    |                     |                             |                          |                            | Name of Zone (If Different from Pool Name)<br><b>Three Forks</b> |                                   |  |
| Date Well Completed (SEE INSTRUCTIONS)<br>4/21/2014                                                                              |                           | Producing Method<br><b>Flowing</b> |                     | Pumping-Size & Type of Pump |                          |                            | Well Status (Producing or Shut-In)<br><b>Producing</b>           |                                   |  |
| Date of Test<br><b>4/23/2014</b>                                                                                                 | Hours Tested<br><b>24</b> | Choke Size<br><b>20 /64</b>        | Production for Test | Oil (Bbls)<br><b>314</b>    | Gas (MCF)<br><b>261</b>  | Water (Bbls)<br><b>596</b> | Oil Gravity-API (Corr.)<br><b>°</b>                              | Disposition of Gas<br><b>Sold</b> |  |
| Flowing Tubing Pressure (PSI)<br><b>650</b>                                                                                      |                           | Flowing Casing Pressure (PSI)      |                     | Calculated 24-Hour Rate     | Oil (Bbls)<br><b>314</b> | Gas (MCF)<br><b>261</b>    | Water (Bbls)<br><b>596</b>                                       | Gas-Oil Ratio<br><b>831</b>       |  |



## GEOLOGICAL MARKERS

[illegible]

## PLUG BACK INFORMATION

[illegible]

## CORES CUT

| Top (Ft) | Bottom (Ft) | Formation | Top (Ft) | Bottom (Ft) | Formation |
|----------|-------------|-----------|----------|-------------|-----------|
|          |             |           |          |             |           |
|          |             |           |          |             |           |
|          |             |           |          |             |           |

### Drill Stem Test

| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
|-------------------------|-----------|----------|-------------|--------------|--------|---------|------------------|------------------|
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |



### Well Specific Stimulations

|                                                                                            |                                            |                                |                                                 |                                 |                                                  |                                |
|--------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------|-------------------------------------------------|---------------------------------|--------------------------------------------------|--------------------------------|
| Date Stimulated<br><b>3/14/2014</b>                                                        | Stimulated Formation<br><b>Three Forks</b> | Top (Ft)<br><b>10830</b>       | Bottom (Ft)<br><b>18434</b>                     | Stimulation Stages<br><b>24</b> | Volume<br><b>64952</b>                           | Volume Units<br><b>Barrels</b> |
| Type Treatment<br><b>Sand Frac</b>                                                         | Acid %                                     | Lbs Proppant<br><b>2829582</b> | Maximum Treatment Pressure (PSI)<br><b>8491</b> |                                 | Maximum Treatment Rate (BBLS/Min)<br><b>31.0</b> |                                |
| Details<br><b>Pumped 89291# 40/70 mesh, 1970321# 20/40 sand and 769970# 20/40 ceramic.</b> |                                            |                                |                                                 |                                 |                                                  |                                |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

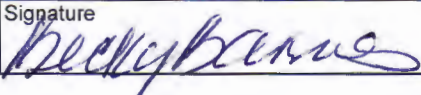
|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

|                 |                      |              |                                  |                    |                                   |              |
|-----------------|----------------------|--------------|----------------------------------|--------------------|-----------------------------------|--------------|
| Date Stimulated | Stimulated Formation | Top (Ft)     | Bottom (Ft)                      | Stimulation Stages | Volume                            | Volume Units |
| Type Treatment  | Acid %               | Lbs Proppant | Maximum Treatment Pressure (PSI) |                    | Maximum Treatment Rate (BBLS/Min) |              |
| Details         |                      |              |                                  |                    |                                   |              |

### ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

|                                                                                                                                |  |                                              |                                                  |
|--------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------|--------------------------------------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. |  | Email Address<br><b>becky.barnes@clr.com</b> | Date<br><b>4/24/2014</b>                         |
| Signature<br>                                |  | Printed Name<br><b>Becky Barnes</b>          | Title<br><b>Regulatory Compliance Specialist</b> |

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## **SYNOPSIS**

|                                 |                                                  |
|---------------------------------|--------------------------------------------------|
| <b>OPERATOR</b>                 | Continental Resources Inc.                       |
| <b>WELL NAME</b>                | Tallahassee 3-16H                                |
| <b>SURFACE LOCATION</b>         | 2585' FNL & 268' FEL: Sec 16, T153N, R101W       |
| <b>BOTTOM HOLE LOCATION</b>     | 232' FSL & 1304' FEL: Sec 21, T153N, R101W       |
| <b>FIELD</b>                    | Williston                                        |
| <b>COUNTY/STATE</b>             | McKenzie Co., North Dakota                       |
| <b>API NUMBER</b>               | 33-053-04853                                     |
| <b>NorAm FILE NUMBER</b>        | NR2691                                           |
| <b>ELEVATIONS</b>               | <b>G.L.</b> = 1920' <b>K.B.</b> = 1940'          |
| <b>SPUD DATE</b>                | November 8, 2013                                 |
| <b>T.D. DATE</b>                | November 21, 2013                                |
| <b>DRILLING CONTRACTOR</b>      | Cyclone 4                                        |
| <b>HOLE SIZE</b>                | 10 3/4" to 1975', 8 3/4" to 10863 , 6" to 18434' |
| <b>CASING SIZE &amp; DEPTH</b>  | 9 5/8" to 1975', 7" to 10829, 5" to              |
| <b>DRILLING MUD COMPANY</b>     | GEO Drilling Fluids Inc                          |
| <b>DRILLING FLUID TYPE</b>      | Invert OBM to 10863', Saltwater Brine to 18434'  |
| <b>DIRECTIONAL COMPANY</b>      | Leam Energy Services                             |
| <b>WIRELINE LOGGING COMPANY</b> | N/A                                              |
| <b>LOG RECORD</b>               | NorAm Wellsite Services                          |
| <b>DRILLING SUPERVISION</b>     | Steve Northern, Biggens, Monty Harris            |
| <b>GEOLOGICAL SUPERVISION</b>   | R. D. Reinert Jr. & Darcy Klessens               |
| <b>MUDLOGGING COMPANY</b>       | NorAm Wellsite Services                          |
| <b>TOTAL DEPTH</b>              | 18434'                                           |

**FORMATION TOPS (ft)**

**KB =1940'**

**GL =1920'**

| FORMATION          | PROGNOSIS |       |       | SAMPLES |       |       | LOGS |     |    |
|--------------------|-----------|-------|-------|---------|-------|-------|------|-----|----|
|                    | MD        | TVD   | SS    | MD      | TVD   | SS    | MD   | TVD | SS |
| Top Charles Salts  | -         | 8329  | -6389 | 8286    | 8285  | -6345 | -    | -   | -  |
| Base Last Salt     | -         | 9025  | -7085 | 8994    | 8993  | -7053 | -    | -   | -  |
| Mission Canyon     | -         | 9250  | -7310 | 9220    | 9219  | -7279 | -    | -   | -  |
| Lodgepole          | -         | 9817  | -7877 | 9786    | 9785  | -7845 | -    | -   | -  |
| False Bakken       | -         |       |       | 10536   | 10483 | -8543 | -    | -   | -  |
| Upper Bakken Shale | -         | 10505 | -8565 | 10548   | 10490 | -8550 | -    | -   | -  |
| Middle Bakken      | -         | 10522 | -8582 | 10578   | 10506 | -8566 | -    | -   | -  |
| Lower Bakken Shale | -         | 10557 | -8617 | 10650   | 10539 | -8599 | -    | -   | -  |
| Three Forks        | -         | 10583 | -8643 | 10748   | 10571 | -8631 | -    | -   | -  |



**DEVIATION SURVEY RECORD (Teledrift and Wireline Data)**

| <b>Depth</b> | <b>Incl</b> | <b>Azim</b> | <b>TVD</b>  | <b>VS</b>   | <b>Coordinates</b> |                 | <b>DLS</b>      |
|--------------|-------------|-------------|-------------|-------------|--------------------|-----------------|-----------------|
| <b>(ft)</b>  | <b>(°)</b>  | <b>(°)</b>  | <b>(ft)</b> | <b>(ft)</b> | <b>N/S (ft)</b>    | <b>E/W (ft)</b> | <b>(°/100')</b> |
| 1938         | 0.70        | 307.10      | 1937.97     | 3.52        | 3.52               | -3.92           | 0.70            |
| 2014         | 0.80        | 280.20      | 2013.96     | -3.21       | 3.89               | -4.81           | 0.48            |
| 2077         | 0.90        | 246.50      | 2076.96     | -2.97       | 3.77               | -5.70           | 0.80            |
| 2170         | 1.10        | 238.70      | 2169.94     | -2.03       | 3.02               | -7.13           | 0.26            |
| 2263         | 1.30        | 236.50      | 2262.92     | -0.78       | 1.97               | -8.77           | 0.22            |
| 2357         | 1.80        | 235.80      | 2356.89     | 0.91        | 0.56               | -10.88          | 0.53            |
| 2450         | 2.40        | 199.00      | 2449.83     | 3.80        | -2.11              | -12.73          | 1.55            |
| 2543         | 2.20        | 181.60      | 2542.76     | 7.48        | -5.73              | -13.41          | 0.78            |
| 2638         | 1.20        | 152.80      | 2637.71     | 10.11       | -8.44              | -13.01          | 1.35            |
| 2732         | 1.00        | 119.70      | 2731.70     | 11.23       | -9.72              | -11.84          | 0.70            |
| 2825         | 1.20        | 134.30      | 2824.68     | 12.11       | -10.80             | -10.44          | 0.37            |
| 2918         | 1.50        | 119.20      | 2917.65     | 13.14       | -12.08             | -8.68           | 0.50            |
| 3011         | 1.80        | 119.00      | 3010.62     | 14.11       | -13.38             | -6.34           | 0.32            |
| 3105         | 1.20        | 84.60       | 3104.59     | 14.42       | -14.00             | -4.07           | 1.12            |
| 3198         | 1.50        | 66.70       | 3197.56     | 13.57       | -13.43             | -1.98           | 0.55            |
| 3291         | 1.10        | 90.20       | 3290.54     | 12.83       | -12.95             | 0.03            | 0.71            |
| 3384         | 1.10        | 92.30       | 3383.52     | 12.63       | -12.99             | 1.81            | 0.04            |
| 3477         | 1.00        | 80.50       | 3476.50     | 12.30       | -12.89             | 3.50            | 0.26            |
| 3571         | 0.90        | 91.60       | 3570.49     | 11.98       | -12.78             | 5.05            | 0.22            |
| 3664         | 0.70        | 94.10       | 3663.48     | 11.87       | -12.84             | 6.35            | 0.22            |
| 3758         | 0.60        | 124.30      | 3757.48     | 12.05       | -13.16             | 7.33            | 0.37            |
| 3851         | 0.50        | 138.60      | 3850.47     | 12.54       | -13.74             | 8.00            | 0.18            |
| 3945         | 0.40        | 127.30      | 3944.47     | 12.97       | -14.24             | 8.53            | 0.14            |
| 4038         | 0.40        | 231.90      | 4037.47     | 13.36       | -14.64             | 8.53            | 0.68            |
| 4132         | 0.30        | 320.00      | 4131.47     | 13.43       | -14.65             | 8.12            | 0.52            |
| 4225         | 0.10        | 245.80      | 4224.47     | 13.31       | -14.50             | 7.89            | 0.31            |
| 4319         | 0.10        | 344.60      | 4318.47     | 13.28       | -14.45             | 7.79            | 0.16            |
| 4412         | 0.30        | 254.00      | 4411.47     | 13.30       | -14.44             | 7.53            | 0.34            |
| 4505         | 0.10        | 258.10      | 4504.46     | 13.42       | -14.53             | 7.22            | 0.22            |
| 4599         | 0.40        | 231.50      | 4598.46     | 13.69       | -14.75             | 6.88            | 0.33            |
| 4692         | 0.30        | 263.70      | 4691.46     | 13.98       | -14.98             | 6.39            | 0.23            |
| 4785         | 0.30        | 351.80      | 4784.46     | 13.81       | -14.76             | 6.11            | 0.45            |
| 4879         | 0.50        | 40.30       | 4878.46     | 13.22       | -14.21             | 6.34            | 0.40            |
| 4972         | 0.40        | 63.10       | 4971.46     | 12.70       | -13.75             | 6.89            | 0.22            |
| 5066         | 0.80        | 44.90       | 5065.45     | 11.99       | -13.14             | 7.65            | 0.47            |
| 5159         | 1.10        | 55.80       | 5158.44     | 10.88       | -12.18             | 8.84            | 0.38            |
| 5253         | 1.10        | 62.60       | 5252.42     | 9.75        | -11.25             | 10.39           | 0.14            |
| 5347         | 0.90        | 60.30       | 5346.41     | 8.79        | -10.47             | 11.83           | 0.22            |
| 5440         | 1.10        | 68.10       | 5439.39     | 7.90        | -9.78              | 13.30           | 0.26            |
| 5533         | 1.20        | 62.80       | 5532.37     | 6.90        | -9.00              | 14.99           | 0.16            |
| 5627         | 0.90        | 51.00       | 5626.36     | 5.80        | -8.08              | 16.44           | 0.39            |

| <b>Depth</b> | <b>Incl</b> | <b>Azim</b> | <b>TVD</b>  | <b>VS</b>   | <b>Coordinates</b> |                 | <b>DLS</b>      |
|--------------|-------------|-------------|-------------|-------------|--------------------|-----------------|-----------------|
| <b>(ft)</b>  | <b>(°)</b>  | <b>(°)</b>  | <b>(ft)</b> | <b>(ft)</b> | <b>N/S (ft)</b>    | <b>E/W (ft)</b> | <b>(°/100')</b> |
| 5720         | 0.80        | 32.60       | 5719.35     | 4.68        | -7.08              | 17.36           | 0.31            |
| 5813         | 0.70        | 79.50       | 5812.34     | 3.91        | -6.43              | 18.27           | 0.65            |
| 5907         | 1.10        | 100.90      | 5906.33     | 3.78        | -6.49              | 19.72           | 0.55            |
| 6000         | 0.40        | 110.60      | 5999.32     | 3.90        | -6.78              | 20.90           | 0.76            |
| 6094         | 0.80        | 289.50      | 6093.32     | 3.84        | -6.67              | 20.59           | 1.28            |
| 6187         | 0.50        | 289.20      | 6186.31     | 3.63        | -6.32              | 19.59           | 0.32            |
| 6281         | 0.50        | 5.70        | 6280.31     | 3.14        | -5.78              | 19.24           | 0.66            |
| 6375         | 0.60        | 43.10       | 6374.31     | 2.33        | -5.01              | 19.62           | 0.39            |
| 6469         | 0.10        | 144.00      | 6468.30     | 1.99        | -4.72              | 20.01           | 0.67            |
| 6562         | 1.10        | 255.60      | 6561.30     | 2.38        | -5.01              | 19.19           | 1.23            |
| 6655         | 1.00        | 255.60      | 6654.28     | 3.02        | -5.43              | 17.54           | 0.11            |
| 6749         | 1.10        | 245.80      | 6748.27     | 3.81        | -6.00              | 15.92           | 0.22            |
| 6843         | 1.10        | 247.20      | 6842.25     | 4.74        | -6.72              | 14.27           | 0.03            |
| 6936         | 1.20        | 254.60      | 6935.23     | 5.58        | -7.33              | 12.50           | 0.19            |
| 7030         | 1.20        | 239.50      | 7029.21     | 6.58        | -8.09              | 10.71           | 0.34            |
| 7123         | 1.80        | 238.60      | 7122.18     | 8.10        | -9.34              | 8.62            | 0.65            |
| 7216         | 1.80        | 246.30      | 7215.13     | 9.78        | -10.69             | 6.04            | 0.26            |
| 7309         | 1.10        | 286.60      | 7308.10     | 10.41       | -11.03             | 3.84            | 1.29            |
| 7404         | 1.10        | 284.30      | 7403.09     | 10.16       | -10.54             | 2.09            | 0.05            |
| 7497         | 1.10        | 286.70      | 7496.07     | 9.92        | -10.06             | 0.37            | 0.05            |
| 7590         | 1.50        | 292.50      | 7589.05     | 9.47        | -9.34              | -1.61           | 0.45            |
| 7683         | 1.60        | 299.40      | 7682.01     | 8.68        | -8.24              | -3.87           | 0.23            |
| 7777         | 1.30        | 305.50      | 7775.98     | 7.70        | -6.97              | -5.88           | 0.36            |
| 7870         | 1.40        | 288.80      | 7868.96     | 6.99        | -5.99              | -7.81           | 0.43            |
| 7963         | 1.10        | 285.20      | 7961.93     | 6.66        | -5.39              | -9.75           | 0.33            |
| 8056         | 0.90        | 279.50      | 8054.92     | 6.52        | -5.04              | -11.33          | 0.24            |
| 8149         | 0.90        | 270.00      | 8147.91     | 6.59        | -4.92              | -12.78          | 0.16            |
| 8243         | 1.00        | 268.30      | 8241.90     | 6.83        | -4.94              | -14.34          | 0.11            |
| 8336         | 1.00        | 272.10      | 8334.88     | 7.04        | -4.94              | -15.96          | 0.07            |
| 8429         | 0.90        | 276.50      | 8427.87     | 7.14        | -4.83              | -17.50          | 0.13            |
| 8522         | 0.90        | 280.40      | 8520.86     | 7.12        | -4.61              | -18.95          | 0.07            |
| 8616         | 1.00        | 257.40      | 8614.84     | 7.37        | -4.66              | -20.47          | 0.42            |
| 8709         | 1.10        | 267.20      | 8707.83     | 7.81        | -4.88              | -22.16          | 0.22            |
| 8803         | 1.00        | 261.10      | 8801.81     | 8.21        | -5.05              | -23.87          | 0.16            |
| 8896         | 0.90        | 264.20      | 8894.80     | 8.62        | -5.25              | -25.40          | 0.12            |
| 8990         | 0.40        | 334.00      | 8988.79     | 8.52        | -5.03              | -26.27          | 0.90            |
| 9083         | 0.80        | 47.70       | 9081.79     | 7.75        | -4.30              | -25.94          | 0.85            |
| 9176         | 0.80        | 63.10       | 9174.78     | 6.88        | -3.57              | -24.88          | 0.23            |
| 9270         | 0.80        | 59.10       | 9268.77     | 6.10        | -2.93              | -23.73          | 0.06            |
| 9363         | 0.50        | 49.80       | 9361.77     | 5.39        | -2.34              | -22.86          | 0.34            |
| 9457         | 0.40        | 48.20       | 9455.76     | 4.84        | -1.86              | -22.30          | 0.11            |
| 9550         | 0.40        | 48.60       | 9548.76     | 4.35        | -1.42              | -21.82          | 0.00            |

| <b>Depth</b> | <b>Incl</b> | <b>Azim</b> | <b>TVD</b>  | <b>VS</b>   | <b>Coordinates</b> |                 | <b>DLS</b>      |
|--------------|-------------|-------------|-------------|-------------|--------------------|-----------------|-----------------|
| <b>(ft)</b>  | <b>(°)</b>  | <b>(°)</b>  | <b>(ft)</b> | <b>(ft)</b> | <b>N/S (ft)</b>    | <b>E/W (ft)</b> | <b>(?/100')</b> |
| 9643         | 0.50        | 51.70       | 9641.76     | 3.81        | -0.96              | -21.26          | 0.11            |
| 9737         | 0.40        | 12.90       | 9735.76     | 3.19        | -0.38              | -20.86          | 0.33            |
| 9830         | 0.30        | 351.40      | 9828.75     | 2.63        | 0.17               | -20.82          | 0.18            |
| 9923         | 0.30        | 308.20      | 9921.75     | 2.27        | 0.56               | -21.05          | 0.24            |
| 10019        | 0.30        | 299.40      | 10017.75    | 2.05        | 0.84               | -21.47          | 0.05            |
| 10112        | 0.40        | 325.20      | 10110.75    | 1.72        | 1.23               | -21.87          | 0.20            |
| 10143        | 1.80        | 252.60      | 10141.74    | 1.85        | 1.17               | -22.39          | 5.56            |
| 10174        | 5.70        | 242.60      | 10172.67    | 2.94        | 0.32               | -24.22          | 12.71           |
| 10205        | 9.10        | 239.30      | 10203.41    | 5.35        | -1.64              | -27.70          | 11.05           |
| 10236        | 12.50       | 234.40      | 10233.86    | 9.18        | -4.85              | -32.54          | 11.35           |
| 10268        | 15.70       | 234.50      | 10264.89    | 14.52       | -9.38              | -38.88          | 10.00           |
| 10299        | 20.40       | 232.60      | 10294.35    | 21.23       | -15.10             | -46.59          | 15.28           |
| 10330        | 25.70       | 233.30      | 10322.87    | 29.77       | -22.40             | -56.28          | 17.12           |
| 10361        | 29.90       | 231.90      | 10350.29    | 40.02       | -31.19             | -67.75          | 13.71           |
| 10392        | 33.20       | 231.00      | 10376.70    | 51.75       | -41.30             | -80.43          | 10.75           |
| 10423        | 36.80       | 230.80      | 10402.09    | 64.71       | -52.52             | -94.23          | 11.62           |
| 10455        | 40.00       | 230.00      | 10427.16    | 79.33       | -65.19             | -109.54         | 10.12           |
| 10485        | 45.40       | 229.80      | 10449.20    | 94.40       | -78.29             | -125.09         | 18.01           |
| 10516        | 49.40       | 230.00      | 10470.18    | 111.31      | -92.98             | -142.55         | 12.91           |
| 10548        | 55.00       | 230.00      | 10489.79    | 130.02      | -109.23            | -161.91         | 17.50           |
| 10579        | 59.40       | 230.00      | 10506.58    | 149.29      | -125.98            | -181.86         | 14.19           |
| 10610        | 61.30       | 229.80      | 10521.91    | 169.26      | -143.33            | -202.47         | 6.15            |
| 10641        | 64.30       | 229.10      | 10536.08    | 189.84      | -161.25            | -223.42         | 9.88            |
| 10672        | 68.90       | 227.80      | 10548.39    | 211.40      | -180.12            | -244.70         | 15.33           |
| 10703        | 72.10       | 227.70      | 10558.74    | 233.77      | -199.77            | -266.33         | 10.33           |
| 10735        | 73.90       | 227.50      | 10568.09    | 257.26      | -220.40            | -288.92         | 5.66            |
| 10766        | 77.20       | 227.10      | 10575.83    | 280.40      | -240.76            | -310.98         | 10.72           |
| 10797        | 82.50       | 226.10      | 10581.29    | 304.15      | -261.72            | -333.14         | 17.39           |
| 10807        | 84.20       | 225.90      | 10582.45    | 311.95      | -268.62            | -340.29         | 17.12           |
| 10889        | 88.00       | 222.60      | 10588.02    | 377.68      | -327.20            | -397.35         | 6.13            |
| 10920        | 89.50       | 221.30      | 10588.70    | 403.30      | -350.25            | -418.07         | 6.40            |
| 11015        | 90.90       | 218.70      | 10588.37    | 483.62      | -423.02            | -479.13         | 3.11            |
| 11110        | 91.80       | 214.70      | 10586.13    | 566.69      | -499.15            | -535.87         | 4.31            |
| 11205        | 89.20       | 213.60      | 10585.30    | 651.76      | -577.76            | -589.20         | 2.97            |
| 11300        | 90.00       | 210.30      | 10585.96    | 738.38      | -658.35            | -639.46         | 3.57            |
| 11395        | 90.40       | 207.60      | 10585.63    | 826.93      | -741.47            | -685.44         | 2.87            |
| 11490        | 90.70       | 205.30      | 10584.72    | 916.89      | -826.52            | -727.75         | 2.44            |
| 11585        | 89.90       | 204.60      | 10584.22    | 1007.63     | -912.65            | -767.82         | 1.12            |
| 11680        | 89.70       | 204.50      | 10584.55    | 1098.57     | -999.06            | -807.29         | 0.24            |
| 11775        | 88.20       | 202.70      | 10586.29    | 1189.92     | -1086.09           | -845.32         | 2.47            |
| 11870        | 89.1        | 201.5       | 10588.532   | 1281.92     | -1174.09           | -881.046        | 1.57866         |
| 11965        | 90.1        | 199.4       | 10589.196   | 1374.58     | -1263.09           | -914.235        | 2.44829         |



| <b>Depth</b> | <b>Incl</b> | <b>Azim</b> | <b>TVD</b>  | <b>VS</b>   | <b>Coordinates</b> |                 | <b>DLS</b>      |
|--------------|-------------|-------------|-------------|-------------|--------------------|-----------------|-----------------|
| <b>(ft)</b>  | <b>(°)</b>  | <b>(°)</b>  | <b>(ft)</b> | <b>(ft)</b> | <b>N/S (ft)</b>    | <b>E/W (ft)</b> | <b>(°/100')</b> |
| 12060        | 89.1        | 197.6       | 10589.859   | 1467.9      | -1353.18           | -944.377        | 2.16744         |
| 12155        | 89.7        | 194.6       | 10590.854   | 1561.88     | -1444.43           | -970.717        | 3.22025         |
| 12250        | 89.8        | 191.8       | 10591.269   | 1656.43     | -1536.91           | -992.408        | 2.94922         |
| 12345        | 90.7        | 189.4       | 10590.854   | 1751.3      | -1630.28           | -1009.88        | 2.69806         |
| 12440        | 91.3        | 186.4       | 10589.196   | 1846.28     | -1724.36           | -1022.94        | 3.21995         |
| 12535        | 88.8        | 184.1       | 10589.113   | 1941.18     | -1818.95           | -1031.63        | 3.57572         |
| 12630        | 89          | 181.6       | 10590.937   | 2035.81     | -1913.8            | -1036.35        | 2.6395          |
| 12725        | 90.40       | 181.80      | 10591.43    | 2130.28     | -2008.76           | -1039.17        | 1.49            |
| 12820        | 90.90       | 179.30      | 10590.36    | 2224.52     | -2103.74           | -1040.08        | 2.68            |
| 12915        | 90.00       | 179.70      | 10589.61    | 2318.54     | -2198.73           | -1039.25        | 1.04            |
| 13010        | 89.30       | 179.00      | 10590.19    | 2412.52     | -2293.72           | -1038.17        | 1.04            |
| 13104        | 88.80       | 180.40      | 10591.75    | 2505.58     | -2387.71           | -1037.68        | 1.58            |
| 13199        | 89.50       | 180.90      | 10593.16    | 2599.85     | -2482.69           | -1038.76        | 0.91            |
| 13295        | 90.80       | 180.90      | 10592.91    | 2695.16     | -2578.67           | -1040.27        | 1.35            |
| 13389        | 91.50       | 179.50      | 10591.02    | 2788.33     | -2672.65           | -1040.59        | 1.66            |
| 13484        | 91.80       | 180.20      | 10588.29    | 2882.39     | -2767.61           | -1040.35        | 0.80            |
| 13579        | 89.20       | 180.00      | 10587.46    | 2976.54     | -2862.60           | -1040.51        | 2.74            |
| 13674        | 89.00       | 180.00      | 10588.95    | 3070.67     | -2957.59           | -1040.51        | 0.21            |
| 13769        | 90.50       | 179.00      | 10589.36    | 3164.68     | -3052.58           | -1039.68        | 1.90            |
| 13864        | 89.70       | 180.40      | 10589.20    | 3258.75     | -3147.57           | -1039.19        | 1.70            |
| 13959        | 89.10       | 180.70      | 10590.19    | 3353.00     | -3242.56           | -1040.10        | 0.71            |
| 14054        | 89.60       | 179.70      | 10591.27    | 3447.17     | -3337.56           | -1040.43        | 1.18            |
| 14149        | 89.70       | 179.70      | 10591.85    | 3541.24     | -3432.55           | -1039.93        | 0.11            |
| 14244        | 90.90       | 179.20      | 10591.35    | 3635.24     | -3527.54           | -1039.02        | 1.37            |
| 14339        | 89.90       | 179.30      | 10590.69    | 3729.20     | -3622.53           | -1037.78        | 1.06            |
| 14434        | 90.40       | 179.00      | 10590.44    | 3823.14     | -3717.52           | -1036.37        | 0.61            |
| 14529        | 90.3        | 178.6       | 10589.86    | 3916.98     | -3812.5            | -1034.38        | 0.434           |
| 14624        | 91.5        | 178.3       | 10588.368   | 4010.73     | -3907.45           | -1031.81        | 1.30202         |
| 14719        | 90.9        | 177.4       | 10586.378   | 4104.3      | -4002.36           | -1028.24        | 1.13842         |
| 14814        | 90.7        | 178.5       | 10585.052   | 4197.9      | -4097.29           | -1024.85        | 1.17677         |
| 14909        | 91.1        | 179.7       | 10583.56    | 4291.82     | -4192.26           | -1023.35        | 1.33134         |
| 15004        | 91.1        | 179.7       | 10581.736   | 4385.87     | -4287.25           | -1022.86        | 1.5E-06         |
| 15099        | 89.60       | 179.30      | 10581.16    | 4479.88     | -4382.24           | -1022.03        | 1.63            |
| 15193        | 89.60       | 180.20      | 10581.81    | 4572.97     | -4476.23           | -1021.62        | 0.96            |
| 15289        | 89.20       | 180.70      | 10582.82    | 4668.19     | -4572.22           | -1022.37        | 0.67            |
| 15383        | 89.30       | 181.30      | 10584.05    | 4761.53     | -4666.20           | -1024.01        | 0.65            |
| 15479        | 90.80       | 180.20      | 10583.96    | 4856.82     | -4762.19           | -1025.27        | 1.94            |
| 15574        | 89.90       | 181.60      | 10583.38    | 4951.14     | -4857.17           | -1026.76        | 1.75            |
| 15668        | 88.80       | 182.50      | 10584.45    | 5044.67     | -4951.10           | -1030.12        | 1.51            |
| 15763        | 90.4        | 182.7       | 10585.113   | 5139.28     | -5046              | -1034.43        | 1.69732         |
| 15858        | 88.5        | 182.8       | 10586.025   | 5233.92     | -5140.88           | -1038.99        | 2.00277         |
| 15953        | 89.3        | 182.7       | 10587.849   | 5328.54     | -5235.75           | -1043.55        | 0.84866         |

| <b>Depth</b> | <b>Incl</b> | <b>Azim</b> | <b>TVD</b>  | <b>VS</b>   | <b>Coordinates</b> |                 | <b>DLS</b>      |
|--------------|-------------|-------------|-------------|-------------|--------------------|-----------------|-----------------|
| <b>(ft)</b>  | <b>(°)</b>  | <b>(°)</b>  | <b>(ft)</b> | <b>(ft)</b> | <b>N/S (ft)</b>    | <b>E/W (ft)</b> | <b>(?/100')</b> |
| 16048        | 90.5        | 182.5       | 10588.015   | 5423.16     | -5330.66           | -1047.86        | 1.28058         |
| 16143        | 88.4        | 182.3       | 10588.926   | 5517.74     | -5425.56           | -1051.83        | 2.22053         |
| 16238        | 89.40       | 182.50      | 10590.75    | 5612.31     | -5520.46           | -1055.81        | 1.07            |
| 16333        | 89.00       | 182.10      | 10592.08    | 5706.87     | -5615.37           | -1059.62        | 0.60            |
| 16428        | 89.00       | 181.80      | 10593.73    | 5801.37     | -5710.30           | -1062.86        | 0.32            |
| 16523        | 92.90       | 181.30      | 10592.16    | 5895.79     | -5805.24           | -1065.42        | 4.14            |
| 16618        | 91.00       | 180.90      | 10588.93    | 5990.10     | -5900.16           | -1067.25        | 2.04            |
| 16712        | 92.30       | 180.20      | 10586.22    | 6083.32     | -5994.11           | -1068.15        | 1.57            |
| 16807        | 91.10       | 180.60      | 10583.40    | 6177.50     | -6089.07           | -1068.81        | 1.33            |
| 16902        | 88.70       | 179.30      | 10583.57    | 6271.61     | -6184.06           | -1068.73        | 2.87            |
| 16997        | 92.10       | 179.50      | 10582.91    | 6365.60     | -6279.04           | -1067.73        | 3.59            |
| 17092        | 89.20       | 180.70      | 10581.83    | 6459.74     | -6374.02           | -1067.90        | 3.30            |
| 17187        | 90.30       | 180.00      | 10582.24    | 6553.95     | -6469.02           | -1068.48        | 1.37            |
| 17282        | 90.10       | 179.20      | 10581.91    | 6647.99     | -6564.01           | -1067.82        | 0.87            |
| 17377        | 88.00       | 178.80      | 10583.49    | 6741.87     | -6658.98           | -1066.16        | 2.25            |
| 17472        | 88.80       | 179.70      | 10586.14    | 6835.79     | -6753.93           | -1064.92        | 1.27            |
| 17567        | 90.30       | 179.20      | 10586.88    | 6929.80     | -6848.92           | -1064.01        | 1.66            |
| 17662        | 90.60       | 178.30      | 10586.14    | 7023.63     | -6943.89           | -1061.93        | 1.00            |
| 17757        | 92.20       | 179.00      | 10583.82    | 7117.41     | -7038.84           | -1059.70        | 1.84            |
| 17852        | 90.40       | 179.00      | 10581.66    | 7211.28     | -7133.79           | -1058.04        | 1.89            |
| 17947        | 91.00       | 178.60      | 10580.50    | 7305.12     | -7228.76           | -1056.05        | 0.76            |
| 18042        | 89.50       | 177.80      | 10580.09    | 7398.80     | -7323.71           | -1053.07        | 1.79            |
| 18137        | 90.00       | 177.60      | 10580.50    | 7492.35     | -7418.64           | -1049.25        | 0.57            |
| 18232        | 91.00       | 177.40      | 10579.67    | 7585.84     | -7513.54           | -1045.11        | 1.07            |
| 18327        | 91.6        | 177.8       | 10577.517   | 7679.33     | -7608.43           | -1041.1         | 0.759           |
| 18385        | 92.3        | 177.6       | 10575.543   | 7736.41     | -7666.35           | -1038.8         | 1.25514         |
| 18434        | 92.3        | 177.6       | 10573.577   | 7784.61     | -7715.27           | -1036.8         | 0               |

**BIT RECORD**

| BIT<br># | TYPE               | SIZE<br>(IN) | IN<br>(FT) | OUT<br>(FT) | TOTAL<br>(FT) | HOURS<br>DRLG | WOB<br>(K) | RPM   |
|----------|--------------------|--------------|------------|-------------|---------------|---------------|------------|-------|
| 1        | Security MM65D     | 8.75         | 1975       | 10064       | 8089          | 78.5          | 40         | 30-60 |
| 2        | Security<br>FDX55M | 8.75         | 10064      | 10863       | 799           | 18.5          | 20-70      | 0-30  |
| 3        | Reed SKHE711M      | 6            | 10863      | 18434       | 7571          | 130           | 15-25      | 40-50 |



**DRILLING FLUID PARAMETERS**

| <b>DATE</b> | <b>WT</b> | <b>VIS</b> | <b>PV</b> | <b>YP</b> | <b>WL</b> | <b>PH</b> | <b>NaCL</b> | <b>Ca</b> | <b>% SOL</b> |
|-------------|-----------|------------|-----------|-----------|-----------|-----------|-------------|-----------|--------------|
| 11/09/2013  | 9.5       | 57         | 12        | 16        | -         | -         | 37k         | 2.9       | 8.83         |
| 11/10/2013  | 9.6       | 51         | 13        | 14        | -         | -         | 41k         | 3.1       | 9.59         |
| 11/11/2013  | 9.9       | 53         | 15        | 11        | -         | -         | 43k         | 4.29      | 10.59        |
| 11/12/2013  | 10.1      | 56         | 14        | 17        | -         | -         | 47k         | 4.68      | 11.97        |
| 11/13/2013  | 10.2      | 49         | 15        | 11        | -         | -         | 42k         | 3.8       | 11.97        |
| 11/14/2013  | 10.4      | 65         | 15        | 10        | -         | -         | 42k         | 3.8       | 13.35        |
| 11/15/2013  | 10.4      | 65         | 15        | 10        | -         | -         | 42k         | 3.8       | 13.35        |
| 11/16/2013  | 10.4      | 65         | 15        | 10        | -         | -         | 42k         | 3.8       | 13.35        |
| 11/17/2013  | 9.7       | 28         | 1         | 1         | -         | 7.5       | 167.5k      | 30,500    | 1.39         |
| 11/18/2013  | 9.7       | 28         | 1         | 1         | -         | 7.5       | 167.5k      | 30,500    | 1.39         |
| 11/19/2013  | 9.7       | 28         | 1         | 1         | -         | 7.5       | 167.5k      | 30,500    | 1.39         |
| 11/20/2013  | 9.8       | 29         | 1         | 1         | -         | 9.0       | 169k        | 32,000    | 0.56         |
| 11/21/2013  | 9.8       | 29         | 1         | 1         | -         | 9.0       | 169k        | 32,000    | 0.56         |
| 11/22/2013  | 9.8       | 29         | 1         | 1         | -         | 8.0       | 162k        | 30,400    | 0.54         |

### **DAILY DRILLING CHRONOLOGY**

| <b>DATE<br/>(m/d/y)</b> | <b>DEPTH @<br/>24.00</b> | <b>PROGRESS<br/>(ft/24 hours)</b> | <b>BREAKDOWN<br/>00:00 – 24:00</b> | <b>RIG ACTIVITY</b>                                                                                                         |
|-------------------------|--------------------------|-----------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 11/10/2013              | 7671                     | 3643                              | 0:00-24:00                         | Drill 7671-8910                                                                                                             |
| 11/11/2013              | 8910                     | 1239                              | 0:00-24:00                         | Drill 8910-9856                                                                                                             |
| 11/12/2013              | 9856                     | 946                               | 0:00-4:00                          | Drill 9856-10064                                                                                                            |
|                         |                          |                                   | 4:00-4:30                          | Problem with drilling assembly,<br>Tag bottom with 20 K, no<br>reactive diff pressure                                       |
|                         |                          |                                   | 4:30-5:30                          | Circulate bottoms up while<br>building slug                                                                                 |
|                         |                          |                                   | 5:30-10:30                         | TOOH for Curve Assembly<br>10064-0                                                                                          |
|                         |                          |                                   | 10:30-12:00                        | Drain motor, pull MWD tool,<br>Break Bit, Lay down old motor/<br>pick up new motor, scribe, M/U<br>new Bit. Insert MWD Tool |
|                         |                          |                                   | 12:00-20:30                        | TIH With Curve Assembly                                                                                                     |
|                         |                          |                                   | 20:30-24:00                        | Build Curve 10064-10157                                                                                                     |
| 11/13/2013              | 10157                    | 301                               | 00:00-16:00                        | Build Curve 10157-10863                                                                                                     |
|                         |                          |                                   | 16:00-24:00                        | Circulate, Wiper Trip, Circulate,<br>TOOH, Lay Down 5" Drill Pipe                                                           |
| 11/14/2013              | 10863                    | 706                               | 00:00-24:00                        | Intermediate Casing Operations                                                                                              |
| 11/15/2013              | 10863                    | 0                                 | 00:00-24:00                        | Intermediate Casing Operations                                                                                              |
| 11/16/2013              | 10863                    | 0                                 | 00:00-01:30                        | Drill 10' new hole, Fit test at<br>10873'MD                                                                                 |
|                         |                          |                                   | 01:30-24:00                        | Drill 10873-12395                                                                                                           |
| 11/17/2013              | 12395                    | 1532                              | 00:00-24:00                        | Drill 12395-13820                                                                                                           |
| 11/18/2013              | 13820                    | 1425                              | 00:00-24:00                        | Drill 13820-14868                                                                                                           |
| 11/19/2013              | 14868                    | 1048                              | 00:00-24:00                        | Drill 14868-16257                                                                                                           |
| 11/20/2013              | 16257                    | 1389                              | 00:00-24:00                        | Drill 16257-17379                                                                                                           |
| 11/21/2013              | 17379                    | 1122                              | 00:00-18:30                        | Drill 17379-18434 (TD Well)                                                                                                 |
|                         |                          |                                   | 18:30-19:30                        | Circulate                                                                                                                   |
|                         |                          |                                   | 19:30-24:00                        | Wiper Trip, 5" Casing<br>Operations                                                                                         |
| 11/22/2013              | 18434'                   | 1055                              |                                    |                                                                                                                             |

### **TOTAL GAS DATA**

| <b>SHOW INTERVAL<br/>FROM – TO (ft)</b> | <b>BKGD<br/>UNITS</b> | <b>SHOW<br/>UNITS</b> | <b>BRIEF DESCRIPTION<br/>(Rock type, porosity, visual show, etc.)</b>                                                                                                                                                                                                                    |
|-----------------------------------------|-----------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8295-9220                               | 20                    | 40                    | SALT: clear, translucent, light brown, cream;<br>LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material;<br>ANHYDRITE: cream to buff                                                                                    |
| 9220-9786                               | 30                    | 56                    | LIMESTONE: light to medium brown, light to medium gray brown, medium to dark gray, tan, slightly mottled, firm to moderately hard, microcrystalline to cryptocrystalline, earthy, argillaceous                                                                                           |
| 9786-10490                              | 70                    | 240                   | LIMESTONE: dark to medium gray brown, dark to medium gray, mottled, firm to moderately hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, trace asphalt staining, earthy texture                                                                               |
| 10490-10578                             | 250                   | 5023                  | LIMESTONE: light to medium brown, light to medium grey, tan, mottled, moderately hard, microcrystalline to cryptocrystalline, argillaceous, silty, partly laminated                                                                                                                      |
| 10578-10650                             | 150                   | 289                   | DOLOMITE: light to medium gray brown, mottled, medium to firm, microcrystalline to cryptocrystalline, shale, sandy, dull yellow fluorescence, flash yellow to green cut                                                                                                                  |
| 10650-18434                             | 300                   | 504                   | DOLOMITE: light brown, tan, light gray, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous |

## LITHOLOGY

Formation tops  
Sample Interval  
(in feet)

### SAMPLE DESCRIPTION

***30' samples were caught in the vertical and curve, and 50' samples caught in the lateral by NorAm Wellsite Services. Logging began at 8250' MD on November 10, 2013 at 10:06 hours.***

|           |                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8250-8280 | LIMESTONE: red orange medium gray brown, trace light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material<br><b><u>CHARLES TOP: 8286' MD, 8285' TVD (-6345)</u></b> |
| 8280-8340 | SALT: clear, translucent, LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, slightly argillaceous                                                                                  |
| 8340-8370 | SALT: clear, translucent; LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff           |
| 8370-8430 | SALT: clear, translucent; trace LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material, trace siltstone material; trace ANHYDRITE: cream to buff     |
| 8430-8460 | SALT: clear, translucent, brown; LIMESTONE: light to medium gray, friable to firm, microcrystalline, slightly argillaceous, trace silica                                                                              |
| 8460-8490 | SALT: clear, translucent, brown; trace LIMESTONE: light to medium gray, friable to firm, microcrystalline, slightly argillaceous                                                                                      |
| 8490-8520 | SALT: clear, translucent, brown; LIMESTONE: light to medium gray, friable to firm, microcrystalline, slightly argillaceous, trace silica                                                                              |
| 8520-8610 | SALT: clear, translucent, light brown, cream, brown; LIMESTONE: light to medium gray, mottled in part, friable to firm, microcrystalline, slightly argillaceous, trace silica                                         |
| 8610-8640 | SALT: clear, translucent; LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material; ANHYDRITE: cream to buff                                           |
| 8640-8670 | SALT: clear, translucent, light brown, cream; LIMESTONE: light to medium gray, firm to brown, microcrystalline, moderately silica, argillaceous, trace sandy material; ANHYDRITE: cream to buff                       |



Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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8670-8700 LIMESTONE: light gray brown, light to medium gray, firm to brown, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; trace SILTSTONE: red to orange, friable to firm, calcareous, material; SALT: clear, translucent, hard

8700-8790 LIMESTONE: light gray brown, light to medium gray, firm to brown, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white, soft, amorphous; SALT: clear, translucent, hard

8790-8850 ANHYDRITE: white, soft, amorphous; LIMESTONE: light gray brown, light to medium gray, firm to brown, microcrystalline, moderately silty, argillaceous; trace SALT: clear, translucent, hard

8850-8880 ANHYDRITE: white, soft, amorphous; LIMESTONE: medium gray brown, light to medium gray, firm to brown, microcrystalline, moderately silty, argillaceous; trace SALT: clear, translucent, hard

8880-8910 ANHYDRITE: white to gray, soft, amorphous; LIMESTONE: light gray brown, light to medium gray, firm to brown, microcrystalline, moderately silty, argillaceous

8910-8940 LIMESTONE: light gray brown to tan, light gray, firm to brown, microcrystalline, moderately silty, argillaceous, ANHYDRITE: white to gray, soft, amorphous, trace SALT: clear, translucent, hard

8940-9000 SALT: clear, translucent, hard; trace LIMESTONE: medium gray, light to medium gray brown, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part

**BASE LAST SALT: 8994' MD, 8993' TVD (-7053)**

9000-9030 SALT: clear, translucent, hard; trace LIMESTONE: light to medium gray, tan to light brown, firm to hard, microcrystalline, moderately silty, trace argillaceous, mottled in part

9030-9060 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part

9060-9090 LIMESTONE: light to medium gray, light brown, tan, off white in part, firm to hard, microcrystalline, silty, trace argillaceous, mottled in part, trace ANHYDRITE: white to gray, soft, amorphous

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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|                                                           |                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9090-9150                                                 | LIMESTONE: medium to dark gray, medium gray brown, off white to cream in part, firm to moderately hard, microcrystalline, silty, argillaceous; ANHYDRITE: white to gray, soft, amorphous                                                                 |
| 9150-9210                                                 | LIMESTONE: light to medium gray, dark gray, light to medium brown, off white to cream in part, firm to moderately hard, microcrystalline to cryptocrystalline, slightly silty, argillaceous; trace ANHYDRITE: white to gray, soft, amorphous, calcareous |
| <b><u>MISSION CANYON: 9220' MD, 9219' TVD (-7279)</u></b> |                                                                                                                                                                                                                                                          |
| 9210-9330                                                 | LIMESTONE: medium to dark gray, medium to dark gray brown, light to medium brown, firm to hard, microcrystalline to cryptocrystalline, moderately silty, very argillaceous, mottled in part                                                              |
| 9330-9390                                                 | LIMESTONE: light to medium brown, medium to dark gray brown, tan, firm to hard, microcrystalline to cryptocrystalline, slightly silty, very argillaceous, mottled                                                                                        |
| 9390-9420                                                 | LIMESTONE: light to medium brown, dark brown, medium to dark gray to gray brown, tan, slightly mottled, firm, cryptocrystalline, earthy, slightly silty, argillaceous                                                                                    |
| 9420-9480                                                 | LIMESTONE: light to medium brown, light to medium gray brown, medium to dark gray, tan, slightly mottled, firm to moderately hard, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous                                           |
| 9480-9540                                                 | LIMESTONE: light to medium brown, light to medium gray brown, medium to dark gray, tan, slightly mottled, firm to moderately hard, microcrystalline to cryptocrystalline, earthy, argillaceous                                                           |
| 9540-9570                                                 | LIMESTONE: light to medium brown, tan, off white, light to medium gray brown, firm to moderately hard, microcrystalline to cryptocrystalline, argillaceous, mottled in part, carbonaceous stain                                                          |
| 9570-9600                                                 | LIMESTONE: light to medium gray brown, medium to dark gray, tan, slightly mottled, firm, cryptocrystalline, earthy, slightly silty, argillaceous                                                                                                         |
| 9600-9630                                                 | LIMESTONE: cream, tan, off white, light to medium brown, light to medium gray brown, medium gray, slightly mottled, firm, cryptocrystalline, earthy, slightly silty, argillaceous, trace chert                                                           |

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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9630-9780 LIMESTONE: medium gray brown, light to medium gray, cream to off white in part, firm to hard, microcrystalline to cryptocrystalline, silty, argillaceous in part, earthy texture

**LODGEPOLE 9786' MD, 9785' TVD (-7845)**

9780-9870 LIMESTONE: light to medium gray to brown, light brown to tan in part, trace medium to dark gray, mottled in part, firm, microcrystalline to cryptocrystalline, earthy, slightly silty, argillaceous

9870-10020 LIMESTONE: dark to medium gray, light to medium brown, very trace tan in part, mottled in part, firm to hard, microcrystalline to cryptocrystalline, slightly to moderately silty, argillaceous, earthy texture

10020-10440 LIMESTONE: dark to medium gray brown, dark to medium gray, mottled, firm to moderately hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, trace asphalt staining, earthy texture

10440-10470 LIMESTONE: light to medium gray, light to medium gray brown, tan, mottled, firm to moderately hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, partly laminated

10470-10500 LIMESTONE: medium to dark grey, light to medium gray to brown, mottled, firm to moderately hard, microcrystalline to cryptocrystalline, moderately silty, argillaceous, partly laminated

10500-10530 LIMESTONE: light to medium brown, light gray, tan, mottled, moderately hard, microcrystalline to cryptocrystalline, slightly argillaceous, laminated; interbedded DOLOMITE: medium to dark grey, mottled, moderately hard, microcrystalline to cryptocrystalline, silty, very argillaceous, partly laminated.

**FALSE BAKKEN 10536' MD, 10483' TVD (-8543)**

**UPPER BAKKEN SHALE: 10548' MD, 10490' TVD (-8550)**

10530-10560 LIMESTONE: light to medium brown, light to medium grey, tan, mottled, moderately hard, microcrystalline to cryptocrystalline, argillaceous, silty, partly laminated

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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**MIDDLE BAKKEN MEMBER: 10578' MD, 10506' TVD (-8566)**

- 10560-10590 DOLOMITE: light to medium gray brown, light to medium brown, moderately, firm to hard microcrystalline to cryptocrystalline, argillaceous, silty, moderately carbonaceous; SHALE: blue, very laminated, hard, no porosity, dull yellow fluorescence, flash yellow to green cut
- 10590-10620 DOLOMITE: light to medium gray brown, mottled, medium to firm, microcrystalline to cryptocrystalline, shale, sandy, dull yellow fluorescence, flash yellow to green cut
- 10620-10650 DOLOMITE: light to medium gray brown, mottled, medium to firm, microcrystalline to cryptocrystalline, shale, sandy, carbonaceous, dull yellow fluorescence, flash yellow to green cut

**LOWER BAKKEN SHALE: 10650' MD, 10539' TVD (-8551)**

- 10650-10680 SHALE: black, firm to medium hard, very laminated, no porosity, dull yellow fluorescence, flash yellow to green cut; LIMESTONE: light to medium grey brown, tan, microcrystalline, cryptocrystalline, carbonaceous, partly lm, dull yellow fluorescence, flash yellow to green cut
- 10680-10710 LIMESTONE: medium to dark gray, black microcrystalline to cryptocrystalline, shale, pyrite, no porosity, dull yellow fluorescence, fast streaming yellow to green fluorescence cut
- 10710-10740 DOLOMITE: dark gray, blue, firm to medium hard, microcrystalline cryptocrystalline, no porosity, laminated, silty, dull yellow fluorescence, flash yellow to green cut

**THREE FORKS: 10748' MD, 10571' TVD (-8583)**

- 10740-10770 DOLOMITE: olive gray, tan, light gray, firm, microcrystalline to cryptocrystalline, rare lamination, argillaceous, no porosity, carbonaceous, light yellow fluorescence, light yellow to green flash; interbedded SHALE: blue, firm to hard, calcareous cement, fast streaming yellow to green cut
- 10770-10830 DOLOMITE: olive gray, tan, pink, blue to grained, firm to moderately hard, microcrystalline to cryptocrystalline, argillaceous, laminated, minor porosity, carbonaceous, pyrite, light to grained dull fluorescence, slow yellow to green flash

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 10830-10863: DOLOMITE: olive gray, tan, pink, blue to grained, firm to moderately hard, microcrystalline to cryptocrystalline, argillaceous, laminated, minor porosity, carbonaceous, pyrite, dull light green fluorescence, slow yellow to green streaming cut
- 10863-11200: DOLOMITE: pink to off white, light gray in part, firm to hard, microcrystalline to cryptocrystalline, calcareous, silty, poor intercrystalline porosity, trace spotty brown oil stain, dull yellow fluorescence, slow yellow to green diff cut; SHALE: light blue to green, hard, sub platy
- 11200-11450 DOLOMITE: pink to off white, tan, light brown, cream, light gray in part, firm to hard, microcrystalline to cryptocrystalline, calcareous, silty, poor intercrystalline porosity, trace dull yellow fluorescence, slow yellow diffuse cut; SHALE: light blue to green, hard, sub platy
- 11450-11600 DOLOMITE: pink to tan, light brown, cream, microcrystalline to cryptocrystalline, firm to hard, poor intercrystalline porosity, no mineral fluorescence, slow streaming yellow to green cut; SHALE: light blue to green, gray, sub platy
- 11600-11650 DOLOMITE: pink to tan, light brown to grey, microcrystalline to cryptocrystalline, calcareous, silty, firm to hard, poor intercrystalline porosity, bright yellow to green mineral fluorescence, flash yellow to green cut; SHALE: light blue to green, gray, sub platy
- 11650-11700 DOLOMITE: pink to tan, light brown to grey, microcrystalline to cryptocrystalline, calcareous, silty, firm to hard, poor intercrystalline porosity, bright yellow to green mineral fluorescence, flash yellow to green cut; SHALE: red, firm to hard, no porosity, argillaceous, calcite cement; SHALE: light blue to green, gray, sub platy
- 11700-11750 DOLOMITE: pink to tan, light brown to grey, microcrystalline to cryptocrystalline, calcareous, silty, firm to hard, poor intercrystalline porosity, bright yellow to green mineral fluorescence, flash yellow to green cut; SHALE: red, firm to hard, no porosity, argillaceous; SHALE: light blue to green, gray, sub platy
- 11750-11850 DOLOMITE: tan, light pink to brown, microcrystalline to cryptocrystalline, firm to hard, calcareous, silty, poor intercrystalline porosity, dull yellow mineral fluorescence, slow orange to brown cut
- 12000-12050 DOLOMITE: tan, light pink to white, microcrystalline to cryptocrystalline, firm to hard, calcareous, intercrystalline porosity, dull yellow mineral fluorescence, no cut; SHALE: light blue to gray to white, firm hard, calcareous, no porosity



Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 12050-12150 DOLOMITE: tan, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, intercrystalline porosity, dull yellow mineral fluorescence, very dull orange to yellow cut; SHALE: light blue to gray, firm, calcareous, laminated, no porosity
- 12150-12200 DOLOMITE: tan, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, intercrystalline porosity, laminated, dull yellow mineral fluorescence, very dull yellow spotted cut, very trace carbonaceous material
- 12200-12250 DOLOMITE: tan, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, intercrystalline porosity, laminated, dull yellow mineral fluorescence, very dull yellow to green cut
- 12250-12350 SHALE: medium to light gray, very fine grained, laminated, firm to hard, calcareous, no porosity, no mineral fluorescence, no cut; trace DOLOMITE: pink to tan, off white, microcrystalline to cryptocrystalline, firm to hard, calcareous, scattered dull yellow fluorescence, very dull yellow diff cut
- 12350-12550 DOLOMITE: pink to tan, off white, microcrystalline to cryptocrystalline, firm to hard, calcareous, scattered dull yellow fluorescence, very dull yellow to green diff cut; SHALE: dull light gray to white, dull light green gray, calcareous, laminated, sub platy in part, no fluorescence, no cut
- 12550-12700 DOLOMITE: pink to tan, off white, microcrystalline to cryptocrystalline, firm to hard, calcareous, scattered dull yellow fluorescence, very dull yellow to green diff cut; SHALE: dull light gray to white, very trace dull light green gray, calcareous, laminated, sub platy in part, no fluorescence, no cut
- 12700-12800 DOLOMITE: tan to light brown, off white, microcrystalline to cryptocrystalline, firm to hard, calcareous, scattered dull yellow fluorescence, very dull yellow to green diff cut; SHALE: dull light gray to white, very trace dull light green gray, calcareous, laminated, sub platy in part, no fluorescence, no cut
- 12800-13000 DOLOMITE: tan to light brown, cream, microcrystalline to cryptocrystalline, firm to hard, calcareous, scattered dull yellow fluorescence, very dull yellow to green diff cut; SHALE: dull light gray, very trace dull light green gray, calcareous, laminated, sub platy in part, no fluorescence, no cut
- 13000-13050 DOLOMITE: tan to light brown, cream, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, laminated, intercrystalline porosity, dull yellow fluorescence, dull yellow to orange cut; trace SHALE: light gray, calcareous, laminated sub platy, no fluorescence, no cut

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 13050-13100 DOLOMITE: tan to light brown, cream, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, laminated, intercrystalline porosity, dull yellow fluorescence, very dull yellow to orange cut; trace SHALE: light gray, calcareous, laminated sub platy, no fluorescence, no cut
- 13100-13250 DOLOMITE: tan to light brown, cream, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, laminated, intercrystalline porosity, dull yellow fluorescence, dull yellow to orange cut; trace SHALE: light gray, calcareous, laminated sub platy, no fluorescence, no cut
- 13250-13300 DOLOMITE: tan to light brown, cream, white, microcrystalline to cryptocrystalline, firm to hard, calcareous, laminated, intercrystalline porosity, scattered dull yellow fluorescence, dull yellow to orange cut; trace SHALE: light gray, calcareous, laminated sub platy, no fluorescence, no cut
- 13300-13450 DOLOMITE: tan to light brown, cream, white, microcrystalline to cryptocrystalline, laminated in part, intercrystalline porosity, no mineral fluorescence, very dull yellow to green cut; SHALE: light blue to gray, laminated, calcareous, sub platy, no fluorescence, no cut
- 13450-13550 DOLOMITE: tan to light brown, pink, white, microcrystalline to cryptocrystalline, laminated in part, calcareous, intercrystalline porosity, no mineral fluorescence, no mineral cut; SHALE: blue to gray, laminated, calcareous, silty, sub platy, no fluorescence, no cut
- 13550-13650 DOLOMITE: tan to light to brown, pink, white, microcrystalline to cryptocrystalline, laminated in part, calcareous, intercrystalline porosity, sample contaminated with lube, no mineral fluorescence, no cut performed; trace SHALE: blue to gray, laminated, calcareous, silty, no fluorescence, no cut
- 13650-13750 DOLOMITE: white, tan to cream, pink, light gray, microcrystalline to cryptocrystalline, laminated in part, calcareous, intercrystalline porosity, sample contaminated with lube, no mineral fluorescence, no cut; trace SHALE: blue to gray, very calcareous, laminated, no fluorescence, no cut
- 13750-13850 DOLOMITE: off white to pink, tan to cream, pink, light gray, microcrystalline to cryptocrystalline, laminated in part, calcareous, trace intercrystalline porosity; trace SHALE: dull blue to gray, very calcareous, laminated, sample contaminated with lube

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 13850-14150 DOLOMITE: pink to tan, cream to off white light gray, firm to moderately hard, microcrystalline to cryptocrystalline, laminated in part, calcareous, trace intercrystalline porosity; trace SHALE: dull blue to gray, very calcareous, laminated, sample contaminated with lube
- 14150-14300 DOLOMITE: tan to light brown, cream to off white, light to medium gray, firm to moderately hard, microcrystalline to cryptocrystalline, laminated in part, calcareous, trace intercrystalline porosity; trace SHALE: dull blue to gray, very calcareous, laminated, sample contaminated with lube
- 14300-14350 DOLOMITE: clean sample, white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, intercrystalline porosity, no mineral fluorescence, very dull green to yellow cut; trace SHALE: dull blue to gray, calcareous, laminated, no apparent porosity, sample contaminated with lube in part
- 14350-14400 DOLOMITE: clean sample, white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, intercrystalline porosity, minor oil staining, no mineral fluorescence, very dull green to yellow cut; trace SHALE: dull blue to gray, calcareous, laminated, no apparent porosity, sample contaminated with lube in part
- 14400-14500 DOLOMITE: white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, intercrystalline porosity, minor oil staining, no mineral fluorescence, very dull green to yellow cut; trace SHALE: dull blue to gray, calcareous, laminated, no apparent porosity, sample contaminated with lube in part
- 14500-14600 DOLOMITE: white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, intercrystalline porosity, minor oil staining, very dull yellow mineral fluorescence, very dull yellow to green cut; trace SHALE: dull blue to gray, calcareous, laminated, sample contaminated with lube in part
- 14600-14700 DOLOMITE: white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, intercrystalline porosity, minor oil staining, very dull yellow mineral fluorescence, very dull yellow to green cut; trace SHALE: blue to gray, light gray, calcareous, laminated, sample contaminated with lube and LCM

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 14700-14750 DOLOMITE: off white to cream, very dull pink to tan firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, spotty light brown oil stain, scattered dull yellow to green fluorescence, yellow to green moderately diff cut; trace SHALE: blue to gray, light gray, calcareous, laminated
- 14750-14850 DOLOMITE: off white to cream, very dull pink to tan firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, spotty light brown oil stain, scattered green fluorescence, yellow to green moderately fast diff cut; trace SHALE: blue to gray, light gray, calcareous, laminated
- 14850-15000 DOLOMITE: off white, light to medium gray, dull pink to tan firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, spotty light brown oil stain, scattered green fluorescence, slow to moderately yellow to green streaming cut; trace SHALE: blue to gray, light gray, calcareous, laminated
- 15000-15050 DOLOMITE: light brown, tan, off white, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, trace intercrystalline porosity, spotty light brown oil stain, yellow to green fluorescence, slow to moderately yellow to green streaming cut; trace SHALE: blue to gray, light gray, calcareous, laminated, trace pyrite
- 15050-15150 DOLOMITE: light to medium gray, light brown, off white, tan, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, trace intercrystalline porosity, spotty light brown oil stain, scattered green fluorescence, yellow to green slow to moderately streaming cut; trace SHALE: blue to gray, light gray, calcareous, laminated, trace pyrite
- 15150-15200 DOLOMITE: light brown, tan, off white, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, trace intercrystalline porosity, spotty light brown oil stain, scattered yellow to green fluorescence, yellow to green slow to moderately streaming cut; trace SHALE: blue to gray, light gray, calcareous, laminated, trace pyrite
- 15200-15300 DOLOMITE: tan, white, cream, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, very laminated in part, trace intercrystalline porosity, spotty carbonaceous material, scattered yellow to green fluorescence, light yellow cut; SHALE: blue to gray, light gray, calcareous, laminated, no porosity, scattered carbonaceous material, trace pyrite

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 15300-15350 DOLOMITE: white, light pink, tan, cream, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, trace intercrystalline porosity, spotty carbonaceous material, trace pyrite, scattered yellow to green fluorescence, light yellow cut; SHALE: blue to gray, light gray, calcareous, laminated, no porosity, scattered carbonaceous material, trace pyrite
- 15350-15450 DOLOMITE: white, light brown, pink, tan, cream, light gray, firm to moderately hard, microcrystalline to cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, spotty carbonaceous material, trace pyrite, dull yellow to green mineral fluorescence, light yellow cut; SHALE: blue to gray, light gray, calcareous, laminated, no porosity, scattered carbonaceous material, trace pyrite
- 15450-15500 DOLOMITE: light brown, cream, pink, off to white, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, scattered carbonaceous material, yellow mineral fluorescence in part, dull light yellow cut; dolomitic SHALE: blue to gray, light gray, calcareous, laminated, no porosity, scattered carbonaceous material, sample contaminated with lube
- 15500-15550 DOLOMITE: cream, pink, off to white, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, trace intercrystalline porosity, scattered carbonaceous material, pyrite, yellow mineral fluorescence in part, dull yellow cut; dolomitic SHALE: blue to gray, light gray, calcareous, laminated, no porosity, scattered carbonaceous material
- 15550-15600 DOLOMITE: cream, pink, off to white, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, cryptic porosity, pyrite, minor carbonaceous material, light yellow mineral fluorescence, cream to white flash cut; SHALE: blue to gray light gray, calcareous, laminated, scattered carbonaceous material, light yellow mineral fluorescence, cream to white flash cut
- 15600-15650 DOLOMITE: cream, off to white, pink, firm to moderately hard, cryptocrystalline, calcareous, pyrite, intercrystalline porosity, minor carbonaceous material, yellow mineral fluorescence, light yellow fluorescence cut, but sample contaminated with lube; SHALE: blue to gray, light gray, calcareous, laminated, scattered carbonaceous material
- 15650-15750 DOLOMITE: cream, off to white, light brown, firm to moderately hard, cryptocrystalline, calcareous, pyrite, intercrystalline porosity, minor carbonaceous material, yellow mineral fluorescence, light yellow flash cut; SHALE: blue to gray, light gray, calcareous, laminated, scattered carbonaceous material



Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 15750-15800 DOLOMITE: cream, off to white, light brown, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, pyrite, intercrystalline porosity, yellow mineral fluorescence, yellow to cream flash cut; SHALE: blue to gray, light gray, very calcareous, laminated, minor carbonaceous material
- 15800-15850 DOLOMITE: cream, off to white, light brown, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, pyrite, intercrystalline porosity, yellow mineral fluorescence, yellow to cream flash cut; SHALE: blue to gray, light gray, very calcareous, laminated, minor carbonaceous material
- 15850-15900 DOLOMITE: cream, off to white, firm to moderately hard, cryptocrystalline, calcareous, laminated in part, pyrite, intercrystalline porosity, yellow mineral fluorescence, yellow to cream flash cut; SHALE: blue to gray, light gray, very calcareous, laminated, minor carbonaceous material
- 15900-16000 DOLOMITE: white, off to white, tan, firm to moderately hard, cryptocrystalline, laminated in part, minor pyrite, intercrystalline porosity, minor carbonaceous material, light yellow mineral fluorescence, yellow to green flash cut; SHALE: blue to gray, light to gray, very calcareous, laminated, minor carbonaceous material, sample contaminated with lube
- 16000-16150 DOLOMITE: white, off to white, tan, pink, firm to moderately hard, microcrystalline to cryptocrystalline, minor pyrite, intercrystalline porosity, carbonaceous material, light yellow mineral fluorescence, blue to white flash cut; SHALE: blue to gray, light to gray, very calcareous, laminated, minor carbonaceous material, sample contaminated with lube
- 16150-16350 DOLOMITE: pink to light brown, cream to white, light gray in part, firm to moderately hard, microcrystalline to cryptocrystalline, very trace pyrite, trace intercrystalline porosity, dull yellow fluorescence, yellow to green moderately streaming cut; SHALE: light blue green to gray, very calcareous, laminated, sample contaminated with lube
- 16350-16400 DOLOMITE: pink to light brown, cream to white, light gray in part, firm to moderately hard, microcrystalline to cryptocrystalline, very trace pyrite, trace intercrystalline porosity, dull yellow fluorescence, yellow to green moderately streaming cut; trace SHALE: light blue green to gray, very calcareous, laminated, sample contaminated with lube

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 16400-16450 DOLOMITE: pink to light brown, cream to white, light to medium gray in part, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, scattered dull yellow diffuse cut; trace SHALE: light blue green to gray, very calcareous, laminated, sample contaminated with lube
- 16450-16550 DOLOMITE: light gray to off white, pink to light brown, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, scattered very dull yellow fluorescence, very dull yellow slow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, very calcareous, laminated, sample contaminated with lube
- 16550-16700 DOLOMITE: light brown, medium brown, tan, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous
- 16700-16750 DOLOMITE: light brown, medium brown, tan, light gray, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous
- 16750-16800 DOLOMITE: light brown, tan, light gray, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous
- 16800-16900 DOLOMITE: light brown, off white, tan, cream, firm to moderately hard, microcrystalline to cryptocrystalline, laminated in part, trace intercrystalline porosity, pyrite, trace carbonaceous material, sample contaminated with lube, no cut taken; SHALE: blue to gray, light gray, laminated, fine grained, very calcareous, cryptic porosity, no cut taken, sample contaminated with lube
- 16900-17000 DOLOMITE: light pink to cream, tan, off white, microcrystalline to cryptocrystalline, laminated in part, calcareous, intercrystalline porosity, pyrite, trace carbonaceous material, light yellow mineral fluorescence, yellow to cream blooming cut; SHALE: blue to gray, light gray, laminated, fine grained to crystalline, very calcareous, cryptic porosity

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 17000-17050 DOLOMITE: light pink to cream, off white, white, tan, microcrystalline to cryptocrystalline, laminated in part, calcareous, intercrystalline porosity, nodular pyrite, trace carbonaceous material, light yellow mineral fluorescence, light yellow to green flash cut; SHALE: blue to gray, light to gray, laminated, fine grained to crystalline, very calcareous, cryptic porosity
- 17050-17150 DOLOMITE: light pink to cream, off white, white, tan, microcrystalline to cryptocrystalline, laminated in part, intercrystalline porosity, minor nodular pyrite, trace carbonaceous material, sample contaminated with lube to no cut performed; SHALE: blue to gray, light to gray, laminated, very fine grained to crystalline, very calcareous, intercrystalline porosity
- 17150-17200 DOLOMITE: light pink to cream, off white, white, tan, microcrystalline to cryptocrystalline, laminated in part, intercrystalline porosity, minor nodular pyrite, trace carbonaceous material, very light yellow mineral fluorescence, light yellow slow cut; SHALE: blue to gray, light to gray, laminated, very fine grained to crystalline, very calcareous, intercrystalline porosity
- 17200-17250 DOLOMITE: light pink to cream, off white, tan, microcrystalline to cryptocrystalline, intercrystalline porosity, minor nodular pyrite, light yellow mineral fluorescence, cream to blue flash cut; SHALE: blue to gray, light gray, laminated in part, very fine grained to crystalline, very calcareous, intercrystalline porosity
- 17250-17300 DOLOMITE: light pink to cream, off white, tan, microcrystalline to cryptocrystalline, intercrystalline porosity, nodular pyrite, light yellow mineral fluorescence, cream to blue flash cut; SHALE: blue to gray, light to gray, laminated in part, very fine grained to crystalline, very calcareous, intercrystalline porosity
- 17300-17400 DOLOMITE: light brown, medium brown, tan, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous
- 17400-17550 DOLOMITE: pink to light brown, tan, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, scattered dull yellow fluorescence, dull yellow to green diffuse cut; trace SHALE: light green to gray, calcareous, laminated

Formation tops  
Sample Interval  
(in feet)

SAMPLE DESCRIPTION

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- 17550-17750 DOLOMITE: light brown, tan, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, scattered dull yellow fluorescence, dull yellow to green diffuse cut; trace SHALE: light green to gray, calcareous
- 17750-17850 DOLOMITE: light brown, medium brown, tan, off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light gray, off white to cream, dull light green to gray, calcareous
- 17850-17950 DOLOMITE: light brown, medium brown, tan, trace off white, firm to moderately hard, microcrystalline to cryptocrystalline, trace intercrystalline porosity, dull yellow fluorescence, dull yellow diffuse cut; trace SHALE: light to medium gray, blue to green, off white to cream, dull light green to gray, calcareous
- 17950-18000 DOLOMITE: cream, light to pink, off white, firm to moderately hard, calcareous, microcrystalline to cryptocrystalline, intercrystalline porosity, no mineral fluorescence, no cut; trace SHALE: light to gray, blue to gray, laminated in part, calcareous, cryptic porosity
- 18000-18400 DOLOMITE: cream, light to pink, off white, firm to moderately hard, clean dolomite, microcrystalline to cryptocrystalline, intercrystalline porosity, no mineral fluorescence, no cut; trace SHALE: light to gray, blue to gray, laminated in part, calcareous, cryptic porosity
- 18400-18434 DOLOMITE: cream, light to pink, off white, firm to moderately hard, clean dolomite, microcrystalline to cryptocrystalline, intercrystalline porosity, no mineral fluorescence, no cut; SHALE: light to gray, blue to gray, laminated in part, calcareous, cryptic porosity

## **GEOLOGICAL SUMMARY & CONCLUSIONS**

Continental Resources, Inc drilled the south bearing Tallahassee 3-16H in Sections 16, and 21 of Township 153N, Range 101W into the First Bench of the Three Forks Member. Cyclone 4 spud the Tallahassee 3-16H Well on November 8th, 2013 from its surface hole location of 2585' FNL & 268' FEL Section 16, Township 153N, Range 101W.

The surface was pre-drilled with 9 inch surface casing set at 1975' MD. From there the 8 3/4" vertical portion was drilled to 10064' MD where the motor had failed. After TOOH for the curve assembly, the well bore rotated to the kick off point of 10137' MD. LEAM provided Directional Drilling as well as the Measurement While Drilling (MWD) services during the entire course of the Well. NorAm Wellsite Services provided both Mud Logging and Geosteering Services.

The curve portion of the Tallahassee 3-16H well bore began on November 12, 2013 5:30 hrs at a kick off point of 10137' MD. The three forks member was intercepted at 10748' MD, 10571' TVD, (-8583SS) and final landing point of 10863' MD, 10586' TVD placed the bit 15' into formation. The curve was completed in roughly 34.5 hrs ending at 10863' MD, 10586' TVD on November 13th, 2013 16:03 hrs.

The Lateral portion of the Tallahassee 3-16H well bore began at 1:57hrs on November 16, 2013 after intermediate 7 inch casing was set at 10829' MD. The Tallahassee 3-16H reached it's total depth (TD) of 18434' MD, 10573.62' TVD on November 21, 2013 at 18:35 hours. The entire lateral portion of the Tallahassee 3-16H well bore was completed in roughly 136.5 hrs with an overall dip of 90.1 degrees. Tallahassee 3-16H bottom hole location is 232' FSL & 1304' FEL Section 21, T153N, R101W. No trips were necessary in the Lateral.

# NEWSCO

## International Energy Services Inc.

Continental Resources  
Company

33326  
Job Number

8/17/2013  
Date

Cyclone 4  
Rig

Tallahassee 3-16H  
Well Name

McKenzie Co., ND  
County & State

Surveyed from depth of: Surface to 1937'

GL to KB: 20'

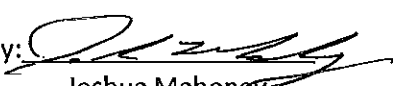
Type of Survey: Nvader/MWD

### True North

Directional Supervisor/Surveyor: David Hopper

The data and calculations for this survey have been checked by me and conform to the standards and procedures set forth by Newsco International Energy Services Inc. This report represents a true and correct directional survey of this well based on the original data obtained at the well site. Wellbore coordinates are calculated using minimum curvature .

Certified by:

  
Joshua Mahoney



# Directional Services U.S.A.

NEWSCO JOB : 33326  
WELL NAME: Tallahassee 3-16H  
RIG: Cyclone 4  
County: Mckenzie

### TARGET DATA

|                |       |
|----------------|-------|
| SENSOR TO BIT: | 47.00 |
| KB TO GL :     | 6.00  |
| Cyclone 4 KB:  | 20.00 |

**All depths are measured from survey tool to Big Rigs KB**

[illegible]



### Survey Certification Sheet

|                                         |                                       |                                              |
|-----------------------------------------|---------------------------------------|----------------------------------------------|
| <u>Continental Resources</u><br>Company | <u>20131228D-ND</u><br>Job Number     | <u>12/04/2013</u><br>Date                    |
| <u>Sec. 16-T153N-R101W</u><br>Lease     | <u>Tallahassee 3-16H</u><br>Well Name | <u>McKenzie County/ ND</u><br>County & State |

Enclosed, please find the survey performed on the referenced well by Leam Drilling Systems, LLC.  
Other information required by your office is as follows:

| Name & Title<br>Of Surveyor | Drainhole<br>Number | Surveyed<br>Depths | Dates<br>Performed    | Type of<br>Survey |
|-----------------------------|---------------------|--------------------|-----------------------|-------------------|
| Randy W. Rakowitz           | OH                  | 2,014' - 18,434'   | 11/08/2013-11/21/2013 | MWD               |

The data and calculations for this survey have been checked and conform to the standards and procedures set forth by LEAM Drilling Systems, LLC. This report represents a true and correct Survey of this well, to the best of our knowledge, based on the original data obtained at the well site.

  
Nora Tucker  
Well Planner

Physical & Mailing: 2010 East Davis, Conroe, TX 77301

(936) 756-7577 • (936) 756-7595 Fax • 1 (800) 756-7504



LEAM Drilling Systems, Inc.  
2027A Airport Rd.  
Conroe, TX 77301  
(936) 569-1315

## Survey Certification Form

State of North Dakota  
McKenzie County

I, Randy W. Rakowitz, an employee of LEAM Drilling Systems, Inc., hereby certify that on the dates beginning on 11/8/2013 and ending on 11/21/2013, I conducted or supervised the taking of an MWD Survey from a beginning depth of 2,014' ft. MD to an ending depth of 18,434' ft. MD; that the depth is true, correct, complete, and within the limitations of the tools as set forth by LEAM Drilling Systems, Inc., that I am authorized and qualified to make this report; that this survey was conducted in reference to True North with a declination of 8.52° with respect to the well Tallahassee 3-16H located in McKenzie County, North Dakota as requested by Continental Resources.

A handwritten signature in black ink, appearing to read 'Randy W. Rakowitz', is written over a solid black horizontal line.

Randy W. Rakowitz  
Sr. MWD Operations Coordinator

# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

|                    |                            |                      |                |
|--------------------|----------------------------|----------------------|----------------|
| <b>Project</b>     | McKenzie County, ND        |                      |                |
| <b>Map System:</b> | US State Plane 1983        | <b>System Datum:</b> | Mean Sea Level |
| <b>Geo Datum:</b>  | North American Datum 1983  |                      |                |
| <b>Map Zone:</b>   | North Dakota Northern Zone |                      |                |

|                              |                                       |                          |                   |
|------------------------------|---------------------------------------|--------------------------|-------------------|
| <b>Site</b>                  | Tallahassee 3, Sec. 16 - T153N R-101W |                          |                   |
| <b>Site Position:</b>        |                                       | <b>Northing:</b>         | 408,123.52 usft   |
| <b>From:</b>                 | Lat/Long                              | <b>Easting:</b>          | 1,193,773.27 usft |
| <b>Position Uncertainty:</b> | 0.00 usft                             | <b>Slot Radius:</b>      | 0 "               |
|                              |                                       | <b>Latitude:</b>         | 48° 4' 30.494 N   |
|                              |                                       | <b>Longitude:</b>        | 103° 40' 12.019 W |
|                              |                                       | <b>Grid Convergence:</b> | -2.36 °           |

|                             |              |                            |                                   |
|-----------------------------|--------------|----------------------------|-----------------------------------|
| <b>Well</b>                 | 3-16H        |                            |                                   |
| <b>Well Position</b>        | <b>+N/-S</b> | 0.00 usft                  | <b>Northing:</b> 408,123.52 usft  |
|                             | <b>+E/-W</b> | 0.00 usft                  | <b>Easting:</b> 1,193,773.27 usft |
| <b>Position Uncertainty</b> | 0.00 usft    | <b>Wellhead Elevation:</b> | usft                              |
|                             |              | <b>Latitude:</b>           | 48° 4' 30.494 N                   |
|                             |              | <b>Longitude:</b>          | 103° 40' 12.019 W                 |
|                             |              | <b>Ground Level:</b>       | 1,920.00 usft                     |

|                  |                   |                    |                        |                      |                            |
|------------------|-------------------|--------------------|------------------------|----------------------|----------------------------|
| <b>Wellbore</b>  | OH                |                    |                        |                      |                            |
| <b>Magnetics</b> | <b>Model Name</b> | <b>Sample Date</b> | <b>Declination (°)</b> | <b>Dip Angle (°)</b> | <b>Field Strength (nT)</b> |
|                  | BGGM2013          | 10/11/13           | 8.52                   | 72.97                | 56,411                     |

|                          |                                |                     |                     |                      |      |
|--------------------------|--------------------------------|---------------------|---------------------|----------------------|------|
| <b>Design</b>            | OH                             |                     |                     |                      |      |
| <b>Audit Notes:</b>      |                                |                     |                     |                      |      |
| <b>Version:</b>          | 1.0                            | <b>Phase:</b>       | ACTUAL              | <b>Tie On Depth:</b> | 0.00 |
| <b>Vertical Section:</b> | <b>Depth From (TVD) (usft)</b> | <b>+N/-S (usft)</b> | <b>+E/-W (usft)</b> | <b>Direction (°)</b> |      |
|                          | 0.00                           | 0.00                | 0.00                | 187.73               |      |

|                       |                  |                          |                  |                    |  |
|-----------------------|------------------|--------------------------|------------------|--------------------|--|
| <b>Survey Program</b> | <b>Date</b>      | 12/04/13                 |                  |                    |  |
| <b>From (usft)</b>    | <b>To (usft)</b> | <b>Survey (Wellbore)</b> | <b>Tool Name</b> | <b>Description</b> |  |
| 143.00                | 1,938.00         | Survey #1 - NEWSCO (OH)  | MWD-ISCWSA       | MWD - Standard     |  |
| 2,014.00              | 18,385.00        | Survey #2 (OH)           | LEAM MWD-ADJ     | MWD - Standard     |  |
| 18,434.00             | 18,434.00        | Survey #3 (OH)           | Project          | Projection         |  |

|                              |                        |                    |                              |                     |                     |                                |                                |                               |                              |  |
|------------------------------|------------------------|--------------------|------------------------------|---------------------|---------------------|--------------------------------|--------------------------------|-------------------------------|------------------------------|--|
| <b>Survey</b>                |                        |                    |                              |                     |                     |                                |                                |                               |                              |  |
| <b>Measured Depth (usft)</b> | <b>Inclination (°)</b> | <b>Azimuth (°)</b> | <b>Vertical Depth (usft)</b> | <b>+N/-S (usft)</b> | <b>+E/-W (usft)</b> | <b>Vertical Section (usft)</b> | <b>Dogleg Rate (°/100usft)</b> | <b>Build Rate (°/100usft)</b> | <b>Turn Rate (°/100usft)</b> |  |
| 1,938.00                     | 0.70                   | 307.10             | 1,937.97                     | 3.52                | -3.92               | -2.96                          | 0.70                           | 0.42                          | -62.08                       |  |
| 2,014.00                     | 0.80                   | 280.20             | 2,013.97                     | 3.89                | -4.82               | -3.21                          | 0.48                           | 0.13                          | -35.39                       |  |
| 2,077.00                     | 0.90                   | 246.50             | 2,076.96                     | 3.77                | -5.70               | -2.97                          | 0.80                           | 0.16                          | -53.49                       |  |
| 2,170.00                     | 1.10                   | 238.70             | 2,169.95                     | 3.02                | -7.14               | -2.03                          | 0.26                           | 0.22                          | -8.39                        |  |
| 2,263.00                     | 1.30                   | 236.50             | 2,262.93                     | 1.97                | -8.78               | -0.77                          | 0.22                           | 0.22                          | -2.37                        |  |
| 2,357.00                     | 1.80                   | 235.80             | 2,356.89                     | 0.55                | -10.89              | 0.92                           | 0.53                           | 0.53                          | -0.74                        |  |
| 2,450.00                     | 2.40                   | 199.00             | 2,449.83                     | -2.11               | -12.73              | 3.80                           | 1.55                           | 0.65                          | -39.57                       |  |
| 2,543.00                     | 2.20                   | 181.60             | 2,542.76                     | -5.74               | -13.42              | 7.49                           | 0.78                           | -0.22                         | -18.71                       |  |
| 2,638.00                     | 1.20                   | 152.80             | 2,637.72                     | -8.44               | -13.01              | 10.12                          | 1.35                           | -1.05                         | -30.32                       |  |

# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

| Survey                |                 |             |                       |              |              |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 2,732.00              | 1.00            | 119.70      | 2,731.70              | -9.72        | -11.85       | 11.23                   | 0.70                    | -0.21                  | -35.21                |
| 2,825.00              | 1.20            | 134.30      | 2,824.68              | -10.81       | -10.45       | 12.11                   | 0.37                    | 0.22                   | 15.70                 |
| 2,918.00              | 1.50            | 119.20      | 2,917.66              | -12.08       | -8.69        | 13.14                   | 0.50                    | 0.32                   | -16.24                |
| 3,011.00              | 1.80            | 119.00      | 3,010.62              | -13.38       | -6.35        | 14.12                   | 0.32                    | 0.32                   | -0.22                 |
| 3,105.00              | 1.20            | 84.60       | 3,104.59              | -14.01       | -4.08        | 14.43                   | 1.12                    | -0.64                  | -36.60                |
| 3,198.00              | 1.50            | 66.70       | 3,197.56              | -13.43       | -1.99        | 13.58                   | 0.55                    | 0.32                   | -19.25                |
| 3,291.00              | 1.10            | 90.20       | 3,290.54              | -12.95       | 0.02         | 12.83                   | 0.71                    | -0.43                  | 25.27                 |
| 3,384.00              | 1.10            | 92.30       | 3,383.52              | -12.99       | 1.81         | 12.63                   | 0.04                    | 0.00                   | 2.26                  |
| 3,477.00              | 1.00            | 80.50       | 3,476.51              | -12.90       | 3.50         | 12.31                   | 0.26                    | -0.11                  | -12.69                |
| 3,571.00              | 0.90            | 91.60       | 3,570.49              | -12.78       | 5.05         | 11.99                   | 0.22                    | -0.11                  | 11.81                 |
| 3,664.00              | 0.70            | 94.10       | 3,663.48              | -12.84       | 6.34         | 11.87                   | 0.22                    | -0.22                  | 2.69                  |
| 3,758.00              | 0.60            | 124.30      | 3,757.48              | -13.16       | 7.32         | 12.06                   | 0.37                    | -0.11                  | 32.13                 |
| 3,851.00              | 0.50            | 138.60      | 3,850.47              | -13.74       | 7.99         | 12.54                   | 0.18                    | -0.11                  | 15.38                 |
| 3,945.00              | 0.40            | 127.30      | 3,944.47              | -14.25       | 8.52         | 12.97                   | 0.14                    | -0.11                  | -12.02                |
| 4,038.00              | 0.40            | 231.90      | 4,037.47              | -14.64       | 8.53         | 13.36                   | 0.68                    | 0.00                   | 112.47                |
| 4,132.00              | 0.30            | 320.00      | 4,131.47              | -14.66       | 8.11         | 13.43                   | 0.52                    | -0.11                  | 93.72                 |
| 4,225.00              | 0.10            | 245.80      | 4,224.47              | -14.50       | 7.88         | 13.31                   | 0.31                    | -0.22                  | -79.78                |
| 4,319.00              | 0.10            | 344.60      | 4,318.47              | -14.46       | 7.78         | 13.28                   | 0.16                    | 0.00                   | 105.11                |
| 4,412.00              | 0.30            | 254.00      | 4,411.47              | -14.45       | 7.53         | 13.30                   | 0.34                    | 0.22                   | -97.42                |
| 4,505.00              | 0.10            | 258.10      | 4,504.47              | -14.53       | 7.21         | 13.43                   | 0.22                    | -0.22                  | 4.41                  |
| 4,599.00              | 0.40            | 231.50      | 4,598.47              | -14.75       | 6.88         | 13.69                   | 0.33                    | 0.32                   | -28.30                |
| 4,692.00              | 0.30            | 263.70      | 4,691.47              | -14.98       | 6.38         | 13.99                   | 0.23                    | -0.11                  | 34.62                 |
| 4,785.00              | 0.30            | 351.80      | 4,784.46              | -14.77       | 6.11         | 13.81                   | 0.45                    | 0.00                   | 94.73                 |
| 4,879.00              | 0.50            | 40.30       | 4,878.46              | -14.21       | 6.34         | 13.23                   | 0.40                    | 0.21                   | 51.60                 |
| 4,972.00              | 0.40            | 63.10       | 4,971.46              | -13.75       | 6.89         | 12.70                   | 0.22                    | -0.11                  | 24.52                 |
| 5,066.00              | 0.80            | 44.90       | 5,065.45              | -13.14       | 7.64         | 11.99                   | 0.47                    | 0.43                   | -19.36                |
| 5,159.00              | 1.10            | 55.80       | 5,158.44              | -12.18       | 8.84         | 10.88                   | 0.38                    | 0.32                   | 11.72                 |
| 5,253.00              | 1.10            | 62.60       | 5,252.42              | -11.26       | 10.39        | 9.76                    | 0.14                    | 0.00                   | 7.23                  |
| 5,347.00              | 0.90            | 60.30       | 5,346.41              | -10.47       | 11.83        | 8.79                    | 0.22                    | -0.21                  | -2.45                 |
| 5,440.00              | 1.10            | 68.10       | 5,439.40              | -9.78        | 13.29        | 7.90                    | 0.26                    | 0.22                   | 8.39                  |
| 5,533.00              | 1.20            | 62.80       | 5,532.38              | -9.00        | 14.99        | 6.90                    | 0.16                    | 0.11                   | -5.70                 |
| 5,627.00              | 0.90            | 51.00       | 5,626.36              | -8.09        | 16.44        | 5.80                    | 0.39                    | -0.32                  | -12.55                |
| 5,720.00              | 0.80            | 32.60       | 5,719.35              | -7.08        | 17.35        | 4.68                    | 0.31                    | -0.11                  | -19.78                |
| 5,813.00              | 0.70            | 79.50       | 5,812.34              | -6.43        | 18.26        | 3.92                    | 0.65                    | -0.11                  | 50.43                 |
| 5,907.00              | 1.10            | 100.90      | 5,906.33              | -6.50        | 19.71        | 3.79                    | 0.55                    | 0.43                   | 22.77                 |
| 6,000.00              | 0.40            | 110.60      | 5,999.32              | -6.78        | 20.89        | 3.91                    | 0.76                    | -0.75                  | 10.43                 |
| 6,094.00              | 0.80            | 289.50      | 6,093.32              | -6.68        | 20.58        | 3.85                    | 1.28                    | 0.43                   | 190.32                |
| 6,187.00              | 0.50            | 289.20      | 6,186.32              | -6.33        | 19.59        | 3.63                    | 0.32                    | -0.32                  | -0.32                 |
| 6,281.00              | 0.50            | 5.70        | 6,280.31              | -5.78        | 19.24        | 3.14                    | 0.66                    | 0.00                   | 81.38                 |
| 6,375.00              | 0.60            | 43.10       | 6,374.31              | -5.02        | 19.62        | 2.33                    | 0.39                    | 0.11                   | 39.79                 |
| 6,469.00              | 0.10            | 144.00      | 6,468.31              | -4.72        | 20.00        | 1.99                    | 0.67                    | -0.53                  | 107.34                |
| 6,562.00              | 1.10            | 255.60      | 6,561.30              | -5.01        | 19.18        | 2.38                    | 1.23                    | 1.08                   | 120.00                |
| 6,655.00              | 1.00            | 255.60      | 6,654.29              | -5.43        | 17.53        | 3.03                    | 0.11                    | -0.11                  | 0.00                  |

# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

| Survey                |                 |             |                       |              |              |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 6,749.00              | 1.10            | 245.80      | 6,748.27              | -6.01        | 15.92        | 3.81                    | 0.22                    | 0.11                   | -10.43                |
| 6,843.00              | 1.10            | 247.20      | 6,842.25              | -6.73        | 14.26        | 4.75                    | 0.03                    | 0.00                   | 1.49                  |
| 6,936.00              | 1.20            | 254.60      | 6,935.23              | -7.33        | 12.50        | 5.58                    | 0.19                    | 0.11                   | 7.96                  |
| 7,030.00              | 1.20            | 239.50      | 7,029.21              | -8.09        | 10.70        | 6.58                    | 0.34                    | 0.00                   | -16.06                |
| 7,123.00              | 1.80            | 238.60      | 7,122.18              | -9.35        | 8.62         | 8.10                    | 0.65                    | 0.65                   | -0.97                 |
| 7,216.00              | 1.80            | 246.30      | 7,215.14              | -10.70       | 6.03         | 9.79                    | 0.26                    | 0.00                   | 8.28                  |
| 7,309.00              | 1.10            | 286.60      | 7,308.11              | -11.03       | 3.84         | 10.41                   | 1.29                    | -0.75                  | 43.33                 |
| 7,404.00              | 1.10            | 284.30      | 7,403.09              | -10.54       | 2.08         | 10.17                   | 0.05                    | 0.00                   | -2.42                 |
| 7,497.00              | 1.10            | 286.70      | 7,496.07              | -10.07       | 0.36         | 9.92                    | 0.05                    | 0.00                   | 2.58                  |
| 7,590.00              | 1.50            | 292.50      | 7,589.05              | -9.34        | -1.62        | 9.48                    | 0.45                    | 0.43                   | 6.24                  |
| 7,683.00              | 1.60            | 299.40      | 7,682.01              | -8.24        | -3.87        | 8.69                    | 0.23                    | 0.11                   | 7.42                  |
| 7,777.00              | 1.30            | 305.50      | 7,775.98              | -6.98        | -5.89        | 7.70                    | 0.36                    | -0.32                  | 6.49                  |
| 7,870.00              | 1.40            | 288.80      | 7,868.96              | -6.00        | -7.82        | 6.99                    | 0.43                    | 0.11                   | -17.96                |
| 7,963.00              | 1.10            | 285.20      | 7,961.94              | -5.40        | -9.76        | 6.66                    | 0.33                    | -0.32                  | -3.87                 |
| 8,056.00              | 0.90            | 279.50      | 8,054.92              | -5.04        | -11.34       | 6.52                    | 0.24                    | -0.22                  | -6.13                 |
| 8,149.00              | 0.90            | 270.00      | 8,147.91              | -4.92        | -12.79       | 6.60                    | 0.16                    | 0.00                   | -10.22                |
| 8,243.00              | 1.00            | 268.30      | 8,241.90              | -4.95        | -14.35       | 6.83                    | 0.11                    | 0.11                   | -1.81                 |
| 8,336.00              | 1.00            | 272.10      | 8,334.88              | -4.94        | -15.97       | 7.04                    | 0.07                    | 0.00                   | 4.09                  |
| 8,429.00              | 0.90            | 276.50      | 8,427.87              | -4.83        | -17.51       | 7.14                    | 0.13                    | -0.11                  | 4.73                  |
| 8,522.00              | 0.90            | 280.40      | 8,520.86              | -4.61        | -18.95       | 7.12                    | 0.07                    | 0.00                   | 4.19                  |
| 8,616.00              | 1.00            | 257.40      | 8,614.85              | -4.66        | -20.48       | 7.37                    | 0.42                    | 0.11                   | -24.47                |
| 8,709.00              | 1.10            | 267.20      | 8,707.83              | -4.88        | -22.16       | 7.82                    | 0.22                    | 0.11                   | 10.54                 |
| 8,803.00              | 1.00            | 261.10      | 8,801.82              | -5.05        | -23.87       | 8.22                    | 0.16                    | -0.11                  | -6.49                 |
| 8,896.00              | 0.90            | 264.20      | 8,894.80              | -5.25        | -25.40       | 8.62                    | 0.12                    | -0.11                  | 3.33                  |
| 8,990.00              | 0.40            | 334.00      | 8,988.80              | -5.03        | -26.28       | 8.52                    | 0.90                    | -0.53                  | 74.26                 |
| 9,083.00              | 0.80            | 47.70       | 9,081.79              | -4.30        | -25.94       | 7.75                    | 0.85                    | 0.43                   | 79.25                 |
| 9,176.00              | 0.80            | 63.10       | 9,174.78              | -3.57        | -24.88       | 6.88                    | 0.23                    | 0.00                   | 16.56                 |
| 9,270.00              | 0.80            | 59.10       | 9,268.78              | -2.94        | -23.73       | 6.10                    | 0.06                    | 0.00                   | -4.26                 |
| 9,363.00              | 0.50            | 49.80       | 9,361.77              | -2.34        | -22.87       | 5.40                    | 0.34                    | -0.32                  | -10.00                |
| 9,457.00              | 0.40            | 48.20       | 9,455.77              | -1.86        | -22.31       | 4.84                    | 0.11                    | -0.11                  | -1.70                 |
| 9,550.00              | 0.40            | 48.60       | 9,548.76              | -1.43        | -21.82       | 4.35                    | 0.00                    | 0.00                   | 0.43                  |
| 9,643.00              | 0.50            | 51.70       | 9,641.76              | -0.96        | -21.26       | 3.81                    | 0.11                    | 0.11                   | 3.33                  |
| 9,737.00              | 0.40            | 12.90       | 9,735.76              | -0.39        | -20.87       | 3.19                    | 0.33                    | -0.11                  | -41.28                |
| 9,830.00              | 0.30            | 351.40      | 9,828.76              | 0.17         | -20.83       | 2.63                    | 0.18                    | -0.11                  | -23.12                |
| 9,923.00              | 0.30            | 308.20      | 9,921.76              | 0.56         | -21.06       | 2.28                    | 0.24                    | 0.00                   | -46.45                |
| 10,019.00             | 0.30            | 299.40      | 10,017.75             | 0.84         | -21.47       | 2.06                    | 0.05                    | 0.00                   | -9.17                 |
| 10,112.00             | 0.40            | 325.20      | 10,110.75             | 1.23         | -21.87       | 1.73                    | 0.20                    | 0.11                   | 27.74                 |
| 10,143.00             | 1.80            | 252.60      | 10,141.75             | 1.17         | -22.40       | 1.85                    | 5.56                    | 4.52                   | -234.19               |
| 10,174.00             | 5.70            | 242.60      | 10,172.68             | 0.32         | -24.23       | 2.95                    | 12.71                   | 12.58                  | -32.26                |
| 10,205.00             | 9.10            | 239.30      | 10,203.41             | -1.65        | -27.71       | 5.36                    | 11.05                   | 10.97                  | -10.65                |
| 10,236.00             | 12.50           | 234.40      | 10,233.86             | -4.85        | -32.54       | 9.18                    | 11.35                   | 10.97                  | -15.81                |
| 10,268.00             | 15.70           | 234.50      | 10,264.89             | -9.38        | -38.88       | 14.53                   | 10.00                   | 10.00                  | 0.31                  |
| 10,299.00             | 20.40           | 232.60      | 10,294.36             | -15.10       | -46.60       | 21.23                   | 15.28                   | 15.16                  | -6.13                 |

# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

| Survey                |                 |             |                       |              |              |                         |                         |                        |                       |  |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| 10,330.00             | 25.70           | 233.30      | 10,322.87             | -22.41       | -56.28       | 29.77                   | 17.12                   | 17.10                  | 2.26                  |  |
| 10,361.00             | 29.90           | 231.90      | 10,350.29             | -31.19       | -67.76       | 40.03                   | 13.71                   | 13.55                  | -4.52                 |  |
| 10,392.00             | 33.20           | 231.00      | 10,376.70             | -41.31       | -80.44       | 51.75                   | 10.75                   | 10.65                  | -2.90                 |  |
| 10,423.00             | 36.80           | 230.80      | 10,402.09             | -52.52       | -94.23       | 64.72                   | 11.62                   | 11.61                  | -0.65                 |  |
| 10,455.00             | 40.00           | 230.00      | 10,427.17             | -65.19       | -109.54      | 79.33                   | 10.12                   | 10.00                  | -2.50                 |  |
| 10,485.00             | 45.40           | 229.80      | 10,449.21             | -78.29       | -125.10      | 94.41                   | 18.01                   | 18.00                  | -0.67                 |  |
| 10,516.00             | 49.40           | 230.00      | 10,470.19             | -92.99       | -142.55      | 111.32                  | 12.91                   | 12.90                  | 0.65                  |  |
| 10,548.00             | 55.00           | 230.00      | 10,489.79             | -109.23      | -161.91      | 130.02                  | 17.50                   | 17.50                  | 0.00                  |  |
| 10,579.00             | 59.40           | 230.00      | 10,506.58             | -125.98      | -181.87      | 149.30                  | 14.19                   | 14.19                  | 0.00                  |  |
| 10,610.00             | 61.30           | 229.80      | 10,521.92             | -143.33      | -202.48      | 169.26                  | 6.15                    | 6.13                   | -0.65                 |  |
| 10,641.00             | 64.30           | 229.10      | 10,536.08             | -161.26      | -223.42      | 189.84                  | 9.88                    | 9.68                   | -2.26                 |  |
| 10,672.00             | 68.90           | 227.80      | 10,548.39             | -180.12      | -244.70      | 211.40                  | 15.33                   | 14.85                  | -4.13                 |  |
| 10,703.00             | 72.10           | 227.70      | 10,558.74             | -199.77      | -266.33      | 233.78                  | 10.33                   | 10.32                  | -0.32                 |  |
| 10,735.00             | 73.90           | 227.50      | 10,568.09             | -220.40      | -288.93      | 257.26                  | 5.66                    | 5.63                   | -0.63                 |  |
| 10,766.00             | 77.20           | 227.10      | 10,575.83             | -240.76      | -310.99      | 280.40                  | 10.72                   | 10.65                  | -1.29                 |  |
| 10,797.00             | 82.50           | 226.10      | 10,581.29             | -261.72      | -333.15      | 304.15                  | 17.39                   | 17.10                  | -3.23                 |  |
| 10,807.00             | 84.20           | 225.90      | 10,582.45             | -268.62      | -340.29      | 311.95                  | 17.12                   | 17.00                  | -2.00                 |  |
| 10,889.00             | 88.00           | 222.60      | 10,588.03             | -327.21      | -397.36      | 377.68                  | 6.13                    | 4.63                   | -4.02                 |  |
| 10,920.00             | 89.50           | 221.30      | 10,588.70             | -350.25      | -418.07      | 403.30                  | 6.40                    | 4.84                   | -4.19                 |  |
| 11,015.00             | 90.90           | 218.70      | 10,588.37             | -423.02      | -479.13      | 483.62                  | 3.11                    | 1.47                   | -2.74                 |  |
| 11,110.00             | 91.80           | 214.70      | 10,586.13             | -499.15      | -535.88      | 566.69                  | 4.31                    | 0.95                   | -4.21                 |  |
| 11,205.00             | 89.20           | 213.60      | 10,585.30             | -577.76      | -589.20      | 651.76                  | 2.97                    | -2.74                  | -1.16                 |  |
| 11,300.00             | 90.00           | 210.30      | 10,585.97             | -658.35      | -639.46      | 738.38                  | 3.57                    | 0.84                   | -3.47                 |  |
| 11,395.00             | 90.40           | 207.60      | 10,585.63             | -741.47      | -685.44      | 826.93                  | 2.87                    | 0.42                   | -2.84                 |  |
| 11,490.00             | 90.70           | 205.30      | 10,584.72             | -826.52      | -727.75      | 916.90                  | 2.44                    | 0.32                   | -2.42                 |  |
| 11,585.00             | 89.90           | 204.60      | 10,584.23             | -912.65      | -767.83      | 1,007.64                | 1.12                    | -0.84                  | -0.74                 |  |
| 11,680.00             | 89.70           | 204.50      | 10,584.56             | -999.06      | -807.30      | 1,098.57                | 0.24                    | -0.21                  | -0.11                 |  |
| 11,775.00             | 88.20           | 202.70      | 10,586.30             | -1,086.10    | -845.32      | 1,189.93                | 2.47                    | -1.58                  | -1.89                 |  |
| 11,870.00             | 89.10           | 201.50      | 10,588.54             | -1,174.09    | -881.05      | 1,281.93                | 1.58                    | 0.95                   | -1.26                 |  |
| 11,965.00             | 90.10           | 199.40      | 10,589.20             | -1,263.10    | -914.24      | 1,374.59                | 2.45                    | 1.05                   | -2.21                 |  |
| 12,060.00             | 89.10           | 197.60      | 10,589.86             | -1,353.18    | -944.38      | 1,467.91                | 2.17                    | -1.05                  | -1.89                 |  |
| 12,155.00             | 89.70           | 194.60      | 10,590.86             | -1,444.44    | -970.72      | 1,561.88                | 3.22                    | 0.63                   | -3.16                 |  |
| 12,250.00             | 89.80           | 191.80      | 10,591.27             | -1,536.92    | -992.41      | 1,656.44                | 2.95                    | 0.11                   | -2.95                 |  |
| 12,345.00             | 90.70           | 189.40      | 10,590.86             | -1,630.29    | -1,009.89    | 1,751.31                | 2.70                    | 0.95                   | -2.53                 |  |
| 12,440.00             | 91.30           | 186.40      | 10,589.20             | -1,724.36    | -1,022.94    | 1,846.28                | 3.22                    | 0.63                   | -3.16                 |  |
| 12,535.00             | 88.80           | 184.10      | 10,589.12             | -1,818.95    | -1,031.63    | 1,941.18                | 3.58                    | -2.63                  | -2.42                 |  |
| 12,630.00             | 89.00           | 181.60      | 10,590.94             | -1,913.81    | -1,036.35    | 2,035.81                | 2.64                    | 0.21                   | -2.63                 |  |
| 12,725.00             | 90.40           | 181.80      | 10,591.44             | -2,008.76    | -1,039.17    | 2,130.28                | 1.49                    | 1.47                   | 0.21                  |  |
| 12,820.00             | 90.90           | 179.30      | 10,590.36             | -2,103.74    | -1,040.08    | 2,224.52                | 2.68                    | 0.53                   | -2.63                 |  |
| 12,915.00             | 90.00           | 179.70      | 10,589.61             | -2,198.73    | -1,039.26    | 2,318.54                | 1.04                    | -0.95                  | 0.42                  |  |
| 13,010.00             | 89.30           | 179.00      | 10,590.19             | -2,293.72    | -1,038.18    | 2,412.52                | 1.04                    | -0.74                  | -0.74                 |  |
| 13,104.00             | 88.80           | 180.40      | 10,591.75             | -2,387.71    | -1,037.69    | 2,505.58                | 1.58                    | -0.53                  | 1.49                  |  |



# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

| Survey                |                 |             |                       |              |              |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 13,199.00             | 89.50           | 180.90      | 10,593.16             | -2,482.69    | -1,038.76    | 2,599.85                | 0.91                    | 0.74                   | 0.53                  |
| 13,295.00             | 90.80           | 180.90      | 10,592.91             | -2,578.68    | -1,040.27    | 2,695.17                | 1.35                    | 1.35                   | 0.00                  |
| 13,389.00             | 91.50           | 179.50      | 10,591.02             | -2,672.65    | -1,040.60    | 2,788.33                | 1.66                    | 0.74                   | -1.49                 |
| 13,484.00             | 91.80           | 180.20      | 10,588.29             | -2,767.61    | -1,040.35    | 2,882.40                | 0.80                    | 0.32                   | 0.74                  |
| 13,579.00             | 89.20           | 180.00      | 10,587.46             | -2,862.60    | -1,040.52    | 2,976.54                | 2.74                    | -2.74                  | -0.21                 |
| 13,674.00             | 89.00           | 180.00      | 10,588.95             | -2,957.59    | -1,040.52    | 3,070.67                | 0.21                    | -0.21                  | 0.00                  |
| 13,769.00             | 90.50           | 179.00      | 10,589.37             | -3,052.58    | -1,039.69    | 3,164.69                | 1.90                    | 1.58                   | -1.05                 |
| 13,864.00             | 89.70           | 180.40      | 10,589.20             | -3,147.58    | -1,039.19    | 3,258.75                | 1.70                    | -0.84                  | 1.47                  |
| 13,959.00             | 89.10           | 180.70      | 10,590.20             | -3,242.57    | -1,040.10    | 3,353.00                | 0.71                    | -0.63                  | 0.32                  |
| 14,054.00             | 89.60           | 179.70      | 10,591.27             | -3,337.56    | -1,040.43    | 3,447.17                | 1.18                    | 0.53                   | -1.05                 |
| 14,149.00             | 89.70           | 179.70      | 10,591.85             | -3,432.56    | -1,039.94    | 3,541.24                | 0.11                    | 0.11                   | 0.00                  |
| 14,244.00             | 90.90           | 179.20      | 10,591.36             | -3,527.55    | -1,039.02    | 3,635.25                | 1.37                    | 1.26                   | -0.53                 |
| 14,339.00             | 89.90           | 179.30      | 10,590.69             | -3,622.54    | -1,037.78    | 3,729.20                | 1.06                    | -1.05                  | 0.11                  |
| 14,434.00             | 90.40           | 179.00      | 10,590.44             | -3,717.52    | -1,036.37    | 3,823.14                | 0.61                    | 0.53                   | -0.32                 |
| 14,529.00             | 90.30           | 178.60      | 10,589.86             | -3,812.50    | -1,034.38    | 3,916.99                | 0.43                    | -0.11                  | -0.42                 |
| 14,624.00             | 91.50           | 178.30      | 10,588.37             | -3,907.45    | -1,031.81    | 4,010.73                | 1.30                    | 1.26                   | -0.32                 |
| 14,719.00             | 90.90           | 177.40      | 10,586.38             | -4,002.36    | -1,028.25    | 4,104.30                | 1.14                    | -0.63                  | -0.95                 |
| 14,814.00             | 90.70           | 178.50      | 10,585.06             | -4,097.29    | -1,024.85    | 4,197.91                | 1.18                    | -0.21                  | 1.16                  |
| 14,909.00             | 91.10           | 179.70      | 10,583.56             | -4,192.27    | -1,023.36    | 4,291.82                | 1.33                    | 0.42                   | 1.26                  |
| 15,004.00             | 91.10           | 179.70      | 10,581.74             | -4,287.25    | -1,022.86    | 4,385.87                | 0.00                    | 0.00                   | 0.00                  |
| 15,099.00             | 89.60           | 179.30      | 10,581.16             | -4,382.24    | -1,022.03    | 4,479.89                | 1.63                    | -1.58                  | -0.42                 |
| 15,193.00             | 89.60           | 180.20      | 10,581.82             | -4,476.24    | -1,021.62    | 4,572.97                | 0.96                    | 0.00                   | 0.96                  |
| 15,289.00             | 89.20           | 180.70      | 10,582.82             | -4,572.23    | -1,022.38    | 4,668.19                | 0.67                    | -0.42                  | 0.52                  |
| 15,383.00             | 89.30           | 181.30      | 10,584.05             | -4,666.20    | -1,024.02    | 4,761.54                | 0.65                    | 0.11                   | 0.64                  |
| 15,479.00             | 90.80           | 180.20      | 10,583.97             | -4,762.19    | -1,025.27    | 4,856.82                | 1.94                    | 1.56                   | -1.15                 |
| 15,574.00             | 89.90           | 181.60      | 10,583.39             | -4,857.18    | -1,026.77    | 4,951.14                | 1.75                    | -0.95                  | 1.47                  |
| 15,668.00             | 88.80           | 182.50      | 10,584.45             | -4,951.11    | -1,030.13    | 5,044.67                | 1.51                    | -1.17                  | 0.96                  |
| 15,763.00             | 90.40           | 182.70      | 10,585.12             | -5,046.00    | -1,034.44    | 5,139.29                | 1.70                    | 1.68                   | 0.21                  |
| 15,858.00             | 88.50           | 182.80      | 10,586.03             | -5,140.89    | -1,038.99    | 5,233.92                | 2.00                    | -2.00                  | 0.11                  |
| 15,953.00             | 89.30           | 182.70      | 10,587.85             | -5,235.76    | -1,043.55    | 5,328.54                | 0.85                    | 0.84                   | -0.11                 |
| 16,048.00             | 90.50           | 182.50      | 10,588.02             | -5,330.66    | -1,047.86    | 5,423.16                | 1.28                    | 1.26                   | -0.21                 |
| 16,143.00             | 88.40           | 182.30      | 10,588.93             | -5,425.57    | -1,051.84    | 5,517.74                | 2.22                    | -2.21                  | -0.21                 |
| 16,238.00             | 89.40           | 182.50      | 10,590.75             | -5,520.46    | -1,055.82    | 5,612.31                | 1.07                    | 1.05                   | 0.21                  |
| 16,333.00             | 89.00           | 182.10      | 10,592.08             | -5,615.38    | -1,059.63    | 5,706.87                | 0.60                    | -0.42                  | -0.42                 |
| 16,428.00             | 89.00           | 181.80      | 10,593.74             | -5,710.31    | -1,062.86    | 5,801.38                | 0.32                    | 0.00                   | -0.32                 |
| 16,523.00             | 92.90           | 181.30      | 10,592.16             | -5,805.24    | -1,065.43    | 5,895.79                | 4.14                    | 4.11                   | -0.53                 |
| 16,618.00             | 91.00           | 180.90      | 10,588.93             | -5,900.16    | -1,067.25    | 5,990.10                | 2.04                    | -2.00                  | -0.42                 |
| 16,712.00             | 92.30           | 180.20      | 10,586.22             | -5,994.12    | -1,068.15    | 6,083.32                | 1.57                    | 1.38                   | -0.74                 |
| 16,807.00             | 91.10           | 180.60      | 10,583.41             | -6,089.07    | -1,068.82    | 6,177.50                | 1.33                    | -1.26                  | 0.42                  |
| 16,902.00             | 88.70           | 179.30      | 10,583.57             | -6,184.06    | -1,068.73    | 6,271.62                | 2.87                    | -2.53                  | -1.37                 |
| 16,997.00             | 92.10           | 179.50      | 10,582.91             | -6,279.04    | -1,067.74    | 6,365.60                | 3.59                    | 3.58                   | 0.21                  |
| 17,092.00             | 89.20           | 180.70      | 10,581.83             | -6,374.02    | -1,067.91    | 6,459.74                | 3.30                    | -3.05                  | 1.26                  |
| 17,187.00             | 90.30           | 180.00      | 10,582.25             | -6,469.02    | -1,068.49    | 6,553.95                | 1.37                    | 1.16                   | -0.74                 |

# LEAM Drilling Systems LLC

## Survey Report

|                  |                       |                                     |                                         |
|------------------|-----------------------|-------------------------------------|-----------------------------------------|
| <b>Company:</b>  | Continental Resources | <b>Local Co-ordinate Reference:</b> | Well 3-16H                              |
| <b>Project:</b>  | McKenzie County, ND   | <b>TVD Reference:</b>               | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Site:</b>     | Tallahassee 3         | <b>MD Reference:</b>                | GL 1920+KB 20 @ 1940.00usft (Cyclone 4) |
| <b>Well:</b>     | 3-16H                 | <b>North Reference:</b>             | True                                    |
| <b>Wellbore:</b> | OH                    | <b>Survey Calculation Method:</b>   | Minimum Curvature                       |
| <b>Design:</b>   | OH                    | <b>Database:</b>                    | EDM 5000.1 Multi User DB                |

| Survey                |                 |             |                       |              |              |                         |                         |                        |                       |  |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| 17,282.00             | 90.10           | 179.20      | 10,581.91             | -6,564.01    | -1,067.82    | 6,647.99                | 0.87                    | -0.21                  | -0.84                 |  |
| 17,377.00             | 88.00           | 178.80      | 10,583.49             | -6,658.98    | -1,066.17    | 6,741.88                | 2.25                    | -2.21                  | -0.42                 |  |
| 17,472.00             | 88.80           | 179.70      | 10,586.14             | -6,753.93    | -1,064.92    | 6,835.80                | 1.27                    | 0.84                   | 0.95                  |  |
| 17,567.00             | 90.30           | 179.20      | 10,586.89             | -6,848.92    | -1,064.01    | 6,929.80                | 1.66                    | 1.58                   | -0.53                 |  |
| 17,662.00             | 90.60           | 178.30      | 10,586.14             | -6,943.90    | -1,061.94    | 7,023.63                | 1.00                    | 0.32                   | -0.95                 |  |
| 17,757.00             | 92.20           | 179.00      | 10,583.82             | -7,038.84    | -1,059.70    | 7,117.41                | 1.84                    | 1.68                   | 0.74                  |  |
| 17,852.00             | 90.40           | 179.00      | 10,581.67             | -7,133.80    | -1,058.04    | 7,211.28                | 1.89                    | -1.89                  | 0.00                  |  |
| 17,947.00             | 91.00           | 178.60      | 10,580.50             | -7,228.77    | -1,056.05    | 7,305.12                | 0.76                    | 0.63                   | -0.42                 |  |
| 18,042.00             | 89.50           | 177.80      | 10,580.09             | -7,323.72    | -1,053.07    | 7,398.81                | 1.79                    | -1.58                  | -0.84                 |  |
| 18,137.00             | 90.00           | 177.60      | 10,580.50             | -7,418.64    | -1,049.26    | 7,492.36                | 0.57                    | 0.53                   | -0.21                 |  |
| 18,232.00             | 91.00           | 177.40      | 10,579.68             | -7,513.54    | -1,045.11    | 7,585.84                | 1.07                    | 1.05                   | -0.21                 |  |
| 18,327.00             | 91.60           | 177.80      | 10,577.52             | -7,608.44    | -1,041.14    | 7,679.33                | 0.76                    | 0.63                   | 0.42                  |  |
| 18,385.00             | 92.30           | 177.60      | 10,575.55             | -7,666.35    | -1,038.81    | 7,736.41                | 1.26                    | 1.21                   | -0.34                 |  |
| 18,434.00             | 92.30           | 177.60      | 10,573.58             | -7,715.27    | -1,036.76    | 7,784.61                | 0.00                    | 0.00                   | 0.00                  |  |
| Projection to Bit     |                 |             |                       |              |              |                         |                         |                        |                       |  |



# SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA

OIL AND GAS DIVISION

600 EAST BOULEVARD DEPT 405

BISMARCK, ND 58505-0840

SFN 5749 (09-2006)



Well File No.

25157

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>August 21, 2013</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                              |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                           |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input checked="" type="checkbox"/> Spill Report  |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Cement to surface</b>                          |

|                                                  |               |                 |              |              |  |
|--------------------------------------------------|---------------|-----------------|--------------|--------------|--|
| Well Name and Number<br><b>Tallahassee 3-16H</b> |               |                 |              |              |  |
| Footages                                         | Qtr-Qtr       | Section         | Township     | Range        |  |
| <b>2585 F N L 268 F E L</b>                      | <b>SENE</b>   | <b>16</b>       | <b>153 N</b> | <b>101 W</b> |  |
| Field                                            | Pool          | County          |              |              |  |
| <b>Baker</b>                                     | <b>Bakken</b> | <b>McKenzie</b> |              |              |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

## DETAILS OF WORK

During the cement job on the Tallahassee 3-16H, we had returns the entire time until the last 9bbls. Cement was not returned to surface. A CBL was scheduled the morning of Monday, August 19th. Cement will be brought to surface on the intermediate casing string.

A 1" top job was performed. We got 1" pipe 200' deep. We got cement to surface and continued pumping and then staging to make sure the cement would not drop.

**\*The Operator is to maintain a pressure gauge on the 9-5/8" X 7" annulus for continuous monitoring and to report any pressure development or pressure management events.**

|                                                  |                                     |                                           |                          |
|--------------------------------------------------|-------------------------------------|-------------------------------------------|--------------------------|
| Company<br><b>Continental Resources, Inc.</b>    |                                     | Telephone Number<br><b>(405) 234-9000</b> |                          |
| Address<br><b>P.O. Box 268870</b>                |                                     |                                           |                          |
| City<br><b>Oklahoma City</b>                     |                                     | State<br><b>OK</b>                        | Zip Code<br><b>73126</b> |
| Signature<br>                                    | Printed Name<br><b>Becky Barnes</b> |                                           |                          |
| Title<br><b>Regulatory Compliance Specialist</b> | Date<br><b>August 22, 2013</b>      |                                           |                          |
| Email Address<br><b>becky.barnes@clr.com</b>     |                                     |                                           |                          |

| FOR STATE USE ONLY                           |                                   |
|----------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received | <input type="checkbox"/> Approved |
| Date<br><b>7-16-2013</b>                     |                                   |
| By<br>                                       |                                   |
| Title<br><b>Richard A. Suggs</b>             | <b>Geologist</b>                  |





# SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (03-2004)



Well File No.  
**25157**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>October 14, 2013</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                               |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                            |

|                                                                      |                                                   |
|----------------------------------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis                          | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair                        | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner                             | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well                                   | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History                        | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon                         | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other <b>Flow back exemption</b> |                                                   |

|                                                  |                       |                           |                      |                          |                       |
|--------------------------------------------------|-----------------------|---------------------------|----------------------|--------------------------|-----------------------|
| Well Name and Number<br><b>Tallahassee 3-16H</b> |                       |                           |                      |                          |                       |
| Footages<br><b>2585 F N L</b>                    | <b>268 F E L</b>      | Qtr-Qtr<br><b>SENE</b>    | Section<br><b>16</b> | Township<br><b>153 N</b> | Range<br><b>101 W</b> |
| Field<br><b>Baker</b>                            | Pool<br><b>Bakken</b> | County<br><b>McKenzie</b> |                      |                          |                       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

## DETAILS OF WORK

Continental Resources, Inc. requests a waiver from the tubing/pkr requirement included in NDIC 43-02-03-21: Casing, Tubing, and Cementing Requirements during the completion period immediately following the upcoming fracture stimulation. The following assurances apply:

- 1) The well is equipped with 26#/ft P-110 7" casing at surface with an API burst rating of 9960 psig for the 26 #/ft casing.
- 2) The frac design will use a safety factor of 0.85 \* API burst rating to determine the max pressure.
- 3) Damage to the casing during the frac would be detected immediately by monitoring equipment.
- 4) The casing is exposed to significantly lower rates and pressures during flow back than during the frac job.
- 5) The frac fluid and formation fluids have very low corrosion and erosion rates
- 6) Production equipment will be installed as soon as possible after the well ceases flowing.
- 7) A 300# gauge will be installed on surface casing during flowback period.

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Company<br><b>Continental Resources, Inc.</b> | Telephone Number<br><b>405-234-9000</b> |
| Address<br><b>P.O. Box 269000</b>             |                                         |
| City<br><b>Oklahoma City</b>                  | State<br><b>OK</b>                      |
| Zip Code<br><b>73126</b>                      |                                         |
| Signature<br>                                 | Printed Name<br><b>Jim Landrigan</b>    |
| Title<br><b>Completion Engineer</b>           | Date<br><b>August 29, 2013</b>          |

| FOR STATE USE ONLY                 |                                              |
|------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received  | <input checked="" type="checkbox"/> Approved |
| Date<br><b>September 4, 2013</b>   |                                              |
| By<br>                             |                                              |
| Title<br><b>PETROLEUM ENGINEER</b> |                                              |

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.  
**25157**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>August 14, 2013</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                              |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                           |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Spud with Small Rig</b>                        |

|                                                  |                       |                           |                          |
|--------------------------------------------------|-----------------------|---------------------------|--------------------------|
| Well Name and Number<br><b>Tallahassee 3-16H</b> |                       |                           |                          |
| Footages<br><b>2585 F N</b>                      | <b>L</b>              | <b>268 F</b>              | <b>E L</b>               |
| Qtr-Qtr<br><b>SENE</b>                           |                       | Section<br><b>16</b>      | Township<br><b>153 N</b> |
| Range<br><b>101 W</b>                            |                       |                           |                          |
| Field<br><b>Bakker</b>                           | Pool<br><b>Bakken</b> | County<br><b>McKenzie</b> |                          |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                                                               |                               |                          |                               |
|---------------------------------------------------------------|-------------------------------|--------------------------|-------------------------------|
| Name of Contractor(s)<br><b>Advanced Energy Services, LLC</b> |                               |                          |                               |
| Address<br><b>P.O. Box 85</b>                                 | City<br><b>South Boardman</b> | State<br><b>Michigan</b> | Zip Code<br><b>49680-0085</b> |

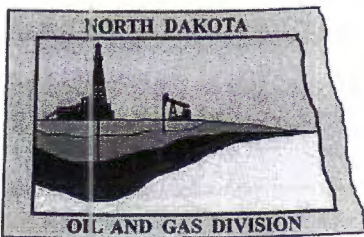
**DETAILS OF WORK**

Continental Resources, Inc. requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Continental Resources, Inc. intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Continental Resources, Inc. understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Continental Resources, Inc. will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

|                                                  |  |                                           |                          |
|--------------------------------------------------|--|-------------------------------------------|--------------------------|
| Company<br><b>Continental Resources, Inc.</b>    |  | Telephone Number<br><b>(405) 234-9000</b> |                          |
| Address<br><b>P.O. Box 268870</b>                |  |                                           |                          |
| City<br><b>Oklahoma City</b>                     |  | State<br><b>OK</b>                        | Zip Code<br><b>73126</b> |
| Signature<br><i>Becky Barnes</i>                 |  | Printed Name<br><b>Becky Barnes</b>       |                          |
| Title<br><b>Regulatory Compliance Specialist</b> |  | Date<br><b>July 30, 2013</b>              |                          |
| Email Address<br><b>becky.barnes@clr.com</b>     |  |                                           |                          |

| FOR STATE USE ONLY                |                                              |
|-----------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received | <input checked="" type="checkbox"/> Approved |
| Date<br><b>8-02-2013</b>          |                                              |
| By<br><i>David Burns</i>          |                                              |
| Title<br><b>David Burns</b>       |                                              |
| <b>Engineering Tech.</b>          |                                              |

CR1 m... 2013 by NDIC Inspector - Richard Penn 701-770-3554 with spud 270



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

25157  
JA

BECKY BARNES  
CONTINENTAL RESOURCES, INC.  
PO BOX 1032  
ENID, OK 73702-1032 USA

Date: 3/18/2013

RE: CORES AND SAMPLES

Well Name: TALLAHASSEE 3-16H Well File No.: 25157  
Location: SENE 16-153-101 County: MCKENZIE  
Permit Type: Development - HORIZONTAL  
Field: BAKER Target Horizon: THREE FORKS

Dear BECKY BARNES:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

**Base of the Last Charles Salt**

Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:  
**ND Geological Survey Core Library  
Campus Road and Cornell  
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Richard A. Suggs  
Geologist





# SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2008)



Well File No.

25157

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <input type="checkbox"/> Notice of Intent                                                                                                       | Approximate Start Date |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed    |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <u>Open Hole Log Waiver</u>                       |

|                                                  |               |                 |              |              |  |
|--------------------------------------------------|---------------|-----------------|--------------|--------------|--|
| Well Name and Number<br><b>Tallahassee 3-16H</b> |               |                 |              |              |  |
| Footages                                         | Qtr-Qtr       | Section         | Township     | Range        |  |
| <b>2585 F N L 268 F E L</b>                      | <b>SENE</b>   | <b>16</b>       | <b>153 N</b> | <b>101 W</b> |  |
| Field                                            | Pool          | County          |              |              |  |
|                                                  | <b>Bakken</b> | <b>McKenzie</b> |              |              |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

## DETAILS OF WORK

Requested variance to not run openhole logs. GR/CBL/CCL will be run from deepest point obtainable to base of surface casing.

Offset logs used will be the Nance Petroleum, Corps of Engineers 31-10, Sec 10-153N-101W, McKenzie County, ND.

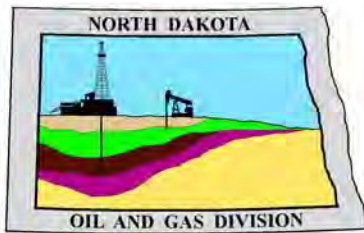
The Gamma Ray Log will be run all the way to surface and all mud logs will be submitted as one digital tiff formatted file and one digital LAS formatted file.

*Approved per logs run on #10710 - Nov 11-9-3A.*

|                                                  |                                       |                                         |  |
|--------------------------------------------------|---------------------------------------|-----------------------------------------|--|
| Company<br><b>Continental Resources, Inc.</b>    |                                       | Telephone Number<br><b>580-233-8955</b> |  |
| Address<br><b>P.O. Box 1032</b>                  |                                       |                                         |  |
| City                                             | State                                 | Zip Code                                |  |
| <b>Enid</b>                                      | <b>OK</b>                             | <b>73702</b>                            |  |
| Signature<br><i>Terry L. Olson</i>               | Printed Name<br><b>Terry L. Olson</b> |                                         |  |
| Title<br><b>Regulatory Compliance Specialist</b> | Date<br><b>June 12, 2012</b>          |                                         |  |
| Email Address<br><b>Terry.Olson@clr.com</b>      |                                       |                                         |  |

| FOR STATE USE ONLY                |                                              |
|-----------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received | <input checked="" type="checkbox"/> Approved |
| Date                              | <b>3/11/2013</b>                             |
| By                                | <i>[Signature]</i>                           |
| Title                             | <b>Richard A. Suggs<br/>Geologist</b>        |





# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

[www.oilgas.nd.gov](http://www.oilgas.nd.gov)

March 11, 2013

Terry L. Olson  
Regulatory Compliance Specialist  
CONTINENTAL RESOURCES, INC.  
P.O. Box 1032  
Enid, OK 73702

**RE: HORIZONTAL WELL  
TALLAHASSEE 3-16H  
SENE Section 16-153N-101W  
McKenzie County  
Well File # 25157**

Dear Terry:

Pursuant to Commission Order No. 21551, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **200' setback** from the north & south boundaries and **500' setback** from the east & west boundaries within the 2560 acre spacing unit consisting of Sections 4, 9, 16, & 21-T153N-R101W. **Tool error is not required pursuant to order.**

**PERMIT STIPULATIONS: The mouse and rat hole must be cemented. A liner must be placed under the location. A spill contingency plan must be in place prior to spudding the wells. No Drilling Pit will be allowed. Due to surficial water adjacent to the well site, a dike is required surrounding the entire location. One horizontal well shall be drilled and completed in the standup 1280-acre spacing unit described as Sections 4 and 9, T153N-R101W, McKenzie County, north Dakota, prior to completing any horizontal well in the 2560-acre spacing unit described as Sections 4, 9, 16, and 21-T153N-R101W McKenzie County, North Dakota. Continental must drill the Columbus Federal 2-16H, 3-16H and Tallahassee 2-16H, 3-16H back to back. CONTINENTAL RESOURCES must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.**

### Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

### Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 232' W. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 7751' S and a minimum legal coordinate from the well head of 232' W.

### Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

### Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

**Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells**

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to [certsurvey@nd.gov](mailto:certsurvey@nd.gov).

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

**Confidential status**

Your request for confidential status of all information furnished to the Director, or his representatives, is hereby granted. Such information, except production runs, shall remain confidential for six months commencing on the date the well is spud.

Confidential status notwithstanding, the Director and his representatives shall have access to all well records wherever located. Your company personnel, or any person performing work for your company shall permit the Director and his representatives to come upon any lease, property, well, or drilling rig operated or controlled by them, complying with all safety rules, and to inspect the records and operation of such wells and to have access at all times to any and all records of wells. The Commission's field personnel periodically inspect producing and drilling wells. Any information regarding such wells shall be made available to them at any time upon request. The information so obtained by the field personnel shall be maintained in strict confidence and shall be available only to the Commission and its staff.

**Surface casing cement**

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

**Logs**

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to [digitallogs@nd.gov](mailto:digitallogs@nd.gov). Thank you for your cooperation.

Sincerely,

David Tabor  
Engineering Technician IV



# APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                |                                      |                                                          |                                             |
|------------------------------------------------|--------------------------------------|----------------------------------------------------------|---------------------------------------------|
| Type of Work<br><b>New Location</b>            | Type of Well<br><b>Oil &amp; Gas</b> | Approximate Date Work Will Start<br><b>6 / 16 / 2012</b> | Confidential Status<br><b>Yes</b>           |
| Operator<br><b>CONTINENTAL RESOURCES, INC.</b> |                                      |                                                          | Telephone Number<br><b>580-233-8955</b>     |
| Address<br><b>P.O. Box 1032</b>                |                                      | City<br><b>Enid</b>                                      | State<br><b>OK</b> Zip Code<br><b>73702</b> |

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

## WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

|                                                                                          |                                                         |                                                                                                         |                                                                                                  |                                             |                                                                                                                 |                                                    |                                     |
|------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------|
| Well Name<br><b>TALLAHASSEE</b>                                                          |                                                         |                                                                                                         |                                                                                                  | Well Number<br><b>3-16H</b>                 |                                                                                                                 |                                                    |                                     |
| Surface Footages<br><b>2585 F N L 268 F E L</b>                                          |                                                         | Qtr-Qtr<br><b>SENE</b>                                                                                  | Section<br><b>16</b>                                                                             | Township<br><b>153 N</b>                    | Range<br><b>101 W</b>                                                                                           | County<br><b>McKenzie</b>                          |                                     |
| Longstring Casing Point Footages<br><b>2676 F S L 598 F E L</b>                          |                                                         | Qtr-Qtr<br><b>NESE</b>                                                                                  | Section<br><b>16</b>                                                                             | Township<br><b>153 N</b>                    | Range<br><b>101 W</b>                                                                                           | County<br><b>McKenzie</b>                          |                                     |
| Longstring Casing Point Coordinates From Well Head<br><b>290 S From WH 330 W From WH</b> |                                                         | Azimuth<br><b>229 °</b>                                                                                 | Longstring Total Depth<br><b>10857 Feet MD 10607 Feet TVD</b>                                    |                                             |                                                                                                                 |                                                    |                                     |
| Bottom Hole Footages From Nearest Section Line<br><b>204 F S L 1320 F E L</b>            |                                                         | Qtr-Qtr<br><b>SESE</b>                                                                                  | Section<br><b>21</b>                                                                             | Township<br><b>153 N</b>                    | Range<br><b>101 W</b>                                                                                           | County<br><b>McKenzie</b>                          |                                     |
| Bottom Hole Coordinates From Well Head<br><b>7747 S From WH 1052 W From WH</b>           |                                                         | KOP Lateral 1<br><b>10166 Feet MD</b>                                                                   | Azimuth Lateral 1<br><b>180 °</b>                                                                |                                             | Estimated Total Depth Lateral 1<br><b>18535 Feet MD 10632 Feet TVD</b>                                          |                                                    |                                     |
| Latitude of Well Head<br><b>48 ° 04 ' 30.49 "</b>                                        | Longitude of Well Head<br><b>-103 ° 40 ' 12.02 "</b>    | NAD Reference<br><b>NAD83</b>                                                                           | Description of (Subject to NDIC Approval)<br><b>Spacing Unit: Sec 4, 9, 16, &amp; 21 153 101</b> |                                             |                                                                                                                 |                                                    |                                     |
| Ground Elevation<br><b>1924 Feet Above S.L.</b>                                          | Acres in Spacing/Drilling Unit<br><b>2560</b>           | Spacing/Drilling Unit Setback Requirement<br><b>200 Feet N/S 500 Feet E/W</b>                           |                                                                                                  |                                             | Industrial Commission Order<br><b>21551</b>                                                                     |                                                    |                                     |
| North Line of Spacing/Drilling Unit<br><b>5279 Feet</b>                                  | South Line of Spacing/Drilling Unit<br><b>5300 Feet</b> | East Line of Spacing/Drilling Unit<br><b>20988 Feet</b>                                                 |                                                                                                  |                                             | West Line of Spacing/Drilling Unit<br><b>20969 Feet</b>                                                         |                                                    |                                     |
| Objective Horizons<br><b>Three Forks</b>                                                 |                                                         |                                                                                                         |                                                                                                  |                                             |                                                                                                                 | Pierre Shale Top<br><b>1841</b>                    |                                     |
| Proposed Surface Casing                                                                  | Size<br><b>9 - 5/8 "</b>                                | Weight<br><b>36 Lb./Ft.</b>                                                                             | Depth<br><b>1940 Feet</b>                                                                        | Cement Volume<br><b>732 Sacks</b>           | <b>NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.</b> |                                                    |                                     |
| Proposed Longstring Casing                                                               | Size<br><b>7 - "</b>                                    | Weight(s)<br><b>26-32 Lb./Ft.</b>                                                                       | Longstring Total Depth<br><b>10857 Feet MD 10607 Feet TVD</b>                                    |                                             | Cement Volume<br><b>859 Sacks</b>                                                                               | Cement Top<br><b>0 Feet</b>                        | Top Dakota Sand<br><b>4826 Feet</b> |
| Base Last Charles Salt (If Applicable)<br><b>9031 Feet</b>                               |                                                         | <b>NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.</b> |                                                                                                  |                                             |                                                                                                                 |                                                    |                                     |
| Proposed Logs<br><b>CBL/GR from deepest depth obtainable to ground surface/mud</b>       |                                                         |                                                                                                         |                                                                                                  |                                             |                                                                                                                 |                                                    |                                     |
| Drilling Mud Type (Vertical Hole - Below Surface Casing)<br><b>Invert</b>                |                                                         |                                                                                                         |                                                                                                  | Drilling Mud Type (Lateral)<br><b>Brine</b> |                                                                                                                 |                                                    |                                     |
| Survey Type in Vertical Portion of Well<br><b>MWD Every 100 Feet</b>                     |                                                         | Survey Frequency: Build Section<br><b>30 Feet</b>                                                       |                                                                                                  | Survey Frequency: Lateral<br><b>90 Feet</b> |                                                                                                                 | Survey Contractor<br><b>LEAM Drilling Services</b> |                                     |

**NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.**

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

**REQUIRED ATTACHMENTS:** Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.  
See Page 2 for Comments section and signature block.

**COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS**

|  |
|--|
|  |
|--|

**Lateral 2**

|                                                                          |                        |                                                                |         |                                                        |                        |
|--------------------------------------------------------------------------|------------------------|----------------------------------------------------------------|---------|--------------------------------------------------------|------------------------|
| KOP Lateral 2<br>Feet MD                                                 | Azimuth Lateral 2<br>° | Estimated Total Depth Lateral 2<br>Feet MD      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH      From WH |                        |
| Formation Entry Point Coordinates From Well Head<br>From WH      From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH      From WH |         |                                                        |                        |
| KOP Footages From Nearest Section Line<br>F      L      F      L         |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |
| Bottom Hole Footages From Nearest Section Line<br>F      L      F      L |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |

**Lateral 3**

|                                                                          |                        |                                                                |         |                                                        |                        |
|--------------------------------------------------------------------------|------------------------|----------------------------------------------------------------|---------|--------------------------------------------------------|------------------------|
| KOP Lateral 3<br>Feet MD                                                 | Azimuth Lateral 3<br>° | Estimated Total Depth Lateral 3<br>Feet MD      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH      From WH |                        |
| Formation Entry Point Coordinates From Well Head<br>From WH      From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH      From WH |         |                                                        |                        |
| KOP Footages From Nearest Section Line<br>F      L      F      L         |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |
| Bottom Hole Footages From Nearest Section Line<br>F      L      F      L |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |

**Lateral 4**

|                                                                          |                        |                                                                |         |                                                        |                        |
|--------------------------------------------------------------------------|------------------------|----------------------------------------------------------------|---------|--------------------------------------------------------|------------------------|
| KOP Lateral 4<br>Feet MD                                                 | Azimuth Lateral 4<br>° | Estimated Total Depth Lateral 4<br>Feet MD      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH      From WH |                        |
| Formation Entry Point Coordinates From Well Head<br>From WH      From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH      From WH |         |                                                        |                        |
| KOP Footages From Nearest Section Line<br>F      L      F      L         |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |
| Bottom Hole Footages From Nearest Section Line<br>F      L      F      L |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |

**Lateral 5**

|                                                                          |                        |                                                                |         |                                                        |                        |
|--------------------------------------------------------------------------|------------------------|----------------------------------------------------------------|---------|--------------------------------------------------------|------------------------|
| KOP Lateral 5<br>Feet MD                                                 | Azimuth Lateral 5<br>° | Estimated Total Depth Lateral 5<br>Feet MD      Feet TVD       |         | KOP Coordinates From Well Head<br>From WH      From WH |                        |
| Formation Entry Point Coordinates From Well Head<br>From WH      From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH      From WH |         |                                                        |                        |
| KOP Footages From Nearest Section Line<br>F      L      F      L         |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |
| Bottom Hole Footages From Nearest Section Line<br>F      L      F      L |                        | Qtr-Qtr                                                        | Section | Township<br>N                                          | Range<br>W      County |

|                                                                                                                           |  |                                       |  |                                                  |
|---------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------|--|--------------------------------------------------|
| I hereby swear or affirm the information provided is true, complete and correct as determined from all available records. |  |                                       |  | Date<br><b>6 / 12 / 2012</b>                     |
| <b>ePermit</b>                                                                                                            |  | Printed Name<br><b>Terry L. Olson</b> |  | Title<br><b>Regulatory Compliance Specialist</b> |
|                                                                                                                           |  |                                       |  |                                                  |

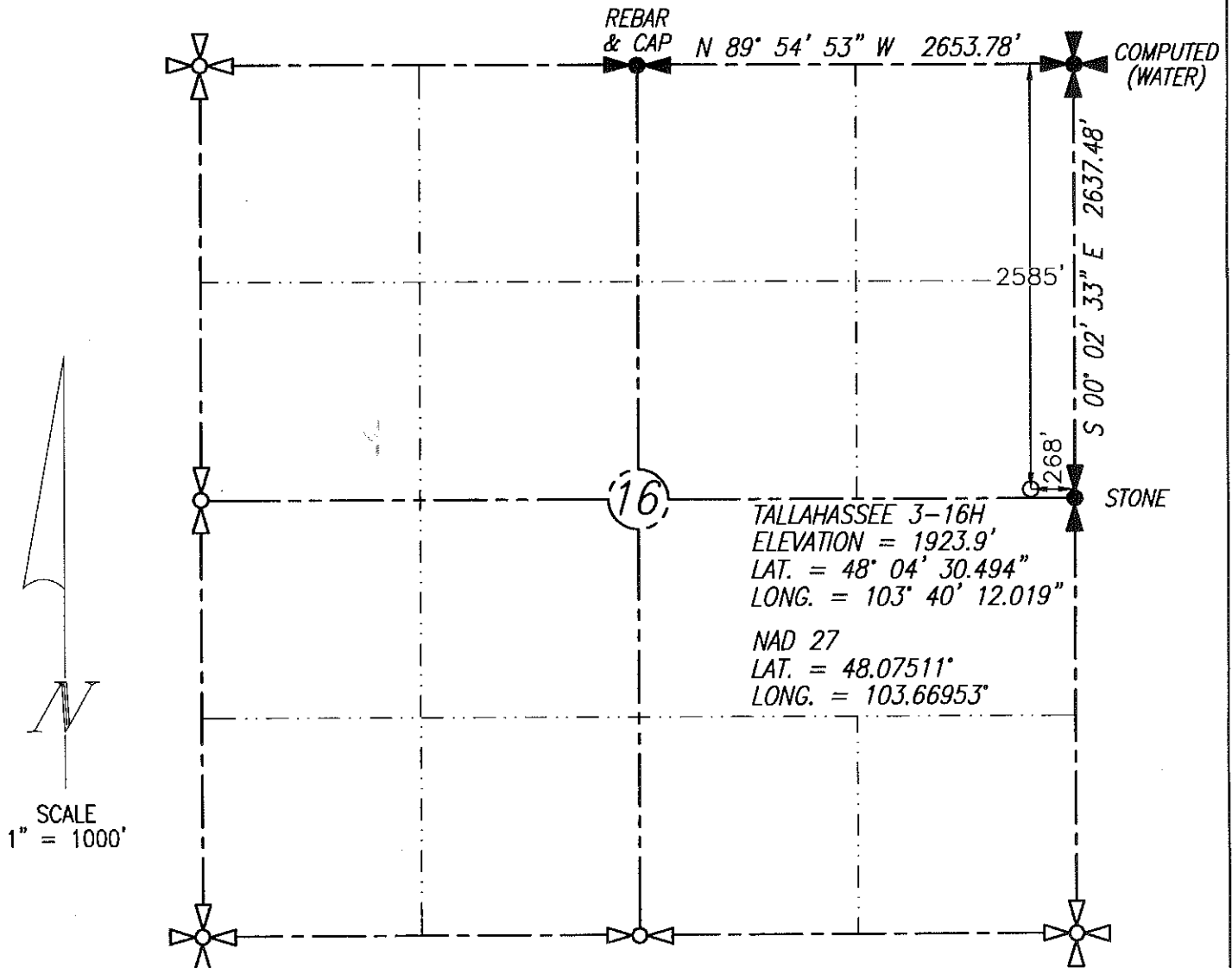
**FOR STATE USE ONLY**

|                                        |                                       |
|----------------------------------------|---------------------------------------|
| Permit and File Number<br><b>25157</b> | API Number<br><b>33 - 053 - 04853</b> |
| Field<br><b>BAKER</b>                  |                                       |
| Pool<br><b>BAKKEN</b>                  | Permit Type<br><b>DEVELOPMENT</b>     |

**FOR STATE USE ONLY**

|                                           |
|-------------------------------------------|
| Date Approved<br><b>3 / 11 / 2013</b>     |
| By<br><b>David Tabor</b>                  |
| Title<br><b>Engineering Technician IV</b> |

WELL LOCATION PLAT  
CONTINENTAL RESOURCES INC.  
TALLAHASSEE 3-16H  
SECTION 16, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA  
2585' FNL & 268' FEL



JOHN PAULSON R.L.S. 3366  
REGISTERED  
LAND  
SURVEYOR  
L.S. 3366  
NORTH DAKOTA

BASIS OF VERTICAL DATUM:  
NAVD 1988 GEOID 09

BOX 357  
BOWMAN, N.D. 58623  
PHONE: 701-523-3340  
FAX: 701-523-5243  
PROJECT NO. 12-10

**GEOLOGIC PROGNOSIS**

Well Name: Tallahassee 3-16H

Rig: Cyclone 20

Prospect: Williston

Target: Three Forks

Spacing: 1280

☐ Pre-Staked

☒ Staked

Rig Grade Elevation: 1920'

KB: 21'

RKB: 1941'

SHL: 2585' FNL & 268' FEL

Sec. 16 - 153N - 101W

McKenzie , ND

BHL: 200' FSL & 1320' FEL

Sec. 21 - 153N - 101W

McKenzie , ND

==>

| FORMATION                | SUBSEA | TVD    |
|--------------------------|--------|--------|
| Pierre Shale             | 100    | 1,841  |
| Greenhorn                | -2,484 | 4,425  |
| Dakota Group (fka Mowry) | -2,885 | 4,826  |
| Base of Dakota Sand      | -3,764 | 5,705  |
| Dunham Salt Top          | -4,792 | 6,733  |
| Dunham Salt Base         | -4,852 | 6,793  |
| Pine Salt Top            | -5,162 | 7,103  |
| Pine Salt Base           | -5,218 | 7,159  |
| Minnekahta               | -5,244 | 7,185  |
| Opeche Salt Top          | NA     |        |
| Opeche Salt Base         | NA     |        |
| Minnelusa Group          | -5,530 | 7,471  |
| Tyler                    | -5,686 | 7,627  |
| Kibbey                   | -6,238 | 8,179  |
| Top Charles              | -6,392 | 8,333  |
| Base Last Charles Salt   | -7,090 | 9,031  |
| Mission Canyon           | -7,314 | 9,255  |
| Lodgepole                | -7,880 | 9,821  |
| Upper Bakken Shale       | -8,570 | 10,511 |
| Middle Bakken Member     | -8,587 | 10,528 |
| Lower Bakken Shale       | -8,629 | 10,570 |
| Three Forks              | -8,646 | 10,587 |
| Three Forks Target       | -8,666 | 10,607 |
| End of Lateral           | -8,691 | 10,632 |

20' into 3forks  
25' Higher



DRILLING PROGRAM

06/08/12

Lease and Well No. Tallahassee 3-16H

MUD PROGRAM

| Depth             | Type        | Weight   | Remarks                                  |
|-------------------|-------------|----------|------------------------------------------|
| 0 ' - 1940 '      | Fresh water | 8.4-8.8  | Add Soap Sticks for Mud Rings            |
| 1940 ' - 6500 '   | Invert      | 9.3-9.5  | 35-50 sec, 10-30 cc's                    |
| 6500 ' - 10857 '  | Invert      | 9.6-10.0 | 40-55 sec, 10-15 cc's O/W 70/30 to 80/20 |
| 10857 ' - 18535 ' | Brine       | 8.7-10.0 | Cuttings pit                             |

TUBULAR PROGRAM

| String Type                                                                                                    | Hole Size | Depth   | Feet    | Casing Diameter | Weight, Grade, Connection  | ERW/ Seamless | Critical Inspection |
|----------------------------------------------------------------------------------------------------------------|-----------|---------|---------|-----------------|----------------------------|---------------|---------------------|
| Surf                                                                                                           | 13 1/2 "  | 1940 '  | 1940 '  | 9 5/8 "         | 9-5/8", 36 #, J-55, STC    | ERW           | BCI & Drift         |
| Float shoe, shoe joint & float collar. Centralize bottom 3 jts and every 4th jt to surface.                    |           |         |         |                 |                            |               |                     |
| Int                                                                                                            | 8 3/4 "   | 80 '    | 80 '    | 7 "             | 7", 32#, P-110 IC, LTC     | ERW           | BCI & Drift         |
|                                                                                                                |           | 4000 '  | 3920 '  | 7 "             | 7", 26#, P-110 IC, LTC     | ERW           | BCI & Drift         |
|                                                                                                                |           | 8130 '  | 4130 '  | 7 "             | 7", 29#, P-110 IC, LTC     | ERW           | BCI & Drift         |
|                                                                                                                |           | 9230 '  | 1100 '  | 7 "             | 7", 32#, P-110 IC, LTC     | ERW           | BCI & Drift         |
|                                                                                                                |           | 10857 ' | 1627 '  | 7 "             | 7", 29#, P-110 IC, LTC     | ERW           | BCI & Drift         |
| Float shoe, shoe joint & float collar. Centralize bottom 3 joints. Centralize thru curve and across all salts. |           |         |         |                 |                            |               |                     |
| Liner                                                                                                          | 6 "       | 18535 ' | 8369 '  | 4 1/2 "         | 4-1/2", 11.6 #, P-110, BTC |               |                     |
| Tubing                                                                                                         |           | 10170 ' | 10170 ' | 2 7/8 "         | 2-7/8", 6.5 #, L-80, EUE   |               |                     |

Notes: Pipe to end up in hole from top to bottom as shown.

CEMENT PROGRAM

| String Type                                                                                 | SHOE/DV Depth | Stage Lead/Tail | Cement Bottom | Cement Top | No Sacks | Cement System                              | Cement Yield | Cement Weight |
|---------------------------------------------------------------------------------------------|---------------|-----------------|---------------|------------|----------|--------------------------------------------|--------------|---------------|
|                                                                                             |               | Lead            | 1360 '        | 0 '        | 431      | 35/65 Poz/Class "C", 3% CaCl, 12% gel      | 2.39         | 12            |
| Surf                                                                                        | 1940          | Tail            | 1940 '        | 1360 '     | 301      | Class "C", 2% CaCl                         | 1.46         | 14.3          |
| (Basis: Gauge hole + 55% excess, tail 30% of length, lead to surface.)                      |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |
| Int                                                                                         | 10857         | Lead            | 7830 '        | 0 '        | 473      | 35/65 Poz/Class "C", 3% KCl, 5 #/sk Silica | 3.21         | 11.3          |
|                                                                                             |               | Tail            | 10857 '       | 7830 '     | 386      | Class "G", 3% KCl, 35% Silica              | 1.59         | 15.6          |
| (Basis: Gauge hole + 35% excess, Tail to 500 ft above top of Charles Salt, Lead to Surface) |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |
|                                                                                             |               |                 |               |            |          |                                            |              |               |

BOP PROGRAM

| Hole Size                                 | Configuration       | Pressure Rating | BOP Test Press | Casing Test Press |
|-------------------------------------------|---------------------|-----------------|----------------|-------------------|
| 13 1/2 "                                  | Circulate Conductor |                 |                |                   |
| 8 3/4 "                                   | RRA                 | 5M              | 5000           |                   |
| 6 "                                       | RRA                 | 5M              | 5000           |                   |
| Utilize commercial testers every 30 days. |                     |                 |                |                   |

LOGGING PROGRAM

| FROM     | TO  | TOOLS                                   |
|----------|-----|-----------------------------------------|
| BLS      | TD  | 2 man mud logger trailer, 30 ft samples |
| Surf csg | TD  | MWD Surveys, 90 ft & 30 ft thru curve   |
| KOP      | TD  | MWD Gamma Ray                           |
| Surface  | DDO | CBL, CCL, Gr                            |

Remarks: 4 1/2 inch Liner to be run. Liner Top @ KOP (10166 ft TVD) and 29 swell packers to be run with liner.  
Liner top to be tested successfully to 4500 psi prior to rigging down.  
Casing Caliper to be run to determine if a frac string is needed.

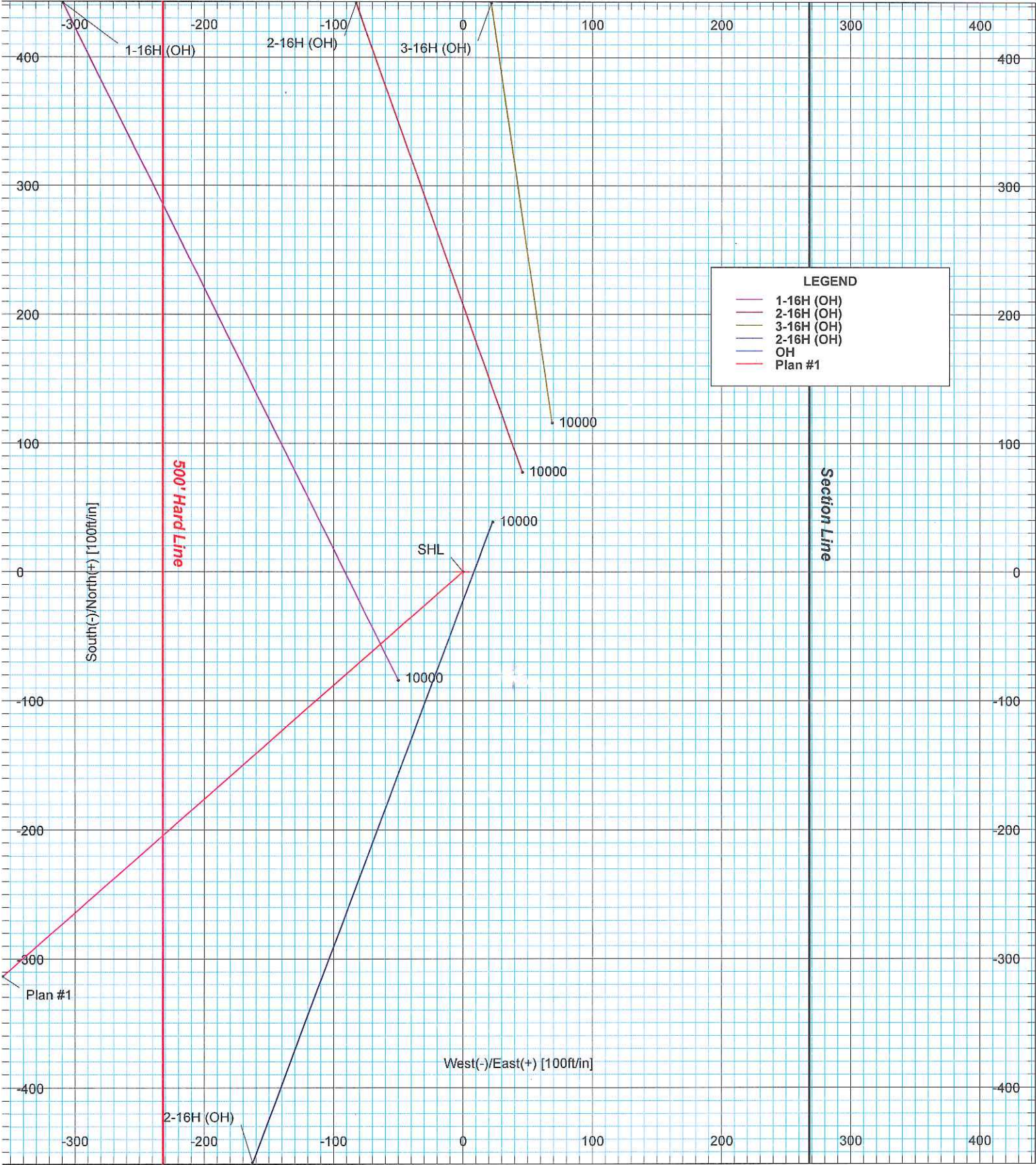
Prepared By: Pat McCollom Date: 04/17/12

Continental Resources

Field: McKenzie County, ND  
Site: Tallahassee 3  
Well: 3-16H  
Wellpath: OH  
Plan: Plan #1



Azimuths to True North  
Magnetic North: 8.65°  
  
Magnetic Field  
Strength: 56675nT  
Dip Angle: 73.09°  
Date: 04/19/2012  
Model: IGRF2010



LEAM DRILLING SYSTEMS, INC.  
2010 East Davis Conroe, Texas 77301  
Phone: 936-756-7577 Fax: 936-756-7595

Plan: Plan #1 (3-16H(OH))  
Created By: Justin Andoe Date: 04/23/2012  
Checked: \_\_\_\_\_ Date: \_\_\_\_\_  
Reviewed: \_\_\_\_\_ Date: \_\_\_\_\_





To: Todd Holweger, NDIC

From: Shawn Svob

Date: 4/5/2012

Re: Continental Resources standard CCL, CBL, 4-1/2" liner running and testing procedures

---

Continental Resources' standard practice for running the cement bond log and casing caliper log is to run both logs immediately after coming out of the hole after TD, prior to running the 4-1/2" liner, to the deepest depth obtainable; however, if there are well control concerns that require us to run the liner sooner, only the CBL will be run and the CCL will be run after setting the liner.

Based on the CCL results, we determine the actual API minimum burst allowance for the 7" casing. If the downgraded API burst pressure is below our minimum required frac pressures, we will run a 4-1/2" frac string; if severe wear or holes are found in the casing, we will run a 5" cemented, to surface, tie back string.

The CBL log is run in order to determine the top of cement, as required by the NDIC. Our current 4-1/2" liner program for a 1280 unit is 30, evenly spaced, stages with 29 swellable packers. The liner shoe is set approximately 180 feet off bottom. The shoe stage below the last packer has 2 joints, a double valved float, one joint, and a ported guide shoe – appx 130 ft. The liner is run using a running tool on the end of 4" DP. The 7" packer/hanger is set about 40 ft above KOP between two casing collars but conditions occasionally occur that require setting higher, either through unexpected failure or in order to isolate casing wear close to KOP. Recently we have tried 40 stage liners and the trend to explore the optimum stage count will continue.. Once the liner is at depth, a ball is dropped through the DP, the ball is pressured up against the setting tool to approximately 2500 psi, and the 7" packer/hanger is set.

A push pull test is done to confirm the hanger has set. Then, a 4500 psi pressure test is completed on the back side of the 4" DP to confirm the packer has set. The setting tool is then backed off and the 4" DP/running tool is laid down.

Immediately after the rotary rig has been moved off the well location, the 7" csg and liner packer/ hanger are tested to the frac pressure. The testers will rig up and test the tubing head to 5000 psi. Next a test plug will be run and set, using wire line, in the top of the 7" packer/hanger. Testers will pressure up to our frac pressure, typically 8500 psi, to confirm the 7" is ready for completion.

Shawn Svob  
Drilling Operations Coordinator

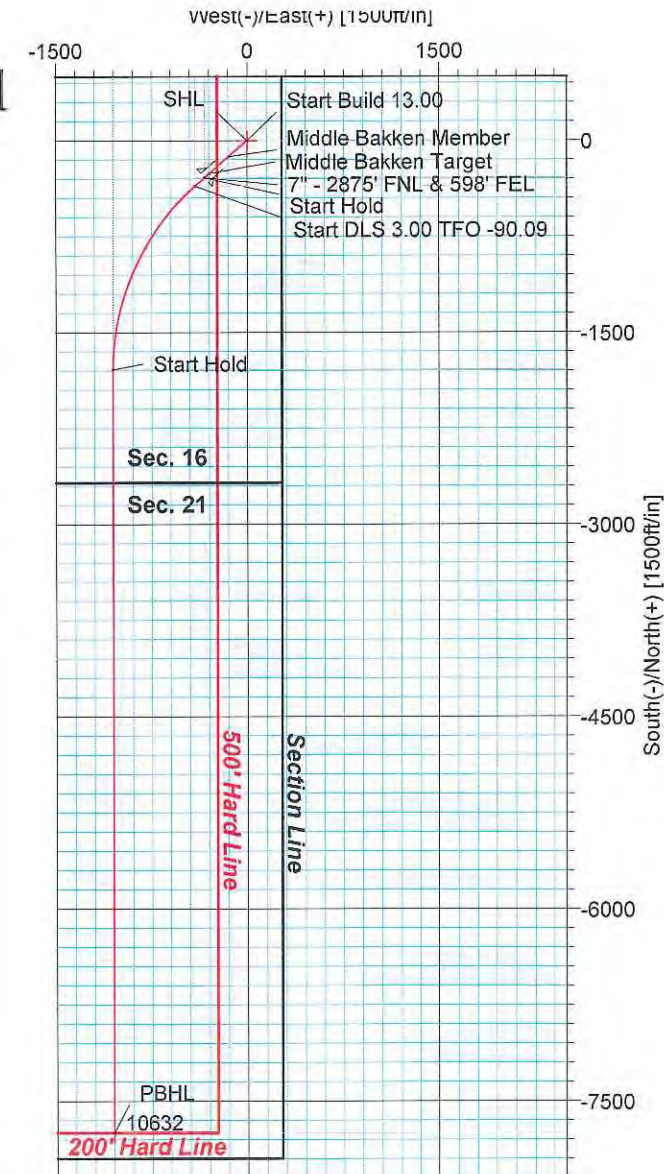
# Continental Resources

Field: McKenzie County, ND  
Site: Tallahassee 3  
Well: 3-16H  
Wellpath: OH  
Plan: Plan #1

**SITE DETAILS**  
Tallahassee 3  
Sec. 16 - T153N R-101W  
SHL 2585' FNL & 268' FEL  
PBHL 200' FSL & 1320' FEL of Sec. 21  
Latitude: 48°04'30.494N  
Longitude: 103°40'12.019W  
Ground Level: 1920.00  
KB: 21.00



Azimuths to True North  
Magnetic North: 8.65°  
Magnetic Field  
Strength: 56675nT  
Dip Angle: 73.09°  
Date: 04/19/2012  
Model: IGRF2010



## TARGET DETAILS

| Name | TVD      | +N/-S    | +E/-W    | Northing  | Easting    | Latitude      | Longitude      | Shape |
|------|----------|----------|----------|-----------|------------|---------------|----------------|-------|
| SHL  | 0.00     | 0.00     | 0.00     | 408124.35 | 1193775.66 | 48°04'30.494N | 103°40'12.019W | Point |
| PBHL | 10632.00 | -7747.00 | -1052.00 | 400427.21 | 1192405.69 | 48°03'14.039N | 103°40'27.505W | Point |

## SECTION DETAILS

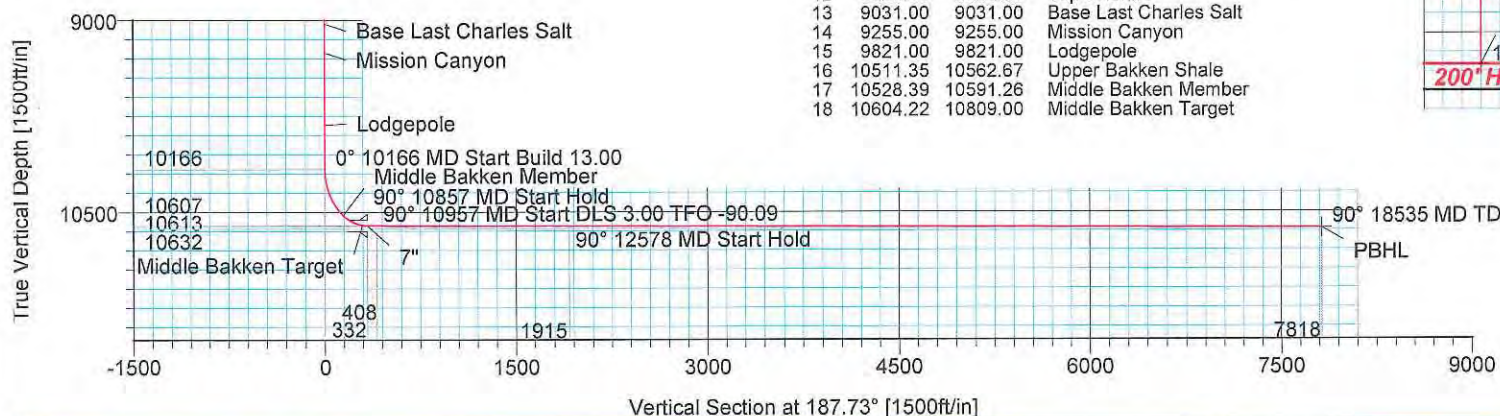
| Sec | MD       | Inc   | Azi    | TVD      | +N/-S    | +E/-W    | DLeg  | TFace  | VSec    | Target |
|-----|----------|-------|--------|----------|----------|----------|-------|--------|---------|--------|
| 1   | 0.00     | 0.00  | 0.00   | 0.00     | 0.00     | 0.00     | 0.00  | 0.00   | -0.00   |        |
| 2   | 10166.27 | 0.00  | 0.00   | 10166.27 | 0.00     | 0.00     | 0.00  | 0.00   | -0.00   |        |
| 3   | 10857.19 | 89.82 | 228.62 | 10607.00 | -290.44  | -329.66  | 13.00 | 228.62 | 332.14  |        |
| 4   | 10957.19 | 89.82 | 228.62 | 10607.32 | -356.55  | -404.69  | 0.00  | 0.00   | 407.74  |        |
| 5   | 12577.81 | 89.82 | 180.00 | 10612.81 | -1789.55 | -1052.00 | 3.00  | -90.09 | 1914.79 |        |
| 6   | 18535.29 | 89.82 | 180.00 | 10632.00 | -7747.00 | -1052.00 | 0.00  | 0.00   | 7818.10 | PBHL   |

## FORMATION TOP DETAILS

| No. | TVDPath  | MDPath   | Formation              |
|-----|----------|----------|------------------------|
| 1   | 1841.00  | 1841.00  | Pierre Shale           |
| 2   | 4425.00  | 4425.00  | Greenhorn              |
| 3   | 4826.00  | 4826.00  | Dakota Group           |
| 4   | 5705.00  | 5705.00  | Base of Dakota Sand    |
| 5   | 6733.00  | 6733.00  | Dunham Salt Top        |
| 6   | 6793.00  | 6793.00  | Dunham Salt Base       |
| 7   | 7103.00  | 7103.00  | Pine Salt Top          |
| 8   | 7159.00  | 7159.00  | Pine Salt Base         |
| 9   | 7185.00  | 7185.00  | Minnekahta             |
| 10  | 7471.00  | 7471.00  | Minnelusa Group        |
| 11  | 8179.00  | 8179.00  | Kibbey                 |
| 12  | 8333.00  | 8333.00  | Top Charles            |
| 13  | 9031.00  | 9031.00  | Base Last Charles Salt |
| 14  | 9255.00  | 9255.00  | Mission Canyon         |
| 15  | 9821.00  | 9821.00  | Lodgepole              |
| 16  | 10511.35 | 10562.67 | Upper Bakken Shale     |
| 17  | 10528.39 | 10591.26 | Middle Bakken Member   |
| 18  | 10604.22 | 10809.00 | Middle Bakken Target   |

## CASING DETAILS

| No. | TVD      | MD       | Name | Size |
|-----|----------|----------|------|------|
| 1   | 10607.00 | 10857.19 | 7"   | 7.00 |



LEAM DRILLING SYSTEMS, INC.  
2010 East Davis, Conroe, Texas 77301  
Phone: 281 756 7577 Fax: 281 756 7505

Plan: Plan #1 (3-16H/OH)  
Created By: Justin Andoe Date: 04/23/2012  
Checked: Date:  
Reviewed: Date:

LEAM Drilling Systems, Inc.  
Planning Report

|           |                       |                            |                              |       |          |       |        |
|-----------|-----------------------|----------------------------|------------------------------|-------|----------|-------|--------|
| Company:  | Continental Resources | Date:                      | 04/23/2012                   | Time: | 17:16:30 | Page: | 1      |
| Field:    | McKenzie County, ND   | Co-ordinate(NE) Reference: | Well: 3-16H, True North      |       |          |       |        |
| Site:     | Tallahassee 3         | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0         |       |          |       |        |
| Well:     | 3-16H                 | Section (VS) Reference:    | Well (0.00N,0.00E,187.73Azi) |       |          |       |        |
| Wellpath: | OH                    | Survey Calculation Method: | Minimum Curvature            |       |          | Db:   | Adapti |

|                                                  |  |  |  |                                       |  |  |  |
|--------------------------------------------------|--|--|--|---------------------------------------|--|--|--|
| Field: McKenzie County, ND                       |  |  |  |                                       |  |  |  |
| Map System:US State Plane Coordinate System 1983 |  |  |  | Map Zone: North Dakota, Northern Zone |  |  |  |
| Geo Datum: GRS 1980                              |  |  |  | Coordinate System: Well Centre        |  |  |  |
| Sys Datum: Mean Sea Level                        |  |  |  | Geomagnetic Model: IGRF2010           |  |  |  |

|                                                                       |            |           |               |                   |                 |  |  |
|-----------------------------------------------------------------------|------------|-----------|---------------|-------------------|-----------------|--|--|
| Site: Tallahassee 3<br>Sec. 16 - T153N R-101W<br>2585' FNL & 268' FEL |            |           |               |                   |                 |  |  |
| Site Position:                                                        |            | Northing: | 408124.35 ft  | Latitude:         | 48 4 30.494 N   |  |  |
| From: Geographic                                                      |            | Easting:  | 1193775.66 ft | Longitude:        | 103 40 12.019 W |  |  |
| Position Uncertainty:                                                 | 0.00 ft    |           |               | North Reference:  | True            |  |  |
| Ground Level:                                                         | 1920.00 ft |           |               | Grid Convergence: | -2.36 deg       |  |  |

|                       |         |         |           |               |            |                 |  |            |  |
|-----------------------|---------|---------|-----------|---------------|------------|-----------------|--|------------|--|
| Well: 3-16H           |         |         |           |               |            |                 |  | Slot Name: |  |
| Well Position:        | +N/-S   | 0.00 ft | Northing: | 408124.35 ft  | Latitude:  | 48 4 30.494 N   |  |            |  |
|                       | +E/-W   | 0.00 ft | Easting : | 1193775.66 ft | Longitnde: | 103 40 12.019 W |  |            |  |
| Position Uncertainty: | 0.00 ft |         |           |               |            |                 |  |            |  |

|                   |                  |  |        |            |                     |                |  |                       |  |
|-------------------|------------------|--|--------|------------|---------------------|----------------|--|-----------------------|--|
| Wellpath: OH      |                  |  |        |            |                     |                |  | Drilled From: Surface |  |
| Cnrrent Datum:    | GL 1920+KB 41    |  | Height | 1941.00 ft | Tie-on Depth:       | 0.00 ft        |  |                       |  |
| Magnetic Data:    | 04/19/2012       |  |        |            | Above System Datum: | Mean Sea Level |  |                       |  |
| Field Strength:   | 56675 nT         |  |        |            | Declination:        | 8.65 deg       |  |                       |  |
| Vertical Section: | Depth From (TVD) |  | +N/-S  |            | Mag Dip Angle:      | 73.09 deg      |  |                       |  |
|                   | ft               |  | ft     |            | +E/-W               | Direction      |  |                       |  |
|                   |                  |  |        |            | ft                  | deg            |  |                       |  |
|                   | 0.00             |  | 0.00   |            | 0.00                | 187.73         |  |                       |  |

|                |  |                           |  |
|----------------|--|---------------------------|--|
| Plan: Plan #1  |  | Date Composed: 04/23/2012 |  |
| Principal: Yes |  | Version: 1                |  |
|                |  | Tied-to: User Defined     |  |

| Plan Section Information |             |             |           |             |             |                  |                    |                   |            |        |
|--------------------------|-------------|-------------|-----------|-------------|-------------|------------------|--------------------|-------------------|------------|--------|
| MD<br>ft                 | Incl<br>deg | Azim<br>deg | TVD<br>ft | +N/-S<br>ft | +E/-W<br>ft | DLS<br>deg/100ft | Build<br>deg/100ft | Turn<br>deg/100ft | TFO<br>deg | Target |
| 0.00                     | 0.00        | 0.00        | 0.00      | 0.00        | 0.00        | 0.00             | 0.00               | 0.00              | 0.00       |        |
| 10166.27                 | 0.00        | 0.00        | 10166.27  | 0.00        | 0.00        | 0.00             | 0.00               | 0.00              | 0.00       |        |
| 10857.19                 | 89.82       | 228.62      | 10607.00  | -290.44     | -329.66     | 13.00            | 13.00              | 0.00              | 228.62     |        |
| 10957.19                 | 89.82       | 228.62      | 10607.32  | -356.55     | -404.69     | 0.00             | 0.00               | 0.00              | 0.00       |        |
| 12577.81                 | 89.82       | 180.00      | 10612.81  | -1789.55    | -1052.00    | 3.00             | 0.00               | -3.00             | -90.09     |        |
| 18535.29                 | 89.82       | 180.00      | 10632.00  | -7747.00    | -1052.00    | 0.00             | 0.00               | 0.00              | 0.00       | PBHL   |

| Survey   |             |             |           |           |           |          |                  |                    |                   |         |
|----------|-------------|-------------|-----------|-----------|-----------|----------|------------------|--------------------|-------------------|---------|
| MD<br>ft | Incl<br>deg | Azim<br>deg | TVD<br>ft | N/S<br>ft | E/W<br>ft | VS<br>ft | DLS<br>deg/100ft | Build<br>deg/100ft | Turn<br>deg/100ft | Comment |
| 10166.27 | 0.00        | 0.00        | 10166.27  | 0.00      | 0.00      | -0.00    | 0.00             | 0.00               | 0.00              |         |
| 10175.00 | 1.13        | 228.62      | 10175.00  | -0.06     | -0.06     | 0.07     | 13.00            | 13.00              | 0.00              |         |
| 10200.00 | 4.38        | 228.62      | 10199.97  | -0.85     | -0.97     | 0.98     | 13.00            | 13.00              | 0.00              |         |
| 10225.00 | 7.63        | 228.62      | 10224.83  | -2.58     | -2.93     | 2.95     | 13.00            | 13.00              | 0.00              |         |
| 10250.00 | 10.88       | 228.62      | 10249.50  | -5.24     | -5.95     | 5.99     | 13.00            | 13.00              | 0.00              |         |
| 10275.00 | 14.13       | 228.62      | 10273.90  | -8.82     | -10.01    | 10.09    | 13.00            | 13.00              | 0.00              |         |
| 10300.00 | 17.38       | 228.62      | 10297.96  | -13.31    | -15.11    | 15.22    | 13.00            | 13.00              | 0.00              |         |
| 10325.00 | 20.63       | 228.62      | 10321.59  | -18.69    | -21.22    | 21.38    | 13.00            | 13.00              | 0.00              |         |
| 10350.00 | 23.88       | 228.62      | 10344.72  | -24.95    | -28.32    | 28.53    | 13.00            | 13.00              | 0.00              |         |
| 10375.00 | 27.13       | 228.62      | 10367.28  | -32.07    | -36.40    | 36.67    | 13.00            | 13.00              | 0.00              |         |
| 10400.00 | 30.38       | 228.62      | 10389.20  | -40.02    | -45.42    | 45.76    | 13.00            | 13.00              | 0.00              |         |
| 10425.00 | 33.63       | 228.62      | 10410.39  | -48.78    | -55.36    | 55.78    | 13.00            | 13.00              | 0.00              |         |
| 10450.00 | 36.88       | 228.62      | 10430.80  | -58.32    | -66.19    | 66.69    | 13.00            | 13.00              | 0.00              |         |
| 10475.00 | 40.13       | 228.62      | 10450.36  | -68.61    | -77.87    | 78.46    | 13.00            | 13.00              | 0.00              |         |
| 10500.00 | 43.38       | 228.62      | 10469.01  | -79.61    | -90.36    | 91.04    | 13.00            | 13.00              | 0.00              |         |

LEAM Drilling Systems, Inc.  
Planning Report

|           |                       |                            |                              |       |          |       |        |
|-----------|-----------------------|----------------------------|------------------------------|-------|----------|-------|--------|
| Company:  | Continental Resources | Date:                      | 04/23/2012                   | Time: | 17:16:30 | Page: | 2      |
| Field:    | McKenzie County, ND   | Co-ordinate(NE) Reference: | Well: 3-16H, True North      |       |          |       |        |
| Site:     | Tallahassee 3         | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0         |       |          |       |        |
| Well:     | 3-16H                 | Section (VS) Reference:    | Well (0.00N,0.00E,187.73Azi) |       |          |       |        |
| Wellpath: | OH                    | Survey Calculation Method: | Minimum Curvature            |       |          | Db:   | Adapti |

Survey

| MD<br>ft | Incl<br>deg | Azim<br>deg | TVD<br>ft | N/S<br>ft | E/W<br>ft | VS<br>ft | DLS<br>deg/100ft | Build<br>deg/100ft | Turn<br>deg/100ft | Comment              |
|----------|-------------|-------------|-----------|-----------|-----------|----------|------------------|--------------------|-------------------|----------------------|
| 10525.00 | 46.63       | 228.62      | 10486.68  | -91.30    | -103.63   | 104.41   | 13.00            | 13.00              | 0.00              |                      |
| 10550.00 | 49.88       | 228.62      | 10503.32  | -103.63   | -117.62   | 118.51   | 13.00            | 13.00              | 0.00              |                      |
| 10562.67 | 51.53       | 228.62      | 10511.35  | -110.11   | -124.98   | 125.92   | 13.00            | 13.00              | 0.00              | Upper Bakken Shale   |
| 10575.00 | 53.13       | 228.62      | 10518.88  | -116.56   | -132.30   | 133.30   | 13.00            | 13.00              | 0.00              |                      |
| 10591.26 | 55.25       | 228.62      | 10528.39  | -125.28   | -142.19   | 143.27   | 13.00            | 13.00              | 0.00              | Middle Bakken Member |
| 10600.00 | 56.38       | 228.62      | 10533.30  | -130.06   | -147.62   | 148.73   | 13.00            | 13.00              | 0.00              |                      |
| 10625.00 | 59.63       | 228.62      | 10546.55  | -144.07   | -163.53   | 164.76   | 13.00            | 13.00              | 0.00              |                      |
| 10650.00 | 62.88       | 228.62      | 10558.57  | -158.56   | -179.97   | 181.33   | 13.00            | 13.00              | 0.00              |                      |
| 10675.00 | 66.13       | 228.62      | 10569.32  | -173.48   | -196.90   | 198.39   | 13.00            | 13.00              | 0.00              |                      |
| 10700.00 | 69.38       | 228.62      | 10578.78  | -188.77   | -214.26   | 215.88   | 13.00            | 13.00              | 0.00              |                      |
| 10725.00 | 72.63       | 228.62      | 10586.92  | -204.40   | -232.00   | 233.75   | 13.00            | 13.00              | 0.00              |                      |
| 10750.00 | 75.88       | 228.62      | 10593.70  | -220.30   | -250.05   | 251.93   | 13.00            | 13.00              | 0.00              |                      |
| 10775.00 | 79.13       | 228.62      | 10599.11  | -236.44   | -268.36   | 270.38   | 13.00            | 13.00              | 0.00              |                      |
| 10800.00 | 82.38       | 228.62      | 10603.12  | -252.75   | -286.87   | 289.04   | 13.00            | 13.00              | 0.00              |                      |
| 10809.00 | 83.55       | 228.62      | 10604.22  | -258.65   | -293.57   | 295.79   | 13.00            | 13.00              | 0.00              | Middle Bakken Target |
| 10825.00 | 85.63       | 228.62      | 10605.73  | -269.18   | -305.53   | 307.83   | 13.00            | 13.00              | 0.00              |                      |
| 10850.00 | 88.88       | 228.62      | 10606.92  | -285.69   | -324.26   | 326.71   | 13.00            | 13.00              | 0.00              |                      |
| 10857.19 | 89.82       | 228.62      | 10607.00  | -290.44   | -329.66   | 332.14   | 13.00            | 13.00              | 0.00              | 7" - 2875' FNL & 598 |
| 10900.00 | 89.82       | 228.62      | 10607.14  | -318.74   | -361.78   | 364.50   | 0.00             | 0.00               | 0.00              |                      |
| 10957.19 | 89.82       | 228.62      | 10607.32  | -356.55   | -404.69   | 407.74   | 0.00             | 0.00               | 0.00              |                      |
| 11000.00 | 89.82       | 227.33      | 10607.45  | -385.20   | -436.49   | 440.41   | 3.00             | 0.00               | -3.00             |                      |
| 11100.00 | 89.81       | 224.33      | 10607.78  | -454.87   | -508.21   | 519.09   | 3.00             | 0.00               | -3.00             |                      |
| 11200.00 | 89.81       | 221.33      | 10608.10  | -528.19   | -576.19   | 600.89   | 3.00             | 0.00               | -3.00             |                      |
| 11300.00 | 89.81       | 218.33      | 10608.44  | -604.97   | -640.24   | 685.59   | 3.00             | 0.00               | -3.00             |                      |
| 11400.00 | 89.80       | 215.33      | 10608.78  | -685.00   | -700.19   | 772.95   | 3.00             | 0.00               | -3.00             |                      |
| 11500.00 | 89.80       | 212.33      | 10609.12  | -768.05   | -755.86   | 862.74   | 3.00             | 0.00               | -3.00             |                      |
| 11600.00 | 89.80       | 209.33      | 10609.46  | -853.91   | -807.11   | 954.71   | 3.00             | 0.00               | -3.00             |                      |
| 11700.00 | 89.80       | 206.33      | 10609.81  | -942.33   | -853.80   | 1048.61  | 3.00             | 0.00               | -3.00             |                      |
| 11800.00 | 89.80       | 203.33      | 10610.16  | -1033.07  | -895.79   | 1144.17  | 3.00             | 0.00               | -3.00             |                      |
| 11900.00 | 89.80       | 200.33      | 10610.51  | -1125.89  | -932.98   | 1241.15  | 3.00             | 0.00               | -3.00             |                      |
| 12000.00 | 89.80       | 197.33      | 10610.86  | -1220.52  | -965.26   | 1339.26  | 3.00             | 0.00               | -3.00             |                      |
| 12100.00 | 89.80       | 194.33      | 10611.20  | -1316.71  | -992.54   | 1438.25  | 3.00             | 0.00               | -3.00             |                      |
| 12200.00 | 89.80       | 191.33      | 10611.55  | -1414.20  | -1014.75  | 1537.84  | 3.00             | 0.00               | -3.00             |                      |
| 12300.00 | 89.81       | 188.33      | 10611.89  | -1512.72  | -1031.83  | 1637.76  | 3.00             | 0.00               | -3.00             |                      |
| 12400.00 | 89.81       | 185.33      | 10612.23  | -1612.00  | -1043.73  | 1737.74  | 3.00             | 0.00               | -3.00             |                      |
| 12500.00 | 89.81       | 182.33      | 10612.56  | -1711.76  | -1050.41  | 1837.50  | 3.00             | 0.00               | -3.00             |                      |
| 12577.81 | 89.82       | 180.00      | 10612.81  | -1789.55  | -1052.00  | 1914.79  | 3.00             | 0.00               | -3.00             |                      |
| 12600.00 | 89.82       | 180.00      | 10612.88  | -1811.74  | -1052.00  | 1936.78  | 0.00             | 0.00               | 0.00              |                      |
| 12700.00 | 89.82       | 180.00      | 10613.20  | -1911.74  | -1052.00  | 2035.87  | 0.00             | 0.00               | 0.00              |                      |
| 12800.00 | 89.82       | 180.00      | 10613.52  | -2011.74  | -1052.00  | 2134.96  | 0.00             | 0.00               | 0.00              |                      |
| 12900.00 | 89.82       | 180.00      | 10613.85  | -2111.74  | -1052.00  | 2234.05  | 0.00             | 0.00               | 0.00              |                      |
| 13000.00 | 89.82       | 180.00      | 10614.17  | -2211.74  | -1052.00  | 2333.14  | 0.00             | 0.00               | 0.00              |                      |
| 13100.00 | 89.82       | 180.00      | 10614.49  | -2311.74  | -1052.00  | 2432.23  | 0.00             | 0.00               | 0.00              |                      |
| 13200.00 | 89.82       | 180.00      | 10614.81  | -2411.74  | -1052.00  | 2531.32  | 0.00             | 0.00               | 0.00              |                      |
| 13300.00 | 89.82       | 180.00      | 10615.13  | -2511.74  | -1052.00  | 2630.41  | 0.00             | 0.00               | 0.00              |                      |
| 13400.00 | 89.82       | 180.00      | 10615.46  | -2611.74  | -1052.00  | 2729.50  | 0.00             | 0.00               | 0.00              |                      |
| 13500.00 | 89.82       | 180.00      | 10615.78  | -2711.74  | -1052.00  | 2828.60  | 0.00             | 0.00               | 0.00              |                      |
| 13600.00 | 89.82       | 180.00      | 10616.10  | -2811.74  | -1052.00  | 2927.69  | 0.00             | 0.00               | 0.00              |                      |
| 13700.00 | 89.82       | 180.00      | 10616.42  | -2911.74  | -1052.00  | 3026.78  | 0.00             | 0.00               | 0.00              |                      |
| 13800.00 | 89.82       | 180.00      | 10616.75  | -3011.74  | -1052.00  | 3125.87  | 0.00             | 0.00               | 0.00              |                      |
| 13900.00 | 89.82       | 180.00      | 10617.07  | -3111.74  | -1052.00  | 3224.96  | 0.00             | 0.00               | 0.00              |                      |
| 14000.00 | 89.82       | 180.00      | 10617.39  | -3211.74  | -1052.00  | 3324.05  | 0.00             | 0.00               | 0.00              |                      |
| 14100.00 | 89.82       | 180.00      | 10617.71  | -3311.73  | -1052.00  | 3423.14  | 0.00             | 0.00               | 0.00              |                      |



LEAM Drilling Systems, Inc.  
Planning Report

|           |                       |                            |                              |       |          |       |            |
|-----------|-----------------------|----------------------------|------------------------------|-------|----------|-------|------------|
| Company:  | Continental Resources | Date:                      | 04/23/2012                   | Time: | 17:16:30 | Page: | 3          |
| Field:    | McKenzie County, ND   | Co-ordinate(NE) Reference: | Well: 3-16H, True North      |       |          |       |            |
| Site:     | Tallahassee 3         | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0         |       |          |       |            |
| Well:     | 3-16H                 | Section (VS) Reference:    | Well (0.00N,0.00E,187.73Azi) |       |          |       |            |
| Wellpath: | OH                    | Survey Calculation Method: | Minimum Curvature            |       |          |       | Db: Adapti |

Survey

| MD<br>ft | Incl<br>deg | Azim<br>deg | TVD<br>ft | N/S<br>ft | E/W<br>ft | VS<br>ft | DLS<br>deg/100ft | Build<br>deg/100ft | Turn<br>deg/100ft | Comment |
|----------|-------------|-------------|-----------|-----------|-----------|----------|------------------|--------------------|-------------------|---------|
| 14200.00 | 89.82       | 180.00      | 10618.03  | -3411.73  | -1052.00  | 3522.23  | 0.00             | 0.00               | 0.00              |         |
| 14300.00 | 89.82       | 180.00      | 10618.36  | -3511.73  | -1052.00  | 3621.32  | 0.00             | 0.00               | 0.00              |         |
| 14400.00 | 89.82       | 180.00      | 10618.68  | -3611.73  | -1052.00  | 3720.41  | 0.00             | 0.00               | 0.00              |         |
| 14500.00 | 89.82       | 180.00      | 10619.00  | -3711.73  | -1052.00  | 3819.50  | 0.00             | 0.00               | 0.00              |         |
| 14600.00 | 89.82       | 180.00      | 10619.32  | -3811.73  | -1052.00  | 3918.59  | 0.00             | 0.00               | 0.00              |         |
| 14700.00 | 89.82       | 180.00      | 10619.64  | -3911.73  | -1052.00  | 4017.68  | 0.00             | 0.00               | 0.00              |         |
| 14800.00 | 89.82       | 180.00      | 10619.97  | -4011.73  | -1052.00  | 4116.78  | 0.00             | 0.00               | 0.00              |         |
| 14900.00 | 89.82       | 180.00      | 10620.29  | -4111.73  | -1052.00  | 4215.87  | 0.00             | 0.00               | 0.00              |         |
| 15000.00 | 89.82       | 180.00      | 10620.61  | -4211.73  | -1052.00  | 4314.96  | 0.00             | 0.00               | 0.00              |         |
| 15100.00 | 89.82       | 180.00      | 10620.93  | -4311.73  | -1052.00  | 4414.05  | 0.00             | 0.00               | 0.00              |         |
| 15200.00 | 89.82       | 180.00      | 10621.26  | -4411.73  | -1052.00  | 4513.14  | 0.00             | 0.00               | 0.00              |         |
| 15300.00 | 89.82       | 180.00      | 10621.58  | -4511.73  | -1052.00  | 4612.23  | 0.00             | 0.00               | 0.00              |         |
| 15400.00 | 89.82       | 180.00      | 10621.90  | -4611.73  | -1052.00  | 4711.32  | 0.00             | 0.00               | 0.00              |         |
| 15500.00 | 89.82       | 180.00      | 10622.22  | -4711.73  | -1052.00  | 4810.41  | 0.00             | 0.00               | 0.00              |         |
| 15600.00 | 89.82       | 180.00      | 10622.54  | -4811.73  | -1052.00  | 4909.50  | 0.00             | 0.00               | 0.00              |         |
| 15700.00 | 89.82       | 180.00      | 10622.87  | -4911.73  | -1052.00  | 5008.59  | 0.00             | 0.00               | 0.00              |         |
| 15800.00 | 89.82       | 180.00      | 10623.19  | -5011.73  | -1052.00  | 5107.68  | 0.00             | 0.00               | 0.00              |         |
| 15900.00 | 89.82       | 180.00      | 10623.51  | -5111.73  | -1052.00  | 5206.77  | 0.00             | 0.00               | 0.00              |         |
| 16000.00 | 89.82       | 180.00      | 10623.83  | -5211.72  | -1052.00  | 5305.86  | 0.00             | 0.00               | 0.00              |         |
| 16100.00 | 89.82       | 180.00      | 10624.15  | -5311.72  | -1052.00  | 5404.96  | 0.00             | 0.00               | 0.00              |         |
| 16200.00 | 89.82       | 180.00      | 10624.48  | -5411.72  | -1052.00  | 5504.05  | 0.00             | 0.00               | 0.00              |         |
| 16300.00 | 89.82       | 180.00      | 10624.80  | -5511.72  | -1052.00  | 5603.14  | 0.00             | 0.00               | 0.00              |         |
| 16400.00 | 89.82       | 180.00      | 10625.12  | -5611.72  | -1052.00  | 5702.23  | 0.00             | 0.00               | 0.00              |         |
| 16500.00 | 89.82       | 180.00      | 10625.44  | -5711.72  | -1052.00  | 5801.32  | 0.00             | 0.00               | 0.00              |         |
| 16600.00 | 89.82       | 180.00      | 10625.77  | -5811.72  | -1052.00  | 5900.41  | 0.00             | 0.00               | 0.00              |         |
| 16700.00 | 89.82       | 180.00      | 10626.09  | -5911.72  | -1052.00  | 5999.50  | 0.00             | 0.00               | 0.00              |         |
| 16800.00 | 89.82       | 180.00      | 10626.41  | -6011.72  | -1052.00  | 6098.59  | 0.00             | 0.00               | 0.00              |         |
| 16900.00 | 89.82       | 180.00      | 10626.73  | -6111.72  | -1052.00  | 6197.68  | 0.00             | 0.00               | 0.00              |         |
| 17000.00 | 89.82       | 180.00      | 10627.05  | -6211.72  | -1052.00  | 6296.77  | 0.00             | 0.00               | 0.00              |         |
| 17100.00 | 89.82       | 180.00      | 10627.38  | -6311.72  | -1052.00  | 6395.86  | 0.00             | 0.00               | 0.00              |         |
| 17200.00 | 89.82       | 180.00      | 10627.70  | -6411.72  | -1052.00  | 6494.95  | 0.00             | 0.00               | 0.00              |         |
| 17300.00 | 89.82       | 180.00      | 10628.02  | -6511.72  | -1052.00  | 6594.04  | 0.00             | 0.00               | 0.00              |         |
| 17400.00 | 89.82       | 180.00      | 10628.34  | -6611.72  | -1052.00  | 6693.14  | 0.00             | 0.00               | 0.00              |         |
| 17500.00 | 89.82       | 180.00      | 10628.66  | -6711.72  | -1052.00  | 6792.23  | 0.00             | 0.00               | 0.00              |         |
| 17600.00 | 89.82       | 180.00      | 10628.99  | -6811.72  | -1052.00  | 6891.32  | 0.00             | 0.00               | 0.00              |         |
| 17700.00 | 89.82       | 180.00      | 10629.31  | -6911.72  | -1052.00  | 6990.41  | 0.00             | 0.00               | 0.00              |         |
| 17800.00 | 89.82       | 180.00      | 10629.63  | -7011.72  | -1052.00  | 7089.50  | 0.00             | 0.00               | 0.00              |         |
| 17900.00 | 89.82       | 180.00      | 10629.95  | -7111.72  | -1052.00  | 7188.59  | 0.00             | 0.00               | 0.00              |         |
| 18000.00 | 89.82       | 180.00      | 10630.28  | -7211.71  | -1052.00  | 7287.68  | 0.00             | 0.00               | 0.00              |         |
| 18100.00 | 89.82       | 180.00      | 10630.60  | -7311.71  | -1052.00  | 7386.77  | 0.00             | 0.00               | 0.00              |         |
| 18200.00 | 89.82       | 180.00      | 10630.92  | -7411.71  | -1052.00  | 7485.86  | 0.00             | 0.00               | 0.00              |         |
| 18300.00 | 89.82       | 180.00      | 10631.24  | -7511.71  | -1052.00  | 7584.95  | 0.00             | 0.00               | 0.00              |         |
| 18400.00 | 89.82       | 180.00      | 10631.56  | -7611.71  | -1052.00  | 7684.04  | 0.00             | 0.00               | 0.00              |         |
| 18500.00 | 89.82       | 180.00      | 10631.89  | -7711.71  | -1052.00  | 7783.13  | 0.00             | 0.00               | 0.00              |         |
| 18535.29 | 89.82       | 180.00      | 10632.00  | -7747.00  | -1052.00  | 7818.10  | 0.00             | 0.00               | 0.00              |         |

Targets

| Name | Description<br>Dip. | Dir. | TVD | +N/-S | +E/-W | Map<br>Northing | Map<br>Easting | <---- Latitude ----> |     |     | <--- Longitude ---> |     |     |
|------|---------------------|------|-----|-------|-------|-----------------|----------------|----------------------|-----|-----|---------------------|-----|-----|
|      |                     |      |     |       |       |                 |                | Deg                  | Min | Sec | Deg                 | Min | Sec |

LEAM Drilling Systems, Inc.  
Planning Report

|           |                       |                            |                              |       |          |       |            |
|-----------|-----------------------|----------------------------|------------------------------|-------|----------|-------|------------|
| Company:  | Continental Resources | Date:                      | 04/23/2012                   | Time: | 17:16:30 | Page: | 4          |
| Field:    | McKenzie County, ND   | Co-ordinate(NE) Reference: | Well: 3-16H, True North      |       |          |       |            |
| Site:     | Tallahassee 3         | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0         |       |          |       |            |
| Well:     | 3-16H                 | Section (VS) Reference:    | Well (0.00N,0.00E,187.73Azi) |       |          |       |            |
| Wellpath: | OH                    | Survey Calculation Method: | Minimum Curvature            |       |          |       | Db: Adapti |

Casing Points

| MD<br>ft | TVD<br>ft | Diameter<br>in | Hole Size<br>in | Name |
|----------|-----------|----------------|-----------------|------|
| 10857.19 | 10607.00  | 7.00           | 8.50            | 7"   |

Formations

| MD<br>ft | TVD<br>ft | Formations             | Lithology | Dip Angle<br>deg | Dip Direction<br>deg |
|----------|-----------|------------------------|-----------|------------------|----------------------|
| 1841.00  | 1841.00   | Pierre Shale           |           | 0.00             | 0.00                 |
| 4425.00  | 4425.00   | Greenhorn              |           | 0.00             | 0.00                 |
| 4826.00  | 4826.00   | Dakota Group           |           | 0.00             | 0.00                 |
| 5705.00  | 5705.00   | Base of Dakota Sand    |           | 0.00             | 0.00                 |
| 6733.00  | 6733.00   | Dunham Salt Top        |           | 0.00             | 0.00                 |
| 6793.00  | 6793.00   | Dunham Salt Base       |           | 0.00             | 0.00                 |
| 7103.00  | 7103.00   | Pine Salt Top          |           | 0.00             | 0.00                 |
| 7159.00  | 7159.00   | Pine Salt Base         |           | 0.00             | 0.00                 |
| 7185.00  | 7185.00   | Minnekahta             |           | 0.00             | 0.00                 |
| 7471.00  | 7471.00   | Minnelusa Group        |           | 0.00             | 0.00                 |
| 8179.00  | 8179.00   | Kibbey                 |           | 0.00             | 0.00                 |
| 8333.00  | 8333.00   | Top Charles            |           | 0.00             | 0.00                 |
| 9031.00  | 9031.00   | Base Last Charles Salt |           | 0.18             | 180.00               |
| 9255.00  | 9255.00   | Mission Canyon         |           | 0.18             | 180.00               |
| 9821.00  | 9821.00   | Lodgepole              |           | 0.18             | 180.00               |
| 10562.67 | 10511.35  | Upper Bakken Shale     |           | 0.18             | 180.00               |
| 10591.26 | 10528.39  | Middle Bakken Member   |           | 0.18             | 180.00               |
| 10809.00 | 10604.22  | Middle Bakken Target   |           | 0.18             | 180.00               |

LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |       |        |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|-------|--------|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page: | 1      |
| Field:              | McKenzie County, ND   |                            |                         |       |          |       |        |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |       |        |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |       |        |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db:   | Adapti |

|                                                              |  |  |  |                |                     |
|--------------------------------------------------------------|--|--|--|----------------|---------------------|
| NO GLOBAL SCAN: Using user defined selection & scan criteria |  |  |  | Reference:     | Plan: Plan #1       |
| Interpolation Method: MD + Stations Interval: 100.00 ft      |  |  |  | Error Model:   | ISCWSA Ellipse      |
| Depth Range: 0.00 to 18535.29 ft                             |  |  |  | Scan Method:   | Closest Approach 3D |
| Maximum Radius: 10000.00 ft                                  |  |  |  | Error Surface: | Ellipse             |

|            |         |                |              |
|------------|---------|----------------|--------------|
| Plan:      | Plan #1 | Date Composed: | 04/23/2012   |
| Principal: | Yes     | Version:       | 1            |
|            |         | Tied-to:       | User Defined |

Summary

| Site               | Offset Wellpath | Reference MD           | Offset MD | Ctr-Ctr Distance | Edge Distance | Separation Factor | Warning |
|--------------------|-----------------|------------------------|-----------|------------------|---------------|-------------------|---------|
|                    | Well            | ft                     | ft        | ft               | ft            |                   |         |
| Columbus Federal 1 | 1-16H           | OH V3 Plan: Plan #2 V1 | 10350.00  | 10343.75         | 51.75         | 25.43             | 1.97    |
| Columbus Federal 2 | 2-16H           | OH V3 Plan: Plan #1 V1 | 10166.27  | 10166.27         | 90.00         | 63.44             | 3.39    |
| Columbus Federal 3 | 3-16H           | OH V1 Plan: Plan #1 V1 | 10100.00  | 10100.00         | 134.95        | 108.56            | 5.11    |
| Tallahassee 2      | 2-16H           | OH V4 Plan: Plan #1 V1 | 10300.00  | 10312.48         | 22.68         | -3.27             | 0.87    |
|                    |                 |                        |           |                  |               |                   | Level 1 |

Site: Columbus Federal 1  
Well: 1-16H  
Wellpath: OH V3 Plan: Plan #2 V1

Inter-Site Error: 0.00 ft

| Reference MD | TVD     | Offset MD | TVD     | Semi-Major Axis Ref | Offset | TFO-HS | Offset Location North | East   | Ctr-Ctr Distance | Edge Distance | Separation Factor | Warning |
|--------------|---------|-----------|---------|---------------------|--------|--------|-----------------------|--------|------------------|---------------|-------------------|---------|
| ft           | ft      | ft        | ft      | ft                  | ft     | deg    | ft                    | ft     | ft               | ft            |                   |         |
| 0.00         | 0.00    | 0.00      | 0.00    | 0.00                | 0.00   | 210.72 | -84.00                | -49.91 | 97.71            |               |                   | No Data |
| 100.00       | 100.00  | 100.00    | 100.00  | 0.00                | 0.21   | 210.72 | -84.00                | -49.91 | 97.71            | 97.50         | 471.98            |         |
| 200.00       | 200.00  | 200.00    | 200.00  | 0.00                | 0.47   | 210.72 | -84.00                | -49.91 | 97.71            | 97.24         | 208.42            |         |
| 300.00       | 300.00  | 300.00    | 300.00  | 0.00                | 0.73   | 210.72 | -84.00                | -49.91 | 97.71            | 96.98         | 133.74            |         |
| 400.00       | 400.00  | 400.00    | 400.00  | 0.00                | 0.99   | 210.72 | -84.00                | -49.91 | 97.71            | 96.72         | 98.46             |         |
| 500.00       | 500.00  | 500.00    | 500.00  | 0.00                | 1.25   | 210.72 | -84.00                | -49.91 | 97.71            | 96.46         | 77.90             |         |
| 600.00       | 600.00  | 600.00    | 600.00  | 0.00                | 1.52   | 210.72 | -84.00                | -49.91 | 97.71            | 96.19         | 64.45             |         |
| 700.00       | 700.00  | 700.00    | 700.00  | 0.00                | 1.78   | 210.72 | -84.00                | -49.91 | 97.71            | 95.93         | 54.96             |         |
| 800.00       | 800.00  | 800.00    | 800.00  | 0.00                | 2.04   | 210.72 | -84.00                | -49.91 | 97.71            | 95.67         | 47.91             |         |
| 900.00       | 900.00  | 900.00    | 900.00  | 0.00                | 2.30   | 210.72 | -84.00                | -49.91 | 97.71            | 95.41         | 42.46             |         |
| 1000.00      | 1000.00 | 1000.00   | 1000.00 | 0.00                | 2.56   | 210.72 | -84.00                | -49.91 | 97.71            | 95.15         | 38.12             |         |
| 1100.00      | 1100.00 | 1100.00   | 1100.00 | 0.00                | 2.82   | 210.72 | -84.00                | -49.91 | 97.71            | 94.88         | 34.59             |         |
| 1200.00      | 1200.00 | 1200.00   | 1200.00 | 0.00                | 3.09   | 210.72 | -84.00                | -49.91 | 97.71            | 94.62         | 31.65             |         |
| 1300.00      | 1300.00 | 1300.00   | 1300.00 | 0.00                | 3.35   | 210.72 | -84.00                | -49.91 | 97.71            | 94.36         | 29.18             |         |
| 1400.00      | 1400.00 | 1400.00   | 1400.00 | 0.00                | 3.61   | 210.72 | -84.00                | -49.91 | 97.71            | 94.10         | 27.06             |         |
| 1500.00      | 1500.00 | 1500.00   | 1500.00 | 0.00                | 3.87   | 210.72 | -84.00                | -49.91 | 97.71            | 93.84         | 25.23             |         |
| 1600.00      | 1600.00 | 1600.00   | 1600.00 | 0.00                | 4.13   | 210.72 | -84.00                | -49.91 | 97.71            | 93.58         | 23.64             |         |
| 1700.00      | 1700.00 | 1700.00   | 1700.00 | 0.00                | 4.40   | 210.72 | -84.00                | -49.91 | 97.71            | 93.31         | 22.23             |         |
| 1800.00      | 1800.00 | 1800.00   | 1800.00 | 0.00                | 4.66   | 210.72 | -84.00                | -49.91 | 97.71            | 93.05         | 20.98             |         |
| 1900.00      | 1900.00 | 1900.00   | 1900.00 | 0.00                | 4.92   | 210.72 | -84.00                | -49.91 | 97.71            | 92.79         | 19.86             |         |
| 2000.00      | 2000.00 | 2000.00   | 2000.00 | 0.00                | 5.18   | 210.72 | -84.00                | -49.91 | 97.71            | 92.53         | 18.86             |         |
| 2100.00      | 2100.00 | 2100.00   | 2100.00 | 0.00                | 5.44   | 210.72 | -84.00                | -49.91 | 97.71            | 92.27         | 17.95             |         |
| 2200.00      | 2200.00 | 2200.00   | 2200.00 | 0.00                | 5.70   | 210.72 | -84.00                | -49.91 | 97.71            | 92.01         | 17.13             |         |
| 2300.00      | 2300.00 | 2300.00   | 2300.00 | 0.00                | 5.97   | 210.72 | -84.00                | -49.91 | 97.71            | 91.74         | 16.38             |         |
| 2400.00      | 2400.00 | 2400.00   | 2400.00 | 0.00                | 6.23   | 210.72 | -84.00                | -49.91 | 97.71            | 91.48         | 15.69             |         |
| 2500.00      | 2500.00 | 2500.00   | 2500.00 | 0.00                | 6.49   | 210.72 | -84.00                | -49.91 | 97.71            | 91.22         | 15.05             |         |
| 2600.00      | 2600.00 | 2600.00   | 2600.00 | 0.00                | 6.75   | 210.72 | -84.00                | -49.91 | 97.71            | 90.96         | 14.47             |         |
| 2700.00      | 2700.00 | 2700.00   | 2700.00 | 0.00                | 7.01   | 210.72 | -84.00                | -49.91 | 97.71            | 90.70         | 13.93             |         |
| 2800.00      | 2800.00 | 2800.00   | 2800.00 | 0.00                | 7.28   | 210.72 | -84.00                | -49.91 | 97.71            | 90.43         | 13.43             |         |
| 2900.00      | 2900.00 | 2900.00   | 2900.00 | 0.00                | 7.54   | 210.72 | -84.00                | -49.91 | 97.71            | 90.17         | 12.96             |         |
| 3000.00      | 3000.00 | 3000.00   | 3000.00 | 0.00                | 7.80   | 210.72 | -84.00                | -49.91 | 97.71            | 89.91         | 12.53             |         |
| 3100.00      | 3100.00 | 3100.00   | 3100.00 | 0.00                | 8.06   | 210.72 | -84.00                | -49.91 | 97.71            | 89.65         | 12.12             |         |
| 3200.00      | 3200.00 | 3200.00   | 3200.00 | 0.00                | 8.32   | 210.72 | -84.00                | -49.91 | 97.71            | 89.39         | 11.74             |         |
| 3300.00      | 3300.00 | 3300.00   | 3300.00 | 0.00                | 8.58   | 210.72 | -84.00                | -49.91 | 97.71            | 89.13         | 11.38             |         |
| 3400.00      | 3400.00 | 3400.00   | 3400.00 | 0.00                | 8.85   | 210.72 | -84.00                | -49.91 | 97.71            | 88.86         | 11.05             |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            | Continental Resources  |         | Date:                      | 04/23/2012      |        | Time:  | 17:17:28                |        | Page:    | 2        |            |         |
|---------------------|------------------------|---------|----------------------------|-----------------|--------|--------|-------------------------|--------|----------|----------|------------|---------|
| Field:              | McKenzie County, ND    |         |                            |                 |        |        |                         |        |          |          |            |         |
| Reference Site:     | Tallahassee 3          |         | Co-ordinate(NE) Reference: |                 |        |        | Well: 3-16H, True North |        |          |          |            |         |
| Reference Well:     | 3-16H                  |         | Vertical (TVD) Reference:  |                 |        |        | GL 1920+KB 41 1941.0    |        |          |          |            |         |
| Reference Wellpath: | OH                     |         |                            |                 |        |        | Db: Adapti              |        |          |          |            |         |
| Site:               | Columbus Federal 1     |         |                            |                 |        |        |                         |        |          |          |            |         |
| Well:               | 1-16H                  |         |                            |                 |        |        |                         |        |          |          |            |         |
| Wellpath:           | OH V3 Plan: Plan #2 V1 |         |                            |                 |        |        | Inter-Site Error:       |        | 0.00     |          | ft         |         |
| Reference           |                        | Offset  |                            | Semi-Major Axis |        |        | Offset Location         |        | Ctr-Ctr  | Edge     | Separation | Warning |
| MD                  | TVD                    | MD      | TVD                        | Ref             | Offset | TFO-HS | North                   | East   | Distance | Distance | Factor     |         |
| ft                  | ft                     | ft      | ft                         | ft              | ft     | deg    | ft                      | ft     | ft       | ft       |            |         |
| 3500.00             | 3500.00                | 3500.00 | 3500.00                    | 0.00            | 9.11   | 210.72 | -84.00                  | -49.91 | 97.71    | 88.60    | 10.73      |         |
| 3600.00             | 3600.00                | 3600.00 | 3600.00                    | 0.00            | 9.37   | 210.72 | -84.00                  | -49.91 | 97.71    | 88.34    | 10.43      |         |
| 3700.00             | 3700.00                | 3700.00 | 3700.00                    | 0.00            | 9.63   | 210.72 | -84.00                  | -49.91 | 97.71    | 88.08    | 10.14      |         |
| 3800.00             | 3800.00                | 3800.00 | 3800.00                    | 0.00            | 9.89   | 210.72 | -84.00                  | -49.91 | 97.71    | 87.82    | 9.88       |         |
| 3900.00             | 3900.00                | 3900.00 | 3900.00                    | 0.00            | 10.16  | 210.72 | -84.00                  | -49.91 | 97.71    | 87.55    | 9.62       |         |
| 4000.00             | 4000.00                | 4000.00 | 4000.00                    | 0.00            | 10.42  | 210.72 | -84.00                  | -49.91 | 97.71    | 87.29    | 9.38       |         |
| 4100.00             | 4100.00                | 4100.00 | 4100.00                    | 0.00            | 10.68  | 210.72 | -84.00                  | -49.91 | 97.71    | 87.03    | 9.15       |         |
| 4200.00             | 4200.00                | 4200.00 | 4200.00                    | 0.00            | 10.94  | 210.72 | -84.00                  | -49.91 | 97.71    | 86.77    | 8.93       |         |
| 4300.00             | 4300.00                | 4300.00 | 4300.00                    | 0.00            | 11.20  | 210.72 | -84.00                  | -49.91 | 97.71    | 86.51    | 8.72       |         |
| 4400.00             | 4400.00                | 4400.00 | 4400.00                    | 0.00            | 11.46  | 210.72 | -84.00                  | -49.91 | 97.71    | 86.25    | 8.52       |         |
| 4500.00             | 4500.00                | 4500.00 | 4500.00                    | 0.00            | 11.73  | 210.72 | -84.00                  | -49.91 | 97.71    | 85.98    | 8.33       |         |
| 4600.00             | 4600.00                | 4600.00 | 4600.00                    | 0.00            | 11.99  | 210.72 | -84.00                  | -49.91 | 97.71    | 85.72    | 8.15       |         |
| 4700.00             | 4700.00                | 4700.00 | 4700.00                    | 0.00            | 12.25  | 210.72 | -84.00                  | -49.91 | 97.71    | 85.46    | 7.98       |         |
| 4800.00             | 4800.00                | 4800.00 | 4800.00                    | 0.00            | 12.51  | 210.72 | -84.00                  | -49.91 | 97.71    | 85.20    | 7.81       |         |
| 4900.00             | 4900.00                | 4900.00 | 4900.00                    | 0.00            | 12.77  | 210.72 | -84.00                  | -49.91 | 97.71    | 84.94    | 7.65       |         |
| 5000.00             | 5000.00                | 5000.00 | 5000.00                    | 0.00            | 13.03  | 210.72 | -84.00                  | -49.91 | 97.71    | 84.67    | 7.50       |         |
| 5100.00             | 5100.00                | 5100.00 | 5100.00                    | 0.00            | 13.30  | 210.72 | -84.00                  | -49.91 | 97.71    | 84.41    | 7.35       |         |
| 5200.00             | 5200.00                | 5200.00 | 5200.00                    | 0.00            | 13.56  | 210.72 | -84.00                  | -49.91 | 97.71    | 84.15    | 7.21       |         |
| 5300.00             | 5300.00                | 5300.00 | 5300.00                    | 0.00            | 13.82  | 210.72 | -84.00                  | -49.91 | 97.71    | 83.89    | 7.07       |         |
| 5400.00             | 5400.00                | 5400.00 | 5400.00                    | 0.00            | 14.08  | 210.72 | -84.00                  | -49.91 | 97.71    | 83.63    | 6.94       |         |
| 5500.00             | 5500.00                | 5500.00 | 5500.00                    | 0.00            | 14.34  | 210.72 | -84.00                  | -49.91 | 97.71    | 83.37    | 6.81       |         |
| 5600.00             | 5600.00                | 5600.00 | 5600.00                    | 0.00            | 14.61  | 210.72 | -84.00                  | -49.91 | 97.71    | 83.10    | 6.69       |         |
| 5700.00             | 5700.00                | 5700.00 | 5700.00                    | 0.00            | 14.87  | 210.72 | -84.00                  | -49.91 | 97.71    | 82.84    | 6.57       |         |
| 5800.00             | 5800.00                | 5800.00 | 5800.00                    | 0.00            | 15.13  | 210.72 | -84.00                  | -49.91 | 97.71    | 82.58    | 6.46       |         |
| 5900.00             | 5900.00                | 5900.00 | 5900.00                    | 0.00            | 15.39  | 210.72 | -84.00                  | -49.91 | 97.71    | 82.32    | 6.35       |         |
| 6000.00             | 6000.00                | 6000.00 | 6000.00                    | 0.00            | 15.65  | 210.72 | -84.00                  | -49.91 | 97.71    | 82.06    | 6.24       |         |
| 6100.00             | 6100.00                | 6100.00 | 6100.00                    | 0.00            | 15.91  | 210.72 | -84.00                  | -49.91 | 97.71    | 81.79    | 6.14       |         |
| 6200.00             | 6200.00                | 6200.00 | 6200.00                    | 0.00            | 16.18  | 210.72 | -84.00                  | -49.91 | 97.71    | 81.53    | 6.04       |         |
| 6300.00             | 6300.00                | 6300.00 | 6300.00                    | 0.00            | 16.44  | 210.72 | -84.00                  | -49.91 | 97.71    | 81.27    | 5.94       |         |
| 6400.00             | 6400.00                | 6400.00 | 6400.00                    | 0.00            | 16.70  | 210.72 | -84.00                  | -49.91 | 97.71    | 81.01    | 5.85       |         |
| 6500.00             | 6500.00                | 6500.00 | 6500.00                    | 0.00            | 16.96  | 210.72 | -84.00                  | -49.91 | 97.71    | 80.75    | 5.76       |         |
| 6600.00             | 6600.00                | 6600.00 | 6600.00                    | 0.00            | 17.22  | 210.72 | -84.00                  | -49.91 | 97.71    | 80.49    | 5.67       |         |
| 6700.00             | 6700.00                | 6700.00 | 6700.00                    | 0.00            | 17.49  | 210.72 | -84.00                  | -49.91 | 97.71    | 80.22    | 5.59       |         |
| 6800.00             | 6800.00                | 6800.00 | 6800.00                    | 0.00            | 17.75  | 210.72 | -84.00                  | -49.91 | 97.71    | 79.96    | 5.51       |         |
| 6900.00             | 6900.00                | 6900.00 | 6900.00                    | 0.00            | 18.01  | 210.72 | -84.00                  | -49.91 | 97.71    | 79.70    | 5.43       |         |
| 7000.00             | 7000.00                | 7000.00 | 7000.00                    | 0.00            | 18.27  | 210.72 | -84.00                  | -49.91 | 97.71    | 79.44    | 5.35       |         |
| 7100.00             | 7100.00                | 7100.00 | 7100.00                    | 0.00            | 18.53  | 210.72 | -84.00                  | -49.91 | 97.71    | 79.18    | 5.27       |         |
| 7200.00             | 7200.00                | 7200.00 | 7200.00                    | 0.00            | 18.79  | 210.72 | -84.00                  | -49.91 | 97.71    | 78.92    | 5.20       |         |
| 7300.00             | 7300.00                | 7300.00 | 7300.00                    | 0.00            | 19.06  | 210.72 | -84.00                  | -49.91 | 97.71    | 78.65    | 5.13       |         |
| 7400.00             | 7400.00                | 7400.00 | 7400.00                    | 0.00            | 19.32  | 210.72 | -84.00                  | -49.91 | 97.71    | 78.39    | 5.06       |         |
| 7500.00             | 7500.00                | 7500.00 | 7500.00                    | 0.00            | 19.58  | 210.72 | -84.00                  | -49.91 | 97.71    | 78.13    | 4.99       |         |
| 7600.00             | 7600.00                | 7600.00 | 7600.00                    | 0.00            | 19.84  | 210.72 | -84.00                  | -49.91 | 97.71    | 77.87    | 4.92       |         |
| 7700.00             | 7700.00                | 7700.00 | 7700.00                    | 0.00            | 20.10  | 210.72 | -84.00                  | -49.91 | 97.71    | 77.61    | 4.86       |         |
| 7800.00             | 7800.00                | 7800.00 | 7800.00                    | 0.00            | 20.37  | 210.72 | -84.00                  | -49.91 | 97.71    | 77.34    | 4.80       |         |
| 7900.00             | 7900.00                | 7900.00 | 7900.00                    | 0.00            | 20.63  | 210.72 | -84.00                  | -49.91 | 97.71    | 77.08    | 4.74       |         |
| 8000.00             | 8000.00                | 8000.00 | 8000.00                    | 0.00            | 20.89  | 210.72 | -84.00                  | -49.91 | 97.71    | 76.82    | 4.68       |         |
| 8100.00             | 8100.00                | 8100.00 | 8100.00                    | 0.00            | 21.15  | 210.72 | -84.00                  | -49.91 | 97.71    | 76.56    | 4.62       |         |
| 8200.00             | 8200.00                | 8200.00 | 8200.00                    | 0.00            | 21.41  | 210.72 | -84.00                  | -49.91 | 97.71    | 76.30    | 4.56       |         |
| 8300.00             | 8300.00                | 8300.00 | 8300.00                    | 0.00            | 21.67  | 210.72 | -84.00                  | -49.91 | 97.71    | 76.04    | 4.51       |         |
| 8400.00             | 8400.00                | 8400.00 | 8400.00                    | 0.00            | 21.94  | 210.72 | -84.00                  | -49.91 | 97.71    | 75.77    | 4.45       |         |
| 8500.00             | 8500.00                | 8500.00 | 8500.00                    | 0.00            | 22.20  | 210.72 | -84.00                  | -49.91 | 97.71    | 75.51    | 4.40       |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company: Continental Resources   |          | Date: 04/23/2012           |          | Time: 17:17:28          |        | Page: 3    |                 |         |          |                   |            |         |    |
|----------------------------------|----------|----------------------------|----------|-------------------------|--------|------------|-----------------|---------|----------|-------------------|------------|---------|----|
| Field: McKenzie County, ND       |          |                            |          |                         |        |            |                 |         |          |                   |            |         |    |
| Reference Site: Tallahassee 3    |          | Co-ordinate(NE) Reference: |          | Well: 3-16H, True North |        |            |                 |         |          |                   |            |         |    |
| Reference Well: 3-16H            |          | Vertical (TVD) Reference:  |          | GL 1920+KB 41 1941.0    |        |            |                 |         |          |                   |            |         |    |
| Reference Wellpath: OH           |          |                            |          |                         |        | Db: Adapti |                 |         |          |                   |            |         |    |
| Site: Columbus Federal 1         |          |                            |          |                         |        |            |                 |         |          |                   |            |         |    |
| Well: 1-16H                      |          |                            |          |                         |        |            |                 |         |          |                   |            |         |    |
| Wellpath: OH V3 Plan: Plan #2 V1 |          |                            |          |                         |        |            |                 |         |          | Inter-Site Error: |            | 0.00    | ft |
| Reference                        |          | Offset                     |          | Semi-Major Axis         |        |            | Offset Location |         | Ctr-Ctr  | Edge              | Separation | Warning |    |
| MD                               | TVD      | MD                         | TVD      | Ref                     | Offset | TFO-HS     | North           | East    | Distance | Distance          | Factor     |         |    |
| ft                               | ft       | ft                         | ft       | ft                      | ft     | deg        | ft              | ft      | ft       | ft                |            |         |    |
| 8600.00                          | 8600.00  | 8600.00                    | 8600.00  | 0.00                    | 22.46  | 210.72     | -84.00          | -49.91  | 97.71    | 75.25             | 4.35       |         |    |
| 8700.00                          | 8700.00  | 8700.00                    | 8700.00  | 0.00                    | 22.72  | 210.72     | -84.00          | -49.91  | 97.71    | 74.99             | 4.30       |         |    |
| 8800.00                          | 8800.00  | 8800.00                    | 8800.00  | 0.00                    | 22.98  | 210.72     | -84.00          | -49.91  | 97.71    | 74.73             | 4.25       |         |    |
| 8900.00                          | 8900.00  | 8900.00                    | 8900.00  | 0.00                    | 23.25  | 210.72     | -84.00          | -49.91  | 97.71    | 74.46             | 4.20       |         |    |
| 9000.00                          | 9000.00  | 9000.00                    | 9000.00  | 0.00                    | 23.51  | 210.72     | -84.00          | -49.91  | 97.71    | 74.20             | 4.16       |         |    |
| 9100.00                          | 9100.00  | 9100.00                    | 9100.00  | 0.00                    | 23.77  | 210.72     | -84.00          | -49.91  | 97.71    | 73.94             | 4.11       |         |    |
| 9200.00                          | 9200.00  | 9200.00                    | 9200.00  | 0.00                    | 24.03  | 210.72     | -84.00          | -49.91  | 97.71    | 73.68             | 4.07       |         |    |
| 9300.00                          | 9300.00  | 9300.00                    | 9300.00  | 0.00                    | 24.29  | 210.72     | -84.00          | -49.91  | 97.71    | 73.42             | 4.02       |         |    |
| 9400.00                          | 9400.00  | 9400.00                    | 9400.00  | 0.00                    | 24.55  | 210.72     | -84.00          | -49.91  | 97.71    | 73.16             | 3.98       |         |    |
| 9500.00                          | 9500.00  | 9500.00                    | 9500.00  | 0.00                    | 24.82  | 210.72     | -84.00          | -49.91  | 97.71    | 72.89             | 3.94       |         |    |
| 9600.00                          | 9600.00  | 9600.00                    | 9600.00  | 0.00                    | 25.08  | 210.72     | -84.00          | -49.91  | 97.71    | 72.63             | 3.90       |         |    |
| 9700.00                          | 9700.00  | 9700.00                    | 9700.00  | 0.00                    | 25.34  | 210.72     | -84.00          | -49.91  | 97.71    | 72.37             | 3.86       |         |    |
| 9800.00                          | 9800.00  | 9800.00                    | 9800.00  | 0.00                    | 25.60  | 210.72     | -84.00          | -49.91  | 97.71    | 72.11             | 3.82       |         |    |
| 9900.00                          | 9900.00  | 9900.00                    | 9900.00  | 0.00                    | 25.86  | 210.72     | -84.00          | -49.91  | 97.71    | 71.85             | 3.78       |         |    |
| 10000.00                         | 10000.00 | 10000.00                   | 10000.00 | 0.00                    | 26.12  | 210.72     | -84.00          | -49.91  | 97.71    | 71.58             | 3.74       |         |    |
| 10100.00                         | 10100.00 | 10100.00                   | 10100.00 | 0.00                    | 26.39  | 210.72     | -84.00          | -49.91  | 97.71    | 71.32             | 3.70       |         |    |
| 10166.27                         | 10166.27 | 10174.42                   | 10174.10 | 0.00                    | 26.58  | 213.71     | -78.71          | -52.52  | 94.95    | 68.44             | 3.58       |         |    |
| 10175.00                         | 10175.00 | 10184.05                   | 10183.58 | 0.00                    | 26.61  | 345.94     | -77.21          | -53.26  | 94.11    | 67.59             | 3.55       |         |    |
| 10200.00                         | 10199.97 | 10211.05                   | 10209.94 | 0.00                    | 26.68  | 348.88     | -72.02          | -55.81  | 90.40    | 63.86             | 3.41       |         |    |
| 10225.00                         | 10224.83 | 10236.96                   | 10234.87 | 0.00                    | 26.75  | 352.80     | -65.68          | -58.94  | 84.96    | 58.38             | 3.20       |         |    |
| 10250.00                         | 10249.50 | 10261.54                   | 10258.10 | 0.00                    | 26.81  | 358.04     | -58.46          | -62.49  | 78.12    | 51.45             | 2.93       |         |    |
| 10275.00                         | 10273.90 | 10284.60                   | 10279.44 | 0.00                    | 26.87  | 5.04       | -50.65          | -66.34  | 70.38    | 43.58             | 2.63       |         |    |
| 10300.00                         | 10297.96 | 10306.02                   | 10298.84 | 0.00                    | 26.93  | 14.22      | -42.50          | -70.36  | 62.49    | 35.57             | 2.32       |         |    |
| 10325.00                         | 10321.59 | 10325.74                   | 10316.28 | 0.00                    | 26.98  | 25.68      | -34.26          | -74.42  | 55.69    | 28.81             | 2.07       |         |    |
| 10350.00                         | 10344.72 | 10343.75                   | 10331.85 | 0.00                    | 27.03  | 38.67      | -26.12          | -78.43  | 51.75    | 25.43             | 1.97       |         |    |
| 10375.00                         | 10367.28 | 10360.10                   | 10345.64 | 0.00                    | 27.07  | 51.49      | -18.25          | -82.30  | 52.60    | 27.52             | 2.10       |         |    |
| 10400.00                         | 10389.20 | 10374.82                   | 10357.78 | 0.00                    | 27.11  | 62.42      | -10.78          | -85.99  | 59.06    | 35.36             | 2.49       |         |    |
| 10425.00                         | 10410.39 | 10388.01                   | 10368.40 | 0.00                    | 27.15  | 70.64      | -3.78           | -89.43  | 70.35    | 47.59             | 3.09       |         |    |
| 10450.00                         | 10430.80 | 10399.74                   | 10377.66 | 0.00                    | 27.18  | 76.23      | 2.69            | -92.62  | 85.11    | 62.78             | 3.81       |         |    |
| 10475.00                         | 10450.36 | 10410.10                   | 10385.67 | 0.00                    | 27.20  | 79.57      | 8.58            | -95.52  | 102.25   | 80.03             | 4.60       |         |    |
| 10500.00                         | 10469.01 | 10419.19                   | 10392.57 | 0.00                    | 27.23  | 81.08      | 13.89           | -98.14  | 121.02   | 98.76             | 5.44       |         |    |
| 10525.00                         | 10486.68 | 10427.09                   | 10398.47 | 0.00                    | 27.25  | 81.08      | 18.60           | -100.46 | 140.96   | 118.57            | 6.30       |         |    |
| 10550.00                         | 10503.32 | 10433.88                   | 10403.47 | 0.00                    | 27.27  | 79.82      | 22.74           | -102.50 | 161.77   | 139.22            | 7.17       |         |    |
| 10575.00                         | 10518.88 | 10439.65                   | 10407.65 | 0.00                    | 27.29  | 77.46      | 26.29           | -104.25 | 183.21   | 160.48            | 8.06       |         |    |
| 10600.00                         | 10533.30 | 10444.45                   | 10411.09 | 0.00                    | 27.30  | 74.17      | 29.29           | -105.73 | 205.14   | 182.21            | 8.95       |         |    |
| 10625.00                         | 10546.55 | 10448.35                   | 10413.86 | 0.00                    | 27.31  | 70.08      | 31.76           | -106.94 | 227.43   | 204.29            | 9.83       |         |    |
| 10650.00                         | 10558.57 | 10451.41                   | 10416.02 | 0.00                    | 27.32  | 65.37      | 33.71           | -107.90 | 249.96   | 226.61            | 10.71      |         |    |
| 10675.00                         | 10569.32 | 10453.68                   | 10417.61 | 0.00                    | 27.32  | 60.24      | 35.16           | -108.62 | 272.66   | 249.10            | 11.57      |         |    |
| 10700.00                         | 10578.78 | 10455.19                   | 10418.66 | 0.00                    | 27.33  | 54.90      | 36.13           | -109.10 | 295.43   | 271.66            | 12.43      |         |    |
| 10725.00                         | 10586.92 | 10455.99                   | 10419.22 | 0.00                    | 27.33  | 49.56      | 36.65           | -109.35 | 318.23   | 294.25            | 13.27      |         |    |
| 10750.00                         | 10593.70 | 10456.12                   | 10419.31 | 0.00                    | 27.33  | 44.41      | 36.73           | -109.39 | 340.97   | 316.78            | 14.10      |         |    |
| 10775.00                         | 10599.11 | 10455.58                   | 10418.94 | 0.00                    | 27.33  | 39.56      | 36.38           | -109.22 | 363.62   | 339.23            | 14.91      |         |    |
| 10800.00                         | 10603.12 | 10454.42                   | 10418.13 | 0.00                    | 27.33  | 35.11      | 35.63           | -108.85 | 386.11   | 361.52            | 15.70      |         |    |
| 10825.00                         | 10605.73 | 10452.65                   | 10416.89 | 0.00                    | 27.32  | 31.07      | 34.50           | -108.29 | 408.39   | 383.61            | 16.48      |         |    |
| 10850.00                         | 10606.92 | 10450.28                   | 10415.22 | 0.00                    | 27.31  | 27.46      | 32.98           | -107.54 | 430.42   | 405.46            | 17.24      |         |    |
| 10857.19                         | 10607.00 | 10449.49                   | 10414.67 | 0.00                    | 27.31  | 26.50      | 32.48           | -107.30 | 436.71   | 411.69            | 17.45      |         |    |
| 10900.00                         | 10607.14 | 10444.69                   | 10411.26 | 0.00                    | 27.30  | 25.32      | 29.45           | -105.80 | 474.47   | 449.17            | 18.76      |         |    |
| 10957.19                         | 10607.32 | 10438.45                   | 10406.78 | 0.00                    | 27.28  | 23.81      | 25.55           | -103.88 | 526.02   | 500.44            | 20.56      |         |    |
| 11000.00                         | 10607.45 | 10433.71                   | 10403.34 | 0.00                    | 27.27  | 19.95      | 22.63           | -102.44 | 565.31   | 539.56            | 21.95      |         |    |
| 11100.00                         | 10607.78 | 10421.68                   | 10394.44 | 0.00                    | 27.24  | 9.62       | 15.37           | -98.87  | 658.94   | 632.91            | 25.32      |         |    |
| 11200.00                         | 10608.10 | 10408.60                   | 10384.52 | 0.00                    | 27.20  | 358.11     | 7.72            | -95.10  | 754.08   | 727.90            | 28.80      |         |    |

LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |       |        |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|-------|--------|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page: | 4      |
| Field:              | McKenzie County, ND   |                            |                         |       |          |       |        |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |       |        |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |       |        |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db:   | Adapti |

Site: Columbus Federal 1  
Well: 1-16H  
Wellpath: OH V3 Plan: Plan #2 V1

Inter-Site Error: 0.00 ft

| Reference |           | Offset   |           | Semi-Major Axis |              |               | Offset Location |            | Ctr-Ctr<br>Distance<br>ft | Edge<br>Distance<br>ft | Separation<br>Factor | Warning |
|-----------|-----------|----------|-----------|-----------------|--------------|---------------|-----------------|------------|---------------------------|------------------------|----------------------|---------|
| MD<br>ft  | TVD<br>ft | MD<br>ft | TVD<br>ft | Ref<br>ft       | Offset<br>ft | TFO-HS<br>deg | North<br>ft     | East<br>ft |                           |                        |                      |         |
| 11300.00  | 10608.44  | 10394.80 | 10373.79  | 0.00            | 27.16        | 346.56        | -0.06           | -91.27     | 849.92                    | 823.63                 | 32.33                |         |
| 11400.00  | 10608.78  | 10380.60 | 10362.46  | 0.00            | 27.13        | 336.03        | -7.75           | -87.48     | 945.91                    | 919.57                 | 35.90                |         |
| 11500.00  | 10609.12  | 10366.29 | 10350.77  | 0.00            | 27.09        | 327.07        | -15.16          | -83.83     | 1041.74                   | 1015.36                | 39.49                |         |
| 11600.00  | 10609.46  | 10352.11 | 10338.94  | 0.00            | 27.05        | 319.72        | -22.16          | -80.38     | 1137.16                   | 1110.76                | 43.07                |         |
| 11700.00  | 10609.81  | 10338.28 | 10327.16  | 0.00            | 27.01        | 313.78        | -28.66          | -77.18     | 1232.00                   | 1205.59                | 46.65                |         |
| 11800.00  | 10610.16  | 10325.00 | 10315.64  | 0.00            | 26.98        | 308.96        | -34.58          | -74.26     | 1326.14                   | 1299.72                | 50.21                |         |
| 11900.00  | 10610.51  | 10312.19 | 10304.34  | 0.00            | 26.94        | 305.02        | -39.99          | -71.59     | 1419.47                   | 1393.06                | 53.74                |         |
| 12000.00  | 10610.86  | 10300.00 | 10293.43  | 0.00            | 26.91        | 301.75        | -44.87          | -69.19     | 1511.90                   | 1485.49                | 57.25                |         |
| 12100.00  | 10611.20  | 10288.78 | 10283.26  | 0.00            | 26.88        | 298.99        | -49.12          | -67.09     | 1603.37                   | 1576.96                | 60.73                |         |
| 12200.00  | 10611.55  | 10275.00 | 10270.61  | 0.00            | 26.85        | 296.81        | -54.02          | -64.68     | 1693.80                   | 1667.39                | 64.14                |         |
| 12300.00  | 10611.89  | 10268.27 | 10264.37  | 0.00            | 26.83        | 294.61        | -56.28          | -63.57     | 1783.12                   | 1756.72                | 67.56                |         |
| 12400.00  | 10612.23  | 10259.08 | 10255.79  | 0.00            | 26.80        | 292.85        | -59.24          | -62.11     | 1871.28                   | 1844.89                | 70.91                |         |
| 12500.00  | 10612.56  | 10250.00 | 10247.25  | 0.00            | 26.78        | 291.32        | -61.99          | -60.75     | 1958.22                   | 1931.83                | 74.21                |         |
| 12577.81  | 10612.81  | 10250.00 | 10247.25  | 0.00            | 26.78        | 289.96        | -61.99          | -60.75     | 2025.00                   | 1998.64                | 76.80                |         |
| 12600.00  | 10612.88  | 10242.71 | 10240.34  | 0.00            | 26.76        | 290.29        | -64.10          | -59.72     | 2043.94                   | 2017.55                | 77.48                |         |
| 12700.00  | 10613.20  | 10235.57 | 10233.54  | 0.00            | 26.74        | 290.62        | -66.05          | -58.75     | 2130.08                   | 2103.70                | 80.75                |         |
| 12800.00  | 10613.52  | 10225.00 | 10223.42  | 0.00            | 26.71        | 291.10        | -68.77          | -57.42     | 2217.32                   | 2190.93                | 84.02                |         |
| 12900.00  | 10613.85  | 10225.00 | 10223.42  | 0.00            | 26.71        | 291.10        | -68.77          | -57.42     | 2305.51                   | 2279.13                | 87.41                |         |
| 13000.00  | 10614.17  | 10225.00 | 10223.42  | 0.00            | 26.71        | 291.10        | -68.77          | -57.42     | 2394.62                   | 2368.26                | 90.83                |         |
| 13100.00  | 10614.49  | 10212.96 | 10211.80  | 0.00            | 26.68        | 291.66        | -71.59          | -56.02     | 2484.43                   | 2458.05                | 94.17                |         |
| 13200.00  | 10614.81  | 10208.45 | 10207.43  | 0.00            | 26.67        | 291.86        | -72.58          | -55.54     | 2574.99                   | 2548.60                | 97.60                |         |
| 13300.00  | 10615.13  | 10200.00 | 10199.20  | 0.00            | 26.65        | 292.25        | -74.32          | -54.68     | 2666.21                   | 2639.81                | 100.99               |         |
| 13400.00  | 10615.46  | 10200.00 | 10199.20  | 0.00            | 26.65        | 292.25        | -74.32          | -54.68     | 2757.97                   | 2731.58                | 104.50               |         |
| 13500.00  | 10615.78  | 10200.00 | 10199.20  | 0.00            | 26.65        | 292.25        | -74.32          | -54.68     | 2850.29                   | 2823.90                | 108.02               |         |
| 13600.00  | 10616.10  | 10200.00 | 10199.20  | 0.00            | 26.65        | 292.25        | -74.32          | -54.68     | 2943.11                   | 2916.73                | 111.56               |         |
| 13700.00  | 10616.42  | 10190.47 | 10189.89  | 0.00            | 26.62        | 292.69        | -76.11          | -53.80     | 3036.30                   | 3009.91                | 115.03               |         |
| 13800.00  | 10616.75  | 10187.59 | 10187.05  | 0.00            | 26.62        | 292.83        | -76.61          | -53.55     | 3129.93                   | 3103.53                | 118.57               |         |
| 13900.00  | 10617.07  | 10184.88 | 10184.40  | 0.00            | 26.61        | 292.95        | -77.07          | -53.32     | 3223.93                   | 3197.53                | 122.11               |         |
| 14000.00  | 10617.39  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3318.32                   | 3291.90                | 125.59               |         |
| 14100.00  | 10617.71  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3412.94                   | 3386.52                | 129.18               |         |
| 14200.00  | 10618.03  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3507.86                   | 3481.44                | 132.79               |         |
| 14300.00  | 10618.36  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3603.05                   | 3576.63                | 136.40               |         |
| 14400.00  | 10618.68  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3698.49                   | 3672.08                | 140.03               |         |
| 14500.00  | 10619.00  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3794.17                   | 3767.76                | 143.66               |         |
| 14600.00  | 10619.32  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3890.07                   | 3863.66                | 147.30               |         |
| 14700.00  | 10619.64  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 3986.17                   | 3959.77                | 150.95               |         |
| 14800.00  | 10619.97  | 10175.00 | 10174.67  | 0.00            | 26.58        | 293.41        | -78.63          | -52.56     | 4082.46                   | 4056.05                | 154.61               |         |
| 14900.00  | 10620.29  | 10165.06 | 10164.85  | 0.00            | 26.56        | 293.86        | -79.99          | -51.89     | 4178.82                   | 4152.40                | 158.18               |         |
| 15000.00  | 10620.61  | 10163.60 | 10163.40  | 0.00            | 26.55        | 293.93        | -80.18          | -51.79     | 4275.42                   | 4249.00                | 161.83               |         |
| 15100.00  | 10620.93  | 10162.20 | 10162.01  | 0.00            | 26.55        | 294.00        | -80.35          | -51.71     | 4372.16                   | 4345.74                | 165.49               |         |
| 15200.00  | 10621.26  | 10160.87 | 10160.69  | 0.00            | 26.55        | 294.06        | -80.51          | -51.63     | 4469.04                   | 4442.62                | 169.15               |         |
| 15300.00  | 10621.58  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 4566.14                   | 4539.71                | 172.72               |         |
| 15400.00  | 10621.90  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 4663.26                   | 4636.82                | 176.39               |         |
| 15500.00  | 10622.22  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 4760.49                   | 4734.06                | 180.07               |         |
| 15600.00  | 10622.54  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 4857.84                   | 4831.40                | 183.76               |         |
| 15700.00  | 10622.87  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 4955.29                   | 4928.85                | 187.45               |         |
| 15800.00  | 10623.19  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 5052.84                   | 5026.41                | 191.14               |         |
| 15900.00  | 10623.51  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 5150.49                   | 5124.05                | 194.84               |         |
| 16000.00  | 10623.83  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 5248.22                   | 5221.79                | 198.54               |         |
| 16100.00  | 10624.15  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 5346.04                   | 5319.61                | 202.25               |         |
| 16200.00  | 10624.48  | 10150.00 | 10149.91  | 0.00            | 26.52        | 294.56        | -81.68          | -51.05     | 5443.94                   | 5417.50                | 205.95               |         |



LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |            |   |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|------------|---|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page:      | 5 |
| Field:              | McKenzie County, ND   |                            |                         |       |          |            |   |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |            |   |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |            |   |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db: Adapti |   |

|           |                        |  |  |  |                   |      |    |
|-----------|------------------------|--|--|--|-------------------|------|----|
| Site:     | Columbus Federal 1     |  |  |  |                   |      |    |
| Well:     | 1-16H                  |  |  |  |                   |      |    |
| Wellpath: | OH V3 Plan: Plan #2 V1 |  |  |  |                   |      |    |
|           |                        |  |  |  | Inter-Site Error: | 0.00 | ft |

| Reference |          | Offset   |          | Semi-Major Axis |        |        | Offset Location |        | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|----------|----------|----------|-----------------|--------|--------|-----------------|--------|----------|----------|------------|---------|
| MD        | TVD      | MD       | TVD      | Ref             | Offset | TFO-HS | North           | East   | Distance | Distance | Factor     |         |
| ft        | ft       | ft       | ft       | ft              | ft     | deg    | ft              | ft     | ft       | ft       |            |         |
| 16300.00  | 10624.80 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 5541.91  | 5515.48  | 209.66     |         |
| 16400.00  | 10625.12 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 5639.95  | 5613.52  | 213.37     |         |
| 16500.00  | 10625.44 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 5738.06  | 5711.63  | 217.09     |         |
| 16600.00  | 10625.77 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 5836.24  | 5809.81  | 220.81     |         |
| 16700.00  | 10626.09 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 5934.47  | 5908.04  | 224.53     |         |
| 16800.00  | 10626.41 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6032.77  | 6006.34  | 228.25     |         |
| 16900.00  | 10626.73 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6131.12  | 6104.69  | 231.97     |         |
| 17000.00  | 10627.05 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6229.52  | 6203.09  | 235.70     |         |
| 17100.00  | 10627.38 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6327.97  | 6301.54  | 239.43     |         |
| 17200.00  | 10627.70 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6426.47  | 6400.04  | 243.16     |         |
| 17300.00  | 10628.02 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6525.02  | 6498.59  | 246.89     |         |
| 17400.00  | 10628.34 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6623.61  | 6597.18  | 250.62     |         |
| 17500.00  | 10628.66 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6722.24  | 6695.81  | 254.35     |         |
| 17600.00  | 10628.99 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6820.91  | 6794.48  | 258.09     |         |
| 17700.00  | 10629.31 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 6919.61  | 6893.19  | 261.83     |         |
| 17800.00  | 10629.63 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 7018.36  | 6991.93  | 265.57     |         |
| 17900.00  | 10629.95 | 10150.00 | 10149.91 | 0.00            | 26.52  | 294.56 | -81.68          | -51.05 | 7117.14  | 7090.71  | 269.31     |         |
| 18000.00  | 10630.28 | 10138.59 | 10138.55 | 0.00            | 26.49  | 295.08 | -82.66          | -50.57 | 7215.82  | 7189.39  | 273.02     |         |
| 18100.00  | 10630.60 | 10138.11 | 10138.07 | 0.00            | 26.49  | 295.11 | -82.69          | -50.56 | 7314.66  | 7288.23  | 276.76     |         |
| 18200.00  | 10630.92 | 10137.64 | 10137.60 | 0.00            | 26.49  | 295.13 | -82.73          | -50.54 | 7413.52  | 7387.09  | 280.50     |         |
| 18300.00  | 10631.24 | 10137.19 | 10137.15 | 0.00            | 26.48  | 295.15 | -82.76          | -50.52 | 7512.42  | 7485.99  | 284.25     |         |
| 18400.00  | 10631.56 | 10136.75 | 10136.71 | 0.00            | 26.48  | 295.17 | -82.79          | -50.51 | 7611.34  | 7584.91  | 287.99     |         |
| 18500.00  | 10631.89 | 10136.32 | 10136.28 | 0.00            | 26.48  | 295.19 | -82.82          | -50.49 | 7710.29  | 7683.86  | 291.73     |         |
| 18535.29  | 10632.00 | 10136.17 | 10136.13 | 0.00            | 26.48  | 295.20 | -82.83          | -50.49 | 7745.22  | 7718.79  | 293.06     |         |

|           |                        |  |  |  |                   |      |    |
|-----------|------------------------|--|--|--|-------------------|------|----|
| Site:     | Columbus Federal 2     |  |  |  |                   |      |    |
| Well:     | 2-16H                  |  |  |  |                   |      |    |
| Wellpath: | OH V3 Plan: Plan #1 V1 |  |  |  |                   |      |    |
|           |                        |  |  |  | Inter-Site Error: | 0.00 | ft |

| Reference |         | Offset  |         | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|---------|---------|---------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD     | MD      | TVD     | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft      | ft      | ft      | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 0.00      | 0.00    | 0.00    | 0.00    | 0.00            | 0.00   | 30.67  | 77.41           | 45.90 | 90.00    |          |            | No Data |
| 100.00    | 100.00  | 100.00  | 100.00  | 0.00            | 0.21   | 30.67  | 77.41           | 45.90 | 90.00    | 89.79    | 434.74     |         |
| 200.00    | 200.00  | 200.00  | 200.00  | 0.00            | 0.47   | 30.67  | 77.41           | 45.90 | 90.00    | 89.53    | 191.97     |         |
| 300.00    | 300.00  | 300.00  | 300.00  | 0.00            | 0.73   | 30.67  | 77.41           | 45.90 | 90.00    | 89.27    | 123.18     |         |
| 400.00    | 400.00  | 400.00  | 400.00  | 0.00            | 0.99   | 30.67  | 77.41           | 45.90 | 90.00    | 89.01    | 90.69      |         |
| 500.00    | 500.00  | 500.00  | 500.00  | 0.00            | 1.25   | 30.67  | 77.41           | 45.90 | 90.00    | 88.75    | 71.76      |         |
| 600.00    | 600.00  | 600.00  | 600.00  | 0.00            | 1.52   | 30.67  | 77.41           | 45.90 | 90.00    | 88.49    | 59.37      |         |
| 700.00    | 700.00  | 700.00  | 700.00  | 0.00            | 1.78   | 30.67  | 77.41           | 45.90 | 90.00    | 88.22    | 50.62      |         |
| 800.00    | 800.00  | 800.00  | 800.00  | 0.00            | 2.04   | 30.67  | 77.41           | 45.90 | 90.00    | 87.96    | 44.13      |         |
| 900.00    | 900.00  | 900.00  | 900.00  | 0.00            | 2.30   | 30.67  | 77.41           | 45.90 | 90.00    | 87.70    | 39.11      |         |
| 1000.00   | 1000.00 | 1000.00 | 1000.00 | 0.00            | 2.56   | 30.67  | 77.41           | 45.90 | 90.00    | 87.44    | 35.11      |         |
| 1100.00   | 1100.00 | 1100.00 | 1100.00 | 0.00            | 2.82   | 30.67  | 77.41           | 45.90 | 90.00    | 87.18    | 31.86      |         |
| 1200.00   | 1200.00 | 1200.00 | 1200.00 | 0.00            | 3.09   | 30.67  | 77.41           | 45.90 | 90.00    | 86.91    | 29.16      |         |
| 1300.00   | 1300.00 | 1300.00 | 1300.00 | 0.00            | 3.35   | 30.67  | 77.41           | 45.90 | 90.00    | 86.65    | 26.88      |         |
| 1400.00   | 1400.00 | 1400.00 | 1400.00 | 0.00            | 3.61   | 30.67  | 77.41           | 45.90 | 90.00    | 86.39    | 24.93      |         |
| 1500.00   | 1500.00 | 1500.00 | 1500.00 | 0.00            | 3.87   | 30.67  | 77.41           | 45.90 | 90.00    | 86.13    | 23.24      |         |
| 1600.00   | 1600.00 | 1600.00 | 1600.00 | 0.00            | 4.13   | 30.67  | 77.41           | 45.90 | 90.00    | 85.87    | 21.77      |         |
| 1700.00   | 1700.00 | 1700.00 | 1700.00 | 0.00            | 4.40   | 30.67  | 77.41           | 45.90 | 90.00    | 85.61    | 20.47      |         |
| 1800.00   | 1800.00 | 1800.00 | 1800.00 | 0.00            | 4.66   | 30.67  | 77.41           | 45.90 | 90.00    | 85.34    | 19.32      |         |
| 1900.00   | 1900.00 | 1900.00 | 1900.00 | 0.00            | 4.92   | 30.67  | 77.41           | 45.90 | 90.00    | 85.08    | 18.30      |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |       |        |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|-------|--------|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page: | 6      |
| Field:              | McKenzie County, ND   |                            |                         |       |          |       |        |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |       |        |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |       |        |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db:   | Adapti |

Site: Columbus Federal 2  
Well: 2-16H  
Wellpath: OH V3 Plan: Plan #1 V1

Inter-Site Error: 0.00 ft

| Reference |         | Offset  |         | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|---------|---------|---------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD     | MD      | TVD     | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft      | ft      | ft      | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 2000.00   | 2000.00 | 2000.00 | 2000.00 | 0.00            | 5.18   | 30.67  | 77.41           | 45.90 | 90.00    | 84.82    | 17.37      |         |
| 2100.00   | 2100.00 | 2100.00 | 2100.00 | 0.00            | 5.44   | 30.67  | 77.41           | 45.90 | 90.00    | 84.56    | 16.54      |         |
| 2200.00   | 2200.00 | 2200.00 | 2200.00 | 0.00            | 5.70   | 30.67  | 77.41           | 45.90 | 90.00    | 84.30    | 15.78      |         |
| 2300.00   | 2300.00 | 2300.00 | 2300.00 | 0.00            | 5.97   | 30.67  | 77.41           | 45.90 | 90.00    | 84.03    | 15.08      |         |
| 2400.00   | 2400.00 | 2400.00 | 2400.00 | 0.00            | 6.23   | 30.67  | 77.41           | 45.90 | 90.00    | 83.77    | 14.45      |         |
| 2500.00   | 2500.00 | 2500.00 | 2500.00 | 0.00            | 6.49   | 30.67  | 77.41           | 45.90 | 90.00    | 83.51    | 13.87      |         |
| 2600.00   | 2600.00 | 2600.00 | 2600.00 | 0.00            | 6.75   | 30.67  | 77.41           | 45.90 | 90.00    | 83.25    | 13.33      |         |
| 2700.00   | 2700.00 | 2700.00 | 2700.00 | 0.00            | 7.01   | 30.67  | 77.41           | 45.90 | 90.00    | 82.99    | 12.83      |         |
| 2800.00   | 2800.00 | 2800.00 | 2800.00 | 0.00            | 7.28   | 30.67  | 77.41           | 45.90 | 90.00    | 82.73    | 12.37      |         |
| 2900.00   | 2900.00 | 2900.00 | 2900.00 | 0.00            | 7.54   | 30.67  | 77.41           | 45.90 | 90.00    | 82.46    | 11.94      |         |
| 3000.00   | 3000.00 | 3000.00 | 3000.00 | 0.00            | 7.80   | 30.67  | 77.41           | 45.90 | 90.00    | 82.20    | 11.54      |         |
| 3100.00   | 3100.00 | 3100.00 | 3100.00 | 0.00            | 8.06   | 30.67  | 77.41           | 45.90 | 90.00    | 81.94    | 11.16      |         |
| 3200.00   | 3200.00 | 3200.00 | 3200.00 | 0.00            | 8.32   | 30.67  | 77.41           | 45.90 | 90.00    | 81.68    | 10.81      |         |
| 3300.00   | 3300.00 | 3300.00 | 3300.00 | 0.00            | 8.58   | 30.67  | 77.41           | 45.90 | 90.00    | 81.42    | 10.48      |         |
| 3400.00   | 3400.00 | 3400.00 | 3400.00 | 0.00            | 8.85   | 30.67  | 77.41           | 45.90 | 90.00    | 81.15    | 10.17      |         |
| 3500.00   | 3500.00 | 3500.00 | 3500.00 | 0.00            | 9.11   | 30.67  | 77.41           | 45.90 | 90.00    | 80.89    | 9.88       |         |
| 3600.00   | 3600.00 | 3600.00 | 3600.00 | 0.00            | 9.37   | 30.67  | 77.41           | 45.90 | 90.00    | 80.63    | 9.61       |         |
| 3700.00   | 3700.00 | 3700.00 | 3700.00 | 0.00            | 9.63   | 30.67  | 77.41           | 45.90 | 90.00    | 80.37    | 9.34       |         |
| 3800.00   | 3800.00 | 3800.00 | 3800.00 | 0.00            | 9.89   | 30.67  | 77.41           | 45.90 | 90.00    | 80.11    | 9.10       |         |
| 3900.00   | 3900.00 | 3900.00 | 3900.00 | 0.00            | 10.16  | 30.67  | 77.41           | 45.90 | 90.00    | 79.85    | 8.86       |         |
| 4000.00   | 4000.00 | 4000.00 | 4000.00 | 0.00            | 10.42  | 30.67  | 77.41           | 45.90 | 90.00    | 79.58    | 8.64       |         |
| 4100.00   | 4100.00 | 4100.00 | 4100.00 | 0.00            | 10.68  | 30.67  | 77.41           | 45.90 | 90.00    | 79.32    | 8.43       |         |
| 4200.00   | 4200.00 | 4200.00 | 4200.00 | 0.00            | 10.94  | 30.67  | 77.41           | 45.90 | 90.00    | 79.06    | 8.23       |         |
| 4300.00   | 4300.00 | 4300.00 | 4300.00 | 0.00            | 11.20  | 30.67  | 77.41           | 45.90 | 90.00    | 78.80    | 8.03       |         |
| 4400.00   | 4400.00 | 4400.00 | 4400.00 | 0.00            | 11.46  | 30.67  | 77.41           | 45.90 | 90.00    | 78.54    | 7.85       |         |
| 4500.00   | 4500.00 | 4500.00 | 4500.00 | 0.00            | 11.73  | 30.67  | 77.41           | 45.90 | 90.00    | 78.27    | 7.68       |         |
| 4600.00   | 4600.00 | 4600.00 | 4600.00 | 0.00            | 11.99  | 30.67  | 77.41           | 45.90 | 90.00    | 78.01    | 7.51       |         |
| 4700.00   | 4700.00 | 4700.00 | 4700.00 | 0.00            | 12.25  | 30.67  | 77.41           | 45.90 | 90.00    | 77.75    | 7.35       |         |
| 4800.00   | 4800.00 | 4800.00 | 4800.00 | 0.00            | 12.51  | 30.67  | 77.41           | 45.90 | 90.00    | 77.49    | 7.19       |         |
| 4900.00   | 4900.00 | 4900.00 | 4900.00 | 0.00            | 12.77  | 30.67  | 77.41           | 45.90 | 90.00    | 77.23    | 7.05       |         |
| 5000.00   | 5000.00 | 5000.00 | 5000.00 | 0.00            | 13.03  | 30.67  | 77.41           | 45.90 | 90.00    | 76.97    | 6.90       |         |
| 5100.00   | 5100.00 | 5100.00 | 5100.00 | 0.00            | 13.30  | 30.67  | 77.41           | 45.90 | 90.00    | 76.70    | 6.77       |         |
| 5200.00   | 5200.00 | 5200.00 | 5200.00 | 0.00            | 13.56  | 30.67  | 77.41           | 45.90 | 90.00    | 76.44    | 6.64       |         |
| 5300.00   | 5300.00 | 5300.00 | 5300.00 | 0.00            | 13.82  | 30.67  | 77.41           | 45.90 | 90.00    | 76.18    | 6.51       |         |
| 5400.00   | 5400.00 | 5400.00 | 5400.00 | 0.00            | 14.08  | 30.67  | 77.41           | 45.90 | 90.00    | 75.92    | 6.39       |         |
| 5500.00   | 5500.00 | 5500.00 | 5500.00 | 0.00            | 14.34  | 30.67  | 77.41           | 45.90 | 90.00    | 75.66    | 6.27       |         |
| 5600.00   | 5600.00 | 5600.00 | 5600.00 | 0.00            | 14.61  | 30.67  | 77.41           | 45.90 | 90.00    | 75.40    | 6.16       |         |
| 5700.00   | 5700.00 | 5700.00 | 5700.00 | 0.00            | 14.87  | 30.67  | 77.41           | 45.90 | 90.00    | 75.13    | 6.05       |         |
| 5800.00   | 5800.00 | 5800.00 | 5800.00 | 0.00            | 15.13  | 30.67  | 77.41           | 45.90 | 90.00    | 74.87    | 5.95       |         |
| 5900.00   | 5900.00 | 5900.00 | 5900.00 | 0.00            | 15.39  | 30.67  | 77.41           | 45.90 | 90.00    | 74.61    | 5.85       |         |
| 6000.00   | 6000.00 | 6000.00 | 6000.00 | 0.00            | 15.65  | 30.67  | 77.41           | 45.90 | 90.00    | 74.35    | 5.75       |         |
| 6100.00   | 6100.00 | 6100.00 | 6100.00 | 0.00            | 15.91  | 30.67  | 77.41           | 45.90 | 90.00    | 74.09    | 5.66       |         |
| 6200.00   | 6200.00 | 6200.00 | 6200.00 | 0.00            | 16.18  | 30.67  | 77.41           | 45.90 | 90.00    | 73.82    | 5.56       |         |
| 6300.00   | 6300.00 | 6300.00 | 6300.00 | 0.00            | 16.44  | 30.67  | 77.41           | 45.90 | 90.00    | 73.56    | 5.47       |         |
| 6400.00   | 6400.00 | 6400.00 | 6400.00 | 0.00            | 16.70  | 30.67  | 77.41           | 45.90 | 90.00    | 73.30    | 5.39       |         |
| 6500.00   | 6500.00 | 6500.00 | 6500.00 | 0.00            | 16.96  | 30.67  | 77.41           | 45.90 | 90.00    | 73.04    | 5.31       |         |
| 6600.00   | 6600.00 | 6600.00 | 6600.00 | 0.00            | 17.22  | 30.67  | 77.41           | 45.90 | 90.00    | 72.78    | 5.23       |         |
| 6700.00   | 6700.00 | 6700.00 | 6700.00 | 0.00            | 17.49  | 30.67  | 77.41           | 45.90 | 90.00    | 72.52    | 5.15       |         |
| 6800.00   | 6800.00 | 6800.00 | 6800.00 | 0.00            | 17.75  | 30.67  | 77.41           | 45.90 | 90.00    | 72.25    | 5.07       |         |
| 6900.00   | 6900.00 | 6900.00 | 6900.00 | 0.00            | 18.01  | 30.67  | 77.41           | 45.90 | 90.00    | 71.99    | 5.00       |         |
| 7000.00   | 7000.00 | 7000.00 | 7000.00 | 0.00            | 18.27  | 30.67  | 77.41           | 45.90 | 90.00    | 71.73    | 4.93       |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company: Continental Resources   |           | Date: 04/23/2012           |           | Time: 17:17:28          |              | Page: 7         |             |            |                |                |         |
|----------------------------------|-----------|----------------------------|-----------|-------------------------|--------------|-----------------|-------------|------------|----------------|----------------|---------|
| Field: McKenzie County, ND       |           | Co-ordinate(NE) Reference: |           | Well: 3-16H, True North |              |                 |             |            |                |                |         |
| Reference Site: Tallahassee 3    |           | Vertical (TVD) Reference:  |           | GL 1920+KB 41 1941.0    |              |                 |             |            |                |                |         |
| Reference Well: 3-16H            |           |                            |           |                         |              | Db: Adapti      |             |            |                |                |         |
| Reference Wellpath: OH           |           |                            |           |                         |              |                 |             |            |                |                |         |
| Site: Columbus Federal 2         |           |                            |           |                         |              |                 |             |            |                |                |         |
| Well: 2-16H                      |           |                            |           |                         |              |                 |             |            |                |                |         |
| Wellpath: OH V3 Plan: Plan #1 V1 |           |                            |           | Inter-Site Error:       |              | 0.00 ft         |             |            |                |                |         |
| Reference                        |           | Offset                     |           | Semi-Major Axis         |              | Offset Location |             | Ctr-Ctr    | Edge           | Separation     | Warning |
| MD<br>ft                         | TVD<br>ft | MD<br>ft                   | TVD<br>ft | Ref<br>ft               | Offset<br>ft | TFO-HS<br>deg   | North<br>ft | East<br>ft | Distance<br>ft | Distance<br>ft |         |
| 7100.00                          | 7100.00   | 7100.00                    | 7100.00   | 0.00                    | 18.53        | 30.67           | 77.41       | 45.90      | 90.00          | 71.47          | 4.86    |
| 7200.00                          | 7200.00   | 7200.00                    | 7200.00   | 0.00                    | 18.79        | 30.67           | 77.41       | 45.90      | 90.00          | 71.21          | 4.79    |
| 7300.00                          | 7300.00   | 7300.00                    | 7300.00   | 0.00                    | 19.06        | 30.67           | 77.41       | 45.90      | 90.00          | 70.94          | 4.72    |
| 7400.00                          | 7400.00   | 7400.00                    | 7400.00   | 0.00                    | 19.32        | 30.67           | 77.41       | 45.90      | 90.00          | 70.68          | 4.66    |
| 7500.00                          | 7500.00   | 7500.00                    | 7500.00   | 0.00                    | 19.58        | 30.67           | 77.41       | 45.90      | 90.00          | 70.42          | 4.60    |
| 7600.00                          | 7600.00   | 7600.00                    | 7600.00   | 0.00                    | 19.84        | 30.67           | 77.41       | 45.90      | 90.00          | 70.16          | 4.54    |
| 7700.00                          | 7700.00   | 7700.00                    | 7700.00   | 0.00                    | 20.10        | 30.67           | 77.41       | 45.90      | 90.00          | 69.90          | 4.48    |
| 7800.00                          | 7800.00   | 7800.00                    | 7800.00   | 0.00                    | 20.37        | 30.67           | 77.41       | 45.90      | 90.00          | 69.64          | 4.42    |
| 7900.00                          | 7900.00   | 7900.00                    | 7900.00   | 0.00                    | 20.63        | 30.67           | 77.41       | 45.90      | 90.00          | 69.37          | 4.36    |
| 8000.00                          | 8000.00   | 8000.00                    | 8000.00   | 0.00                    | 20.89        | 30.67           | 77.41       | 45.90      | 90.00          | 69.11          | 4.31    |
| 8100.00                          | 8100.00   | 8100.00                    | 8100.00   | 0.00                    | 21.15        | 30.67           | 77.41       | 45.90      | 90.00          | 68.85          | 4.26    |
| 8200.00                          | 8200.00   | 8200.00                    | 8200.00   | 0.00                    | 21.41        | 30.67           | 77.41       | 45.90      | 90.00          | 68.59          | 4.20    |
| 8300.00                          | 8300.00   | 8300.00                    | 8300.00   | 0.00                    | 21.67        | 30.67           | 77.41       | 45.90      | 90.00          | 68.33          | 4.15    |
| 8400.00                          | 8400.00   | 8400.00                    | 8400.00   | 0.00                    | 21.94        | 30.67           | 77.41       | 45.90      | 90.00          | 68.06          | 4.10    |
| 8500.00                          | 8500.00   | 8500.00                    | 8500.00   | 0.00                    | 22.20        | 30.67           | 77.41       | 45.90      | 90.00          | 67.80          | 4.05    |
| 8600.00                          | 8600.00   | 8600.00                    | 8600.00   | 0.00                    | 22.46        | 30.67           | 77.41       | 45.90      | 90.00          | 67.54          | 4.01    |
| 8700.00                          | 8700.00   | 8700.00                    | 8700.00   | 0.00                    | 22.72        | 30.67           | 77.41       | 45.90      | 90.00          | 67.28          | 3.96    |
| 8800.00                          | 8800.00   | 8800.00                    | 8800.00   | 0.00                    | 22.98        | 30.67           | 77.41       | 45.90      | 90.00          | 67.02          | 3.92    |
| 8900.00                          | 8900.00   | 8900.00                    | 8900.00   | 0.00                    | 23.25        | 30.67           | 77.41       | 45.90      | 90.00          | 66.76          | 3.87    |
| 9000.00                          | 9000.00   | 9000.00                    | 9000.00   | 0.00                    | 23.51        | 30.67           | 77.41       | 45.90      | 90.00          | 66.49          | 3.83    |
| 9100.00                          | 9100.00   | 9100.00                    | 9100.00   | 0.00                    | 23.77        | 30.67           | 77.41       | 45.90      | 90.00          | 66.23          | 3.79    |
| 9200.00                          | 9200.00   | 9200.00                    | 9200.00   | 0.00                    | 24.03        | 30.67           | 77.41       | 45.90      | 90.00          | 65.97          | 3.75    |
| 9300.00                          | 9300.00   | 9300.00                    | 9300.00   | 0.00                    | 24.29        | 30.67           | 77.41       | 45.90      | 90.00          | 65.71          | 3.70    |
| 9400.00                          | 9400.00   | 9400.00                    | 9400.00   | 0.00                    | 24.55        | 30.67           | 77.41       | 45.90      | 90.00          | 65.45          | 3.67    |
| 9500.00                          | 9500.00   | 9500.00                    | 9500.00   | 0.00                    | 24.82        | 30.67           | 77.41       | 45.90      | 90.00          | 65.18          | 3.63    |
| 9600.00                          | 9600.00   | 9600.00                    | 9600.00   | 0.00                    | 25.08        | 30.67           | 77.41       | 45.90      | 90.00          | 64.92          | 3.59    |
| 9700.00                          | 9700.00   | 9700.00                    | 9700.00   | 0.00                    | 25.34        | 30.67           | 77.41       | 45.90      | 90.00          | 64.66          | 3.55    |
| 9800.00                          | 9800.00   | 9800.00                    | 9800.00   | 0.00                    | 25.60        | 30.67           | 77.41       | 45.90      | 90.00          | 64.40          | 3.52    |
| 9900.00                          | 9900.00   | 9900.00                    | 9900.00   | 0.00                    | 25.86        | 30.67           | 77.41       | 45.90      | 90.00          | 64.14          | 3.48    |
| 10000.00                         | 10000.00  | 10000.00                   | 10000.00  | 0.00                    | 26.12        | 30.67           | 77.41       | 45.90      | 90.00          | 63.88          | 3.44    |
| 10100.00                         | 10100.00  | 10100.00                   | 10100.00  | 0.00                    | 26.39        | 30.67           | 77.41       | 45.90      | 90.00          | 63.61          | 3.41    |
| 10166.27                         | 10166.27  | 10166.27                   | 10166.27  | 0.00                    | 26.56        | 30.67           | 77.41       | 45.90      | 90.00          | 63.44          | 3.39    |
| 10175.00                         | 10175.00  | 10173.99                   | 10173.98  | 0.00                    | 26.58        | 162.02          | 77.48       | 45.88      | 90.13          | 63.55          | 3.39    |
| 10200.00                         | 10199.97  | 10195.99                   | 10195.96  | 0.00                    | 26.64        | 161.71          | 78.36       | 45.57      | 91.96          | 65.34          | 3.46    |
| 10225.00                         | 10224.83  | 10217.65                   | 10217.54  | 0.00                    | 26.69        | 161.05          | 80.24       | 44.91      | 95.92          | 69.30          | 3.60    |
| 10250.00                         | 10249.50  | 10238.76                   | 10238.43  | 0.00                    | 26.75        | 160.12          | 83.03       | 43.93      | 101.99         | 75.38          | 3.83    |
| 10275.00                         | 10273.90  | 10259.09                   | 10258.40  | 0.00                    | 26.80        | 158.98          | 86.60       | 42.67      | 110.09         | 83.52          | 4.14    |
| 10300.00                         | 10297.96  | 10278.47                   | 10277.26  | 0.00                    | 26.85        | 157.71          | 90.82       | 41.18      | 120.16         | 93.65          | 4.53    |
| 10325.00                         | 10321.59  | 10296.78                   | 10294.88  | 0.00                    | 26.90        | 156.33          | 95.51       | 39.53      | 132.08         | 105.65         | 5.00    |
| 10350.00                         | 10344.72  | 10313.91                   | 10311.17  | 0.00                    | 26.95        | 154.87          | 100.52      | 37.76      | 145.73         | 119.40         | 5.53    |
| 10375.00                         | 10367.28  | 10329.80                   | 10326.07  | 0.00                    | 26.99        | 153.32          | 105.71      | 35.94      | 160.97         | 134.74         | 6.14    |
| 10400.00                         | 10389.20  | 10344.41                   | 10339.60  | 0.00                    | 27.03        | 151.65          | 110.92      | 34.11      | 177.67         | 151.54         | 6.80    |
| 10425.00                         | 10410.39  | 10357.74                   | 10351.77  | 0.00                    | 27.06        | 149.82          | 116.03      | 32.30      | 195.66         | 169.64         | 7.52    |
| 10450.00                         | 10430.80  | 10369.80                   | 10362.64  | 0.00                    | 27.09        | 147.78          | 120.96      | 30.57      | 214.82         | 188.90         | 8.29    |
| 10475.00                         | 10450.36  | 10380.62                   | 10372.26  | 0.00                    | 27.12        | 145.44          | 125.62      | 28.93      | 235.01         | 209.17         | 9.10    |
| 10500.00                         | 10469.01  | 10390.23                   | 10380.71  | 0.00                    | 27.15        | 142.71          | 129.94      | 27.40      | 256.08         | 230.32         | 9.94    |
| 10525.00                         | 10486.68  | 10398.69                   | 10388.06  | 0.00                    | 27.17        | 139.45          | 133.89      | 26.01      | 277.93         | 252.23         | 10.81   |
| 10550.00                         | 10503.32  | 10406.04                   | 10394.39  | 0.00                    | 27.19        | 135.48          | 137.43      | 24.76      | 300.42         | 274.77         | 11.71   |
| 10575.00                         | 10518.88  | 10412.35                   | 10399.76  | 0.00                    | 27.21        | 130.56          | 140.55      | 23.67      | 323.45         | 297.84         | 12.63   |
| 10600.00                         | 10533.30  | 10417.66                   | 10404.24  | 0.00                    | 27.22        | 124.40          | 143.23      | 22.72      | 346.93         | 321.33         | 13.55   |
| 10625.00                         | 10546.55  | 10422.02                   | 10407.90  | 0.00                    | 27.23        | 116.64          | 145.46      | 21.94      | 370.74         | 345.15         | 14.49   |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company: Continental Resources   |          | Date: 04/23/2012           |          | Time: 17:17:28          |        | Page: 8    |        |            |          |          |        |         |
|----------------------------------|----------|----------------------------|----------|-------------------------|--------|------------|--------|------------|----------|----------|--------|---------|
| Field: McKenzie County, ND       |          |                            |          |                         |        |            |        |            |          |          |        |         |
| Reference Site: Tallahassee 3    |          | Co-ordinate(NE) Reference: |          | Well: 3-16H, True North |        |            |        |            |          |          |        |         |
| Reference Well: 3-16H            |          | Vertical (TVD) Reference:  |          | GL 1920+KB 41 1941.0    |        |            |        |            |          |          |        |         |
| Reference Wellpath: OH           |          |                            |          |                         |        | Db: Adapti |        |            |          |          |        |         |
| Site: Columbus Federal 2         |          |                            |          |                         |        |            |        |            |          |          |        |         |
| Well: 2-16H                      |          |                            |          |                         |        |            |        |            |          |          |        |         |
| Wellpath: OH V3 Plan: Plan #1 V1 |          |                            |          | Inter-Site Error:       |        | 0.00 ft    |        |            |          |          |        |         |
| Reference                        | Offset   | Semi-Major Axis            |          | Offset Location         |        | Ctr-Ctr    | Edge   | Separation |          |          |        |         |
| MD                               | TVD      | MD                         | TVD      | Ref                     | Offset | TFO-HS     | North  | East       | Distance | Distance | Factor | Warning |
| ft                               | ft       | ft                         | ft       | ft                      | ft     | deg        | ft     | ft         | ft       | ft       |        |         |
| 10650.00                         | 10558.57 | 10425.48                   | 10410.79 | 0.00                    | 27.24  | 106.98     | 147.27 | 21.30      | 394.82   | 369.22   | 15.42  |         |
| 10675.00                         | 10569.32 | 10428.09                   | 10412.96 | 0.00                    | 27.25  | 95.40      | 148.64 | 20.82      | 419.06   | 393.44   | 16.36  |         |
| 10700.00                         | 10578.78 | 10429.89                   | 10414.45 | 0.00                    | 27.25  | 82.47      | 149.59 | 20.48      | 443.40   | 417.76   | 17.29  |         |
| 10725.00                         | 10586.92 | 10430.92                   | 10415.30 | 0.00                    | 27.26  | 69.38      | 150.14 | 20.29      | 467.76   | 442.08   | 18.22  |         |
| 10750.00                         | 10593.70 | 10431.22                   | 10415.54 | 0.00                    | 27.26  | 57.40      | 150.30 | 20.23      | 492.08   | 466.35   | 19.13  |         |
| 10775.00                         | 10599.11 | 10430.81                   | 10415.21 | 0.00                    | 27.26  | 47.27      | 150.08 | 20.31      | 516.28   | 490.51   | 20.03  |         |
| 10800.00                         | 10603.12 | 10429.73                   | 10414.31 | 0.00                    | 27.25  | 39.10      | 149.50 | 20.51      | 540.32   | 514.49   | 20.92  |         |
| 10825.00                         | 10605.73 | 10428.00                   | 10412.88 | 0.00                    | 27.25  | 32.63      | 148.59 | 20.84      | 564.12   | 538.24   | 21.79  |         |
| 10850.00                         | 10606.92 | 10425.64                   | 10410.93 | 0.00                    | 27.24  | 27.50      | 147.35 | 21.27      | 587.65   | 561.70   | 22.65  |         |
| 10857.19                         | 10607.00 | 10424.86                   | 10410.27 | 0.00                    | 27.24  | 26.23      | 146.94 | 21.42      | 594.36   | 568.39   | 22.89  |         |
| 10900.00                         | 10607.14 | 10420.09                   | 10406.29 | 0.00                    | 27.23  | 25.21      | 144.47 | 22.29      | 634.36   | 608.30   | 24.34  |         |
| 10957.19                         | 10607.32 | 10413.96                   | 10401.12 | 0.00                    | 27.21  | 23.93      | 141.36 | 23.38      | 688.24   | 662.07   | 26.30  |         |
| 11000.00                         | 10607.45 | 10409.43                   | 10397.28 | 0.00                    | 27.20  | 19.42      | 139.10 | 24.18      | 728.89   | 702.65   | 27.78  |         |
| 11100.00                         | 10607.78 | 10398.54                   | 10387.94 | 0.00                    | 27.17  | 7.67       | 133.82 | 26.04      | 824.81   | 798.46   | 31.30  |         |
| 11200.00                         | 10608.10 | 10387.43                   | 10378.27 | 0.00                    | 27.14  | 355.03     | 128.67 | 27.85      | 921.50   | 895.08   | 34.88  |         |
| 11300.00                         | 10608.44 | 10375.00                   | 10367.28 | 0.00                    | 27.11  | 342.73     | 123.17 | 29.79      | 1018.48  | 992.00   | 38.47  |         |
| 11400.00                         | 10608.78 | 10365.36                   | 10358.66 | 0.00                    | 27.08  | 331.89     | 119.12 | 31.22      | 1115.40  | 1088.89  | 42.09  |         |
| 11500.00                         | 10609.12 | 10354.71                   | 10349.02 | 0.00                    | 27.05  | 322.84     | 114.84 | 32.72      | 1212.03  | 1185.50  | 45.69  |         |
| 11600.00                         | 10609.46 | 10344.46                   | 10339.64 | 0.00                    | 27.03  | 315.52     | 110.93 | 34.10      | 1308.19  | 1281.65  | 49.29  |         |
| 11700.00                         | 10609.81 | 10334.68                   | 10330.61 | 0.00                    | 27.00  | 309.66     | 107.40 | 35.34      | 1403.73  | 1377.18  | 52.88  |         |
| 11800.00                         | 10610.16 | 10325.00                   | 10321.59 | 0.00                    | 26.98  | 304.96     | 104.09 | 36.51      | 1498.53  | 1471.97  | 56.43  |         |
| 11900.00                         | 10610.51 | 10316.70                   | 10313.80 | 0.00                    | 26.95  | 301.09     | 101.40 | 37.46      | 1592.48  | 1565.93  | 59.97  |         |
| 12000.00                         | 10610.86 | 10308.54                   | 10306.08 | 0.00                    | 26.93  | 297.92     | 98.89  | 38.34      | 1685.51  | 1658.95  | 63.47  |         |
| 12100.00                         | 10611.20 | 10300.00                   | 10297.96 | 0.00                    | 26.91  | 295.34     | 96.41  | 39.22      | 1777.51  | 1750.95  | 66.92  |         |
| 12200.00                         | 10611.55 | 10293.82                   | 10292.05 | 0.00                    | 26.89  | 293.05     | 94.71  | 39.81      | 1868.42  | 1841.86  | 70.35  |         |
| 12300.00                         | 10611.89 | 10287.23                   | 10285.72 | 0.00                    | 26.88  | 291.14     | 92.98  | 40.42      | 1958.16  | 1931.60  | 73.73  |         |
| 12400.00                         | 10612.23 | 10275.00                   | 10273.90 | 0.00                    | 26.85  | 289.82     | 90.00  | 41.47      | 2046.69  | 2020.11  | 77.02  |         |
| 12500.00                         | 10612.56 | 10275.00                   | 10273.90 | 0.00                    | 26.85  | 288.10     | 90.00  | 41.47      | 2133.84  | 2107.28  | 80.34  |         |
| 12577.81                         | 10612.81 | 10275.00                   | 10273.90 | 0.00                    | 26.85  | 286.93     | 90.00  | 41.47      | 2200.74  | 2174.19  | 82.89  |         |
| 12600.00                         | 10612.88 | 10275.00                   | 10273.90 | 0.00                    | 26.85  | 286.93     | 90.00  | 41.47      | 2219.73  | 2193.19  | 83.62  |         |
| 12700.00                         | 10613.20 | 10265.45                   | 10264.62 | 0.00                    | 26.82  | 287.36     | 87.90  | 42.21      | 2305.94  | 2279.39  | 86.83  |         |
| 12800.00                         | 10613.52 | 10261.09                   | 10260.36 | 0.00                    | 26.81  | 287.56     | 87.00  | 42.53      | 2393.20  | 2366.65  | 90.12  |         |
| 12900.00                         | 10613.85 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2481.45  | 2454.87  | 93.36  |         |
| 13000.00                         | 10614.17 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2570.43  | 2543.86  | 96.74  |         |
| 13100.00                         | 10614.49 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2660.20  | 2633.64  | 100.14 |         |
| 13200.00                         | 10614.81 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2750.68  | 2724.12  | 103.57 |         |
| 13300.00                         | 10615.13 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2841.80  | 2815.24  | 107.02 |         |
| 13400.00                         | 10615.46 | 10250.00                   | 10249.50 | 0.00                    | 26.78  | 288.07     | 84.90  | 43.27      | 2933.49  | 2906.94  | 110.49 |         |
| 13500.00                         | 10615.78 | 10238.79                   | 10238.47 | 0.00                    | 26.75  | 288.58     | 83.03  | 43.93      | 3025.58  | 2999.01  | 113.88 |         |
| 13600.00                         | 10616.10 | 10236.44                   | 10236.14 | 0.00                    | 26.74  | 288.68     | 82.67  | 44.05      | 3118.22  | 3091.65  | 117.36 |         |
| 13700.00                         | 10616.42 | 10234.24                   | 10233.97 | 0.00                    | 26.74  | 288.78     | 82.35  | 44.17      | 3211.29  | 3184.72  | 120.85 |         |
| 13800.00                         | 10616.75 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3304.81  | 3278.22  | 124.28 |         |
| 13900.00                         | 10617.07 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3398.62  | 3372.03  | 127.82 |         |
| 14000.00                         | 10617.39 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3492.77  | 3466.18  | 131.37 |         |
| 14100.00                         | 10617.71 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3587.23  | 3560.65  | 134.93 |         |
| 14200.00                         | 10618.03 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3681.99  | 3655.41  | 138.50 |         |
| 14300.00                         | 10618.36 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3777.02  | 3750.44  | 142.08 |         |
| 14400.00                         | 10618.68 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3872.30  | 3845.72  | 145.68 |         |
| 14500.00                         | 10619.00 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 3967.81  | 3941.23  | 149.28 |         |
| 14600.00                         | 10619.32 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 4063.54  | 4036.96  | 152.88 |         |
| 14700.00                         | 10619.64 | 10225.00                   | 10224.83 | 0.00                    | 26.71  | 289.20     | 81.10  | 44.61      | 4159.46  | 4132.89  | 156.50 |         |

## Anticollision Report

|                            |                       |                                   |                         |              |          |              |        |
|----------------------------|-----------------------|-----------------------------------|-------------------------|--------------|----------|--------------|--------|
| <b>Company:</b>            | Continental Resources | <b>Date:</b>                      | 04/23/2012              | <b>Time:</b> | 17:17:28 | <b>Page:</b> | 9      |
| <b>Field:</b>              | McKenzie County, ND   |                                   |                         |              |          |              |        |
| <b>Reference Site:</b>     | Tallahassee 3         | <b>Co-ordinate(NE) Reference:</b> | Well: 3-16H, True North |              |          |              |        |
| <b>Reference Well:</b>     | 3-16H                 | <b>Vertical (TVD) Reference:</b>  | GL 1920+KB 41 1941.0    |              |          |              |        |
| <b>Reference Wellpath:</b> | OH                    |                                   |                         |              |          | <b>Db:</b>   | Adapti |

Site: Columbus Federal 2  
Well: 2-16H  
Wellpath: OH V3 Plan: Plan #1 V1

**Inter-Site Error:** 0.00 ft

| Reference |          | Offset   |          | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|----------|----------|----------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD      | MD       | TVD      | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft       | ft       | ft       | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 14800.00  | 10619.97 | 10225.00 | 10224.83 | 0.00            | 26.71  | 289.20 | 81.10           | 44.61 | 4255.58  | 4229.00  | 160.12     |         |
| 14900.00  | 10620.29 | 10225.00 | 10224.83 | 0.00            | 26.71  | 289.20 | 81.10           | 44.61 | 4351.87  | 4325.29  | 163.75     |         |
| 15000.00  | 10620.61 | 10214.65 | 10214.55 | 0.00            | 26.69  | 289.67 | 79.92           | 45.02 | 4448.21  | 4421.62  | 167.31     |         |
| 15100.00  | 10620.93 | 10213.61 | 10213.52 | 0.00            | 26.68  | 289.72 | 79.81           | 45.06 | 4544.79  | 4518.21  | 170.94     |         |
| 15200.00  | 10621.26 | 10212.61 | 10212.53 | 0.00            | 26.68  | 289.77 | 79.71           | 45.10 | 4641.52  | 4614.93  | 174.58     |         |
| 15300.00  | 10621.58 | 10211.66 | 10211.58 | 0.00            | 26.68  | 289.81 | 79.62           | 45.13 | 4738.38  | 4711.79  | 178.22     |         |
| 15400.00  | 10621.90 | 10210.75 | 10210.67 | 0.00            | 26.68  | 289.85 | 79.53           | 45.16 | 4835.37  | 4808.78  | 181.86     |         |
| 15500.00  | 10622.22 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 4932.57  | 4905.97  | 185.43     |         |
| 15600.00  | 10622.54 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5029.77  | 5003.17  | 189.08     |         |
| 15700.00  | 10622.87 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5127.08  | 5100.48  | 192.74     |         |
| 15800.00  | 10623.19 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5224.49  | 5197.89  | 196.41     |         |
| 15900.00  | 10623.51 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5322.00  | 5295.40  | 200.07     |         |
| 16000.00  | 10623.83 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5419.59  | 5392.99  | 203.74     |         |
| 16100.00  | 10624.15 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5517.28  | 5490.68  | 207.42     |         |
| 16200.00  | 10624.48 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5615.04  | 5588.44  | 211.10     |         |
| 16300.00  | 10624.80 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5712.88  | 5686.28  | 214.78     |         |
| 16400.00  | 10625.12 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5810.80  | 5784.20  | 218.46     |         |
| 16500.00  | 10625.44 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 5908.78  | 5882.18  | 222.14     |         |
| 16600.00  | 10625.77 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6006.83  | 5980.23  | 225.83     |         |
| 16700.00  | 10626.09 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6104.95  | 6078.35  | 229.52     |         |
| 16800.00  | 10626.41 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6203.12  | 6176.52  | 233.21     |         |
| 16900.00  | 10626.73 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6301.35  | 6274.75  | 236.91     |         |
| 17000.00  | 10627.05 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6399.64  | 6373.04  | 240.61     |         |
| 17100.00  | 10627.38 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6497.98  | 6471.38  | 244.30     |         |
| 17200.00  | 10627.70 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6596.37  | 6569.77  | 248.00     |         |
| 17300.00  | 10628.02 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6694.80  | 6668.21  | 251.71     |         |
| 17400.00  | 10628.34 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6793.29  | 6766.69  | 255.41     |         |
| 17500.00  | 10628.66 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6891.81  | 6865.21  | 259.12     |         |
| 17600.00  | 10628.99 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 6990.38  | 6963.78  | 262.82     |         |
| 17700.00  | 10629.31 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7088.99  | 7062.39  | 266.53     |         |
| 17800.00  | 10629.63 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7187.63  | 7161.04  | 270.24     |         |
| 17900.00  | 10629.95 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7286.32  | 7259.72  | 273.95     |         |
| 18000.00  | 10630.28 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7385.04  | 7358.44  | 277.67     |         |
| 18100.00  | 10630.60 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7483.79  | 7457.19  | 281.38     |         |
| 18200.00  | 10630.92 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7582.58  | 7555.98  | 285.10     |         |
| 18300.00  | 10631.24 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7681.39  | 7654.80  | 288.81     |         |
| 18400.00  | 10631.56 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7780.24  | 7753.64  | 292.53     |         |
| 18500.00  | 10631.89 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7879.12  | 7852.52  | 296.25     |         |
| 18535.29  | 10632.00 | 10200.00 | 10199.97 | 0.00            | 26.65  | 290.34 | 78.63           | 45.48 | 7914.01  | 7887.42  | 297.56     |         |

**Site:** Columbus Federal 3  
**Well:** 3-16H  
**Wellpath:** OH V1 Plan: Plan #1 V1

**Inter-Site Error:** 0.00 ft

| Reference |        | Offset |        | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|--------|--------|--------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD    | MD     | TVD    | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft     | ft     | ft     | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 0.00      | 0.00   | 0.00   | 0.00   | 0.00            | 0.00   | 30.71  | 116.02          | 68.92 | 134.95   |          |            | No Data |
| 100.00    | 100.00 | 100.00 | 100.00 | 0.00            | 0.21   | 30.71  | 116.02          | 68.92 | 134.95   | 134.74   | 651.86     |         |
| 200.00    | 200.00 | 200.00 | 200.00 | 0.00            | 0.47   | 30.71  | 116.02          | 68.92 | 134.95   | 134.48   | 287.85     |         |
| 300.00    | 300.00 | 300.00 | 300.00 | 0.00            | 0.73   | 30.71  | 116.02          | 68.92 | 134.95   | 134.22   | 184.70     |         |
| 400.00    | 400.00 | 400.00 | 400.00 | 0.00            | 0.99   | 30.71  | 116.02          | 68.92 | 134.95   | 133.96   | 135.98     |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            | Continental Resources  |                 | Date:                      | 04/23/2012              |        | Time:      | 17:17:28 |       | Page:             | 10       |        |         |
|---------------------|------------------------|-----------------|----------------------------|-------------------------|--------|------------|----------|-------|-------------------|----------|--------|---------|
| Field:              | McKenzie County, ND    |                 |                            |                         |        |            |          |       |                   |          |        |         |
| Reference Site:     | Tallahassee 3          |                 | Co-ordinate(NE) Reference: | Well: 3-16H, True North |        |            |          |       |                   |          |        |         |
| Reference Well:     | 3-16H                  |                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |        |            |          |       |                   |          |        |         |
| Reference Wellpath: | OH                     |                 |                            |                         |        |            |          |       | Db:               | Adapti   |        |         |
| Site:               | Columbus Federal 3     |                 |                            |                         |        |            |          |       |                   |          |        |         |
| Well:               | 3-16H                  |                 |                            |                         |        |            |          |       |                   |          |        |         |
| Wellpath:           | OH V1 Plan: Plan #1 V1 |                 |                            |                         |        |            |          |       | Inter-Site Error: | 0.00     | ft     |         |
| Reference           | Offset                 | Semi-Major Axis | Offset Location            | Ctr-Ctr                 | Edge   | Separation |          |       |                   |          |        |         |
| MD                  | TVD                    | MD              | TVD                        | Ref                     | Offset | TFO-HS     | North    | East  | Distance          | Distance | Factor | Warning |
| ft                  | ft                     | ft              | ft                         | ft                      | ft     | deg        | ft       | ft    | ft                | ft       |        |         |
| 500.00              | 500.00                 | 500.00          | 500.00                     | 0.00                    | 1.25   | 30.71      | 116.02   | 68.92 | 134.95            | 133.69   | 107.60 |         |
| 600.00              | 600.00                 | 600.00          | 600.00                     | 0.00                    | 1.52   | 30.71      | 116.02   | 68.92 | 134.95            | 133.43   | 89.02  |         |
| 700.00              | 700.00                 | 700.00          | 700.00                     | 0.00                    | 1.78   | 30.71      | 116.02   | 68.92 | 134.95            | 133.17   | 75.91  |         |
| 800.00              | 800.00                 | 800.00          | 800.00                     | 0.00                    | 2.04   | 30.71      | 116.02   | 68.92 | 134.95            | 132.91   | 66.16  |         |
| 900.00              | 900.00                 | 900.00          | 900.00                     | 0.00                    | 2.30   | 30.71      | 116.02   | 68.92 | 134.95            | 132.65   | 58.64  |         |
| 1000.00             | 1000.00                | 1000.00         | 1000.00                    | 0.00                    | 2.56   | 30.71      | 116.02   | 68.92 | 134.95            | 132.39   | 52.65  |         |
| 1100.00             | 1100.00                | 1100.00         | 1100.00                    | 0.00                    | 2.82   | 30.71      | 116.02   | 68.92 | 134.95            | 132.12   | 47.77  |         |
| 1200.00             | 1200.00                | 1200.00         | 1200.00                    | 0.00                    | 3.09   | 30.71      | 116.02   | 68.92 | 134.95            | 131.86   | 43.72  |         |
| 1300.00             | 1300.00                | 1300.00         | 1300.00                    | 0.00                    | 3.35   | 30.71      | 116.02   | 68.92 | 134.95            | 131.60   | 40.30  |         |
| 1400.00             | 1400.00                | 1400.00         | 1400.00                    | 0.00                    | 3.61   | 30.71      | 116.02   | 68.92 | 134.95            | 131.34   | 37.38  |         |
| 1500.00             | 1500.00                | 1500.00         | 1500.00                    | 0.00                    | 3.87   | 30.71      | 116.02   | 68.92 | 134.95            | 131.08   | 34.85  |         |
| 1600.00             | 1600.00                | 1600.00         | 1600.00                    | 0.00                    | 4.13   | 30.71      | 116.02   | 68.92 | 134.95            | 130.82   | 32.64  |         |
| 1700.00             | 1700.00                | 1700.00         | 1700.00                    | 0.00                    | 4.40   | 30.71      | 116.02   | 68.92 | 134.95            | 130.55   | 30.70  |         |
| 1800.00             | 1800.00                | 1800.00         | 1800.00                    | 0.00                    | 4.66   | 30.71      | 116.02   | 68.92 | 134.95            | 130.29   | 28.97  |         |
| 1900.00             | 1900.00                | 1900.00         | 1900.00                    | 0.00                    | 4.92   | 30.71      | 116.02   | 68.92 | 134.95            | 130.03   | 27.43  |         |
| 2000.00             | 2000.00                | 2000.00         | 2000.00                    | 0.00                    | 5.18   | 30.71      | 116.02   | 68.92 | 134.95            | 129.77   | 26.05  |         |
| 2100.00             | 2100.00                | 2100.00         | 2100.00                    | 0.00                    | 5.44   | 30.71      | 116.02   | 68.92 | 134.95            | 129.51   | 24.79  |         |
| 2200.00             | 2200.00                | 2200.00         | 2200.00                    | 0.00                    | 5.70   | 30.71      | 116.02   | 68.92 | 134.95            | 129.24   | 23.66  |         |
| 2300.00             | 2300.00                | 2300.00         | 2300.00                    | 0.00                    | 5.97   | 30.71      | 116.02   | 68.92 | 134.95            | 128.98   | 22.62  |         |
| 2400.00             | 2400.00                | 2400.00         | 2400.00                    | 0.00                    | 6.23   | 30.71      | 116.02   | 68.92 | 134.95            | 128.72   | 21.67  |         |
| 2500.00             | 2500.00                | 2500.00         | 2500.00                    | 0.00                    | 6.49   | 30.71      | 116.02   | 68.92 | 134.95            | 128.46   | 20.79  |         |
| 2600.00             | 2600.00                | 2600.00         | 2600.00                    | 0.00                    | 6.75   | 30.71      | 116.02   | 68.92 | 134.95            | 128.20   | 19.99  |         |
| 2700.00             | 2700.00                | 2700.00         | 2700.00                    | 0.00                    | 7.01   | 30.71      | 116.02   | 68.92 | 134.95            | 127.94   | 19.24  |         |
| 2800.00             | 2800.00                | 2800.00         | 2800.00                    | 0.00                    | 7.28   | 30.71      | 116.02   | 68.92 | 134.95            | 127.67   | 18.55  |         |
| 2900.00             | 2900.00                | 2900.00         | 2900.00                    | 0.00                    | 7.54   | 30.71      | 116.02   | 68.92 | 134.95            | 127.41   | 17.90  |         |
| 3000.00             | 3000.00                | 3000.00         | 3000.00                    | 0.00                    | 7.80   | 30.71      | 116.02   | 68.92 | 134.95            | 127.15   | 17.30  |         |
| 3100.00             | 3100.00                | 3100.00         | 3100.00                    | 0.00                    | 8.06   | 30.71      | 116.02   | 68.92 | 134.95            | 126.89   | 16.74  |         |
| 3200.00             | 3200.00                | 3200.00         | 3200.00                    | 0.00                    | 8.32   | 30.71      | 116.02   | 68.92 | 134.95            | 126.63   | 16.21  |         |
| 3300.00             | 3300.00                | 3300.00         | 3300.00                    | 0.00                    | 8.58   | 30.71      | 116.02   | 68.92 | 134.95            | 126.36   | 15.72  |         |
| 3400.00             | 3400.00                | 3400.00         | 3400.00                    | 0.00                    | 8.85   | 30.71      | 116.02   | 68.92 | 134.95            | 126.10   | 15.25  |         |
| 3500.00             | 3500.00                | 3500.00         | 3500.00                    | 0.00                    | 9.11   | 30.71      | 116.02   | 68.92 | 134.95            | 125.84   | 14.82  |         |
| 3600.00             | 3600.00                | 3600.00         | 3600.00                    | 0.00                    | 9.37   | 30.71      | 116.02   | 68.92 | 134.95            | 125.58   | 14.40  |         |
| 3700.00             | 3700.00                | 3700.00         | 3700.00                    | 0.00                    | 9.63   | 30.71      | 116.02   | 68.92 | 134.95            | 125.32   | 14.01  |         |
| 3800.00             | 3800.00                | 3800.00         | 3800.00                    | 0.00                    | 9.89   | 30.71      | 116.02   | 68.92 | 134.95            | 125.06   | 13.64  |         |
| 3900.00             | 3900.00                | 3900.00         | 3900.00                    | 0.00                    | 10.16  | 30.71      | 116.02   | 68.92 | 134.95            | 124.79   | 13.29  |         |
| 4000.00             | 4000.00                | 4000.00         | 4000.00                    | 0.00                    | 10.42  | 30.71      | 116.02   | 68.92 | 134.95            | 124.53   | 12.95  |         |
| 4100.00             | 4100.00                | 4100.00         | 4100.00                    | 0.00                    | 10.68  | 30.71      | 116.02   | 68.92 | 134.95            | 124.27   | 12.64  |         |
| 4200.00             | 4200.00                | 4200.00         | 4200.00                    | 0.00                    | 10.94  | 30.71      | 116.02   | 68.92 | 134.95            | 124.01   | 12.33  |         |
| 4300.00             | 4300.00                | 4300.00         | 4300.00                    | 0.00                    | 11.20  | 30.71      | 116.02   | 68.92 | 134.95            | 123.75   | 12.05  |         |
| 4400.00             | 4400.00                | 4400.00         | 4400.00                    | 0.00                    | 11.46  | 30.71      | 116.02   | 68.92 | 134.95            | 123.48   | 11.77  |         |
| 4500.00             | 4500.00                | 4500.00         | 4500.00                    | 0.00                    | 11.73  | 30.71      | 116.02   | 68.92 | 134.95            | 123.22   | 11.51  |         |
| 4600.00             | 4600.00                | 4600.00         | 4600.00                    | 0.00                    | 11.99  | 30.71      | 116.02   | 68.92 | 134.95            | 122.96   | 11.26  |         |
| 4700.00             | 4700.00                | 4700.00         | 4700.00                    | 0.00                    | 12.25  | 30.71      | 116.02   | 68.92 | 134.95            | 122.70   | 11.02  |         |
| 4800.00             | 4800.00                | 4800.00         | 4800.00                    | 0.00                    | 12.51  | 30.71      | 116.02   | 68.92 | 134.95            | 122.44   | 10.79  |         |
| 4900.00             | 4900.00                | 4900.00         | 4900.00                    | 0.00                    | 12.77  | 30.71      | 116.02   | 68.92 | 134.95            | 122.18   | 10.56  |         |
| 5000.00             | 5000.00                | 5000.00         | 5000.00                    | 0.00                    | 13.03  | 30.71      | 116.02   | 68.92 | 134.95            | 121.91   | 10.35  |         |
| 5100.00             | 5100.00                | 5100.00         | 5100.00                    | 0.00                    | 13.30  | 30.71      | 116.02   | 68.92 | 134.95            | 121.65   | 10.15  |         |
| 5200.00             | 5200.00                | 5200.00         | 5200.00                    | 0.00                    | 13.56  | 30.71      | 116.02   | 68.92 | 134.95            | 121.39   | 9.95   |         |
| 5300.00             | 5300.00                | 5300.00         | 5300.00                    | 0.00                    | 13.82  | 30.71      | 116.02   | 68.92 | 134.95            | 121.13   | 9.76   |         |
| 5400.00             | 5400.00                | 5400.00         | 5400.00                    | 0.00                    | 14.08  | 30.71      | 116.02   | 68.92 | 134.95            | 120.87   | 9.58   |         |
| 5500.00             | 5500.00                | 5500.00         | 5500.00                    | 0.00                    | 14.34  | 30.71      | 116.02   | 68.92 | 134.95            | 120.60   | 9.41   |         |



LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            |           | Continental Resources  |           | Date:                      |              | 04/23/2012              |                 | Time:      |                           | 17:17:28               |                      | Page:   |  | 11     |  |
|---------------------|-----------|------------------------|-----------|----------------------------|--------------|-------------------------|-----------------|------------|---------------------------|------------------------|----------------------|---------|--|--------|--|
| Field:              |           | McKenzie County, ND    |           |                            |              |                         |                 |            |                           |                        |                      |         |  |        |  |
| Reference Site:     |           | Tallahassee 3          |           | Co-ordinate(NE) Reference: |              | Well: 3-16H, True North |                 |            |                           |                        |                      |         |  |        |  |
| Reference Well:     |           | 3-16H                  |           | Vertical (TVD) Reference:  |              | GL 1920+KB 41 1941.0    |                 |            |                           |                        |                      |         |  |        |  |
| Reference Wellpath: |           | OH                     |           |                            |              |                         |                 |            |                           |                        |                      | Db:     |  | Adapti |  |
| Site:               |           | Columbus Federal 3     |           |                            |              |                         |                 |            |                           |                        |                      |         |  |        |  |
| Well:               |           | 3-16H                  |           |                            |              |                         |                 |            |                           |                        |                      |         |  |        |  |
| Wellpath:           |           | OH V1 Plan: Plan #1 V1 |           |                            |              |                         |                 |            |                           | Inter-Site Error:      |                      | 0.00    |  | ft     |  |
| Reference           |           | Offset                 |           | Semi-Major Axis            |              |                         | Offset Location |            | Ctr-Ctr<br>Distance<br>ft | Edge<br>Distance<br>ft | Separation<br>Factor | Warning |  |        |  |
| MD<br>ft            | TVD<br>ft | MD<br>ft               | TVD<br>ft | Ref<br>ft                  | Offset<br>ft | TFO-HS<br>deg           | North<br>ft     | East<br>ft |                           |                        |                      |         |  |        |  |
| 5600.00             | 5600.00   | 5600.00                | 5600.00   | 0.00                       | 14.61        | 30.71                   | 116.02          | 68.92      | 134.95                    | 120.34                 | 9.24                 |         |  |        |  |
| 5700.00             | 5700.00   | 5700.00                | 5700.00   | 0.00                       | 14.87        | 30.71                   | 116.02          | 68.92      | 134.95                    | 120.08                 | 9.08                 |         |  |        |  |
| 5800.00             | 5800.00   | 5800.00                | 5800.00   | 0.00                       | 15.13        | 30.71                   | 116.02          | 68.92      | 134.95                    | 119.82                 | 8.92                 |         |  |        |  |
| 5900.00             | 5900.00   | 5900.00                | 5900.00   | 0.00                       | 15.39        | 30.71                   | 116.02          | 68.92      | 134.95                    | 119.56                 | 8.77                 |         |  |        |  |
| 6000.00             | 6000.00   | 6000.00                | 6000.00   | 0.00                       | 15.65        | 30.71                   | 116.02          | 68.92      | 134.95                    | 119.30                 | 8.62                 |         |  |        |  |
| 6100.00             | 6100.00   | 6100.00                | 6100.00   | 0.00                       | 15.91        | 30.71                   | 116.02          | 68.92      | 134.95                    | 119.03                 | 8.48                 |         |  |        |  |
| 6200.00             | 6200.00   | 6200.00                | 6200.00   | 0.00                       | 16.18        | 30.71                   | 116.02          | 68.92      | 134.95                    | 118.77                 | 8.34                 |         |  |        |  |
| 6300.00             | 6300.00   | 6300.00                | 6300.00   | 0.00                       | 16.44        | 30.71                   | 116.02          | 68.92      | 134.95                    | 118.51                 | 8.21                 |         |  |        |  |
| 6400.00             | 6400.00   | 6400.00                | 6400.00   | 0.00                       | 16.70        | 30.71                   | 116.02          | 68.92      | 134.95                    | 118.25                 | 8.08                 |         |  |        |  |
| 6500.00             | 6500.00   | 6500.00                | 6500.00   | 0.00                       | 16.96        | 30.71                   | 116.02          | 68.92      | 134.95                    | 117.99                 | 7.96                 |         |  |        |  |
| 6600.00             | 6600.00   | 6600.00                | 6600.00   | 0.00                       | 17.22        | 30.71                   | 116.02          | 68.92      | 134.95                    | 117.73                 | 7.83                 |         |  |        |  |
| 6700.00             | 6700.00   | 6700.00                | 6700.00   | 0.00                       | 17.49        | 30.71                   | 116.02          | 68.92      | 134.95                    | 117.46                 | 7.72                 |         |  |        |  |
| 6800.00             | 6800.00   | 6800.00                | 6800.00   | 0.00                       | 17.75        | 30.71                   | 116.02          | 68.92      | 134.95                    | 117.20                 | 7.60                 |         |  |        |  |
| 6900.00             | 6900.00   | 6900.00                | 6900.00   | 0.00                       | 18.01        | 30.71                   | 116.02          | 68.92      | 134.95                    | 116.94                 | 7.49                 |         |  |        |  |
| 7000.00             | 7000.00   | 7000.00                | 7000.00   | 0.00                       | 18.27        | 30.71                   | 116.02          | 68.92      | 134.95                    | 116.68                 | 7.39                 |         |  |        |  |
| 7100.00             | 7100.00   | 7100.00                | 7100.00   | 0.00                       | 18.53        | 30.71                   | 116.02          | 68.92      | 134.95                    | 116.42                 | 7.28                 |         |  |        |  |
| 7200.00             | 7200.00   | 7200.00                | 7200.00   | 0.00                       | 18.79        | 30.71                   | 116.02          | 68.92      | 134.95                    | 116.15                 | 7.18                 |         |  |        |  |
| 7300.00             | 7300.00   | 7300.00                | 7300.00   | 0.00                       | 19.06        | 30.71                   | 116.02          | 68.92      | 134.95                    | 115.89                 | 7.08                 |         |  |        |  |
| 7400.00             | 7400.00   | 7400.00                | 7400.00   | 0.00                       | 19.32        | 30.71                   | 116.02          | 68.92      | 134.95                    | 115.63                 | 6.99                 |         |  |        |  |
| 7500.00             | 7500.00   | 7500.00                | 7500.00   | 0.00                       | 19.58        | 30.71                   | 116.02          | 68.92      | 134.95                    | 115.37                 | 6.89                 |         |  |        |  |
| 7600.00             | 7600.00   | 7600.00                | 7600.00   | 0.00                       | 19.84        | 30.71                   | 116.02          | 68.92      | 134.95                    | 115.11                 | 6.80                 |         |  |        |  |
| 7700.00             | 7700.00   | 7700.00                | 7700.00   | 0.00                       | 20.10        | 30.71                   | 116.02          | 68.92      | 134.95                    | 114.85                 | 6.71                 |         |  |        |  |
| 7800.00             | 7800.00   | 7800.00                | 7800.00   | 0.00                       | 20.37        | 30.71                   | 116.02          | 68.92      | 134.95                    | 114.58                 | 6.63                 |         |  |        |  |
| 7900.00             | 7900.00   | 7900.00                | 7900.00   | 0.00                       | 20.63        | 30.71                   | 116.02          | 68.92      | 134.95                    | 114.32                 | 6.54                 |         |  |        |  |
| 8000.00             | 8000.00   | 8000.00                | 8000.00   | 0.00                       | 20.89        | 30.71                   | 116.02          | 68.92      | 134.95                    | 114.06                 | 6.46                 |         |  |        |  |
| 8100.00             | 8100.00   | 8100.00                | 8100.00   | 0.00                       | 21.15        | 30.71                   | 116.02          | 68.92      | 134.95                    | 113.80                 | 6.38                 |         |  |        |  |
| 8200.00             | 8200.00   | 8200.00                | 8200.00   | 0.00                       | 21.41        | 30.71                   | 116.02          | 68.92      | 134.95                    | 113.54                 | 6.30                 |         |  |        |  |
| 8300.00             | 8300.00   | 8300.00                | 8300.00   | 0.00                       | 21.67        | 30.71                   | 116.02          | 68.92      | 134.95                    | 113.27                 | 6.23                 |         |  |        |  |
| 8400.00             | 8400.00   | 8400.00                | 8400.00   | 0.00                       | 21.94        | 30.71                   | 116.02          | 68.92      | 134.95                    | 113.01                 | 6.15                 |         |  |        |  |
| 8500.00             | 8500.00   | 8500.00                | 8500.00   | 0.00                       | 22.20        | 30.71                   | 116.02          | 68.92      | 134.95                    | 112.75                 | 6.08                 |         |  |        |  |
| 8600.00             | 8600.00   | 8600.00                | 8600.00   | 0.00                       | 22.46        | 30.71                   | 116.02          | 68.92      | 134.95                    | 112.49                 | 6.01                 |         |  |        |  |
| 8700.00             | 8700.00   | 8700.00                | 8700.00   | 0.00                       | 22.72        | 30.71                   | 116.02          | 68.92      | 134.95                    | 112.23                 | 5.94                 |         |  |        |  |
| 8800.00             | 8800.00   | 8800.00                | 8800.00   | 0.00                       | 22.98        | 30.71                   | 116.02          | 68.92      | 134.95                    | 111.97                 | 5.87                 |         |  |        |  |
| 8900.00             | 8900.00   | 8900.00                | 8900.00   | 0.00                       | 23.25        | 30.71                   | 116.02          | 68.92      | 134.95                    | 111.70                 | 5.81                 |         |  |        |  |
| 9000.00             | 9000.00   | 9000.00                | 9000.00   | 0.00                       | 23.51        | 30.71                   | 116.02          | 68.92      | 134.95                    | 111.44                 | 5.74                 |         |  |        |  |
| 9100.00             | 9100.00   | 9100.00                | 9100.00   | 0.00                       | 23.77        | 30.71                   | 116.02          | 68.92      | 134.95                    | 111.18                 | 5.68                 |         |  |        |  |
| 9200.00             | 9200.00   | 9200.00                | 9200.00   | 0.00                       | 24.03        | 30.71                   | 116.02          | 68.92      | 134.95                    | 110.92                 | 5.62                 |         |  |        |  |
| 9300.00             | 9300.00   | 9300.00                | 9300.00   | 0.00                       | 24.29        | 30.71                   | 116.02          | 68.92      | 134.95                    | 110.66                 | 5.56                 |         |  |        |  |
| 9400.00             | 9400.00   | 9400.00                | 9400.00   | 0.00                       | 24.55        | 30.71                   | 116.02          | 68.92      | 134.95                    | 110.39                 | 5.50                 |         |  |        |  |
| 9500.00             | 9500.00   | 9500.00                | 9500.00   | 0.00                       | 24.82        | 30.71                   | 116.02          | 68.92      | 134.95                    | 110.13                 | 5.44                 |         |  |        |  |
| 9600.00             | 9600.00   | 9600.00                | 9600.00   | 0.00                       | 25.08        | 30.71                   | 116.02          | 68.92      | 134.95                    | 109.87                 | 5.38                 |         |  |        |  |
| 9700.00             | 9700.00   | 9700.00                | 9700.00   | 0.00                       | 25.34        | 30.71                   | 116.02          | 68.92      | 134.95                    | 109.61                 | 5.33                 |         |  |        |  |
| 9800.00             | 9800.00   | 9800.00                | 9800.00   | 0.00                       | 25.60        | 30.71                   | 116.02          | 68.92      | 134.95                    | 109.35                 | 5.27                 |         |  |        |  |
| 9900.00             | 9900.00   | 9900.00                | 9900.00   | 0.00                       | 25.86        | 30.71                   | 116.02          | 68.92      | 134.95                    | 109.09                 | 5.22                 |         |  |        |  |
| 10000.00            | 10000.00  | 10000.00               | 10000.00  | 0.00                       | 26.12        | 30.71                   | 116.02          | 68.92      | 134.95                    | 108.82                 | 5.17                 |         |  |        |  |
| 10100.00            | 10100.00  | 10100.00               | 10100.00  | 0.00                       | 26.39        | 30.71                   | 116.02          | 68.92      | 134.95                    | 108.56                 | 5.11                 |         |  |        |  |
| 10166.27            | 10166.27  | 10153.72               | 10153.60  | 0.00                       | 26.53        | 29.93                   | 118.99          | 68.50      | 137.88                    | 111.45                 | 5.22                 |         |  |        |  |
| 10175.00            | 10175.00  | 10160.65               | 10160.48  | 0.00                       | 26.55        | 161.06                  | 119.84          | 68.37      | 138.82                    | 112.40                 | 5.25                 |         |  |        |  |
| 10200.00            | 10199.97  | 10180.24               | 10179.84  | 0.00                       | 26.60        | 160.24                  | 122.83          | 67.94      | 143.01                    | 116.64                 | 5.42                 |         |  |        |  |
| 10225.00            | 10224.83  | 10199.26               | 10198.48  | 0.00                       | 26.65        | 159.31                  | 126.54          | 67.41      | 149.38                    | 123.09                 | 5.68                 |         |  |        |  |
| 10250.00            | 10249.50  | 10217.53               | 10216.22  | 0.00                       | 26.69        | 158.28                  | 130.85          | 66.79      | 157.86                    | 131.68                 | 6.03                 |         |  |        |  |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            |          | Continental Resources  |          |                 | Date: 04/23/2012           |        |                 | Time: 17:17:28          |          |          | Page: 12   |         |    |
|---------------------|----------|------------------------|----------|-----------------|----------------------------|--------|-----------------|-------------------------|----------|----------|------------|---------|----|
| Field:              |          | McKenzie County, ND    |          |                 |                            |        |                 |                         |          |          |            |         |    |
| Reference Site:     |          | Taliahassee 3          |          |                 | Co-ordinate(NE) Reference: |        |                 | Well: 3-16H, True North |          |          |            |         |    |
| Reference Well:     |          | 3-16H                  |          |                 | Vertical (TVD) Reference:  |        |                 | GL 1920+KB 41 1941.0    |          |          |            |         |    |
| Reference Wellpath: |          | OH                     |          |                 |                            |        |                 |                         |          |          | Db: Adapti |         |    |
| Site:               |          | Columbus Federal 3     |          |                 |                            |        |                 |                         |          |          |            |         |    |
| Well:               |          | 3-16H                  |          |                 |                            |        |                 |                         |          |          |            |         |    |
| Wellpath:           |          | OH V1 Plan: Plan #1 V1 |          |                 |                            |        |                 | Inter-Site Error:       |          |          | 0.00       |         | ft |
| Reference           |          | Offset                 |          | Semi-Major Axis |                            |        | Offset Location |                         | Ctr-Ctr  | Edge     | Separation | Warning |    |
| MD                  | TVD      | MD                     | TVD      | Ref             | Offset                     | TFO-HS | North           | East                    | Distance | Distance | Factor     |         |    |
| ft                  | ft       | ft                     | ft       | ft              | ft                         | deg    | ft              | ft                      | ft       | ft       |            |         |    |
| 10275.00            | 10273.90 | 10234.89               | 10232.90 | 0.00            | 26.74                      | 157.17 | 135.62          | 66.10                   | 168.34   | 142.28   | 6.46       |         |    |
| 10300.00            | 10297.96 | 10251.23               | 10248.41 | 0.00            | 26.78                      | 155.99 | 140.70          | 65.37                   | 180.70   | 154.78   | 6.97       |         |    |
| 10325.00            | 10321.59 | 10266.46               | 10262.69 | 0.00            | 26.82                      | 154.71 | 145.95          | 64.61                   | 194.79   | 169.02   | 7.56       |         |    |
| 10350.00            | 10344.72 | 10280.53               | 10275.71 | 0.00            | 26.86                      | 153.32 | 151.22          | 63.85                   | 210.47   | 184.85   | 8.22       |         |    |
| 10375.00            | 10367.28 | 10293.40               | 10287.47 | 0.00            | 26.89                      | 151.78 | 156.41          | 63.10                   | 227.58   | 202.12   | 8.94       |         |    |
| 10400.00            | 10389.20 | 10305.08               | 10298.00 | 0.00            | 26.92                      | 150.03 | 161.40          | 62.38                   | 245.99   | 220.67   | 9.71       |         |    |
| 10425.00            | 10410.39 | 10315.58               | 10307.35 | 0.00            | 26.95                      | 148.03 | 166.13          | 61.70                   | 265.53   | 240.34   | 10.54      |         |    |
| 10450.00            | 10430.80 | 10324.92               | 10315.57 | 0.00            | 26.98                      | 145.67 | 170.52          | 61.07                   | 286.08   | 261.00   | 11.41      |         |    |
| 10475.00            | 10450.36 | 10333.15               | 10322.74 | 0.00            | 27.00                      | 142.83 | 174.53          | 60.49                   | 307.48   | 282.51   | 12.31      |         |    |
| 10500.00            | 10469.01 | 10340.32               | 10328.91 | 0.00            | 27.02                      | 139.37 | 178.12          | 59.98                   | 329.63   | 304.74   | 13.24      |         |    |
| 10525.00            | 10486.68 | 10346.46               | 10334.15 | 0.00            | 27.03                      | 135.07 | 181.28          | 59.52                   | 352.40   | 327.57   | 14.19      |         |    |
| 10550.00            | 10503.32 | 10351.63               | 10338.54 | 0.00            | 27.05                      | 129.62 | 184.00          | 59.13                   | 375.67   | 350.89   | 15.16      |         |    |
| 10575.00            | 10518.88 | 10355.89               | 10342.12 | 0.00            | 27.06                      | 122.67 | 186.28          | 58.80                   | 399.35   | 374.60   | 16.14      |         |    |
| 10600.00            | 10533.30 | 10359.28               | 10344.95 | 0.00            | 27.07                      | 113.82 | 188.12          | 58.54                   | 423.33   | 398.60   | 17.12      |         |    |
| 10625.00            | 10546.55 | 10361.85               | 10347.10 | 0.00            | 27.08                      | 102.80 | 189.52          | 58.33                   | 447.54   | 422.81   | 18.10      |         |    |
| 10650.00            | 10558.57 | 10363.64               | 10348.59 | 0.00            | 27.08                      | 89.85  | 190.51          | 58.19                   | 471.87   | 447.14   | 19.07      |         |    |
| 10675.00            | 10569.32 | 10364.71               | 10349.47 | 0.00            | 27.08                      | 76.02  | 191.11          | 58.10                   | 496.27   | 471.51   | 20.04      |         |    |
| 10700.00            | 10578.78 | 10365.09               | 10349.79 | 0.00            | 27.08                      | 62.85  | 191.32          | 58.07                   | 520.65   | 495.85   | 21.00      |         |    |
| 10725.00            | 10586.92 | 10364.82               | 10349.56 | 0.00            | 27.08                      | 51.51  | 191.17          | 58.10                   | 544.94   | 520.10   | 21.93      |         |    |
| 10750.00            | 10593.70 | 10363.93               | 10348.82 | 0.00            | 27.08                      | 42.33  | 190.67          | 58.17                   | 569.09   | 544.19   | 22.86      |         |    |
| 10775.00            | 10599.11 | 10362.45               | 10347.60 | 0.00            | 27.08                      | 35.13  | 189.86          | 58.28                   | 593.02   | 568.07   | 23.76      |         |    |
| 10800.00            | 10603.12 | 10360.42               | 10345.91 | 0.00            | 27.07                      | 29.51  | 188.74          | 58.45                   | 616.70   | 591.67   | 24.64      |         |    |
| 10825.00            | 10605.73 | 10357.85               | 10343.76 | 0.00            | 27.06                      | 25.09  | 187.34          | 58.65                   | 640.05   | 614.95   | 25.50      |         |    |
| 10850.00            | 10606.92 | 10354.78               | 10341.19 | 0.00            | 27.06                      | 21.57  | 185.68          | 58.89                   | 663.03   | 637.85   | 26.34      |         |    |
| 10857.19            | 10607.00 | 10353.81               | 10340.37 | 0.00            | 27.05                      | 20.69  | 185.16          | 58.96                   | 669.56   | 644.37   | 26.57      |         |    |
| 10900.00            | 10607.14 | 10350.00               | 10337.16 | 0.00            | 27.04                      | 20.14  | 183.14          | 59.25                   | 708.55   | 683.24   | 28.00      |         |    |
| 10957.19            | 10607.32 | 10340.64               | 10329.18 | 0.00            | 27.02                      | 18.84  | 178.29          | 59.95                   | 761.12   | 735.65   | 29.89      |         |    |
| 11000.00            | 10607.45 | 10335.29               | 10324.58 | 0.00            | 27.00                      | 15.09  | 175.59          | 60.34                   | 800.84   | 775.28   | 31.34      |         |    |
| 11100.00            | 10607.78 | 10325.00               | 10315.64 | 0.00            | 26.98                      | 5.90   | 170.55          | 61.07                   | 894.75   | 869.04   | 34.80      |         |    |
| 11200.00            | 10608.10 | 10311.11               | 10303.38 | 0.00            | 26.94                      | 355.83 | 164.09          | 62.00                   | 989.64   | 963.80   | 38.29      |         |    |
| 11300.00            | 10608.44 | 10300.00               | 10293.43 | 0.00            | 26.91                      | 346.04 | 159.19          | 62.70                   | 1085.04  | 1059.10  | 41.83      |         |    |
| 11400.00            | 10608.78 | 10288.88               | 10283.35 | 0.00            | 26.88                      | 336.87 | 154.55          | 63.37                   | 1180.58  | 1154.57  | 45.40      |         |    |
| 11500.00            | 10609.12 | 10275.00               | 10270.61 | 0.00            | 26.85                      | 328.82 | 149.10          | 64.16                   | 1276.00  | 1249.92  | 48.91      |         |    |
| 11600.00            | 10609.46 | 10269.03               | 10265.08 | 0.00            | 26.83                      | 321.58 | 146.88          | 64.48                   | 1371.07  | 1344.97  | 52.52      |         |    |
| 11700.00            | 10609.81 | 10260.04               | 10256.69 | 0.00            | 26.81                      | 315.55 | 143.68          | 64.94                   | 1465.66  | 1439.53  | 56.07      |         |    |
| 11800.00            | 10610.16 | 10250.00               | 10247.25 | 0.00            | 26.78                      | 310.58 | 140.30          | 65.43                   | 1559.63  | 1533.45  | 59.57      |         |    |
| 11900.00            | 10610.51 | 10250.00               | 10247.25 | 0.00            | 26.78                      | 305.80 | 140.30          | 65.43                   | 1652.87  | 1626.71  | 63.18      |         |    |
| 12000.00            | 10610.86 | 10236.64               | 10234.57 | 0.00            | 26.74                      | 302.58 | 136.14          | 66.03                   | 1745.18  | 1718.97  | 66.58      |         |    |
| 12100.00            | 10611.20 | 10225.00               | 10223.42 | 0.00            | 26.71                      | 299.79 | 132.83          | 66.50                   | 1836.60  | 1810.35  | 69.95      |         |    |
| 12200.00            | 10611.55 | 10225.00               | 10223.42 | 0.00            | 26.71                      | 296.82 | 132.83          | 66.50                   | 1926.93  | 1900.69  | 73.44      |         |    |
| 12300.00            | 10611.89 | 10225.00               | 10223.42 | 0.00            | 26.71                      | 294.27 | 132.83          | 66.50                   | 2016.20  | 1989.98  | 76.89      |         |    |
| 12400.00            | 10612.23 | 10212.70               | 10211.55 | 0.00            | 26.68                      | 292.68 | 129.64          | 66.96                   | 2104.16  | 2077.90  | 80.11      |         |    |
| 12500.00            | 10612.56 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 291.34 | 126.70          | 67.38                   | 2190.96  | 2164.64  | 83.26      |         |    |
| 12577.81            | 10612.81 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 290.00 | 126.70          | 67.38                   | 2257.46  | 2231.15  | 85.82      |         |    |
| 12600.00            | 10612.88 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 290.00 | 126.70          | 67.38                   | 2276.34  | 2250.04  | 86.54      |         |    |
| 12700.00            | 10613.20 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 290.00 | 126.70          | 67.38                   | 2362.13  | 2335.84  | 89.84      |         |    |
| 12800.00            | 10613.52 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 290.00 | 126.70          | 67.38                   | 2449.01  | 2422.72  | 93.18      |         |    |
| 12900.00            | 10613.85 | 10200.00               | 10199.20 | 0.00            | 26.65                      | 290.00 | 126.70          | 67.38                   | 2536.84  | 2510.57  | 96.56      |         |    |
| 13000.00            | 10614.17 | 10188.43               | 10187.88 | 0.00            | 26.62                      | 290.50 | 124.33          | 67.73                   | 2625.40  | 2599.09  | 99.78      |         |    |
| 13100.00            | 10614.49 | 10185.38               | 10184.89 | 0.00            | 26.61                      | 290.64 | 123.75          | 67.81                   | 2714.80  | 2688.48  | 103.14     |         |    |
| 13200.00            | 10614.81 | 10175.00               | 10174.67 | 0.00            | 26.58                      | 291.09 | 121.95          | 68.07                   | 2804.97  | 2778.61  | 106.40     |         |    |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            |          | Continental Resources  |          | Date:                      |        | 04/23/2012 |                 | Time: |          | 17:17:28             |            | Page:   |  | 13     |  |
|---------------------|----------|------------------------|----------|----------------------------|--------|------------|-----------------|-------|----------|----------------------|------------|---------|--|--------|--|
| Field:              |          | McKenzie County, ND    |          |                            |        |            |                 |       |          |                      |            |         |  |        |  |
| Reference Site:     |          | Tallahassee 3          |          | Co-ordinate(NE) Reference: |        |            |                 | Well: |          | 3-16H, True North    |            |         |  |        |  |
| Reference Well:     |          | 3-16H                  |          | Vertical (TVD) Reference:  |        |            |                 |       |          | GL 1920+KB 41 1941.0 |            |         |  |        |  |
| Reference Wellpath: |          | OH                     |          |                            |        |            |                 |       |          |                      |            | Db:     |  | Adapti |  |
| Site:               |          | Columbus Federal 3     |          |                            |        |            |                 |       |          |                      |            |         |  |        |  |
| Well:               |          | 3-16H                  |          |                            |        |            |                 |       |          |                      |            |         |  |        |  |
| Wellpath:           |          | OH V1 Plan: Plan #1 V1 |          |                            |        |            |                 |       |          | Inter-Site Error:    |            | 0.00    |  | ft     |  |
| Reference           |          | Offset                 |          | Semi-Major Axis            |        |            | Offset Location |       | Ctr-Ctr  | Edge                 | Separation |         |  |        |  |
| MD                  | TVD      | MD                     | TVD      | Ref                        | Offset | TFO-HS     | North           | East  | Distance | Distance             | Factor     | Warning |  |        |  |
| ft                  | ft       | ft                     | ft       | ft                         | ft     | deg        | ft              | ft    | ft       | ft                   |            |         |  |        |  |
| 13300.00            | 10615.13 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 2895.66  | 2869.31              | 109.86     |         |  |        |  |
| 13400.00            | 10615.46 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 2986.95  | 2960.60              | 113.34     |         |  |        |  |
| 13500.00            | 10615.78 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3078.78  | 3052.43              | 116.84     |         |  |        |  |
| 13600.00            | 10616.10 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3171.11  | 3144.76              | 120.35     |         |  |        |  |
| 13700.00            | 10616.42 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3263.88  | 3237.54              | 123.89     |         |  |        |  |
| 13800.00            | 10616.75 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3357.08  | 3330.73              | 127.44     |         |  |        |  |
| 13900.00            | 10617.07 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3450.65  | 3424.31              | 131.00     |         |  |        |  |
| 14000.00            | 10617.39 | 10175.00               | 10174.67 | 0.00                       | 26.58  | 291.09     | 121.95          | 68.07 | 3544.57  | 3518.23              | 134.58     |         |  |        |  |
| 14100.00            | 10617.71 | 10163.70               | 10163.50 | 0.00                       | 26.55  | 291.58     | 120.25          | 68.31 | 3638.67  | 3612.30              | 137.98     |         |  |        |  |
| 14200.00            | 10618.03 | 10162.15               | 10161.96 | 0.00                       | 26.55  | 291.65     | 120.04          | 68.34 | 3733.18  | 3706.80              | 141.55     |         |  |        |  |
| 14300.00            | 10618.36 | 10160.67               | 10160.50 | 0.00                       | 26.55  | 291.71     | 119.85          | 68.37 | 3827.95  | 3801.58              | 145.13     |         |  |        |  |
| 14400.00            | 10618.68 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 3923.09  | 3896.68              | 148.55     |         |  |        |  |
| 14500.00            | 10619.00 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4018.33  | 3991.92              | 152.16     |         |  |        |  |
| 14600.00            | 10619.32 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4113.80  | 4087.39              | 155.78     |         |  |        |  |
| 14700.00            | 10619.64 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4209.48  | 4183.07              | 159.41     |         |  |        |  |
| 14800.00            | 10619.97 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4305.35  | 4278.94              | 163.04     |         |  |        |  |
| 14900.00            | 10620.29 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4401.41  | 4375.00              | 166.68     |         |  |        |  |
| 15000.00            | 10620.61 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4497.64  | 4471.23              | 170.33     |         |  |        |  |
| 15100.00            | 10620.93 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4594.03  | 4567.63              | 173.99     |         |  |        |  |
| 15200.00            | 10621.26 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4690.57  | 4664.17              | 177.65     |         |  |        |  |
| 15300.00            | 10621.58 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4787.26  | 4760.85              | 181.31     |         |  |        |  |
| 15400.00            | 10621.90 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4884.08  | 4857.67              | 184.98     |         |  |        |  |
| 15500.00            | 10622.22 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 4981.02  | 4954.62              | 188.65     |         |  |        |  |
| 15600.00            | 10622.54 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5078.08  | 5051.68              | 192.33     |         |  |        |  |
| 15700.00            | 10622.87 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5175.26  | 5148.85              | 196.02     |         |  |        |  |
| 15800.00            | 10623.19 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5272.54  | 5246.14              | 199.70     |         |  |        |  |
| 15900.00            | 10623.51 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5369.92  | 5343.52              | 203.39     |         |  |        |  |
| 16000.00            | 10623.83 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5467.39  | 5440.99              | 207.09     |         |  |        |  |
| 16100.00            | 10624.15 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5564.96  | 5538.55              | 210.79     |         |  |        |  |
| 16200.00            | 10624.48 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5662.60  | 5636.20              | 214.49     |         |  |        |  |
| 16300.00            | 10624.80 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5760.33  | 5733.93              | 218.19     |         |  |        |  |
| 16400.00            | 10625.12 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5858.14  | 5831.74              | 221.90     |         |  |        |  |
| 16500.00            | 10625.44 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 5956.02  | 5929.62              | 225.61     |         |  |        |  |
| 16600.00            | 10625.77 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 6053.97  | 6027.57              | 229.32     |         |  |        |  |
| 16700.00            | 10626.09 | 10150.00               | 10149.91 | 0.00                       | 26.52  | 292.18     | 118.58          | 68.56 | 6151.98  | 6125.58              | 233.04     |         |  |        |  |
| 16800.00            | 10626.41 | 10138.58               | 10138.54 | 0.00                       | 26.49  | 292.67     | 117.50          | 68.71 | 6249.91  | 6223.50              | 236.64     |         |  |        |  |
| 16900.00            | 10626.73 | 10138.05               | 10138.01 | 0.00                       | 26.49  | 292.69     | 117.46          | 68.72 | 6348.03  | 6321.62              | 240.35     |         |  |        |  |
| 17000.00            | 10627.05 | 10137.53               | 10137.49 | 0.00                       | 26.49  | 292.72     | 117.42          | 68.72 | 6446.21  | 6419.80              | 244.06     |         |  |        |  |
| 17100.00            | 10627.38 | 10137.03               | 10136.99 | 0.00                       | 26.48  | 292.74     | 117.38          | 68.73 | 6544.45  | 6518.03              | 247.78     |         |  |        |  |
| 17200.00            | 10627.70 | 10136.54               | 10136.51 | 0.00                       | 26.48  | 292.76     | 117.34          | 68.73 | 6642.73  | 6616.32              | 251.50     |         |  |        |  |
| 17300.00            | 10628.02 | 10136.07               | 10136.03 | 0.00                       | 26.48  | 292.78     | 117.30          | 68.74 | 6741.07  | 6714.66              | 255.22     |         |  |        |  |
| 17400.00            | 10628.34 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 6839.58  | 6813.16              | 258.83     |         |  |        |  |
| 17500.00            | 10628.66 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 6938.00  | 6911.58              | 262.55     |         |  |        |  |
| 17600.00            | 10628.99 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7036.47  | 7010.04              | 266.28     |         |  |        |  |
| 17700.00            | 10629.31 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7134.97  | 7108.55              | 270.01     |         |  |        |  |
| 17800.00            | 10629.63 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7233.52  | 7207.10              | 273.74     |         |  |        |  |
| 17900.00            | 10629.95 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7332.11  | 7305.68              | 277.47     |         |  |        |  |
| 18000.00            | 10630.28 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7430.73  | 7404.31              | 281.20     |         |  |        |  |
| 18100.00            | 10630.60 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7529.39  | 7502.97              | 284.94     |         |  |        |  |
| 18200.00            | 10630.92 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7628.09  | 7601.67              | 288.67     |         |  |        |  |
| 18300.00            | 10631.24 | 10125.00               | 10124.99 | 0.00                       | 26.45  | 293.26     | 116.60          | 68.84 | 7726.82  | 7700.39              | 292.41     |         |  |        |  |

LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |       |        |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|-------|--------|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page: | 14     |
| Field:              | McKenzie County, ND   |                            |                         |       |          |       |        |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |       |        |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |       |        |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db:   | Adapti |

|           |                        |                   |      |    |  |  |  |
|-----------|------------------------|-------------------|------|----|--|--|--|
| Site:     | Columbus Federal 3     |                   |      |    |  |  |  |
| Well:     | 3-16H                  |                   |      |    |  |  |  |
| Wellpath: | OH V1 Plan: Plan #1 V1 | Inter-Site Error: | 0.00 | ft |  |  |  |

| Reference |          | Offset   |          | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|----------|----------|----------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD      | MD       | TVD      | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft       | ft       | ft       | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 18400.00  | 10631.56 | 10125.00 | 10124.99 | 0.00            | 26.45  | 293.26 | 116.60          | 68.84 | 7825.58  | 7799.16  | 296.15     |         |
| 18500.00  | 10631.89 | 10125.00 | 10124.99 | 0.00            | 26.45  | 293.26 | 116.60          | 68.84 | 7924.37  | 7897.95  | 299.88     |         |
| 18535.29  | 10632.00 | 10125.00 | 10124.99 | 0.00            | 26.45  | 293.26 | 116.60          | 68.84 | 7959.24  | 7932.82  | 301.20     |         |

|           |                        |                   |      |    |  |  |  |
|-----------|------------------------|-------------------|------|----|--|--|--|
| Site:     | Tallahassee 2          |                   |      |    |  |  |  |
| Well:     | 2-16H                  |                   |      |    |  |  |  |
| Wellpath: | OH V4 Plan: Plan #1 V1 | Inter-Site Error: | 0.00 | ft |  |  |  |

| Reference |         | Offset  |         | Semi-Major Axis |        |        | Offset Location |       | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|---------|---------|---------|-----------------|--------|--------|-----------------|-------|----------|----------|------------|---------|
| MD        | TVD     | MD      | TVD     | Ref             | Offset | TFO-HS | North           | East  | Distance | Distance | Factor     |         |
| ft        | ft      | ft      | ft      | ft              | ft     | deg    | ft              | ft    | ft       | ft       |            |         |
| 0.00      | 0.00    | 0.00    | 0.00    | 0.00            | 0.00   | 30.67  | 38.71           | 22.95 | 45.00    |          |            | No Data |
| 100.00    | 100.00  | 100.00  | 100.00  | 0.00            | 0.21   | 30.67  | 38.71           | 22.95 | 45.00    | 44.79    | 217.37     |         |
| 200.00    | 200.00  | 200.00  | 200.00  | 0.00            | 0.47   | 30.67  | 38.71           | 22.95 | 45.00    | 44.53    | 95.99      |         |
| 300.00    | 300.00  | 300.00  | 300.00  | 0.00            | 0.73   | 30.67  | 38.71           | 22.95 | 45.00    | 44.27    | 61.59      |         |
| 400.00    | 400.00  | 400.00  | 400.00  | 0.00            | 0.99   | 30.67  | 38.71           | 22.95 | 45.00    | 44.01    | 45.34      |         |
| 500.00    | 500.00  | 500.00  | 500.00  | 0.00            | 1.25   | 30.67  | 38.71           | 22.95 | 45.00    | 43.75    | 35.88      |         |
| 600.00    | 600.00  | 600.00  | 600.00  | 0.00            | 1.52   | 30.67  | 38.71           | 22.95 | 45.00    | 43.48    | 29.68      |         |
| 700.00    | 700.00  | 700.00  | 700.00  | 0.00            | 1.78   | 30.67  | 38.71           | 22.95 | 45.00    | 43.22    | 25.31      |         |
| 800.00    | 800.00  | 800.00  | 800.00  | 0.00            | 2.04   | 30.67  | 38.71           | 22.95 | 45.00    | 42.96    | 22.06      |         |
| 900.00    | 900.00  | 900.00  | 900.00  | 0.00            | 2.30   | 30.67  | 38.71           | 22.95 | 45.00    | 42.70    | 19.55      |         |
| 1000.00   | 1000.00 | 1000.00 | 1000.00 | 0.00            | 2.56   | 30.67  | 38.71           | 22.95 | 45.00    | 42.44    | 17.56      |         |
| 1100.00   | 1100.00 | 1100.00 | 1100.00 | 0.00            | 2.82   | 30.67  | 38.71           | 22.95 | 45.00    | 42.18    | 15.93      |         |
| 1200.00   | 1200.00 | 1200.00 | 1200.00 | 0.00            | 3.09   | 30.67  | 38.71           | 22.95 | 45.00    | 41.91    | 14.58      |         |
| 1300.00   | 1300.00 | 1300.00 | 1300.00 | 0.00            | 3.35   | 30.67  | 38.71           | 22.95 | 45.00    | 41.65    | 13.44      |         |
| 1400.00   | 1400.00 | 1400.00 | 1400.00 | 0.00            | 3.61   | 30.67  | 38.71           | 22.95 | 45.00    | 41.39    | 12.46      |         |
| 1500.00   | 1500.00 | 1500.00 | 1500.00 | 0.00            | 3.87   | 30.67  | 38.71           | 22.95 | 45.00    | 41.13    | 11.62      |         |
| 1600.00   | 1600.00 | 1600.00 | 1600.00 | 0.00            | 4.13   | 30.67  | 38.71           | 22.95 | 45.00    | 40.87    | 10.89      |         |
| 1700.00   | 1700.00 | 1700.00 | 1700.00 | 0.00            | 4.40   | 30.67  | 38.71           | 22.95 | 45.00    | 40.60    | 10.24      |         |
| 1800.00   | 1800.00 | 1800.00 | 1800.00 | 0.00            | 4.66   | 30.67  | 38.71           | 22.95 | 45.00    | 40.34    | 9.66       |         |
| 1900.00   | 1900.00 | 1900.00 | 1900.00 | 0.00            | 4.92   | 30.67  | 38.71           | 22.95 | 45.00    | 40.08    | 9.15       |         |
| 2000.00   | 2000.00 | 2000.00 | 2000.00 | 0.00            | 5.18   | 30.67  | 38.71           | 22.95 | 45.00    | 39.82    | 8.69       |         |
| 2100.00   | 2100.00 | 2100.00 | 2100.00 | 0.00            | 5.44   | 30.67  | 38.71           | 22.95 | 45.00    | 39.56    | 8.27       |         |
| 2200.00   | 2200.00 | 2200.00 | 2200.00 | 0.00            | 5.70   | 30.67  | 38.71           | 22.95 | 45.00    | 39.30    | 7.89       |         |
| 2300.00   | 2300.00 | 2300.00 | 2300.00 | 0.00            | 5.97   | 30.67  | 38.71           | 22.95 | 45.00    | 39.03    | 7.54       |         |
| 2400.00   | 2400.00 | 2400.00 | 2400.00 | 0.00            | 6.23   | 30.67  | 38.71           | 22.95 | 45.00    | 38.77    | 7.23       |         |
| 2500.00   | 2500.00 | 2500.00 | 2500.00 | 0.00            | 6.49   | 30.67  | 38.71           | 22.95 | 45.00    | 38.51    | 6.93       |         |
| 2600.00   | 2600.00 | 2600.00 | 2600.00 | 0.00            | 6.75   | 30.67  | 38.71           | 22.95 | 45.00    | 38.25    | 6.66       |         |
| 2700.00   | 2700.00 | 2700.00 | 2700.00 | 0.00            | 7.01   | 30.67  | 38.71           | 22.95 | 45.00    | 37.99    | 6.42       |         |
| 2800.00   | 2800.00 | 2800.00 | 2800.00 | 0.00            | 7.28   | 30.67  | 38.71           | 22.95 | 45.00    | 37.72    | 6.19       |         |
| 2900.00   | 2900.00 | 2900.00 | 2900.00 | 0.00            | 7.54   | 30.67  | 38.71           | 22.95 | 45.00    | 37.46    | 5.97       |         |
| 3000.00   | 3000.00 | 3000.00 | 3000.00 | 0.00            | 7.80   | 30.67  | 38.71           | 22.95 | 45.00    | 37.20    | 5.77       |         |
| 3100.00   | 3100.00 | 3100.00 | 3100.00 | 0.00            | 8.06   | 30.67  | 38.71           | 22.95 | 45.00    | 36.94    | 5.58       |         |
| 3200.00   | 3200.00 | 3200.00 | 3200.00 | 0.00            | 8.32   | 30.67  | 38.71           | 22.95 | 45.00    | 36.68    | 5.41       |         |
| 3300.00   | 3300.00 | 3300.00 | 3300.00 | 0.00            | 8.58   | 30.67  | 38.71           | 22.95 | 45.00    | 36.42    | 5.24       |         |
| 3400.00   | 3400.00 | 3400.00 | 3400.00 | 0.00            | 8.85   | 30.67  | 38.71           | 22.95 | 45.00    | 36.15    | 5.09       |         |
| 3500.00   | 3500.00 | 3500.00 | 3500.00 | 0.00            | 9.11   | 30.67  | 38.71           | 22.95 | 45.00    | 35.89    | 4.94       |         |
| 3600.00   | 3600.00 | 3600.00 | 3600.00 | 0.00            | 9.37   | 30.67  | 38.71           | 22.95 | 45.00    | 35.63    | 4.80       |         |
| 3700.00   | 3700.00 | 3700.00 | 3700.00 | 0.00            | 9.63   | 30.67  | 38.71           | 22.95 | 45.00    | 35.37    | 4.67       |         |
| 3800.00   | 3800.00 | 3800.00 | 3800.00 | 0.00            | 9.89   | 30.67  | 38.71           | 22.95 | 45.00    | 35.11    | 4.55       |         |
| 3900.00   | 3900.00 | 3900.00 | 3900.00 | 0.00            | 10.16  | 30.67  | 38.71           | 22.95 | 45.00    | 34.85    | 4.43       |         |
| 4000.00   | 4000.00 | 4000.00 | 4000.00 | 0.00            | 10.42  | 30.67  | 38.71           | 22.95 | 45.00    | 34.58    | 4.32       |         |

LEAM Drilling Systems, Inc.  
Anticollision Report

|                                  |         |                            |         |                           |        |                 |       |                  |               |                   |         |
|----------------------------------|---------|----------------------------|---------|---------------------------|--------|-----------------|-------|------------------|---------------|-------------------|---------|
| Company: Continental Resources   |         | Date: 04/23/2012           |         | Time: 17:17:28            |        | Page: 15        |       |                  |               |                   |         |
| Field: McKenzie County, ND       |         |                            |         |                           |        |                 |       |                  |               |                   |         |
| Reference Site: Tallahassee 3    |         | Co-ordinate(NE) Reference: |         | Well: 3-16H, True North   |        |                 |       |                  |               |                   |         |
| Reference Well: 3-16H            |         | Vertical (TVD) Reference:  |         | GL 1920+KB 41 1941.0      |        |                 |       |                  |               |                   |         |
| Reference Wellpath: OH           |         |                            |         |                           |        | Db: Adapti      |       |                  |               |                   |         |
|                                  |         |                            |         |                           |        |                 |       |                  |               |                   |         |
| Site: Tallahassee 2              |         |                            |         |                           |        |                 |       |                  |               |                   |         |
| Well: 2-16H                      |         |                            |         |                           |        |                 |       |                  |               |                   |         |
| Wellpath: OH V4 Plan: Plan #1 V1 |         |                            |         | Inter-Site Error: 0.00 ft |        |                 |       |                  |               |                   |         |
| Reference                        |         | Offset                     |         | Semi-Major Axis           |        | Offset Location |       | Ctr-Ctr Distance | Edge Distance | Separation Factor | Warning |
| MD                               | TVD     | MD                         | TVD     | Ref                       | Offset | TFO-HS          | North |                  |               |                   |         |
| ft                               | ft      | ft                         | ft      | ft                        | ft     | deg             | ft    | ft               | ft            | ft                |         |
| 4100.00                          | 4100.00 | 4100.00                    | 4100.00 | 0.00                      | 10.68  | 30.67           | 38.71 | 22.95            | 45.00         | 34.32             | 4.21    |
| 4200.00                          | 4200.00 | 4200.00                    | 4200.00 | 0.00                      | 10.94  | 30.67           | 38.71 | 22.95            | 45.00         | 34.06             | 4.11    |
| 4300.00                          | 4300.00 | 4300.00                    | 4300.00 | 0.00                      | 11.20  | 30.67           | 38.71 | 22.95            | 45.00         | 33.80             | 4.02    |
| 4400.00                          | 4400.00 | 4400.00                    | 4400.00 | 0.00                      | 11.46  | 30.67           | 38.71 | 22.95            | 45.00         | 33.54             | 3.93    |
| 4500.00                          | 4500.00 | 4500.00                    | 4500.00 | 0.00                      | 11.73  | 30.67           | 38.71 | 22.95            | 45.00         | 33.27             | 3.84    |
| 4600.00                          | 4600.00 | 4600.00                    | 4600.00 | 0.00                      | 11.99  | 30.67           | 38.71 | 22.95            | 45.00         | 33.01             | 3.75    |
| 4700.00                          | 4700.00 | 4700.00                    | 4700.00 | 0.00                      | 12.25  | 30.67           | 38.71 | 22.95            | 45.00         | 32.75             | 3.67    |
| 4800.00                          | 4800.00 | 4800.00                    | 4800.00 | 0.00                      | 12.51  | 30.67           | 38.71 | 22.95            | 45.00         | 32.49             | 3.60    |
| 4900.00                          | 4900.00 | 4900.00                    | 4900.00 | 0.00                      | 12.77  | 30.67           | 38.71 | 22.95            | 45.00         | 32.23             | 3.52    |
| 5000.00                          | 5000.00 | 5000.00                    | 5000.00 | 0.00                      | 13.03  | 30.67           | 38.71 | 22.95            | 45.00         | 31.97             | 3.45    |
| 5100.00                          | 5100.00 | 5100.00                    | 5100.00 | 0.00                      | 13.30  | 30.67           | 38.71 | 22.95            | 45.00         | 31.70             | 3.38    |
| 5200.00                          | 5200.00 | 5200.00                    | 5200.00 | 0.00                      | 13.56  | 30.67           | 38.71 | 22.95            | 45.00         | 31.44             | 3.32    |
| 5300.00                          | 5300.00 | 5300.00                    | 5300.00 | 0.00                      | 13.82  | 30.67           | 38.71 | 22.95            | 45.00         | 31.18             | 3.26    |
| 5400.00                          | 5400.00 | 5400.00                    | 5400.00 | 0.00                      | 14.08  | 30.67           | 38.71 | 22.95            | 45.00         | 30.92             | 3.20    |
| 5500.00                          | 5500.00 | 5500.00                    | 5500.00 | 0.00                      | 14.34  | 30.67           | 38.71 | 22.95            | 45.00         | 30.66             | 3.14    |
| 5600.00                          | 5600.00 | 5600.00                    | 5600.00 | 0.00                      | 14.61  | 30.67           | 38.71 | 22.95            | 45.00         | 30.39             | 3.08    |
| 5700.00                          | 5700.00 | 5700.00                    | 5700.00 | 0.00                      | 14.87  | 30.67           | 38.71 | 22.95            | 45.00         | 30.13             | 3.03    |
| 5800.00                          | 5800.00 | 5800.00                    | 5800.00 | 0.00                      | 15.13  | 30.67           | 38.71 | 22.95            | 45.00         | 29.87             | 2.97    |
| 5900.00                          | 5900.00 | 5900.00                    | 5900.00 | 0.00                      | 15.39  | 30.67           | 38.71 | 22.95            | 45.00         | 29.61             | 2.92    |
| 6000.00                          | 6000.00 | 6000.00                    | 6000.00 | 0.00                      | 15.65  | 30.67           | 38.71 | 22.95            | 45.00         | 29.35             | 2.87    |
| 6100.00                          | 6100.00 | 6100.00                    | 6100.00 | 0.00                      | 15.91  | 30.67           | 38.71 | 22.95            | 45.00         | 29.09             | 2.83    |
| 6200.00                          | 6200.00 | 6200.00                    | 6200.00 | 0.00                      | 16.18  | 30.67           | 38.71 | 22.95            | 45.00         | 28.82             | 2.78    |
| 6300.00                          | 6300.00 | 6300.00                    | 6300.00 | 0.00                      | 16.44  | 30.67           | 38.71 | 22.95            | 45.00         | 28.56             | 2.74    |
| 6400.00                          | 6400.00 | 6400.00                    | 6400.00 | 0.00                      | 16.70  | 30.67           | 38.71 | 22.95            | 45.00         | 28.30             | 2.69    |
| 6500.00                          | 6500.00 | 6500.00                    | 6500.00 | 0.00                      | 16.96  | 30.67           | 38.71 | 22.95            | 45.00         | 28.04             | 2.65    |
| 6600.00                          | 6600.00 | 6600.00                    | 6600.00 | 0.00                      | 17.22  | 30.67           | 38.71 | 22.95            | 45.00         | 27.78             | 2.61    |
| 6700.00                          | 6700.00 | 6700.00                    | 6700.00 | 0.00                      | 17.49  | 30.67           | 38.71 | 22.95            | 45.00         | 27.51             | 2.57    |
| 6800.00                          | 6800.00 | 6800.00                    | 6800.00 | 0.00                      | 17.75  | 30.67           | 38.71 | 22.95            | 45.00         | 27.25             | 2.54    |
| 6900.00                          | 6900.00 | 6900.00                    | 6900.00 | 0.00                      | 18.01  | 30.67           | 38.71 | 22.95            | 45.00         | 26.99             | 2.50    |
| 7000.00                          | 7000.00 | 7000.00                    | 7000.00 | 0.00                      | 18.27  | 30.67           | 38.71 | 22.95            | 45.00         | 26.73             | 2.46    |
| 7100.00                          | 7100.00 | 7100.00                    | 7100.00 | 0.00                      | 18.53  | 30.67           | 38.71 | 22.95            | 45.00         | 26.47             | 2.43    |
| 7200.00                          | 7200.00 | 7200.00                    | 7200.00 | 0.00                      | 18.79  | 30.67           | 38.71 | 22.95            | 45.00         | 26.21             | 2.39    |
| 7300.00                          | 7300.00 | 7300.00                    | 7300.00 | 0.00                      | 19.06  | 30.67           | 38.71 | 22.95            | 45.00         | 25.94             | 2.36    |
| 7400.00                          | 7400.00 | 7400.00                    | 7400.00 | 0.00                      | 19.32  | 30.67           | 38.71 | 22.95            | 45.00         | 25.68             | 2.33    |
| 7500.00                          | 7500.00 | 7500.00                    | 7500.00 | 0.00                      | 19.58  | 30.67           | 38.71 | 22.95            | 45.00         | 25.42             | 2.30    |
| 7600.00                          | 7600.00 | 7600.00                    | 7600.00 | 0.00                      | 19.84  | 30.67           | 38.71 | 22.95            | 45.00         | 25.16             | 2.27    |
| 7700.00                          | 7700.00 | 7700.00                    | 7700.00 | 0.00                      | 20.10  | 30.67           | 38.71 | 22.95            | 45.00         | 24.90             | 2.24    |
| 7800.00                          | 7800.00 | 7800.00                    | 7800.00 | 0.00                      | 20.37  | 30.67           | 38.71 | 22.95            | 45.00         | 24.63             | 2.21    |
| 7900.00                          | 7900.00 | 7900.00                    | 7900.00 | 0.00                      | 20.63  | 30.67           | 38.71 | 22.95            | 45.00         | 24.37             | 2.18    |
| 8000.00                          | 8000.00 | 8000.00                    | 8000.00 | 0.00                      | 20.89  | 30.67           | 38.71 | 22.95            | 45.00         | 24.11             | 2.15    |
| 8100.00                          | 8100.00 | 8100.00                    | 8100.00 | 0.00                      | 21.15  | 30.67           | 38.71 | 22.95            | 45.00         | 23.85             | 2.13    |
| 8200.00                          | 8200.00 | 8200.00                    | 8200.00 | 0.00                      | 21.41  | 30.67           | 38.71 | 22.95            | 45.00         | 23.59             | 2.10    |
| 8300.00                          | 8300.00 | 8300.00                    | 8300.00 | 0.00                      | 21.67  | 30.67           | 38.71 | 22.95            | 45.00         | 23.33             | 2.08    |
| 8400.00                          | 8400.00 | 8400.00                    | 8400.00 | 0.00                      | 21.94  | 30.67           | 38.71 | 22.95            | 45.00         | 23.06             | 2.05    |
| 8500.00                          | 8500.00 | 8500.00                    | 8500.00 | 0.00                      | 22.20  | 30.67           | 38.71 | 22.95            | 45.00         | 22.80             | 2.03    |
| 8600.00                          | 8600.00 | 8600.00                    | 8600.00 | 0.00                      | 22.46  | 30.67           | 38.71 | 22.95            | 45.00         | 22.54             | 2.00    |
| 8700.00                          | 8700.00 | 8700.00                    | 8700.00 | 0.00                      | 22.72  | 30.67           | 38.71 | 22.95            | 45.00         | 22.28             | 1.98    |
| 8800.00                          | 8800.00 | 8800.00                    | 8800.00 | 0.00                      | 22.98  | 30.67           | 38.71 | 22.95            | 45.00         | 22.02             | 1.96    |
| 8900.00                          | 8900.00 | 8900.00                    | 8900.00 | 0.00                      | 23.25  | 30.67           | 38.71 | 22.95            | 45.00         | 21.76             | 1.94    |
| 9000.00                          | 9000.00 | 9000.00                    | 9000.00 | 0.00                      | 23.51  | 30.67           | 38.71 | 22.95            | 45.00         | 21.49             | 1.91    |
| 9100.00                          | 9100.00 | 9100.00                    | 9100.00 | 0.00                      | 23.77  | 30.67           | 38.71 | 22.95            | 45.00         | 21.23             | 1.89    |

LEAM Drilling Systems, Inc.  
Anticollision Report

| Company:            | Continental Resources  |          | Date:                      | 04/23/2012      |        | Time:  | 17:17:28             |         | Page:             | 16       |            |         |
|---------------------|------------------------|----------|----------------------------|-----------------|--------|--------|----------------------|---------|-------------------|----------|------------|---------|
| Field:              | McKenzie County, ND    |          |                            |                 |        |        |                      |         |                   |          |            |         |
| Reference Site:     | Tallahassee 3          |          | Co-ordinate(NE) Reference: |                 |        | Well:  | 3-16H, True North    |         |                   |          |            |         |
| Reference Well:     | 3-16H                  |          | Vertical (TVD) Reference:  |                 |        |        | GL 1920+KB 41 1941.0 |         |                   |          |            |         |
| Reference Wellpath: | OH                     |          |                            |                 |        |        |                      |         | Db:               | Adapti   |            |         |
| Site:               | Tallahassee 2          |          |                            |                 |        |        |                      |         |                   |          |            |         |
| Well:               | 2-16H                  |          |                            |                 |        |        |                      |         |                   |          |            |         |
| Wellpath:           | OH V4 Plan: Plan #1 V1 |          |                            |                 |        |        |                      |         | Inter-Site Error: | 0.00     | ft         |         |
| Reference           |                        | Offset   |                            | Semi-Major Axis |        |        | Offset Location      |         | Ctr-Ctr           | Edge     | Separation | Warning |
| MD                  | TVD                    | MD       | TVD                        | Ref             | Offset | TFO-HS | North                | East    | Distance          | Distance | Factor     |         |
| ft                  | ft                     | ft       | ft                         | ft              | ft     | deg    | ft                   | ft      | ft                | ft       |            |         |
| 9200.00             | 9200.00                | 9200.00  | 9200.00                    | 0.00            | 24.03  | 30.67  | 38.71                | 22.95   | 45.00             | 20.97    | 1.87       |         |
| 9300.00             | 9300.00                | 9300.00  | 9300.00                    | 0.00            | 24.29  | 30.67  | 38.71                | 22.95   | 45.00             | 20.71    | 1.85       |         |
| 9400.00             | 9400.00                | 9400.00  | 9400.00                    | 0.00            | 24.55  | 30.67  | 38.71                | 22.95   | 45.00             | 20.45    | 1.83       |         |
| 9500.00             | 9500.00                | 9500.00  | 9500.00                    | 0.00            | 24.82  | 30.67  | 38.71                | 22.95   | 45.00             | 20.18    | 1.81       |         |
| 9600.00             | 9600.00                | 9600.00  | 9600.00                    | 0.00            | 25.08  | 30.67  | 38.71                | 22.95   | 45.00             | 19.92    | 1.79       |         |
| 9700.00             | 9700.00                | 9700.00  | 9700.00                    | 0.00            | 25.34  | 30.67  | 38.71                | 22.95   | 45.00             | 19.66    | 1.78       |         |
| 9800.00             | 9800.00                | 9800.00  | 9800.00                    | 0.00            | 25.60  | 30.67  | 38.71                | 22.95   | 45.00             | 19.40    | 1.76       |         |
| 9900.00             | 9900.00                | 9900.00  | 9900.00                    | 0.00            | 25.86  | 30.67  | 38.71                | 22.95   | 45.00             | 19.14    | 1.74       |         |
| 10000.00            | 10000.00               | 10000.00 | 10000.00                   | 0.00            | 26.12  | 30.67  | 38.71                | 22.95   | 45.00             | 18.88    | 1.72       |         |
| 10100.00            | 10100.00               | 10100.00 | 10100.00                   | 0.00            | 26.39  | 30.67  | 38.71                | 22.95   | 45.00             | 18.61    | 1.71       |         |
| 10166.27            | 10166.27               | 10172.81 | 10172.51                   | 0.00            | 26.58  | 32.12  | 33.43                | 20.98   | 39.96             | 13.66    | 1.52       |         |
| 10175.00            | 10175.00               | 10182.25 | 10181.81                   | 0.00            | 26.60  | 164.08 | 31.93                | 20.42   | 38.59             | 12.35    | 1.47       | Level 3 |
| 10200.00            | 10199.97               | 10209.05 | 10208.01                   | 0.00            | 26.67  | 166.81 | 26.65                | 18.45   | 34.61             | 8.56     | 1.33       | Level 3 |
| 10225.00            | 10224.83               | 10235.50 | 10233.48                   | 0.00            | 26.74  | 171.58 | 19.98                | 15.97   | 30.68             | 4.86     | 1.19       | Level 2 |
| 10250.00            | 10249.50               | 10261.56 | 10258.12                   | 0.00            | 26.81  | 179.07 | 12.03                | 13.00   | 27.05             | 1.42     | 1.06       | Level 2 |
| 10275.00            | 10273.90               | 10287.23 | 10281.85                   | 0.00            | 26.88  | 189.99 | 2.88                 | 9.58    | 24.16             | -1.46    | 0.94       | Level 1 |
| 10300.00            | 10297.96               | 10312.48 | 10304.60                   | 0.00            | 26.94  | 204.32 | -7.37                | 5.76    | 22.68             | -3.27    | 0.87       | Level 1 |
| 10325.00            | 10321.59               | 10337.31 | 10326.32                   | 0.00            | 27.01  | 220.30 | -18.63               | 1.55    | 23.26             | -3.28    | 0.88       | Level 1 |
| 10350.00            | 10344.72               | 10361.71 | 10346.99                   | 0.00            | 27.07  | 235.05 | -30.79               | -2.98   | 26.10             | -0.89    | 0.97       | Level 1 |
| 10375.00            | 10367.28               | 10385.69 | 10366.56                   | 0.00            | 27.14  | 246.79 | -43.77               | -7.83   | 30.88             | 3.75     | 1.14       | Level 2 |
| 10400.00            | 10389.20               | 10409.25 | 10385.02                   | 0.00            | 27.20  | 255.46 | -57.47               | -12.94  | 37.11             | 10.05    | 1.37       | Level 3 |
| 10425.00            | 10410.39               | 10432.40 | 10402.38                   | 0.00            | 27.26  | 261.77 | -71.81               | -18.29  | 44.37             | 17.48    | 1.65       |         |
| 10450.00            | 10430.80               | 10455.14 | 10418.63                   | 0.00            | 27.33  | 266.42 | -86.72               | -23.86  | 52.41             | 25.70    | 1.96       |         |
| 10475.00            | 10450.36               | 10477.50 | 10433.79                   | 0.00            | 27.39  | 269.94 | -102.12              | -29.60  | 61.05             | 34.52    | 2.30       |         |
| 10500.00            | 10469.01               | 10499.49 | 10447.86                   | 0.00            | 27.45  | 272.67 | -117.95              | -35.51  | 70.18             | 43.79    | 2.66       |         |
| 10525.00            | 10486.68               | 10521.13 | 10460.87                   | 0.00            | 27.51  | 274.87 | -134.15              | -41.56  | 79.72             | 53.43    | 3.03       |         |
| 10550.00            | 10503.32               | 10542.45 | 10472.83                   | 0.00            | 27.57  | 276.69 | -150.67              | -47.72  | 89.60             | 63.38    | 3.42       |         |
| 10575.00            | 10518.88               | 10563.45 | 10483.78                   | 0.00            | 27.63  | 278.23 | -167.46              | -53.99  | 99.78             | 73.60    | 3.81       |         |
| 10600.00            | 10533.30               | 10584.16 | 10493.73                   | 0.00            | 27.68  | 279.58 | -184.48              | -60.34  | 110.21            | 84.03    | 4.21       |         |
| 10625.00            | 10546.55               | 10604.62 | 10502.70                   | 0.00            | 27.74  | 280.77 | -201.70              | -66.77  | 120.86            | 94.65    | 4.61       |         |
| 10650.00            | 10558.57               | 10624.84 | 10510.72                   | 0.00            | 27.80  | 281.85 | -219.09              | -73.25  | 131.69            | 105.43   | 5.02       |         |
| 10675.00            | 10569.32               | 10644.84 | 10517.82                   | 0.00            | 27.86  | 282.84 | -236.61              | -79.79  | 142.67            | 116.33   | 5.42       |         |
| 10700.00            | 10578.78               | 10664.66 | 10524.00                   | 0.00            | 27.92  | 283.76 | -254.25              | -86.38  | 153.77            | 127.34   | 5.82       |         |
| 10725.00            | 10586.92               | 10684.32 | 10529.29                   | 0.00            | 27.97  | 284.63 | -271.99              | -93.00  | 164.96            | 138.41   | 6.21       |         |
| 10750.00            | 10593.70               | 10703.84 | 10533.70                   | 0.00            | 28.03  | 285.44 | -289.81              | -99.64  | 176.22            | 149.54   | 6.61       |         |
| 10775.00            | 10599.11               | 10723.26 | 10537.25                   | 0.00            | 28.09  | 286.22 | -307.69              | -106.32 | 187.51            | 160.70   | 6.99       |         |
| 10800.00            | 10603.12               | 10742.59 | 10539.94                   | 0.00            | 28.15  | 286.96 | -325.62              | -113.01 | 198.82            | 171.86   | 7.37       |         |
| 10825.00            | 10605.73               | 10761.87 | 10541.79                   | 0.00            | 28.21  | 287.66 | -343.60              | -119.72 | 210.12            | 183.00   | 7.75       |         |
| 10850.00            | 10606.92               | 10781.12 | 10542.80                   | 0.00            | 28.26  | 288.33 | -361.60              | -126.44 | 221.38            | 194.10   | 8.11       |         |
| 10857.19            | 10607.00               | 10786.65 | 10542.94                   | 0.00            | 28.28  | 288.52 | -366.79              | -128.37 | 224.61            | 197.28   | 8.22       |         |
| 10900.00            | 10607.14               | 10823.03 | 10542.91                   | 0.00            | 28.39  | 287.11 | -400.87              | -141.09 | 244.07            | 216.49   | 8.85       |         |
| 10957.19            | 10607.32               | 10873.45 | 10542.75                   | 0.00            | 28.55  | 285.51 | -448.11              | -158.72 | 270.28            | 242.39   | 9.69       |         |
| 11000.00            | 10607.45               | 10911.42 | 10542.62                   | 0.00            | 28.67  | 284.36 | -483.68              | -172.00 | 289.58            | 261.49   | 10.31      |         |
| 11100.00            | 10607.78               | 11001.76 | 10542.32                   | 0.00            | 28.96  | 282.34 | -568.32              | -203.58 | 331.60            | 303.08   | 11.63      |         |
| 11200.00            | 10608.10               | 11094.22 | 10542.01                   | 0.00            | 29.27  | 280.96 | -654.95              | -235.91 | 369.09            | 340.18   | 12.77      |         |
| 11300.00            | 10608.44               | 11188.55 | 10541.70                   | 0.00            | 29.59  | 279.98 | -743.32              | -268.89 | 401.87            | 372.59   | 13.72      |         |
| 11400.00            | 10608.78               | 11268.37 | 10541.43                   | 0.00            | 29.87  | 279.33 | -818.56              | -295.51 | 431.44            | 401.85   | 14.58      |         |
| 11500.00            | 10609.12               | 11346.19 | 10541.17                   | 0.00            | 30.14  | 278.79 | -892.92              | -318.42 | 459.96            | 430.08   | 15.40      |         |
| 11600.00            | 10609.46               | 11423.35 | 10540.91                   | 0.00            | 30.41  | 278.34 | -967.52              | -338.14 | 487.38            | 457.23   | 16.16      |         |
| 11700.00            | 10609.81               | 11500.00 | 10540.64                   | 0.00            | 30.68  | 277.96 | -1042.35             | -354.72 | 513.68            | 483.24   | 16.88      |         |
| 11800.00            | 10610.16               | 11575.91 | 10540.39                   | 0.00            | 30.94  | 277.63 | -1117.05             | -368.17 | 538.80            | 508.09   | 17.55      |         |



LEAM Drilling Systems, Inc.  
Anticollision Report

| Company: Continental Resources   |          | Date: 04/23/2012           |          | Time: 17:17:28          |        | Page: 17   |          |            |          |          |        |         |
|----------------------------------|----------|----------------------------|----------|-------------------------|--------|------------|----------|------------|----------|----------|--------|---------|
| Field: McKenzie County, ND       |          |                            |          |                         |        |            |          |            |          |          |        |         |
| Reference Site: Tallahassee 3    |          | Co-ordinate(NE) Reference: |          | Well: 3-16H, True North |        |            |          |            |          |          |        |         |
| Reference Well: 3-16H            |          | Vertical (TVD) Reference:  |          | GL 1920+KB 41 1941.0    |        |            |          |            |          |          |        |         |
| Reference Wellpath: OH           |          |                            |          |                         |        | Db: Adapti |          |            |          |          |        |         |
| Site: Tallahassee 2              |          |                            |          |                         |        |            |          |            |          |          |        |         |
| Well: 2-16H                      |          |                            |          |                         |        |            |          |            |          |          |        |         |
| Wellpath: OH V4 Plan: Plan #1 V1 |          |                            |          | Inter-Site Error:       |        | 0.00       | ft       |            |          |          |        |         |
| Reference                        | Offset   | Semi-Major Axis            |          | Offset Location         |        | Ctr-Ctr    | Edge     | Separation |          |          |        |         |
| MD                               | TVD      | MD                         | TVD      | Ref                     | Offset | TFO-HS     | North    | East       | Distance | Distance | Factor | Warning |
| ft                               | ft       | ft                         | ft       | ft                      | ft     | deg        | ft       | ft         | ft       | ft       |        |         |
| 11900.00                         | 10610.51 | 11651.38                   | 10540.13 | 0.00                    | 31.21  | 277.34     | -1191.80 | -378.59    | 562.71   | 531.74   | 18.17  |         |
| 12000.00                         | 10610.86 | 11726.38                   | 10539.87 | 0.00                    | 31.47  | 277.10     | -1266.42 | -386.02    | 585.38   | 554.15   | 18.75  |         |
| 12100.00                         | 10611.20 | 11800.00                   | 10539.62 | 0.00                    | 31.73  | 276.89     | -1339.90 | -390.45    | 606.78   | 575.30   | 19.28  |         |
| 12200.00                         | 10611.55 | 11875.10                   | 10539.37 | 0.00                    | 31.99  | 276.70     | -1414.97 | -392.05    | 626.87   | 595.14   | 19.75  |         |
| 12300.00                         | 10611.89 | 11972.60                   | 10539.04 | 0.00                    | 32.36  | 276.54     | -1512.48 | -392.05    | 643.91   | 611.81   | 20.06  |         |
| 12400.00                         | 10612.23 | 12071.87                   | 10538.70 | 0.00                    | 32.74  | 276.45     | -1611.75 | -392.05    | 655.81   | 623.32   | 20.18  |         |
| 12500.00                         | 10612.56 | 12171.64                   | 10538.37 | 0.00                    | 33.12  | 276.43     | -1711.51 | -392.05    | 662.53   | 629.65   | 20.15  |         |
| 12577.81                         | 10612.81 | 12249.42                   | 10538.11 | 0.00                    | 33.43  | 276.46     | -1789.30 | -392.05    | 664.16   | 630.97   | 20.01  |         |
| 12600.00                         | 10612.88 | 12271.61                   | 10538.03 | 0.00                    | 33.52  | 276.47     | -1811.49 | -392.05    | 664.18   | 630.90   | 19.96  |         |
| 12700.00                         | 10613.20 | 12371.61                   | 10537.69 | 0.00                    | 33.92  | 276.53     | -1911.49 | -392.05    | 664.26   | 630.58   | 19.72  |         |
| 12800.00                         | 10613.52 | 12471.61                   | 10537.36 | 0.00                    | 34.33  | 276.58     | -2011.48 | -392.05    | 664.33   | 630.25   | 19.49  |         |
| 12900.00                         | 10613.85 | 12571.61                   | 10537.02 | 0.00                    | 34.74  | 276.64     | -2111.48 | -392.05    | 664.41   | 629.91   | 19.26  |         |
| 13000.00                         | 10614.17 | 12671.60                   | 10536.68 | 0.00                    | 35.16  | 276.70     | -2211.48 | -392.05    | 664.48   | 629.57   | 19.03  |         |
| 13100.00                         | 10614.49 | 12771.60                   | 10536.34 | 0.00                    | 35.58  | 276.75     | -2311.48 | -392.05    | 664.56   | 629.23   | 18.81  |         |
| 13200.00                         | 10614.81 | 12871.60                   | 10536.01 | 0.00                    | 36.01  | 276.81     | -2411.47 | -392.05    | 664.64   | 628.88   | 18.59  |         |
| 13300.00                         | 10615.13 | 12971.60                   | 10535.67 | 0.00                    | 36.44  | 276.87     | -2511.47 | -392.05    | 664.72   | 628.53   | 18.37  |         |
| 13400.00                         | 10615.46 | 13071.59                   | 10535.33 | 0.00                    | 36.88  | 276.92     | -2611.47 | -392.05    | 664.80   | 628.17   | 18.15  |         |
| 13500.00                         | 10615.78 | 13171.59                   | 10534.99 | 0.00                    | 37.32  | 276.98     | -2711.46 | -392.05    | 664.88   | 627.81   | 17.94  |         |
| 13600.00                         | 10616.10 | 13271.59                   | 10534.66 | 0.00                    | 37.77  | 277.04     | -2811.46 | -392.05    | 664.96   | 627.44   | 17.73  |         |
| 13700.00                         | 10616.42 | 13371.59                   | 10534.32 | 0.00                    | 38.22  | 277.09     | -2911.46 | -392.05    | 665.04   | 627.08   | 17.52  |         |
| 13800.00                         | 10616.75 | 13471.59                   | 10533.98 | 0.00                    | 38.68  | 277.15     | -3011.46 | -392.05    | 665.12   | 626.70   | 17.31  |         |
| 13900.00                         | 10617.07 | 13571.58                   | 10533.64 | 0.00                    | 39.14  | 277.20     | -3111.45 | -392.05    | 665.20   | 626.33   | 17.11  |         |
| 14000.00                         | 10617.39 | 13671.58                   | 10533.31 | 0.00                    | 39.60  | 277.26     | -3211.45 | -392.04    | 665.29   | 625.95   | 16.91  |         |
| 14100.00                         | 10617.71 | 13771.58                   | 10532.97 | 0.00                    | 40.06  | 277.32     | -3311.45 | -392.04    | 665.37   | 625.57   | 16.72  |         |
| 14200.00                         | 10618.03 | 13871.58                   | 10532.63 | 0.00                    | 40.53  | 277.37     | -3411.45 | -392.04    | 665.46   | 625.19   | 16.53  |         |
| 14300.00                         | 10618.36 | 13971.58                   | 10532.29 | 0.00                    | 41.01  | 277.43     | -3511.44 | -392.04    | 665.54   | 624.80   | 16.34  |         |
| 14400.00                         | 10618.68 | 14071.57                   | 10531.96 | 0.00                    | 41.48  | 277.49     | -3611.44 | -392.04    | 665.63   | 624.41   | 16.15  |         |
| 14500.00                         | 10619.00 | 14171.57                   | 10531.62 | 0.00                    | 41.96  | 277.54     | -3711.44 | -392.04    | 665.71   | 624.02   | 15.97  |         |
| 14600.00                         | 10619.32 | 14271.57                   | 10531.28 | 0.00                    | 42.45  | 277.60     | -3811.43 | -392.04    | 665.80   | 623.63   | 15.79  |         |
| 14700.00                         | 10619.64 | 14371.57                   | 10530.94 | 0.00                    | 42.93  | 277.65     | -3911.43 | -392.04    | 665.89   | 623.23   | 15.61  |         |
| 14800.00                         | 10619.97 | 14471.56                   | 10530.61 | 0.00                    | 43.42  | 277.71     | -4011.43 | -392.04    | 665.98   | 622.83   | 15.44  |         |
| 14900.00                         | 10620.29 | 14571.56                   | 10530.27 | 0.00                    | 43.91  | 277.77     | -4111.43 | -392.04    | 666.07   | 622.43   | 15.27  |         |
| 15000.00                         | 10620.61 | 14671.56                   | 10529.93 | 0.00                    | 44.40  | 277.82     | -4211.42 | -392.04    | 666.16   | 622.03   | 15.10  |         |
| 15100.00                         | 10620.93 | 14771.56                   | 10529.59 | 0.00                    | 44.90  | 277.88     | -4311.42 | -392.04    | 666.25   | 621.63   | 14.93  |         |
| 15200.00                         | 10621.26 | 14871.56                   | 10529.26 | 0.00                    | 45.40  | 277.94     | -4411.42 | -392.04    | 666.34   | 621.22   | 14.77  |         |
| 15300.00                         | 10621.58 | 14971.55                   | 10528.92 | 0.00                    | 45.90  | 277.99     | -4511.41 | -392.04    | 666.43   | 620.81   | 14.61  |         |
| 15400.00                         | 10621.90 | 15071.55                   | 10528.58 | 0.00                    | 46.41  | 278.05     | -4611.41 | -392.04    | 666.52   | 620.40   | 14.45  |         |
| 15500.00                         | 10622.22 | 15171.55                   | 10528.24 | 0.00                    | 46.91  | 278.10     | -4711.41 | -392.04    | 666.61   | 619.99   | 14.30  |         |
| 15600.00                         | 10622.54 | 15271.55                   | 10527.91 | 0.00                    | 47.42  | 278.16     | -4811.41 | -392.04    | 666.71   | 619.58   | 14.15  |         |
| 15700.00                         | 10622.87 | 15371.54                   | 10527.57 | 0.00                    | 47.93  | 278.22     | -4911.40 | -392.04    | 666.80   | 619.17   | 14.00  |         |
| 15800.00                         | 10623.19 | 15471.54                   | 10527.23 | 0.00                    | 48.44  | 278.27     | -5011.40 | -392.04    | 666.90   | 618.75   | 13.85  |         |
| 15900.00                         | 10623.51 | 15571.54                   | 10526.89 | 0.00                    | 48.96  | 278.33     | -5111.40 | -392.04    | 666.99   | 618.34   | 13.71  |         |
| 16000.00                         | 10623.83 | 15671.54                   | 10526.56 | 0.00                    | 49.47  | 278.38     | -5211.40 | -392.04    | 667.09   | 617.92   | 13.57  |         |
| 16100.00                         | 10624.15 | 15771.54                   | 10526.22 | 0.00                    | 49.99  | 278.44     | -5311.39 | -392.04    | 667.18   | 617.50   | 13.43  |         |
| 16200.00                         | 10624.48 | 15871.53                   | 10525.88 | 0.00                    | 50.51  | 278.50     | -5411.39 | -392.04    | 667.28   | 617.08   | 13.29  |         |
| 16300.00                         | 10624.80 | 15971.53                   | 10525.54 | 0.00                    | 51.03  | 278.55     | -5511.39 | -392.04    | 667.38   | 616.66   | 13.16  |         |
| 16400.00                         | 10625.12 | 16071.53                   | 10525.21 | 0.00                    | 51.56  | 278.61     | -5611.38 | -392.04    | 667.48   | 616.24   | 13.03  |         |
| 16500.00                         | 10625.44 | 16171.53                   | 10524.87 | 0.00                    | 52.08  | 278.66     | -5711.38 | -392.04    | 667.58   | 615.81   | 12.90  |         |
| 16600.00                         | 10625.77 | 16271.53                   | 10524.53 | 0.00                    | 52.61  | 278.72     | -5811.38 | -392.04    | 667.68   | 615.39   | 12.77  |         |
| 16700.00                         | 10626.09 | 16371.52                   | 10524.19 | 0.00                    | 53.14  | 278.78     | -5911.38 | -392.04    | 667.78   | 614.96   | 12.64  |         |
| 16800.00                         | 10626.41 | 16471.52                   | 10523.86 | 0.00                    | 53.67  | 278.83     | -6011.37 | -392.04    | 667.88   | 614.54   | 12.52  |         |

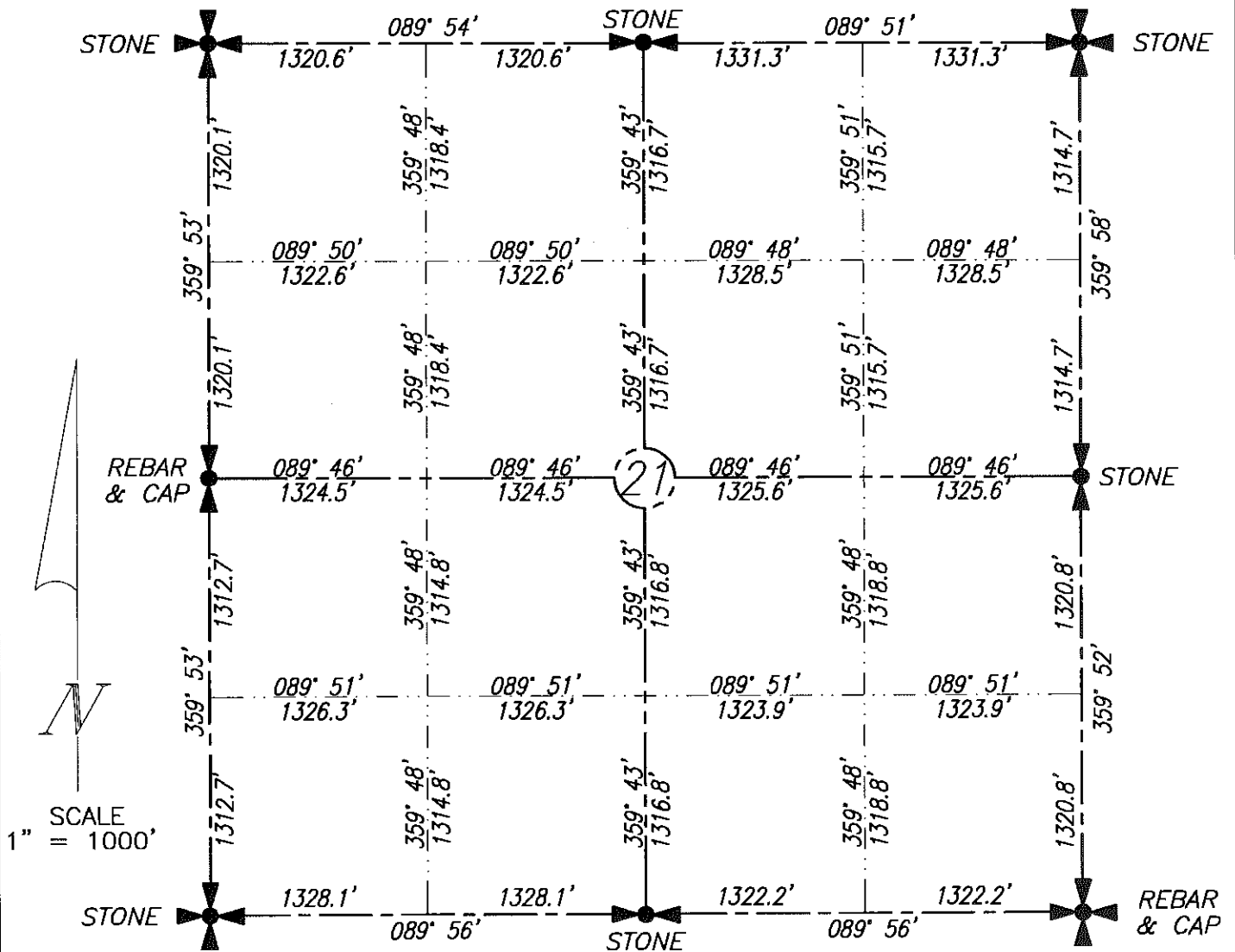
LEAM Drilling Systems, Inc.  
Anticollision Report

|                     |                       |                            |                         |       |          |       |        |
|---------------------|-----------------------|----------------------------|-------------------------|-------|----------|-------|--------|
| Company:            | Continental Resources | Date:                      | 04/23/2012              | Time: | 17:17:28 | Page: | 18     |
| Field:              | McKenzie County, ND   |                            |                         |       |          |       |        |
| Reference Site:     | Tallahassee 3         | Co-ordinate(NE) Reference: | Well: 3-16H, True North |       |          |       |        |
| Reference Well:     | 3-16H                 | Vertical (TVD) Reference:  | GL 1920+KB 41 1941.0    |       |          |       |        |
| Reference Wellpath: | OH                    |                            |                         |       |          | Db:   | Adapti |

|           |                        |                           |  |  |  |  |  |  |  |  |  |
|-----------|------------------------|---------------------------|--|--|--|--|--|--|--|--|--|
| Site:     | Tallahassee 2          |                           |  |  |  |  |  |  |  |  |  |
| Well:     | 2-16H                  |                           |  |  |  |  |  |  |  |  |  |
| Wellpath: | OH V4 Plan: Plan #1 V1 | Inter-Site Error: 0.00 ft |  |  |  |  |  |  |  |  |  |

| Reference |          | Offset   |          | Semi-Major Axis |        |        | Offset Location |         | Ctr-Ctr  | Edge     | Separation | Warning |
|-----------|----------|----------|----------|-----------------|--------|--------|-----------------|---------|----------|----------|------------|---------|
| MD        | TVD      | MD       | TVD      | Ref             | Offset | TFO-HS | North           | East    | Distance | Distance | Factor     |         |
| ft        | ft       | ft       | ft       | ft              | ft     | deg    | ft              | ft      | ft       | ft       |            |         |
| 16900.00  | 10626.73 | 16571.52 | 10523.52 | 0.00            | 54.20  | 278.89 | -6111.37        | -392.04 | 667.98   | 614.11   | 12.40      |         |
| 17000.00  | 10627.05 | 16671.52 | 10523.18 | 0.00            | 54.73  | 278.94 | -6211.37        | -392.04 | 668.08   | 613.69   | 12.28      |         |
| 17100.00  | 10627.38 | 16771.51 | 10522.84 | 0.00            | 55.26  | 279.00 | -6311.37        | -392.04 | 668.19   | 613.26   | 12.16      |         |
| 17200.00  | 10627.70 | 16871.51 | 10522.51 | 0.00            | 55.80  | 279.06 | -6411.36        | -392.04 | 668.29   | 612.83   | 12.05      |         |
| 17300.00  | 10628.02 | 16971.51 | 10522.17 | 0.00            | 56.34  | 279.11 | -6511.36        | -392.04 | 668.39   | 612.40   | 11.94      |         |
| 17400.00  | 10628.34 | 17071.51 | 10521.83 | 0.00            | 56.87  | 279.17 | -6611.36        | -392.04 | 668.50   | 611.97   | 11.83      |         |
| 17500.00  | 10628.66 | 17171.51 | 10521.49 | 0.00            | 57.41  | 279.22 | -6711.35        | -392.04 | 668.61   | 611.54   | 11.72      |         |
| 17600.00  | 10628.99 | 17271.50 | 10521.16 | 0.00            | 57.95  | 279.28 | -6811.35        | -392.04 | 668.71   | 611.11   | 11.61      |         |
| 17700.00  | 10629.31 | 17371.50 | 10520.82 | 0.00            | 58.49  | 279.34 | -6911.35        | -392.04 | 668.82   | 610.68   | 11.50      |         |
| 17800.00  | 10629.63 | 17471.50 | 10520.48 | 0.00            | 59.04  | 279.39 | -7011.35        | -392.04 | 668.93   | 610.25   | 11.40      |         |
| 17900.00  | 10629.95 | 17571.50 | 10520.14 | 0.00            | 59.58  | 279.45 | -7111.34        | -392.04 | 669.03   | 609.82   | 11.30      |         |
| 18000.00  | 10630.28 | 17671.49 | 10519.81 | 0.00            | 60.13  | 279.50 | -7211.34        | -392.04 | 669.14   | 609.38   | 11.20      |         |
| 18100.00  | 10630.60 | 17771.49 | 10519.47 | 0.00            | 60.67  | 279.56 | -7311.34        | -392.04 | 669.25   | 608.95   | 11.10      |         |
| 18200.00  | 10630.92 | 17871.49 | 10519.13 | 0.00            | 61.22  | 279.61 | -7411.34        | -392.04 | 669.36   | 608.52   | 11.00      |         |
| 18300.00  | 10631.24 | 17971.49 | 10518.79 | 0.00            | 61.77  | 279.67 | -7511.33        | -392.04 | 669.47   | 608.08   | 10.91      |         |
| 18400.00  | 10631.56 | 18071.49 | 10518.46 | 0.00            | 62.31  | 279.73 | -7611.33        | -392.04 | 669.58   | 607.65   | 10.81      |         |
| 18500.00  | 10631.89 | 18171.48 | 10518.12 | 0.00            | 62.86  | 279.78 | -7711.33        | -392.04 | 669.70   | 607.22   | 10.72      |         |
| 18535.29  | 10632.00 | 18206.77 | 10518.00 | 0.00            | 63.06  | 279.80 | -7746.61        | -392.04 | 669.74   | 607.06   | 10.69      |         |

HORIZONTAL SECTION PLAT  
CONTINENTAL RESOURCES INC.  
TALLAHASSEE 3-16H  
SECTION 21, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

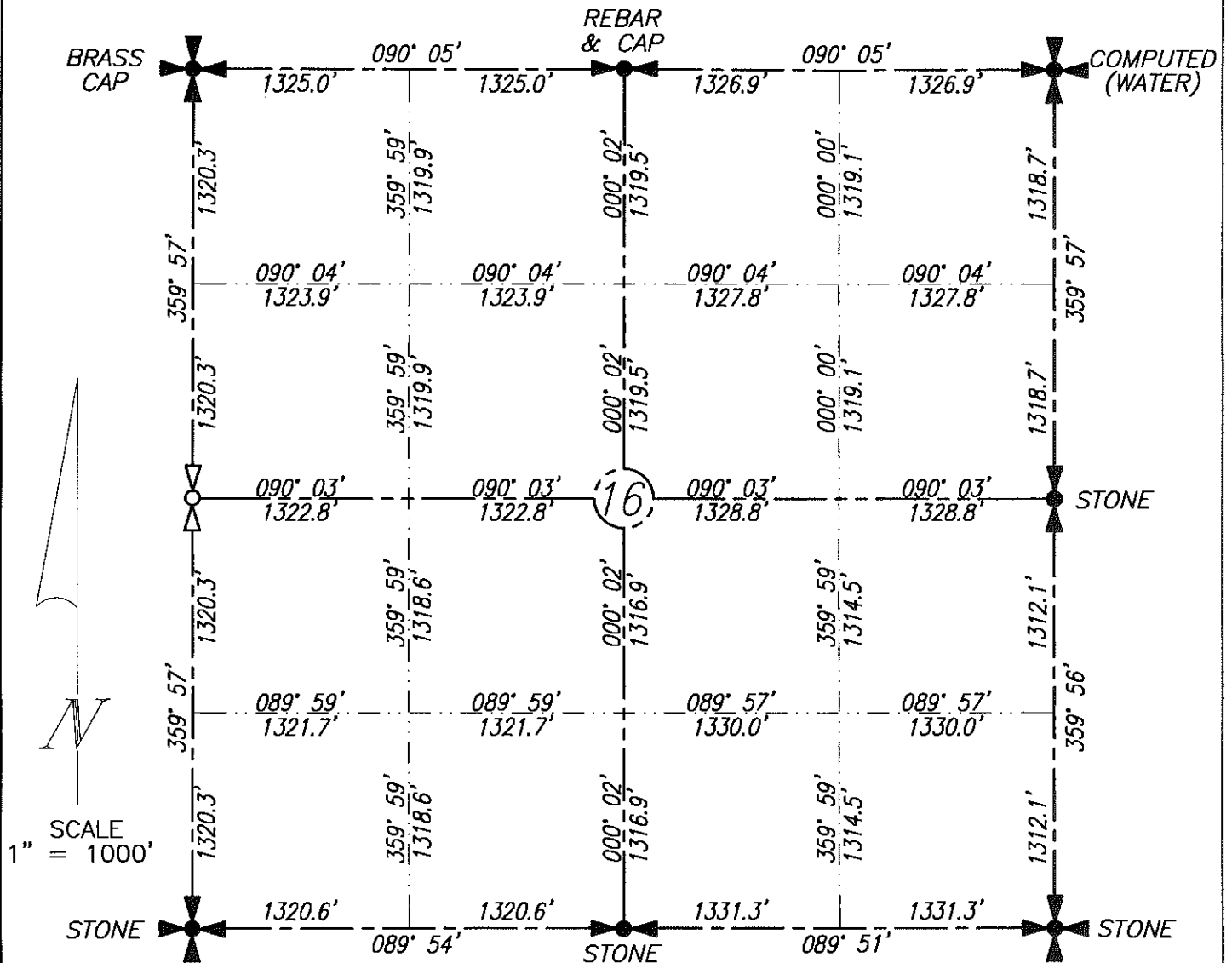


ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD  
DISTANCES TO ALL OTHERS ARE CALCULATED.  
ALL BEARINGS SHOWN ARE ASSUMED.

JOHN PAULSON  
I HEREBY CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS  
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE  
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF  
MY KNOWLEDGE AND BELIEF  
JOHN PAULSON R.L.S. 3366

**BROSZ ENGINEERING INC.**  
BOX 357  
BOWMAN, N.D. 58623  
PHONE: 701-523-3340  
FAX: 701-523-5243  
PROJECT NO. 12-10

# HORIZONTAL SECTION PLAT CONTINENTAL RESOURCES INC. TALLAHASSEE 3-16H SECTION 16, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA



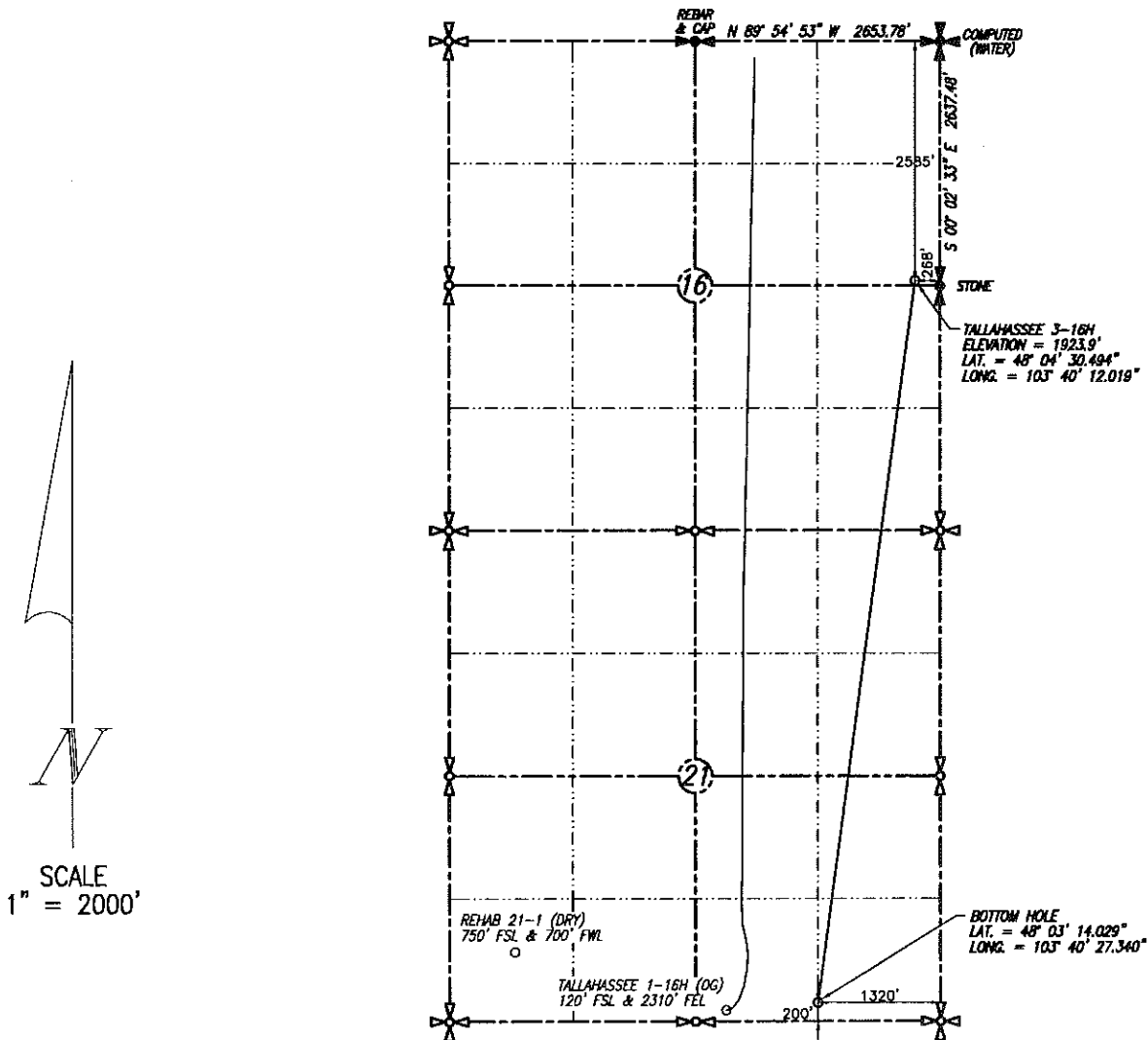
ALL CORNERS SHOWN ON THIS PLAT WERE FOUND IN THE FIELD  
DISTANCES TO ALL OTHERS ARE CALCULATED.  
ALL BEARINGS SHOWN ARE ASSUMED.

JOHN PAULSON  
REGISTERED SURVEYOR  
I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS  
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE  
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF  
MY KNOWLEDGE AND BELIEF  
JOHN PAULSON R.L.S. 3366

**BROSZ ENGINEERING INC.**  
BOX 357  
BOWMAN, N.D. 58623  
PHONE: 701-523-3340  
FAX: 701-523-5243  
PROJECT NO. 12-10

BOTTOM HOLE LOCATION PLAT  
 CONTINENTAL RESOURCES INC.  
 TALLAHASSEE 3-16H  
 SECTION 16, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 2585' FNL & 268' FEL

REVISED: 4-16-2012



I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS  
 WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE  
 CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF  
 MY KNOWLEDGE AND BELIEF

JOHN PAULSON  
 4-16-12

JOHN PAULSON REG. 3366

LAND  
 SURVEYOR  
 L.S. 3366

DATE STAKED: 1-4-2012

BASIS OF VERTICAL DATUM:  
 NAVD 1988 GEOID 09

PERSON AUTHORIZING SURVEY;  
 CHAD NEWBY

EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

**BROSZ ENGINEERING INC.**

BOX 357

BOWMAN, N.D. 58623

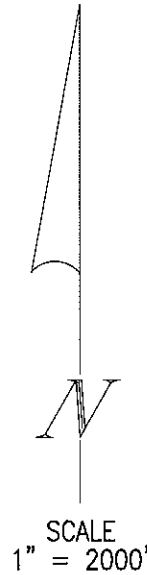
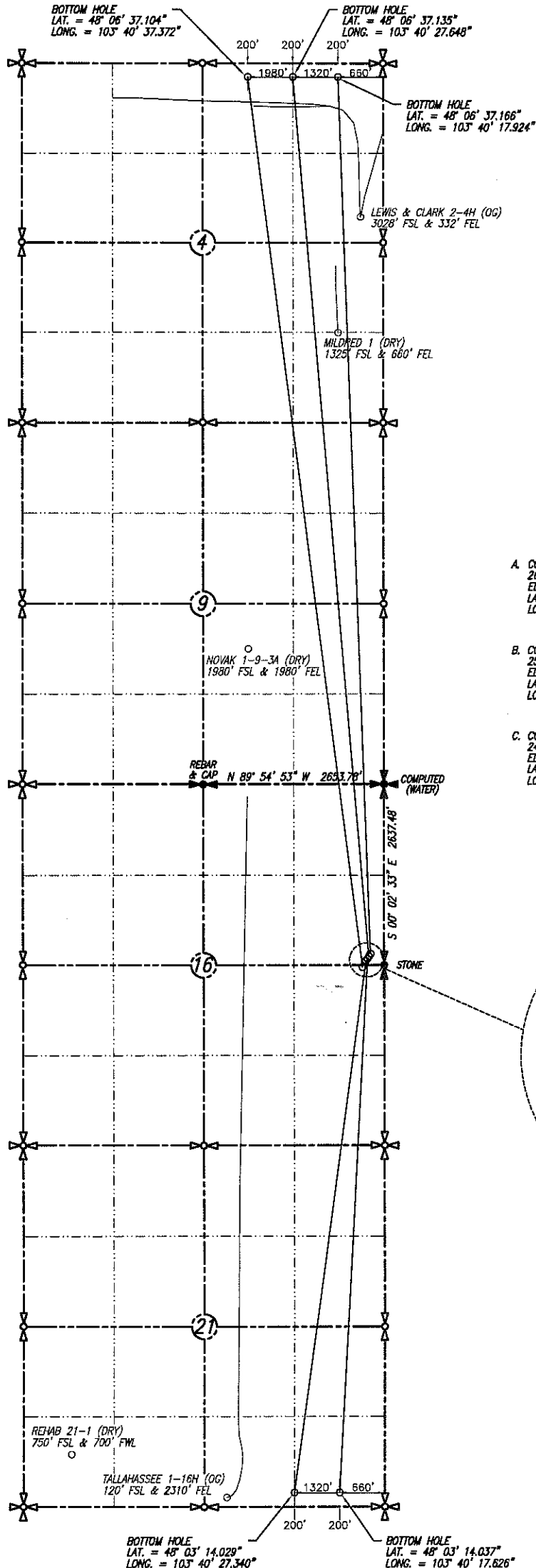
PHONE: 701-523-3340

FAX: 701-523-5243

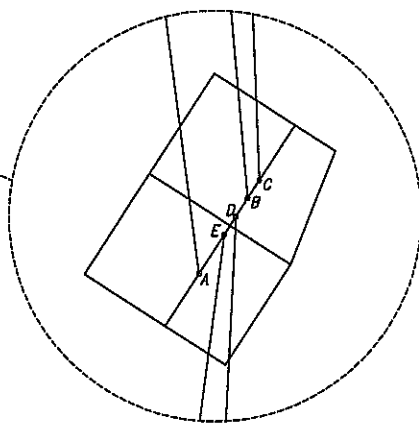
PROJECT NO. 12-10

REVISED: 4-16-2012

# BOTTOM HOLE LOCATION PLAT CONTINENTAL RESOURCES INC. COLUMBUS FEDERAL 1, 2, & 3 – TALLAHASSEE 2 & 3 ECO PAD SECTION 16, T153N, R101W MCKENZIE COUNTY, NORTH DAKOTA



- |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>A. COLUMBUS FEDERAL 1-16H<br/>2669' FNL &amp; 318' FEL<br/>ELEVATION = 1927.5'<br/>LAT. = 48° 04' 29.665"<br/>LONG. = 103° 40' 12.754"</p> <p>B. COLUMBUS FEDERAL 2-16H<br/>2508' FNL &amp; 222' FEL<br/>ELEVATION = 1921.0'<br/>LAT. = 48° 04' 31.258"<br/>LONG. = 103° 40' 11.343"</p> <p>C. COLUMBUS FEDERAL 3-16H<br/>2469' FNL &amp; 199' FEL<br/>ELEVATION = 1920.0'<br/>LAT. = 48° 04' 31.639"<br/>LONG. = 103° 40' 11.004"</p> | <p>D. TALLAHASSEE 2-16H<br/>2547' FNL &amp; 245' FEL<br/>ELEVATION = 1921.7'<br/>LAT. = 48° 04' 30.876"<br/>LONG. = 103° 40' 11.681"</p> <p>E. TALLAHASSEE 3-16H<br/>2585' FNL &amp; 268' FEL<br/>ELEVATION = 1923.9'<br/>LAT. = 48° 04' 30.494"<br/>LONG. = 103° 40' 12.019"</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



PERSON AUTHORIZING SURVEY;  
CHAD NEWBY

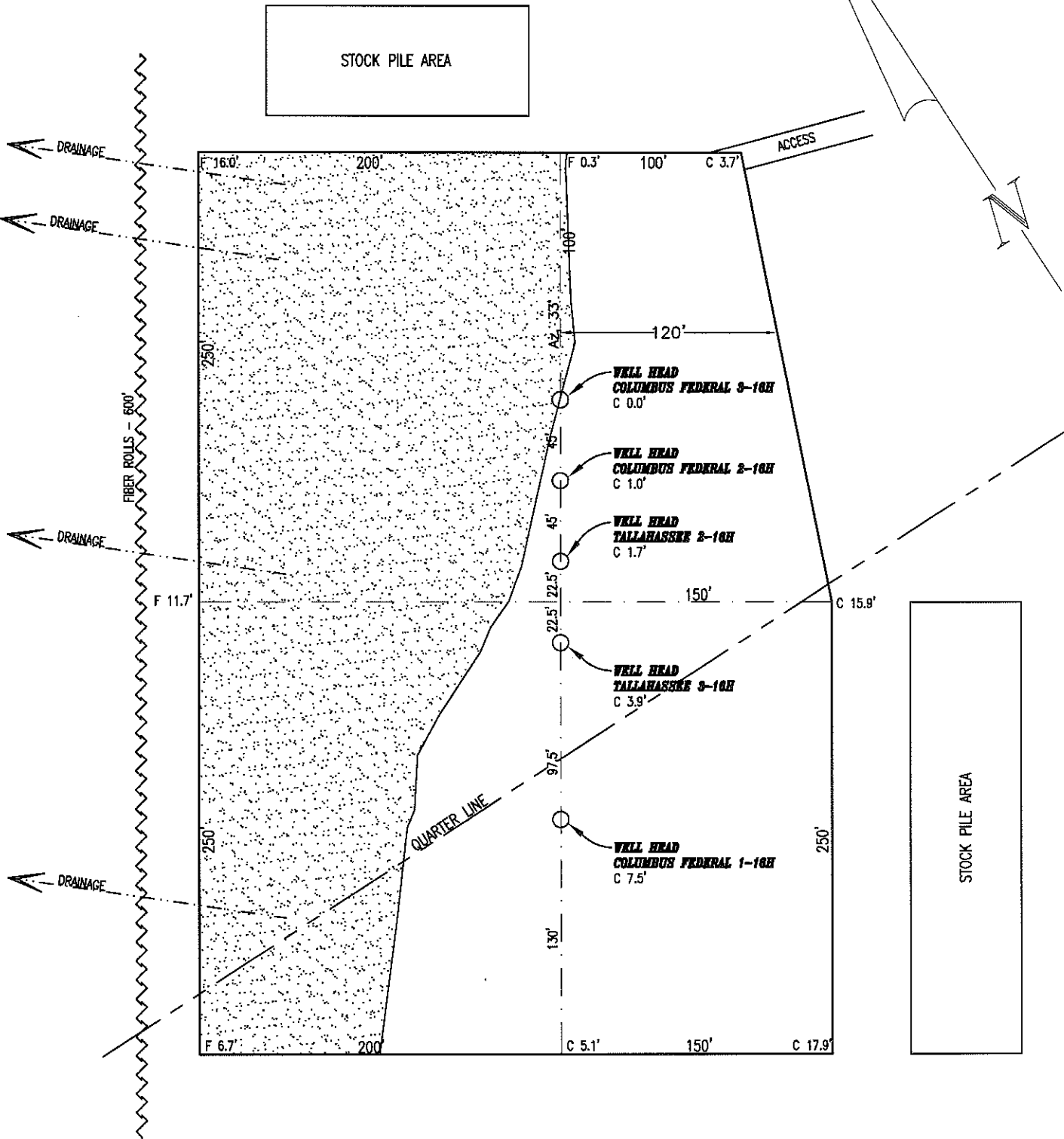
EXPLANATION AREA: NAD83(CORS96)

BASIS OF BEARING: TRUE NORTH

**BROSZ ENGINEERING INC.**

BOX 357  
BOWMAN, N.D. 58623  
PHONE: 701-523-3340  
FAX: 701-523-5243  
PROJECT NO. 12-10





CONTINENTAL RESOURCES INC.  
PO BOX 1032  
ENID, OKLAHOMA 73702

DRILLING RIG LAYOUT  
TALLAHASSEE 3-16H  
SECTION 16, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

ESTIMATED EARTH QUANTITIES

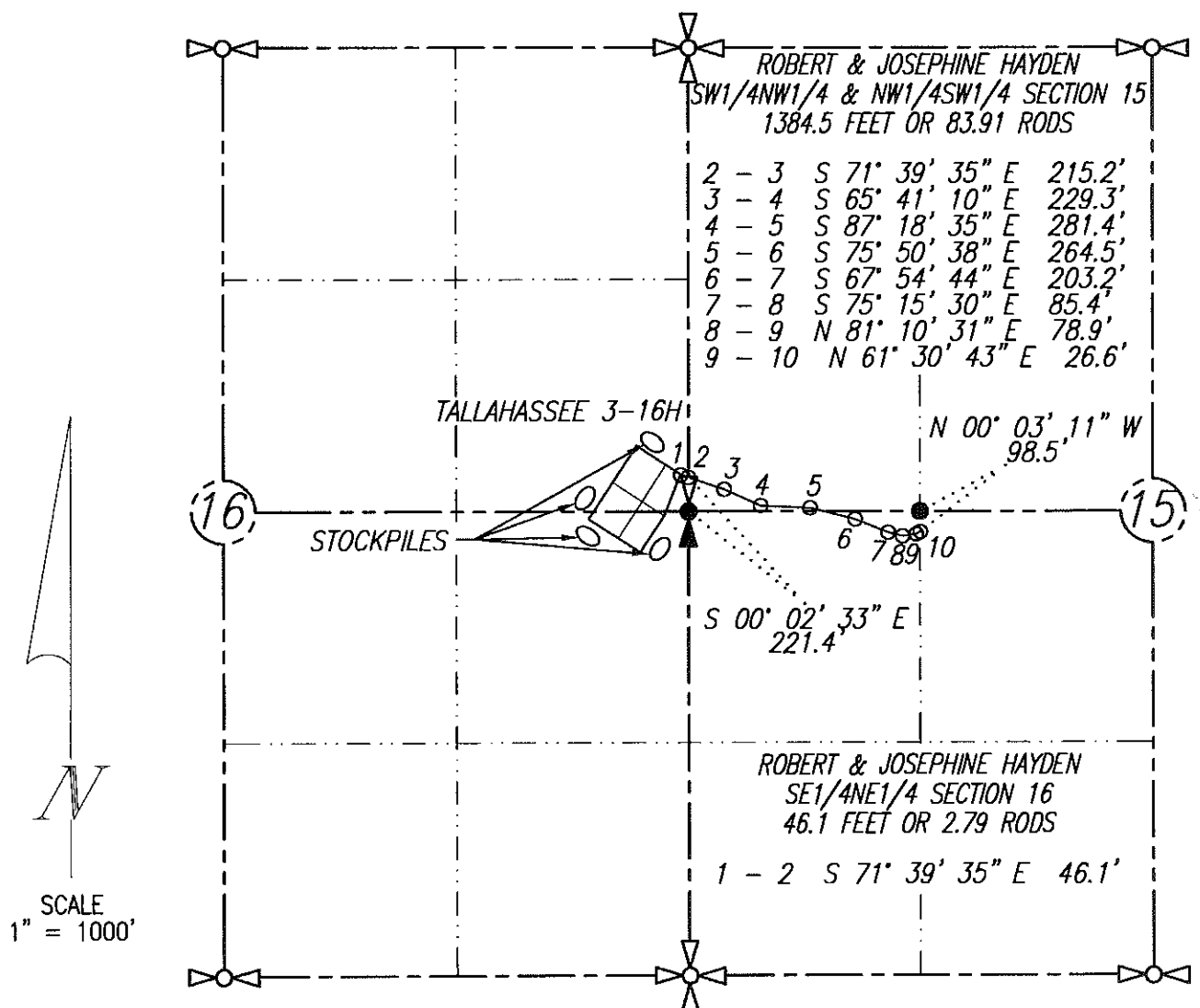
|             |        |             |
|-------------|--------|-------------|
| TOP-SOIL:   | 3,125  | CUBIC YARDS |
| SUB-SOIL:   | 28,719 | CUBIC YARDS |
| <hr/>       |        |             |
| TOTAL CUT:  | 31,844 | CUBIC YARDS |
| TOTAL FILL: | 27,716 | CUBIC YARDS |

Use excess materials in access road fill

ALL INDICATED  
CUTS & FILLS  
ARE STAKED  
GRADE ELEVA-  
TIONS.  
BACKSLOPES  
ASSUMED  
AT 1 1/2 : 1 %

Ground Elevation at Well Head: 1923.9 ft. ASL  
Finished Rig Grade Elevation: 1920.0 ft. ASL

DRILL SITE LOCATION & ROAD LOCATION PLAT  
CONTINENTAL RESOURCES INC.  
TALLAHASSEE 3-16H  
SECTIONS 15 & 16, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA



NOTE: THERE IS A PIPELINE AND POWERLINE EASEMENT  
LYING ON BOTH SIDES OF THE WELL ROAD EASEMENT.

I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS  
WORK PERFORMED BY ME OR UNDER MY RESPONSIBLE  
CHARGE, AND IS TRUE AND CORRECT TO THE BEST OF  
MY KNOWLEDGE AND BELIEF.

PERSON AUTHORIZING SURVEY;  
CHAD NEWBY

BASIS OF BEARING: TRUE NORTH

**BROSZ ENGINEERING INC.**

BOX 357  
BOWMAN, N.D. 58623  
PHONE: 701-523-3340  
FAX: 701-523-5243

PROJECT NO. 12-10

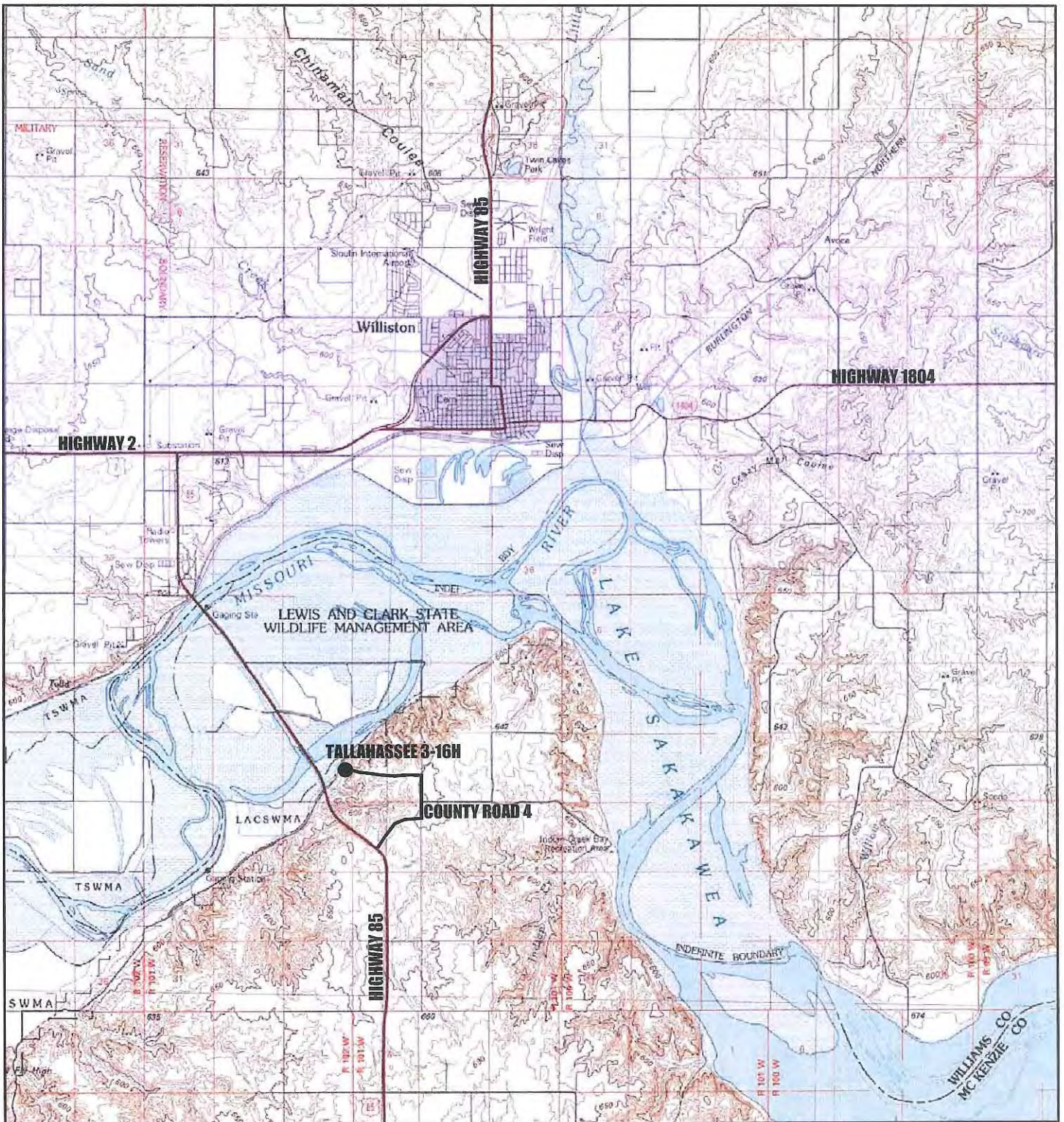
JOHN R. MILLER L.S. 3366

LAND  
SURVEYOR  
L.S. 3366

DATE STAKED: 1-4-2012

NORTH DAKOTA





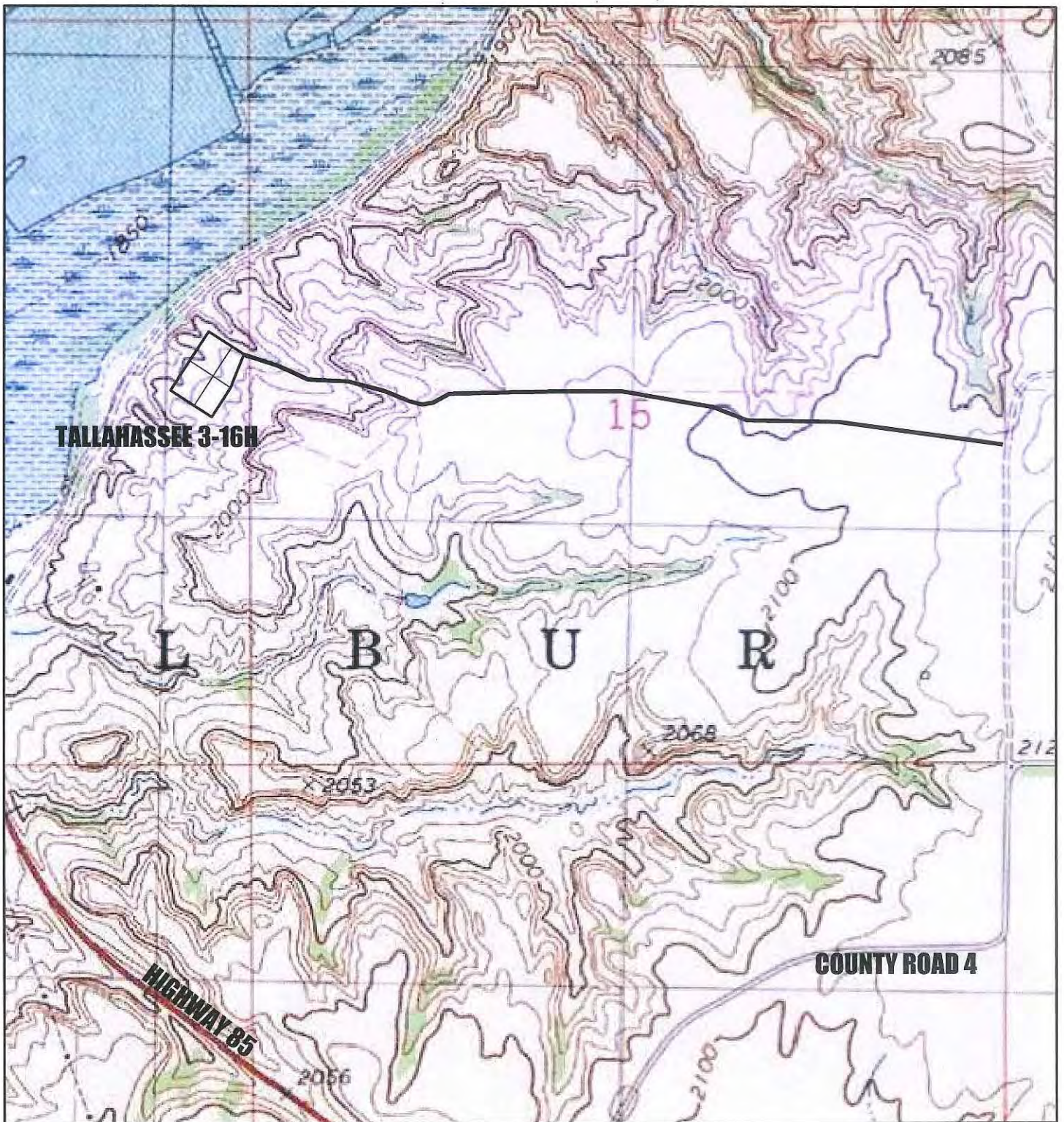
**CONTINENTAL RESOURCES INC.**

VICINITY MAP



TALLAHASSEE 3-16H  
SECTION 16, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA





CONTINENTAL RESOURCES INC.

WELL LOCATION



TALLAHASSEE 3-16H  
SECTION 16, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

## **Tabor, David**

---

**From:** Tabor, David  
**Sent:** Wednesday, March 06, 2013 2:25 PM  
**To:** Tabor, David  
**Subject:** FW: Columbus Federal 3-16H and pad.

---

**From:** Robert Sandbo [<mailto:Robert.Sandbo@clr.com>]  
**Sent:** Friday, March 01, 2013 4:48 PM  
**To:** Tabor, David  
**Subject:** RE: Columbus Federal 3-16H and pad.

Thanks David. I got the email. Here is our schedule for the pad as of right now:

Columbus Federal 1-16H to spud on 3/24/2013 (may spud with the small rig around one week earlier if possible). We had planned on drilling the Columbus 1-16H first and then come back in a year or so and drill the other 4 but if we get the permits for all and the order on the 1280 allows us to drill them all back to back, we will most likely go ahead and drill them all at once. We will take a look at the 1280 order that covers the Columbus 1-16H and see if we can batch drill them.

You should have everything on the 3-16H and the affidavit sometime Monday.

Thanks for the help,

**Bob Sandbo**  
Regulatory Compliance Supervisor

Continental Resources, Inc.  
20 N. Broadway  
OKC, OK 73102  
**P:** 405-234-9020  
**F:** 405-234-9562  
**C:** 405-708-0691  
[robert.sandbo@clr.com](mailto:robert.sandbo@clr.com)  
[www.clr.com](http://www.clr.com)





March 7, 2013

Industrial Commission of North Dakota  
Oil & Gas Division  
600 East Boulevard, Dept 405  
Bismarck, North Dakota 58505

Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H & 3-16H1, and the Tallahassee 2-21H & 3-21H1

Township 153N, Range 101W

Section 16, NE/4 SE/4 & SE/4 NE/4 McKenzie County, North Dakota.

The Columbus Federal and Tallahassee well(s) are located in an environmentally sensitive area in close proximity to the Missouri River. Therefore, Continental Resources Inc. would like to propose the following automatic shut down equipment and level sensing monitoring equipment be deployed on the site to aid in the prevention of any accidental release or safety issue. A schematic diagram and pictures of an existing tank battery with emergency shutdown equipment has been included with this affidavit.

- 1) Tank Side (oil & water tanks)
  - a. ABB Levelmaster dual float
  - b. High level switch as backup
  - c. Battery box with solar backup
- 2) Treater & Separator
  - a. Temperature device in each vessel
  - b. Pressure transducer in each vessel
  - c. Level switch in each vessel
  - d. Battery box with solar backup
- 3) Wellhead
  - a. TotalFlow Controller
  - b. Emergency Shutdown Valve Package
  - c. Battery box with solar backup
  - d. Pressure transducer on casing & tubing
- 4) Flare
  - a. Scrubber pot high level switch
  - b. Monitoring flare for low temp output – alarm if flare goes out



- 5) System Automation through the proposed equipment will provide an independent control system on all equipment on site which will be able to shut the well(s) in should any of the other equipment be incapacitated or functioning improperly.

Don Kennedy  
Don Kennedy, Sr. Production Engineer  
Continental Resources, Inc.

[illegible]

On the 7<sup>th</sup> day of March, 2013, before me, a Notary Public in and for said County and State, personally appeared **Don Kennedy**, known to me to be the Sr. Production Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.

Becky Barnes  
Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023

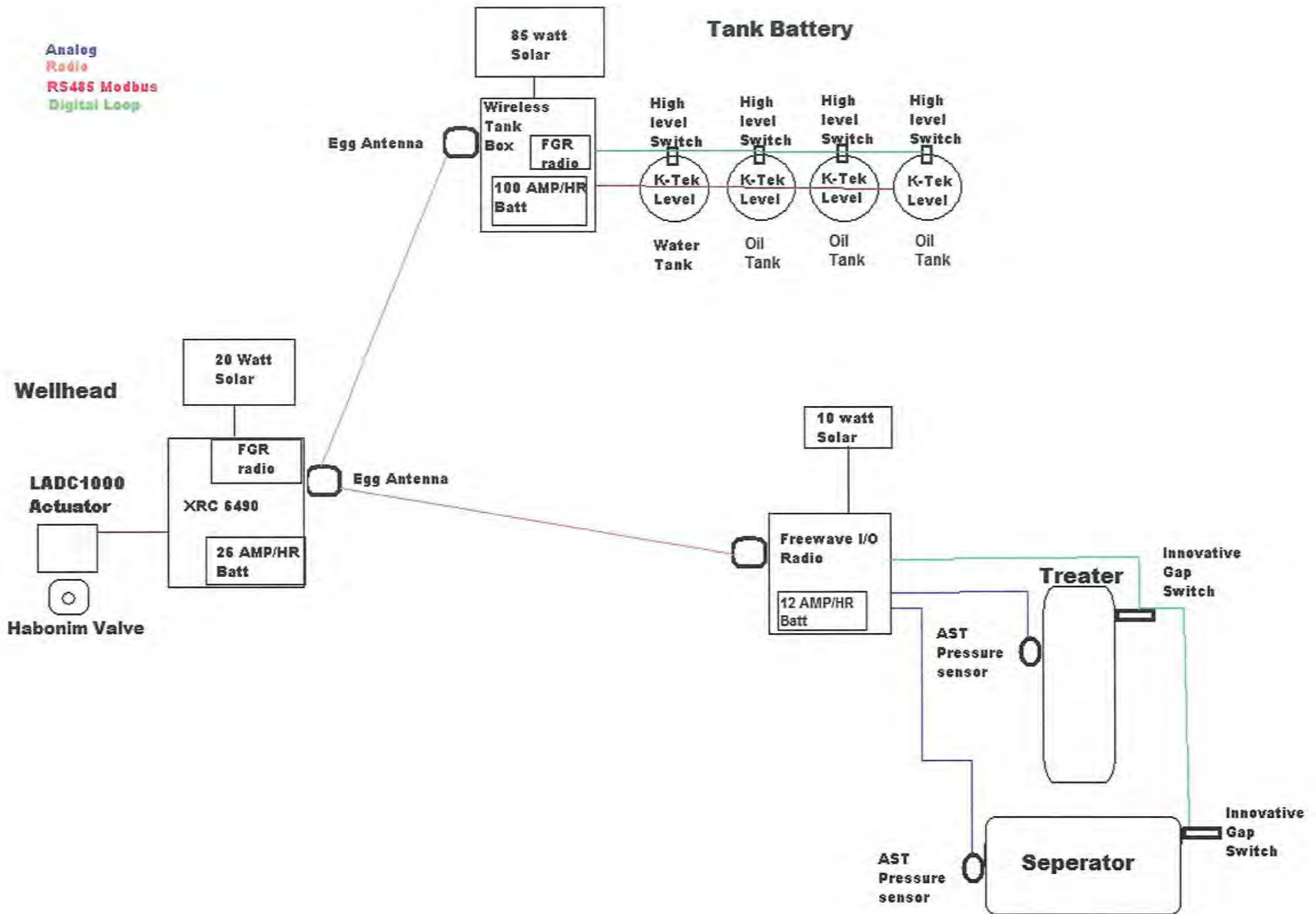


Sincerely,

**CONTINENTAL RESOURCES, INC.**

Becky Barnes  
Regulatory Compliance Specialist

Analog  
Radio  
RS485 Modbus  
Digital Loop







U003 Gap Switch and  
AST 4600 Transducer  
monitor pressures and  
liquid content of Flare  
and Sales Lines,  
transmitted to XRC via  
FreeWave Radio.

### ABB TotalFlow XRC 6490

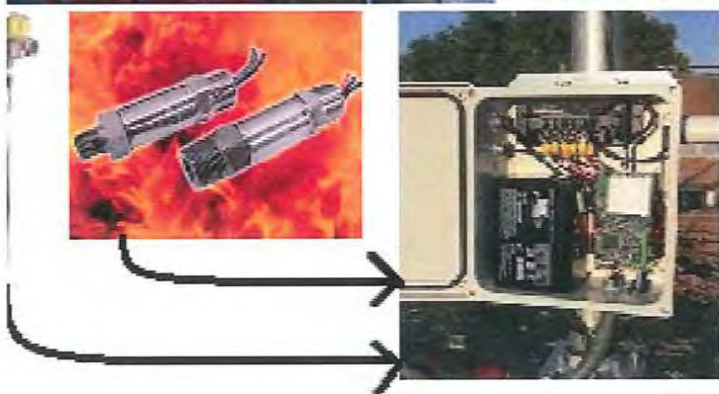


All well information is  
passed to your  
SCADA system via  
FreeWave Network  
(Future)

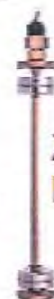
Winn-Marion's Well Head  
Kit with FGRIO Radio and  
Power Supply



Tank Level information is  
passed from Tank Battery to  
XRC via FreeWave Radio



K-Tek MT5100  
Guided Wave Radar



Z-Bend High  
Level Switch



ESD Valve Package  
Standard Port Ball Valve  
rated to 6000 psi topped  
with a 12 VDC Actuator  
w/ Battery Backup



Winn-Marion, Inc.



BECKY BARNES  
NOTARY  
#11006023  
EXP. 7/5/2015  
PUBLIC  
STATE OF OKLAHOMA

## CLR Spill Trailer Inventory

(To be Checked After Each Use)

| Supplies                           | Quantity | Actual | Supplies                                 | Quantity | Actual |
|------------------------------------|----------|--------|------------------------------------------|----------|--------|
| Personal Protection                |          |        | Miscellaneous                            |          |        |
| Trauma/1st Aid Kit                 | 1        |        | EnviroClean (5-gal units of concentrate) | 2        |        |
| Eye Wash                           | 1        |        | Duct Tape (Case)                         | 3        |        |
| Hand Cleaner                       | 2        |        | Flashlights                              | 6        |        |
| Nitrile Gloves (L & XL Case)       | 2        |        | Flood Lights                             | 2        |        |
| FRC Rain Coat - Extra Large        | 3        |        | Extension Cord 50' 12-gauge              | 5        |        |
| FRC Rain Coat - Large              | 3        |        | 55-gal. Drums w/lids                     | 2        |        |
| Rubber Safety Toed Boots - Size 10 | 2        |        | Large Trash Cans                         | 2        |        |
| Rubber Safety Toed Boots - Size 11 | 2        |        | HD Drum Liners - boxes                   | 2        |        |
| Rubber Safety Toed Boots - Size 12 | 2        |        | Hoses - Kit (Blue & Green)               | 5        |        |
| FRC Tyvex Suits - Case XL          | 1        |        | Plastic Buckets                          | 5        |        |
| Neoprene Chest Waders - L          | 1        |        | Propane Cylinders - 20-lb.               | 2        |        |
| Neoprene Chest Waders - L          | 1        |        | Propane Weed Burner W/Hose               | 1        |        |
| Containment                        |          |        | Pump - Trash                             | 2        |        |
| Absorbent (sphag)                  | 10       |        | Pump - 115V Water Transfer               | 2        |        |
| Absorbent Boom 3" x 10'            | 2        |        | Gas Powered Generator (3-5K Watt)        | 1        |        |
| Absorbent Boom 5" x 10'            | 10       |        | Misc. Ratchet Straps                     | 6        |        |
| Absorbent Boom 8" x 10'            | 8        |        | Rope 1/2" x 100'                         | 2        |        |
| Containment Boom - Fast Water      | 3        |        | Rope 1/4" x 50'                          | 4        |        |
| Absorbent Pads (Hydrocarbon)       | 10       |        | Rope 3/8" x 100'                         | 2        |        |
| Absorbent Pads (Universal)         | 5        |        | Shop Towels - box                        | 2        |        |
| Absorbent Pillows 18" x 18" box    | 3        |        | Caulking Gun                             | 2        |        |
| Absorbent Pom Pom Cube             | 7        |        | Silicon Tubes                            | 10       |        |
| Absorbent Sweep - 16" x 100' - Bag | 5        |        | Metal Stakes/Spikes                      | 8        |        |
| Miscellaneous                      |          |        | Metal T-Posts                            | 6        |        |
| Antifreeze                         | 2        |        | Bungee Cords                             | 3        |        |
| Push Broom                         | 2        |        | Wire - 25' roll - smooth                 | 1        |        |
| Shovels                            | 2        |        | Fire Extinguisher                        | 1        |        |
| Rake                               | 5        |        | Equipment Hooks                          | 6        |        |
| Squeegees                          | 2        |        | Shelving                                 | 4        |        |
| Scoop                              | 2        |        | Drawers                                  | 1        |        |
| Spark Resistant Scoop              | 1        |        | Misc. Building Supplies                  | 1        |        |



Continental Resources, Inc. (CRI) respectfully submits the following information concerning the drilling of the Columbus Federal 1-16H, 2-16H and 3-16H and the Tallahassee 2-16H and 3-16H, Sec. 16, T153N, R101W, Williams County, North Dakota.

CRI would like to propose the following safeguards and precautions to be taken while drilling the Columbus Federal-Tallahassee wells to prevent any contamination to freshwater sources during the drilling and completion of the wells:

- 1) During construction of the location, the entire location will be constructed per NDIC permit stipulations, ensuring any spills or runoff which occur on location do not penetrate the fresh ground water and are contained on the surface of the location. These modifications include, but are not limited to, the inclusion of a liner being placed under the location, and cementing of the rathole and mousehole.
- 2) Drainage will be re-routed to avoid the location, and erosion controls will be employed, as appropriate, around the site to reduce erosion and the resulting sediment contamination to freshwater runoff due to weather events.
- 3) The earthen berm, constructed to keep any freshwater runoff off the location, will also eliminate any spills from leaving the location
- 4) No reserve pit or dry cuttings pit will be utilized on location.
- 5) The conductor will be drilled to a depth of 80', and 20" pipe will be run to depth and cemented to surface.
- 6) During drilling operations, a freshwater protection string of 13-3/8" 48# H40 casing will be set to a depth of 500' and cemented to surface to protect the shallow freshwater zones. Standard 9-5/8" 36# J-55 surface casing will be set 100' into the Pierre Shale, to a depth of 1940', and cemented to surface.
- 7) A frac string will be used to protect the intermediate casing during hydraulic fracturing of the well.
- 8) CRI is submitting a comprehensive rig specific Spill Prevention Containment and Countermeasure Plan to prepare for any event which may occur during drilling and completion operations. A spill trailer will be located on location for spill response, if necessary.

CRI believes adequate planning and precautions are being taken to prevent any contamination to ground water and surface waters.

  
Sarah Madden, Project Development Engineer  
Continental Resources, Inc.

STATE OF OKLAHOMA )  
 )ss:  
COUNTY OF OKLAHOMA)

On the 7th day of March 2013, before me, a Notary Public in and for said County and State, personally appeared **Sarah Madden**, known to me to be a Project Development Engineer of Continental Resources, Inc., the Corporation that executed the within instrument, and acknowledged to me that such Corporation executed the same.

  
Notary Public

Oklahoma County, Oklahoma

My Commission Expires: 7/5/2015

Commission No.: 11006023





**SPILL PREVENTION CONTAINMENT  
AND COUNTERMEASURE PLAN**

**FOR**

**CYCLONE DRILLING, INC.  
P.O. BOX 908  
GILLETTE, WYOMING 82717-908**

**PREPARED BY;**

**TOP LINE ENGINEERING, LLC  
12635 HWY 200, P.O. BOX 884  
SIDNEY, MONTANA 59270  
PH; (701)570-2844**



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**APPENDIX**

TYPICAL DRILL RIG LOCATION PLAN

DISCHARGE REPORTING FORM

DISCHARGE SOURCE INSPECTION RECORD

## **SPILL PREVENTION CONTAINMENT AND COUNTERMEASURE PLAN (SPCC)**

**PER 40 CFR 112.7 DATED: SEPTEMBER 14, 2012**

This plan will follow the guidelines and format of 40 CFR part 112.7 and will replace the self-certified plan currently in affect.

**112.7(a)(1) Discussion of facilities' conformance with the requirements listed in this part.** The storage tanks on site store petroleum based drilling mud, diesel fuel, the invert tanks being the base for the drilling mud, salt water tanks, fresh water tanks, crude oil storage tanks and empty tanks for emergency storage. In addition to the storage tanks, other operations that can produce contaminants are the drilling operation itself and the cuttings drying operation. It is the purpose of this plan to define the prevention containment and countermeasure plan for discharges from any of these sources on the site. All the actions taken in this regard are per approved policies of this part of 40 CFR 112. Universally and in setting up the drilling operation on the site prevention measures are taken by grading the site to contain discharges, building in effective valves and other related operational and control mechanisms to prevent discharges. This equipment is installed and tested prior to beginning drilling operations. The tanks and equipment are installed within secondary containment berms and containment trenches and sumps are installed around the facilities prior to beginning drilling operations with capacity to contain with freeboard a rupture of the largest vessel associated with the containment. In the event of a discharge the sites are equipped with vacuums that are capable of picking up the spilled material and loading it into containers where it can be transferred to an approved disposal location. In the event of a discharge larger than can be handled by Cyclone and other manpower on location or that leaks offsite and potentially into navigable waters of the US, the operator, under his directions will arrange for trained contract personnel to immediately contain and clean-up the discharge.

**112.7(a)(2) This plan will comply with all applicable items of this part. With no deviations anticipated.**

**112.7(a)(3) The attached sketch shows a typical layout of the drill rig sites depicting the location, size and contents of the tanks. There are no underground tanks nor are there any mobile or portable containers associated with these drilling locations.** The site is prepared in advance of the drill rig coming onto the site. The sites are designed and constructed to contain spills and tank ruptures on the site. The drill rig and associated tanks and equipment are laid out in an efficient and effective manner to streamline the drilling operation. Frack tanks are located in such a manner that they have easy access to the mud tanks with valves located on both ends of transfer lines. The drilling mud tanks are plumbed to the drill rig with valves appropriately located to

stop flow from the tanks to the rig or from the rig to the tanks. The cuttings from the drilling operation are screened from the recycled drilling mud and directed to a container where they are dried using fly ash or some other equally effective absorbent material and are then transferred to the lined cuttings disposal pit onsite. The fuel tank is tied into the fuel lines providing fuel flow to the generators. Blowout preventers are placed over the drill hole casing and tested prior to beginning the drilling.

**112.7(a)(3)(i) The type of oil in each fixed container and its storage capacity. No mobile or portable containers anticipated.**

| <b>SOURCE</b>                    | <b>MAJOR TYPE OF FAILURE</b> | <b>TOTAL CAP. (BBLs.) (Type Fluid)</b>                     | <b>RATE (BBLs/MIN)</b>     | <b>DIRECTION OF FLOW</b>                              | <b>SECONDARY CONTAINMENT</b> |
|----------------------------------|------------------------------|------------------------------------------------------------|----------------------------|-------------------------------------------------------|------------------------------|
| Diesel Fuel Tank                 | Leak or rupture              | 14,000 Maximum (Diesel)                                    | Depends on size            | Dams and/or Trenches                                  | Site Perimeter Containment   |
| Pumps & Engines                  | Leakage                      | ½ - gal/hr. (lube oil)                                     | ½ - 1 gal/hr.              | Drip Pans or Trenches                                 | Site Perimeter Containment   |
| Crude Oil Tanks                  | Rupture or Leak              | 400 BBLs                                                   | Depends on Type of Failure | Contained within Berm                                 | Tank Battery Berm            |
| Mud Tanks, Vibrator Rotary Hoses | Leak or rupture              | 450 BBLs Maximum per tank (Petroleum based drilling fluid) | Varies                     | Drip Pans for Hoses, Trench & Sump or Berms for Tanks | Site Perimeter Containment   |
| Wellhead Blowout Preventer       | Blowout or kicking well      | Depends on severity (Well bore fluids)                     | Will vary                  | Away from well head to Trench and Sump                | Site Perimeter Containment   |

**112.7(a)(3)(ii) Discharge Prevention Measures including procedures for routine handling of products (loading, unloading and facility transfers, etc.);** Before the drill rig moves on, the site is prepared for discharges associated with tank ruptures, as the site is graded to contain all discharges and storm runoff. When the rig is moved onto the site and set up additional discharge prevention and containment measures are taken. Equipment inspection records



are reviewed and the equipment is inspected following written procedures, the written procedures are attached in the appendix of the plan. A dated record of the inspection, signed by the appropriate supervisor or inspector is recorded and filed with time sheets or safety meeting minutes, when the drill rig changes locations and notes of deficiencies or acceptance are included. Operation and maintenance of equipment is designed to prevent discharges. Personnel are trained and made aware of the discharge prevention procedures and applicable pollution control laws, rules and regulations. The tool pusher in charge of the drill rig will have in his possession a copy of the Cyclone Drilling SPCC Plan and it is his responsibility to instruct the employees and make them aware of the discharge prevention measures contained in the plan. Scheduled briefings and refreshers are to be conducted with all the employees prior to and during rig up operations, such briefings are to be held during regularly scheduled safety meetings.

Employees are assigned to make routine inspections of valves, hose connections and other fluid connections for leaks. These inspections are made and documented on a monthly basis at a minimum. These leaks are provided drip pans and reported to the appropriate personnel for repair. If repairs cannot be readily made then regular emptying of the drip pans is required until the repairs can be made. Inspections of trenches and sumps are made on a regular basis to insure that they are free flowing and functional. The containment berms are also inspected periodically to insure their stability and function.

**112.7(a)(3)(iii) Discharge or drainage controls such as secondary containment around containers and other structures, equipment, and procedures for the control of the discharge;** Containment berms are placed around the tanks containing crude oil, the invert drilling fluid, salt water tanks and the empty tanks to be used for emergency storage, as secondary containment. The containment berms are of adequate height to contain a rupture and total failure of the largest tank contained within it amounting to 400 BBLS requiring a containment capacity of 3,000 cu. ft. Similarly, berms are built around the crude oil storage tanks on the site and requiring the same containment volume. Berms or trenches and sumps are constructed around the drill rig itself, the drilling mud tanks and diesel fuel tank, the sumps and trenches are sized to contain the volume of the largest tank in the case of a rupture and complete failure. Additionally, drip pans are placed under leaking valves, hose joints and other sources of small leaks until the item can be replaced or repaired to not leak. Regular inspections of these facilities are made to insure that they will function per plan.

**112.7(a)(3)(iv) Countermeasures for discharge discovery, response, and cleanup (both the facility's capability and those that might be required of a contractor);** Regular inspections may reveal a discharge such as a leaking valve or hose joint. These instances can readily be cleaned up by employees of Cyclone Drilling using on site vacuums and containers. Larger spills up to and

beyond the reporting limits can also be handled by Cyclone Drilling employees. Spills that exceed the limitations of the onsite cleanup equipment or in the event of a spill migrating offsite, will require the tool pushers' notification of the operator/owner of the site and they will contact the clean-up contractor they have contracted with to contain and clean-up such events.

**112.7(a)(3)(v) Methods of disposal of recovered materials in accordance with applicable legal requirements:** When spills occur the fluids will be either soaked up with absorptive material and placed in a container for transporting to an approved disposal site or vacuumed up and placed in a storage container for transportation to an approved disposal site. It is understood by all Cyclone Drilling employees that there is not an approved disposal site on or around the drilling rig to dispose of these recovered materials or fluids, other than drying drilling cuttings and disposing of them in the cuttings pit.

**112.7(a)(3)(vi) Contact list and phone numbers for the facility response coordinator, National Response Center, cleanup contractors with whom operator has an agreement for response, and all appropriate Federal, State, and local agencies who must be contacted in the event of a discharge as described in § 112.1(b).** Facility Response Coordinator for Cyclone Drilling HSE Dept. (Ph. 307-682-4161) is responsible for contacting the operators environmental group so they can contact the appropriate Federal, State and Local personnel in the event of a discharge as described in § 112.1(b).

**112.7(a)(4) Information and procedures to enable a person reporting a discharge as described in § 112.1(b) to relate information on the exact address or location and phone number of the facility; the date and time of the discharge, the type of material discharged; estimates of the total quantity discharged; estimates of the quantity discharged as described in § 112.1(b); the source of the discharge: a description of all affected media; the cause of the discharge; any damages or injuries caused by the discharge; actions being used to stop, remove, and mitigate the effects of the discharge; whether an evacuation may be needed; and, the names of individuals and/or organizations who have also been contacted.** In the event of a discharge as described in §112.1(b) it is the responsibility of the tool pusher to relate the above information to the operators environmental group so they can contact, any emergency organizations and provide the reporting necessary to the Federal, State and Local agencies that require reporting of such a discharge. The tool pusher on the rig is the person in charge and either he or the drilling superintendent have the responsibility of reporting the answers to the above questions. These questions are spelled out on a "Discharge Reporting Form" that each has at his disposal in the appendix of "Cyclone Drilling Oil Spill Contingency Plan" In reporting a discharge a list of all individuals and or organizations that received the report will be listed.



**112.7(a)(5) Plan procedures to be used when a discharge occurs:**

- 1) Immediately extinguish any heater or fire that may ignite the spill.
- 2) No smoking during spill control operations.
- 3) Close all associated valves.
- 4) Direct discharge to ditches or drains that will carry the discharge to a safe holding sump or reserve pit.
- 5) Distribute hill, fibertex, gel, barite, or any other absorptive material available as required to contain the discharge not entering the ditch, sump or reserve pit.
- 6) Inspect area to ensure that all of the discharge is contained in ditches, sumps or reserve pits. Add ditches or diversion structures as required to contain the discharge onsite.
- 7) Start jet or sump pumps and transfer discharged material from sumps to reserve pit or holding tanks.
- 8) After discharge is stopped, collect all used hulls, fibertex and similar absorptive material for disposal per instruction from the drilling superintendent or the tool pusher. No oil is to remain in the ditches or sumps that may create a fire hazard.
- 9) Hold collected discharged material for hauling and disposal in an approved location.
- 10) If discharge migrates from the drill rig location Cyclone Drilling personnel will use these same procedures stop and recover the discharge. Additionally the tool pusher will notify the owners representative to notify the contract clean-up company for clean-up and mitigation of the offsite discharge migration.
- 11) Fill out Site Discharge Reporting Form and distribute to appropriate agencies and personnel. (Form included in Appendix of Plan)

**112.7(a)(5)(b) Flow diagram of potential discharges from significant sources as a result of facilities failure:** Flow diagrams and quantities included in appendix of the plan.

**112.7(a)(5)(c) Site discharge containment system:** The site discharge containment system for the Cyclone Drilling Rigs is a combination of containment berms, confinement trenches, sumps and reserve pits. Drip pans will be used for small discharges until repairs are made or equipment replaced. (See typical site plan included in appendix.)

**112.7(a)(5)(d) Not applicable as all measures spelled out in this part are practical as describe in each section.**

**112.7(a)(5)(e) Inspections, tests and records:** Written procedures are provided on the "Discharge Source Inspection Record", these inspections are recorded at least on a monthly basis or whenever a leak or discharge is detected and reported, using the "Discharge Source Inspection Record" in the appendix. These records are signed by the tool pusher onsite and maintained by Cyclone Drilling for a period of 3 years.

**112.7(a)(5)(f) Personnel Training and Discharge Prevention Procedures:**

The personnel working on the drilling crews are trained in the maintenance and operation of all the equipment to prevent discharges, the discharge procedures and general facility operations. They are also made aware of the contents of the SPCC Plan. The tool pusher on each crew is the person on the location who is accountable for discharge prevention and who reports to the Cyclone Drilling Management. The tool pusher is responsible for conducting discharge prevention briefings and assuring an adequate understanding of the Cyclone Drilling SPCC Plan for the facility and any recent developments of new precautionary measures due to failures or malfunctions are implemented.

**112.7(a)(5)(g) Site Security:** All persons entering the site are required to check in with the tool pusher of the on duty crew. Unauthorized persons are not allowed on the site. Unauthorized visitors are easily detected, questioned regarding their presents and appropriately escorted to take care of their business.

**112.7(a)(5)(h) Facility Tank Truck Loading/Unloading:** The tank truck loading and unloading areas on the Cyclone Drilling Sites are located adjacent to the frack tank or crude oil tank batteries the area adjacent to the tank battery, where tank trucks are loaded and unloaded will be graded to drain into the containment trenches and associated sump, surrounding the drill rig, mud tanks, fuel tank, etc. Wheel chocks are used to prevent the truck from moving while connected to the tanks during loading/unloading operations. Before departure the vehicle is inspected for leaks in the lowermost drains and outlets and if they are any detected the associated valves are tightened or adjusted to prevent discharge while in transit. All above ground containers are checked and evaluated for risk of discharge or failure and as necessary appropriate action shall be taken.

## APPENDIX





June 12, 2012

Industrial Commission of North Dakota  
Oil & Gas Division  
600 East Boulevard, Dept 405  
Bismarck, North Dakota 58505

Re: Tallahassee 3-16H

Continental Resources, Inc., would like to request all filings and information regarding the above captioned well be considered "Tight Hole".

Please charge the Continental Resources, Inc., credit card that is on file with your agency for the application fee of this well.

Thank you for your prompt attention to this matter. If you have any questions, you may contact me at 580-548-5139 or email the following [Terry.Olson@clr.com](mailto:Terry.Olson@clr.com).

Sincerely,

**CONTINENTAL RESOURCES, INC.**

  
Terry L. Olson  
Regulatory Compliance Specialist